MECHANICS APPLIED TO AEROSPACE ENGINEERING - PART I

Topic 5: Relative Motion - Problem Sets

Department of Continuum Mechanics and Aeronautical Engineering • BSc in Aerospace Engineering, UC3M

Source: sources/cuatrimestre-1/03-engineering-mechanics/problemas/Problems.pdf (2026 Edition)

Statements and solutions: Twenty problems of the official collection that need a moving reference frame. Each card gives the statement (with the PDF page) and a worked solution in the order of the course: reference frames, bases and coordinates; the change of basis and the position vector, deduced from drawings of each plane and each rotation; kinematics with the relative-motion formulas (Notes Eqs. 3.26–3.27); forces and equations of motion with the inertia forces (Notes Eqs. 4.62–4.63); and the answers to every part with unit and limit checks. Problems 7, 25, 27, 33, 34, 36, 37, 40, 41 and 47 have an official key entry, compared in each card; the other results are derived here and checked symbolically.

PROBLEM 07

Skater Towed by a Vehicle: Rotating Frame

Original statement · Problems.pdf, p. 8

Rotating Frame + Inertia Forces

A skater B is being towed by a vehicle A on a perfectly smooth horizontal plane, where we define an inertial reference frame Oxy. The vehicle is moving along a circumference of radius a and center at O with constant angular velocity ω. The skater is attached to the vehicle by a bar without mass and constant length $l = a$. Initially, the vehicle is located at $(a, 0)$ and the skater is at rest at $(2a, 0)$. Defining a rotating reference frame Ox_1y_1 with origin at O and axis Ox_1 along OA, and calling ψ the angle between the axis Ox_1 and the bar AB, determine:

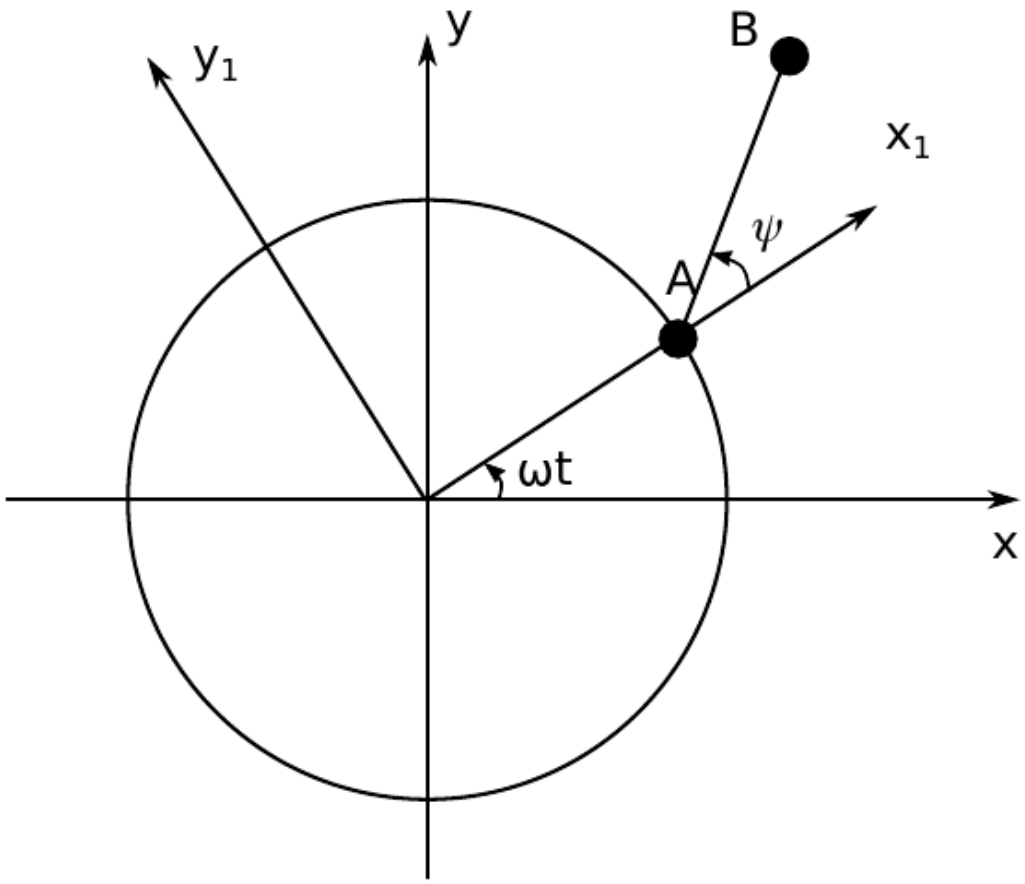


Figure: Inertial axes Oxy, rotating axes Ox1y1 turned by ωt, vehicle A on the circle of radius a, bar AB at angle ψ from Ox1, skater B (Problems.pdf, p. 8).

(a) The relative acceleration of the skater with respect to Ox_1y_1 as a function of ψ and its time derivatives.

(b) The centripetal and Coriolis acceleration terms also as a function of ψ and its time derivatives.

(c) Reduce the evolution of ψ to integrals of the form $\int [\dots]\, d\psi = \int [\dots]\, dt$ (consider that $\dot\psi^2 = \int 2\dot\psi\ddot\psi\, dt$).

(d) Determine the tension on the bar.

(e) If the rigid bar is substituted by a string which cannot bear compression, determine if/when the string will become loose.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** the vehicle A moves on the circle of radius a centred at O with constant angular rate ω. The skater B (mass m) is joined to A by a massless rigid bar of length $l = a$ and slides on a smooth horizontal plane.
- **Why a rotating frame.** In the frame Ox_1y_1 that turns with the line OA, the vehicle is at rest at $(a, 0)$, so the skater moves on a fixed circle of radius a around A: one coordinate ψ describes the motion and the constraint is fixed in that frame. The price is that Ox_1y_1 is non-inertial, so we must add the inertia forces of Notes Eq. 4.62 to Newton's law (Notes Eq. 4.63).
- **Hypotheses:**
  1. Oxy is inertial; the plane is smooth (its reaction is vertical and cancels the weight, so neither enters the horizontal dynamics).
  2. The bar is massless, so the force it exerts on B is directed along the bar: $-T\,\mathbf{u}_{AB}$, with $T > 0$ meaning tension.
  3. ω is constant.
- **Fixed (inertial) frame:** $S_0 = Oxy = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, basis $\mathcal{B}_0$, with $\mathbf{k}_0$ vertical (normal to the plane of motion).
- **Moving frame:** $S_1 = Ox_1y_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, basis $\mathcal{B}_1$, with $\mathbf{i}_1$ along OA and $\mathbf{k}_1 = \mathbf{k}_0$. The origin of S_1 is O (it plays the role of the generic origin "A" of Notes Ch. 3, which here is the fixed point O; the letter A is kept for the vehicle as in the statement) and the particle is $P \equiv B$.
- **Bar basis:** $\{\mathbf{u}_{AB}, \mathbf{u}_{\perp}, \mathbf{k}_1\}$, the unit vector from A to B and the unit vector normal to the bar in the plane; they are deduced in the next section.
- **Coordinate and degrees of freedom:** B moves in the plane (2 coordinates) with one constraint $|AB| = a$: **1 DOF**, the angle ψ from Ox_1 to the bar AB, measured at A counter-clockwise. The angle ωt of S_1 is prescribed.
- **Initial conditions:** A at $(a, 0)$ and B at rest at $(2a, 0)$, so $\psi(0) = 0$ and $\mathbf{v}_0^B(0) = \mathbf{0}$ (this gives $\dot{\psi}(0)$ in part (c)).

Symbol	Meaning	SI unit
m	mass of the skater B	kg
a	radius of the vehicle's circle and bar length	m
ω	angular rate of the vehicle (constant)	rad/s
ψ	angle from Ox_1 to the bar AB	rad
T	tension of the bar (positive when pulling B toward A)	N

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from above

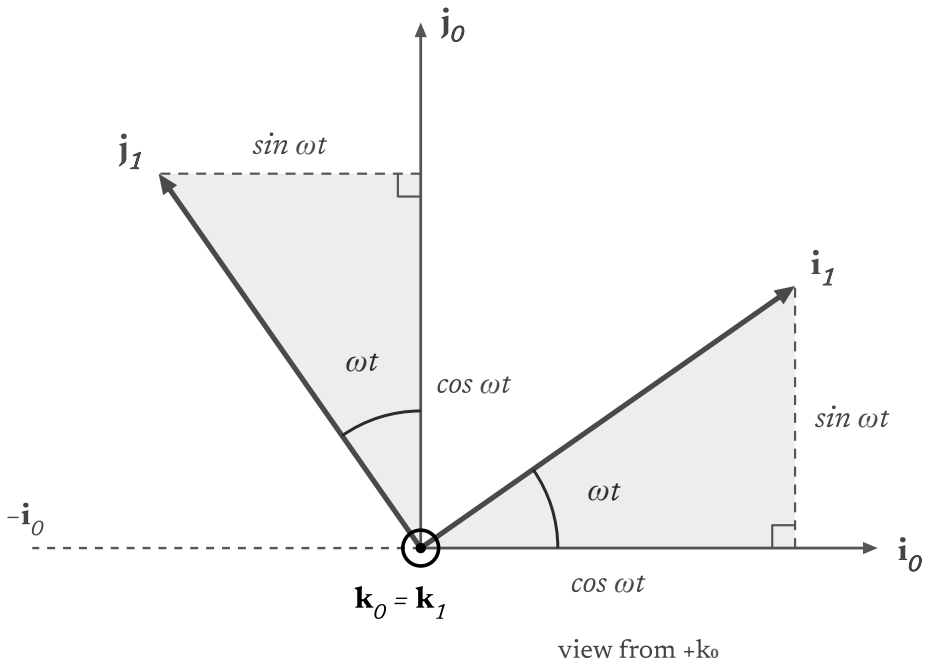


Figure 1. View from $+\mathbf{k}_0$ ($\mathbf{k}_0 = \mathbf{k}_1$ out of the paper, $\odot$). $\mathbf{i}_1$ is $\mathbf{i}_0$ turned counter-clockwise by ωt: legs $\cos\omega t$ along $\mathbf{i}_0$ and $\sin\omega t$ along $\mathbf{j}_0$. $\mathbf{j}_1$ turns the same angle from $\mathbf{j}_0$, so its opposite leg $\sin\omega t$ points along $-\mathbf{i}_0$: hence $\mathbf{j}_1 = -\sin\omega t\,\mathbf{i}_0 + \cos\omega t\,\mathbf{j}_0$.

The rotation is about the vertical $\mathbf{k}_0$, so it is drawn looking down from $+\mathbf{k}_0$: $\mathbf{k}_0 = \mathbf{k}_1$ points out of the paper, towards the reader ($\odot$). With $\mathbf{i}_0$ to the right, the right-hand rule $\mathbf{k}_0 = \mathbf{i}_0 \times \mathbf{j}_0$ puts $\mathbf{j}_0$ up the page (fingers from $\mathbf{i}_0$ to $\mathbf{j}_0$, counter-clockwise, thumb towards the reader). The vehicle starts on Ox and turns counter-clockwise, so after a time t the angle from $\mathbf{i}_0$ to $\mathbf{i}_1$ is ωt :

- $\mathbf{i}_1$ is the hypotenuse (length 1) of a right triangle whose leg adjacent to ωt lies along $\mathbf{i}_0$ and measures $\cos\omega t$, and whose opposite leg lies along $+\mathbf{j}_0$ and measures $\sin\omega t$: $\mathbf{i}_1 = \cos\omega t\,\mathbf{i}_0 + \sin\omega t\,\mathbf{j}_0$.
- $\mathbf{j}_1$ is $\mathbf{i}_1$ turned a further 90° counter-clockwise (so that $\mathbf{k}_1 = \mathbf{i}_1 \times \mathbf{j}_1$ still points towards the reader). It has turned the same angle ωt from $\mathbf{j}_0$, and turning counter-clockwise from "up" moves its tip to the left: its leg along $\mathbf{j}_0$ (adjacent) is $\cos\omega t$ and its leg along $-\mathbf{i}_0$ (opposite) is $\sin\omega t$. **That is the origin of the minus sign:** $\mathbf{j}_1 = -\sin\omega t\,\mathbf{i}_0 + \cos\omega t\,\mathbf{j}_0$.
- $\mathbf{k}_1 = \mathbf{k}_0$, the axis of the rotation.

The columns of $[{}_0R_1]$ are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Checks: $\det[{}_0R_1] = \cos\omega t\cos\omega t - (-\sin\omega t)\sin\omega t = \cos^2\omega t + \sin^2\omega t = 1$; the columns are unit vectors ($\cos^2\omega t + \sin^2\omega t = 1$) and mutually orthogonal ($-\cos\omega t\sin\omega t + \sin\omega t\cos\omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.

Motion of S_1. Simple rotation about $\mathbf{k}_0 = \mathbf{k}_1$ (Slide 16):

$$\boldsymbol{\omega}_{10} = \omega\,\mathbf{k}_1, \qquad \boldsymbol{\alpha}_{10} = \left.\frac{d\boldsymbol{\omega}_{10}}{dt}\right|_0 = \dot{\omega}\,\mathbf{k}_1 = \mathbf{0}, \qquad \mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$$

Check with Poisson's formula: by the chain rule $\left.\frac{d\mathbf{i}_1}{dt}\right|_0 = -\omega\sin\omega t\,\mathbf{i}_0 + \omega\cos\omega t\,\mathbf{j}_0 = \omega\,\mathbf{j}_1$, and $\boldsymbol{\omega}_{10}\times\mathbf{i}_1 = \omega\,\mathbf{k}_1\times\mathbf{i}_1 = \omega\,\mathbf{j}_1$.

The position vector in the rotating plane

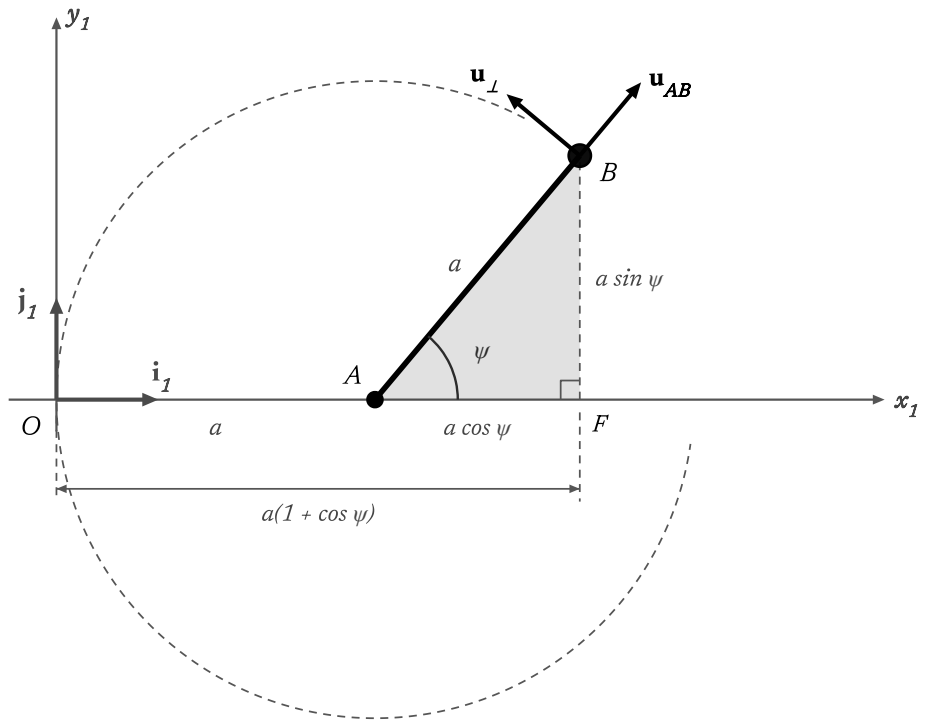


Figure 2. Plane of S_1 ($\mathbf{i}_1$ along OA, $\mathbf{k}_1$ out of the paper). $\mathbf{r}_1^B = OA + AB$: in the shaded right triangle AFB the hypotenuse is $|AB| = a$, the leg adjacent to ψ is $a\cos\psi$ along $\mathbf{i}_1$ and the opposite leg is $a\sin\psi$ along $\mathbf{j}_1$. The dashed circle is the path of B relative to S_1; $\mathbf{u}_{AB}, \mathbf{u}_\perp$ in orange.

In S_1 the vehicle is at rest at $A = (a, 0)$, so the plane is drawn with $\mathbf{i}_1$ to the right and $\mathbf{j}_1$ up. To go from O to B we pass through A, because the distance from A is the one fixed by the bar (Notes Eq. 3.16 with $\mathbf{r}_0^O = \mathbf{0}$):

$$\mathbf{r}_1^B = \overrightarrow{OA} + \overrightarrow{AB}$$

- $\overrightarrow{OA} = a\,\mathbf{i}_1$: A lies on Ox_1 at distance a from O.
- $\overrightarrow{AB}$ has length a and makes the angle ψ with $\mathbf{i}_1$, measured counter-clockwise at A. Let F be the foot of the perpendicular from B to the line Ox_1. In the shaded right triangle AFB (right angle at F) the hypotenuse is $|AB| = a$, the leg adjacent to ψ is $AF = a\cos\psi$ along $\mathbf{i}_1$ and the leg opposite to ψ is $FB = a\sin\psi$ along $\mathbf{j}_1$. So $\overrightarrow{AB} = a\cos\psi\,\mathbf{i}_1 + a\sin\psi\,\mathbf{j}_1$.

Adding the two vectors:

$$\mathbf{r}_1^B = a\,\mathbf{i}_1 + a\,(\cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1) = a(1+\cos\psi)\,\mathbf{i}_1 + a\sin\psi\,\mathbf{j}_1$$

These are the dashed components of the figure: $|OF| = a + a\cos\psi$ along $\mathbf{i}_1$ and $|FB| = a\sin\psi$ along $\mathbf{j}_1$. For $\psi < 0$ (the case of the motion, as part (c) shows) $\sin\psi < 0$ and B is below Ox_1; the same formula holds. Here ψ is the standard polar angle of B about A (measured from $\mathbf{i}_1$, counter-clockwise), so no shift of the angle is needed. Checks: $\psi = 0$ gives $2a\,\mathbf{i}_1$, the initial position; $\psi = -\pi$ gives $\mathbf{0}$, the point O.

Bar basis. The unit vector along the bar is $\mathbf{u}_{AB} = \overrightarrow{AB}/a = \cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1$ (from A to B). Differentiating with respect to ψ ($\frac{d}{d\psi}\cos\psi = -\sin\psi$, $\frac{d}{d\psi}\sin\psi = \cos\psi$) gives the unit vector normal to the bar, in the sense of increasing ψ:

$$\mathbf{u}_\perp = \frac{d\mathbf{u}_{AB}}{d\psi} = -\sin\psi\,\mathbf{i}_1 + \cos\psi\,\mathbf{j}_1 = \mathbf{k}_1 \times \mathbf{u}_{AB}$$

(the last form with $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$ and $\mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$). Their matrix $\begin{pmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{pmatrix}$ (completed with $\mathbf{k}_1$) is again a simple rotation: $\det = \cos^2\psi + \sin^2\psi = 1$, and RR^T has diagonal entries $\cos^2\psi + \sin^2\psi = 1$ and off-diagonal entries $\cos\psi\sin\psi - \sin\psi\cos\psi = 0$, so $RR^T = I$. Also $\frac{d\mathbf{u}_\perp}{d\psi} = -\cos\psi\,\mathbf{i}_1 - \sin\psi\,\mathbf{j}_1 = -\mathbf{u}_{AB}$.

Kinematics

(a) Relative acceleration $\mathbf{a}_1^B$

> **Pedagogical Justification:** the relative quantities are derivatives in S_1, where $\mathbf{i}_1, \mathbf{j}_1$ are constant, so only the components are differentiated.

Relative velocity, with the chain rule $\frac{d}{dt}\cos\psi = -\sin\psi\,\dot\psi$ and $\frac{d}{dt}\sin\psi = \cos\psi\,\dot\psi$:

$$\mathbf{v}_1^B = \left.\frac{d\mathbf{r}_1^B}{dt}\right|_1 = -a\dot\psi\sin\psi\,\mathbf{i}_1 + a\dot\psi\cos\psi\,\mathbf{j}_1 = a\dot\psi\,\mathbf{u}_\perp$$

Relative acceleration, with the product rule $\frac{d}{dt}(\dot\psi\sin\psi) = \ddot\psi\sin\psi + \dot\psi^2\cos\psi$ and $\frac{d}{dt}(\dot\psi\cos\psi) = \ddot\psi\cos\psi - \dot\psi^2\sin\psi$:

$$\boxed{\mathbf{a}_1^B = -a\left(\ddot\psi\sin\psi + \dot\psi^2\cos\psi\right)\mathbf{i}_1 + a\left(\ddot\psi\cos\psi - \dot\psi^2\sin\psi\right)\mathbf{j}_1 = a\ddot\psi\,\mathbf{u}_\perp - a\dot\psi^2\,\mathbf{u}_{AB}}$$

(the last form follows by grouping the $\ddot\psi$ and $\dot\psi^2$ terms: circular motion of radius a about A, tangential plus normal part).

(b) Centripetal and Coriolis terms

> **Pedagogical Justification:** they are the two non-zero rotation terms of Notes Eq. 3.27 (Slide 25); the Euler term vanishes because $\boldsymbol{\alpha}_{10} = \mathbf{0}$ and the origin term because $\mathbf{a}_0^O = \mathbf{0}$.

Cross products used: $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$.

Centripetal term.

$$\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B = \omega\mathbf{k}_1 \times [a(1+\cos\psi)\mathbf{i}_1 + a\sin\psi\,\mathbf{j}_1] = a\omega(1+\cos\psi)\,\mathbf{j}_1 - a\omega\sin\psi\,\mathbf{i}_1$$

$$\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B\right) = \omega\mathbf{k}_1 \times [-a\omega\sin\psi\,\mathbf{i}_1 + a\omega(1+\cos\psi)\,\mathbf{j}_1] = -a\omega^2\sin\psi\,\mathbf{j}_1 - a\omega^2(1+\cos\psi)\,\mathbf{i}_1$$

$$\boxed{\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B\right) = -a\omega^2 \left[(1+\cos\psi)\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1\right] = -\omega^2\,\mathbf{r}_1^B}$$

It equals $-\omega^2 \mathbf{r}_\perp$ with $\mathbf{r}_\perp = \mathbf{r}_1^B$ because $\mathbf{r}_1^B$ lies in the plane normal to $\boldsymbol{\omega}_{10}$ (Slide 30).

Coriolis term.

$$2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^B = 2\omega\mathbf{k}_1 \times \left(-a\dot{\psi}\sin\psi\,\mathbf{i}_1 + a\dot{\psi}\cos\psi\,\mathbf{j}_1\right) = -2a\omega\dot{\psi}\sin\psi\,\mathbf{j}_1 - 2a\omega\dot{\psi}\cos\psi\,\mathbf{i}_1$$

$$\boxed{2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^B = -2a\omega\dot{\psi}\left(\cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1\right) = -2a\omega\dot{\psi}\,\mathbf{u}_{AB}}$$

Absolute acceleration (Notes Eq. 3.27):

$$\mathbf{a}_0^B = \underbrace{\mathbf{a}_0^O}_{\mathbf{0}} + \mathbf{a}_1^B + \underbrace{\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^B}_{\mathbf{0}} - \omega^2\mathbf{r}_1^B - 2a\omega\dot{\psi}\,\mathbf{u}_{AB}$$

Forces and Equations of Motion

(c) Forces on the skater in S_1

Pedagogical Justification: in S_1 Newton's law holds with the inertia forces (Notes Eqs. 4.62–4.63): $m\mathbf{a}_1^B = \mathbf{F} + \mathbf{F}_i$. Here $\mathbf{F} = -T\,\mathbf{u}_{AB}$ (horizontal part), $\mathbf{F}_A = \mathbf{0}$, $\mathbf{F}_E = \mathbf{0}$, and

$$\mathbf{F}_C = -m\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B\right) = ma\omega^2\left[(1+\cos\psi)\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1\right], \qquad \mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^B = 2ma\omega\dot{\psi}\,\mathbf{u}_{AB}$$

(c, continued) Equation of motion

The unknown T disappears if we project on $\mathbf{u}_\perp$, the only direction along which the bar exerts no force.

Projections on $\mathbf{u}_\perp = -\sin\psi\,\mathbf{i}_1 + \cos\psi\,\mathbf{j}_1$:

- $m\mathbf{a}_1^B \cdot \mathbf{u}_\perp = ma\ddot{\psi}$ (from the grouped form of part (a), since $\mathbf{u}_{AB} \cdot \mathbf{u}_\perp = 0$);
- $-T\,\mathbf{u}_{AB} \cdot \mathbf{u}_\perp = 0$ and $\mathbf{F}_{COR} \cdot \mathbf{u}_\perp = 0$;
- $\mathbf{F}_C \cdot \mathbf{u}_\perp = ma\omega^2\left[-(1+\cos\psi)\sin\psi + \sin\psi\cos\psi\right] = -ma\omega^2\sin\psi$.

$$ma\ddot{\psi} = -ma\omega^2\sin\psi \quad\Longrightarrow\quad \boxed{\ddot{\psi} + \omega^2\sin\psi = 0}$$

In S_1 the skater behaves as a pendulum of length a hanging from A: about A the centrifugal force $m\omega^2(\mathbf{OA} + \mathbf{AB})$ has the same moment as the uniform field $ma\omega^2\,\mathbf{i}_1$ (because $\mathbf{AB} \times \mathbf{AB} = \mathbf{0}$), so the "effective gravity" has strength $a\omega^2$ and points along $+\mathbf{i}_1$.

(d) Equation for the tension

> **Pedagogical Justification:** project $m\mathbf{a}_1^B = -T\mathbf{u}_{AB} + \mathbf{F}_C + \mathbf{F}_{COR}$ on $\mathbf{u}_{AB}$, the direction in which the bar force acts.

Projections on $\mathbf{u}_{AB} = \cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1$:

- $m\mathbf{a}_1^B \cdot \mathbf{u}_{AB} = -ma\dot{\psi}^2$;
- $\mathbf{F}_C \cdot \mathbf{u}_{AB} = ma\omega^2\left[(1+\cos\psi)\cos\psi + \sin^2\psi\right] = ma\omega^2(1+\cos\psi)$;
- $\mathbf{F}_{COR} \cdot \mathbf{u}_{AB} = 2ma\omega\dot{\psi}$.

$$-ma\dot{\psi}^2 = -T + ma\omega^2(1+\cos\psi) + 2ma\omega\dot{\psi} \quad\Longrightarrow\quad T = ma\left[\dot{\psi}^2 + 2\omega\dot{\psi} + \omega^2(1+\cos\psi)\right]$$

Answers and Checks

(a) Relative acceleration

$\mathbf{a}_1^B = -a\left(\ddot{\psi}\sin\psi + \dot{\psi}^2\cos\psi\right)\mathbf{i}_1 + a\left(\ddot{\psi}\cos\psi - \dot{\psi}^2\sin\psi\right)\mathbf{j}_1 = a\ddot{\psi}\,\mathbf{u}_\perp - a\dot{\psi}^2\,\mathbf{u}_{AB}$.

(b) Centripetal and Coriolis terms

$\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B\right) = -a\omega^2\left[(1+\cos\psi)\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1\right]$ and $2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^B = -2a\omega\dot{\psi}\left(\cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1\right)$. Both agree with the official key (Problems.pdf, p. 72), where the centripetal term is called "centrifugal acceleration".

(c) First integral, initial rate and quadrature

First integral (hint of the statement). Multiply $\ddot{\psi} + \omega^2 \sin\psi = 0$ by $2\dot{\psi}$. By the chain rule $\frac{d}{dt}\left(\dot{\psi}^2\right) = 2\dot{\psi}\ddot{\psi}$ and $\frac{d}{dt}\cos\psi = -\sin\psi\,\dot{\psi}$, so

$$2\dot{\psi}\ddot{\psi} = -2\omega^2 \sin\psi\,\dot{\psi} \quad\Longleftrightarrow\quad \frac{d}{dt}\left(\dot{\psi}^2\right) = 2\omega^2 \frac{d}{dt}\left(\cos\psi\right)$$

Integrate from 0 to t and apply Barrow's rule $[F]_0^t = F(t) - F(0)$:

On the left, $\int_0^t 2\dot{\psi}\ddot{\psi}\,dt' = \left[\dot{\psi}^2\right]_0^t$. On the right, the change of variable $\psi = \psi(t')$, $d\psi = \dot{\psi}\,dt'$, turns $\int_0^t \left(-2\omega^2\sin\psi\right)\dot{\psi}\,dt'$ into $\int_{\psi(0)}^{\psi(t)} \left(-2\omega^2\sin\psi\right)d\psi$, whose antiderivative is $2\omega^2\cos\psi$:

$$\dot{\psi}^2(t) - \dot{\psi}^2(0) = \int_{\psi(0)}^{\psi(t)} \left(-2\omega^2\sin\psi\right)d\psi = 2\omega^2\left[\cos\psi\right]_{\psi(0)}^{\psi(t)} = 2\omega^2\left(\cos\psi - 1\right)$$

Initial rate. By Notes Eq. 3.26, $\mathbf{v}_0^B = \mathbf{v}_0^O + \mathbf{v}_1^B + \boldsymbol{\omega}_{10}\times\mathbf{r}_1^B$. At $t=0$: $\psi = 0$, $\mathbf{r}_1^B = 2a\,\mathbf{i}_1$, $\mathbf{v}_1^B = a\dot{\psi}(0)\,\mathbf{j}_1$, $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^B = 2a\omega\,\mathbf{j}_1$, and $\mathbf{v}_0^B = \mathbf{0}$:

$$\mathbf{0} = \left[a\dot{\psi}(0) + 2a\omega\right]\mathbf{j}_1 \quad\Longrightarrow\quad \dot{\psi}(0) = -2\omega$$

Hence, with $1+\cos\psi = 2\cos^2(\psi/2)$,

$$\dot{\psi}^2 = 4\omega^2 + 2\omega^2(\cos\psi - 1) = 2\omega^2(1+\cos\psi) = 4\omega^2\cos^2\frac{\psi}{2}$$

For $-\pi < \psi < \pi$ the factor $\cos(\psi/2) > 0$, so $\dot{\psi}$ cannot vanish nor change sign there; $\dot{\psi}(0) = -2\omega < 0$ selects

$$\boxed{\dot{\psi} = -2\omega\cos\frac{\psi}{2}, \qquad -\pi < \psi \le 0}$$

which agrees with the official key (Problems.pdf, p. 72).

Energy check in S_1 (Slides 29–30). The origin is fixed and $\boldsymbol{\omega}_{10}$ is constant, so $\mathbf{F}_C$ has the potential $V_1 = -\frac{1}{2}m\omega^2 r_\perp^2$, $\mathbf{F}_{COR}$ does no work, and the bar force does no work in S_1 ($-T\mathbf{u}_{AB}\cdot a\dot{\psi}\mathbf{u}_\perp = 0$). With $r_\perp^2 = a^2\left[(1+\cos\psi)^2 + \sin^2\psi\right] = 2a^2(1+\cos\psi)$:

$$E_1^B = \frac{1}{2}ma^2\dot\psi^2 - ma^2\omega^2(1+\cos\psi) = \text{const} = \frac{1}{2}ma^2(4\omega^2) - 2ma^2\omega^2 = 0$$

which gives again $\dot\psi^2 = 2\omega^2(1+\cos\psi)$.

Quadrature. Separate variables in $\frac{d\psi}{dt} = -2\omega\cos\frac{\psi}{2}$:

$$\boxed{\int_0^{\psi} \frac{d\psi'}{2\cos(\psi'/2)} = -\int_0^t \omega\, dt'}$$

The left integral is elementary. Substitute $w = \psi'/2$, $d\psi' = 2\,dw$, limits $0 \to \psi/2$; the antiderivative of $\sec w$ is $\ln|\sec w + \tan w|$ (its derivative is $\frac{\sec w\tan w + \sec^2 w}{\sec w + \tan w} = \sec w$). By Barrow's rule:

$$\int_0^{\psi/2} \frac{dw}{\cos w} = \left[\ln|\sec w + \tan w|\right]_0^{\psi/2} = \ln\left|\sec\frac{\psi}{2} + \tan\frac{\psi}{2}\right| - \ln 1 = -\omega t$$

so $\psi(t)$ is given implicitly by $\sec\frac{\psi}{2} + \tan\frac{\psi}{2} = e^{-\omega t}$; as $t\to\infty$, $\psi \to -\pi$, i.e. B tends to the centre O ($\mathbf{r}_1^B \to \mathbf{0}$) without ever reaching it.

(d) Tension on the bar

Substitute the first integral in $T = ma\left[\dot\psi^2 + 2\omega\dot\psi + \omega^2(1+\cos\psi)\right]$, with $c \equiv \cos(\psi/2)$: $\dot\psi^2 = 4\omega^2c^2$, $\dot\psi = -2\omega c$, $1+\cos\psi = 2c^2$:

$$T = ma\omega^2\left(4c^2 - 4c + 2c^2\right) = ma\omega^2\left(6c^2 - 4c\right)$$

$$\boxed{T = 2ma\omega^2\cos\frac{\psi}{2}\left(3\cos\frac{\psi}{2} - 2\right)}$$

which agrees with the official key (Problems.pdf, p. 72). At $t=0$: $T(0) = 2ma\omega^2 > 0$.

(e) Does a string become loose?

> **Pedagogical Justification:** a string can only pull, so it stays taut while $T \geq 0$; it goes slack at the first instant where T would become negative.

As ψ decreases from 0 toward $-\pi$, $c = \cos(\psi/2)$ decreases monotonically from 1 toward 0. The factor c stays positive, and $3c - 2$ changes sign at $c = 2/3$:

$$\cos\frac{\psi}{2} = \frac{2}{3} \quad \Longrightarrow \quad \boxed{\psi_{\text{loose}} = -2\arccos\frac{2}{3} \approx -1.682 \text{ rad} \approx -96.38^\circ}$$

(official key: $\psi = -2\arccos(2/3)$). **When:** at $\psi/2 = -\arccos(2/3)$, $\sec(\psi/2) = 3/2$ and $\tan(\psi/2) = -\sqrt{1 - 4/9}/(2/3) = -\sqrt{5}/2$, so from part (c)

$$\omega t_{\text{loose}} = -\ln\left(\frac{3}{2} - \frac{\sqrt{5}}{2}\right) = \ln\frac{2}{3 - \sqrt{5}} = \ln\frac{3 + \sqrt{5}}{2} \approx 0.9624, \qquad t_{\text{loose}} \approx \frac{0.962}{\omega}$$

So yes: the string goes slack after about $0.96/\omega$ seconds, when the bar has turned 96.4° clockwise from Ox_1.

Checks

- **Reading in S_1.** The rotating observer sees B as a pendulum of length a hung from A in an effective field of strength $a\omega^2$ pointing away from O (the centrifugal force), whose "lowest" point is $\psi = 0$. The skater starts there with relative speed $2a\omega$, which is exactly the separatrix energy ($E_1^B = 0$): it creeps toward the "inverted" position $\psi = -\pi$ (where B would sit at O) in infinite time.
- **Why T drops.** The Coriolis force $2ma\omega\dot\psi\,\mathbf{u}_{AB}$ points toward A when $\dot\psi < 0$, and the centrifugal pull along the bar, $ma\omega^2(1 + \cos\psi)$, shrinks as B swings in; when the required centripetal force $ma\dot\psi^2$ is supplied by these inertia forces alone the bar is unloaded ($T = 0$).
- **Limit $\omega \to 0$:** $\dot\psi \to 0, T \to 0$ (nothing moves).
- **Orthogonality:** $\mathbf{u}_{AB} \cdot \mathbf{u}_\perp = -\cos\psi\sin\psi + \sin\psi\cos\psi = 0$, $|\mathbf{u}_{AB}| = |\mathbf{u}_\perp| = 1$.
- **Dimensions:** $[\mathbf{a}_1^B] = [a\ddot\psi] = [a\dot\psi^2] = \mathrm{m\,s^{-2}}$; $[\omega^2 a] = [\omega\dot\psi a] = \mathrm{m\,s^{-2}}$; $[T] = [ma\omega^2] = \mathrm{kg\,m\,s^{-2}} = \mathrm{N}$; $[\omega t] = 1$.
- **Verification (sympy):** the inertial double derivative of $[{}_0R_1]\{\mathbf{r}_1^B\}$ equals $\mathbf{a}_1^B - \omega^2\mathbf{r}_1^B - 2a\omega\dot\psi\mathbf{u}_{AB}$; the EOM, $\dot\psi(0) = -2\omega$, $dE_1^B/dt = 0$ along the EOM, the tension and ψ_{loose} are reproduced; an RK4 integration with $\omega = 1$ reaches $\cos(\psi/2) = 2/3$ at $t = 0.9625$ s, matching $\ln\frac{3+\sqrt{5}}{2} = 0.9624$.

- **Check against the official key:** the key of `Problems.pdf` (p. 72) gives the centripetal and Coriolis terms, $\dot{\psi} = -2\omega\cos(\psi/2)$, $T = 2ma\omega^2\cos(\psi/2)\,(3\cos(\psi/2) - 2)$ and $\psi = -2\arccos(2/3)$ for the string going slack. All of them coincide with the results above (no discrepancy).

Source: Problems.pdf (PDF page 8); key on PDF page 72

PROBLEM 08

Kinematics, Two Trains on Tracks

Original statement · Problems.pdf, p. 8

Relative Motion

Two trains, A and B, are moving along parallel tracks. The separation between the tracks is a. The absolute velocity of each train is v_A and v_B, respectively. What is the relative velocity of each train with respect to the other?

Solve the same problem considering that the train B is now moving along a circular track of radius R. The minimum distance between the circular track and the straight track of A is a.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical System:** Two trains A and B move on separate tracks with distance of closest separation a.
- Case 1: Two parallel straight tracks with constant speeds v_A and v_B.
- Case 2: Train A on a straight track, Train B on a circular track of radius R.
- **Parameters & Units:** Separation a $[\mathrm{m}]$, radius R $[\mathrm{m}]$, speeds v_A, v_B $[\mathrm{m/s}]$.
- **Hypotheses:** the tracks are in a plane; at $t = 0$ train A is at the origin and train B is at the point of its track closest to the track of A; the speeds are constant.
- **Inertial Cartesian Frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}\}$, with $\mathbf{i}_0$ along the track of A, $\mathbf{j}_0$ towards the track of B and $\mathbf{k} = \mathbf{i}_0 \times \mathbf{j}_0$ normal to the plane of the tracks.
- **Frame of train B (Case 2, rotating observer):** $S_B = \{B; \mathbf{i}_B, \mathbf{j}_B, \mathbf{k}\}$, attached to the train, with $\mathbf{i}_B$ along its direction of motion (the tangent to the track) and $\mathbf{j}_B$ pointing towards the centre C of the track. It turns with the train by the angle $\omega_B t$ about $\mathbf{k}$, $\omega_B = v_B/R$.
- **Coordinates and degrees of freedom.** The coordinates are the arc positions along the tracks (one per train). Both motions are prescribed, $x_A = v_A t$ and $\theta_B = v_B t/R$, so no degree of freedom is unknown (0 DOF): the problem is purely kinematic. θ_B is the angle of CB measured at the centre C from the downward vertical $-\mathbf{j}_0$, positive counter-clockwise (towards $+\mathbf{i}_0$).

- **Observers.** In Case 1 the relative velocity is the same for any observer that translates with a train. In Case 2 an observer attached to B turns with it at $\omega_B = v_B/R$, so the velocity of A seen by this observer needs the rotating frame S_B.

Change of Basis and Position Vector

Position of each train

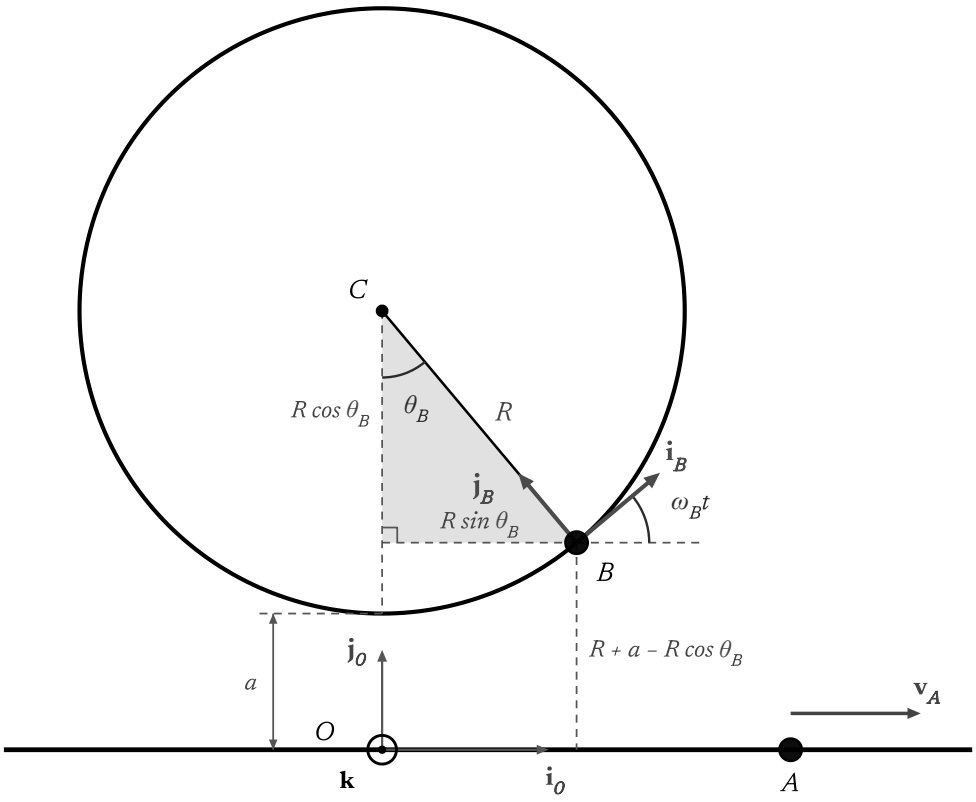


Figure 1. Plane of the tracks ($\mathbf{k}$ out of the paper). $\mathbf{r}_B = OC + CB$: in the shaded right triangle CFB the hypotenuse is $|CB| = R$, the leg opposite θ_B is $R\sin\theta_B$ (along $\mathbf{i}_0$) and the adjacent leg is $R\cos\theta_B$ (along $-\mathbf{j}_0$). The tangent $\mathbf{i}_B$ makes the same angle $\theta_B = \omega_B t$ with the horizontal; $\mathbf{j}_B$ points to C.

The plane of the tracks is drawn from above: $\mathbf{k}$ points out of the paper, towards the reader ($\odot$), $\mathbf{i}_0$ to the right and $\mathbf{j}_0$ up (the right-hand rule $\mathbf{i}_0 \times \mathbf{j}_0 = \mathbf{k}$: fingers from $\mathbf{i}_0$ to $\mathbf{j}_0$ counter-clockwise, thumb towards the reader).

Case 1. Train A runs on the line $y = 0$ and train B on the parallel line $y = a$:

$$\mathbf{r}_A(t) = v_A t\,\mathbf{i}_0, \qquad \mathbf{r}_B(t) = (x_{B0} + v_B t)\,\mathbf{i}_0 + a\,\mathbf{j}_0$$

Case 2. Train A is still on the x-axis, $\mathbf{r}_A = v_A t\,\mathbf{i}_0$. The circular track has its centre C at $(0, R+a)$, so that its lowest point is at height $(R+a) - R = a$ above the track of A, the given minimum distance. To reach B we pass through the centre, because the distance from C is the one that stays constant:

$$\mathbf{r}_B = \overrightarrow{OC} + \overrightarrow{CB}$$

- $\overrightarrow{OC} = (R+a)\,\mathbf{j}_0$.
- $\overrightarrow{CB}$ has length R and makes the angle $\theta_B = \omega_B t$ with the downward vertical. Let F be the point of the vertical through C at the height of B. In the shaded right triangle CFB (right angle at F) the hypotenuse is $|CB| = R$; the leg opposite to θ_B is $FB = R\sin\omega_B t$, horizontal, along $+\mathbf{i}_0$; the leg adjacent to θ_B is $CF = R\cos\omega_B t$, vertical and pointing down, along $-\mathbf{j}_0$. So $\overrightarrow{CB} = R\sin\omega_B t\,\mathbf{i}_0 - R\cos\omega_B t\,\mathbf{j}_0$.

Adding:

$$\mathbf{r}_B(t) = R\sin(\omega_B t)\,\mathbf{i}_0 + \left[R + a - R\cos(\omega_B t)\right]\mathbf{j}_0$$

Check: $t = 0$ gives $a\,\mathbf{j}_0$, the lowest point of the circle, closest to the track of A. The angle θ_B is measured from $-\mathbf{j}_0$ and not from $\mathbf{i}_0$: the usual polar angle about C is $\varphi = \theta_B - \pi/2$, because $\cos\left(\theta_B - \frac{\pi}{2}\right) = \cos\theta_B \cos\frac{\pi}{2} + \sin\theta_B \sin\frac{\pi}{2} = \sin\theta_B$ and $\sin\left(\theta_B - \frac{\pi}{2}\right) = \sin\theta_B \cos\frac{\pi}{2} - \cos\theta_B \sin\frac{\pi}{2} = -\cos\theta_B$, which turns $R(\cos\varphi\,\mathbf{i}_0 + \sin\varphi\,\mathbf{j}_0)$ into the $\overrightarrow{CB}$ above. It is chosen so that $\theta_B = 0$ is the starting point.

The basis of train B

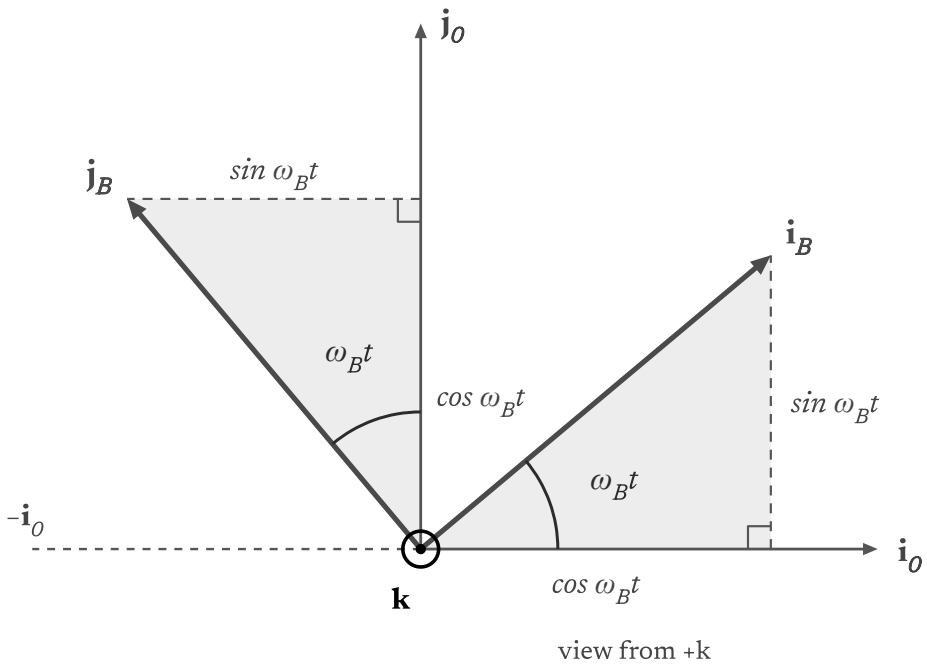


Figure 2. View from $+\mathbf{k}$ ($\odot$). $\mathbf{i}_B$ is $\mathbf{i}_0$ turned counter-clockwise by $\omega_B t$: legs $\cos\omega_B t$ along $\mathbf{i}_0$ and $\sin\omega_B t$ along $\mathbf{j}_0$. $\mathbf{j}_B$ turns the same angle from $\mathbf{j}_0$, so its opposite leg $\sin\omega_B t$ points along $-\mathbf{i}_0$ (the minus sign of $[{}_0R_B]$).

$\mathbf{i}_B$ is the unit tangent to the track in the sense of motion. Differentiating $\overrightarrow{CB}$ with respect to θ_B ($\frac{d}{d\theta}\sin\theta = \cos\theta$, $\frac{d}{d\theta}(-\cos\theta) = \sin\theta$) and dividing by its length $R\sqrt{\cos^2\theta_B + \sin^2\theta_B} = R$:

$$\mathbf{i}_B = \cos\omega_B t\,\mathbf{i}_0 + \sin\omega_B t\,\mathbf{j}_0$$

so $\mathbf{i}_B$ is $\mathbf{i}_0$ turned counter-clockwise by $\omega_B t$ (seen from $+\mathbf{k}$, $\odot$): leg $\cos\omega_B t$ along $\mathbf{i}_0$ (adjacent) and leg $\sin\omega_B t$ along $\mathbf{j}_0$ (opposite). $\mathbf{j}_B = \mathbf{k} \times \mathbf{i}_B$ is $\mathbf{i}_B$ turned a further $90°$; it has turned the same angle $\omega_B t$ from $\mathbf{j}_0$, and turning counter-clockwise from "up" moves its tip towards $-\mathbf{i}_0$, so its opposite leg is $-\sin\omega_B t\,\mathbf{i}_0$:

$$\mathbf{j}_B = -\sin\omega_B t\,\mathbf{i}_0 + \cos\omega_B t\,\mathbf{j}_0 = -\frac{\overrightarrow{CB}}{R}$$

(comparing with $\overrightarrow{CB} = R\sin\omega_B t\,\mathbf{i}_0 - R\cos\omega_B t\,\mathbf{j}_0$), so $\mathbf{j}_B$ points from B towards the centre C, as drawn. With the columns of $\mathbf{i}_B, \mathbf{j}_B, \mathbf{k}$ in $\mathcal{B}_0$:

$$[{}_0R_B] = \begin{pmatrix} \cos\omega_B t & -\sin\omega_B t & 0 \\ \sin\omega_B t & \cos\omega_B t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_B] = \cos\omega_B t\cos\omega_B t - (-\sin\omega_B t)\sin\omega_B t = \cos^2\omega_B t + \sin^2\omega_B t = 1$$

and $[{}_0R_B][{}_0R_B]^T$ has diagonal entries $\cos^2\omega_B t + \sin^2\omega_B t = 1$ (and 1) and off-diagonal entries $\cos\omega_B t\sin\omega_B t - \sin\omega_B t\cos\omega_B t = 0$, so it is I. The angular velocity of S_B is $\boldsymbol{\omega}_{B0} = \omega_B\,\mathbf{k}$ (Poisson: $\frac{d\mathbf{i}_B}{dt} = -\omega_B\sin\omega_B t\,\mathbf{i}_0 + \omega_B\cos\omega_B t\,\mathbf{j}_0 = \omega_B\,\mathbf{j}_B = \omega_B\,\mathbf{k}\times\mathbf{i}_B$).

Kinematics

Case 1: Parallel Tracks

Differentiating the positions (the speeds are constant):

$$\mathbf{v}_A = v_A\,\mathbf{i}_0, \qquad \mathbf{v}_B = v_B\,\mathbf{i}_0$$

- **Relative Velocity:**

$$\mathbf{v}_{B/A} = \mathbf{v}_B - \mathbf{v}_A = (v_B - v_A)\mathbf{i}_0$$

$$\mathbf{v}_{A/B} = -\mathbf{v}_{B/A} = (v_A - v_B)\mathbf{i}_0$$

Case 2: Train B on a Circular Track of Radius R

- Velocity of train B, with the chain rule $\frac{d}{dt}\sin(\omega_B t) = \omega_B\cos(\omega_B t)$ and $\frac{d}{dt}\cos(\omega_B t) = -\omega_B\sin(\omega_B t)$:

$$\mathbf{v}_B = \frac{d\mathbf{r}_B}{dt} = R\omega_B\cos(\omega_B t)\mathbf{i}_0 + R\omega_B\sin(\omega_B t)\mathbf{j}_0 = v_B\cos\left(\frac{v_B t}{R}\right)\mathbf{i}_0 + v_B\sin\left(\frac{v_B t}{R}\right)\mathbf{j}_0 = v_B\,\mathbf{i}_B$$

- **Relative Velocity $\mathbf{v}_{B/A}$:**

$$\mathbf{v}_{B/A} = \mathbf{v}_B - \mathbf{v}_A = \left[v_B\cos\left(\frac{v_B t}{R}\right) - v_A\right]\mathbf{i}_0 + v_B\sin\left(\frac{v_B t}{R}\right)\mathbf{j}_0$$

- **Magnitude of Relative Velocity** (expanding the square and using $\cos^2 + \sin^2 = 1$):

$$v_{\text{rel}} = \sqrt{\left[v_B\cos\left(\frac{v_B t}{R}\right) - v_A\right]^2 + v_B^2\sin^2\left(\frac{v_B t}{R}\right)} = \sqrt{v_B^2 + v_A^2 - 2v_A v_B\cos\left(\frac{v_B t}{R}\right)}$$

Case 2, velocity of A with respect to an observer attached to B

The frame S_B of the change-of-basis section turns with $\boldsymbol{\omega}_{B0} = \omega_B\mathbf{k}$. Notes Eq. 3.26 gives $\mathbf{v}_{A/B} = \mathbf{v}_A - \mathbf{v}_B - \boldsymbol{\omega}_{B0}\times(\mathbf{r}_A - \mathbf{r}_B)$. With $\mathbf{r}_A - \mathbf{r}_B = (v_A t - R\sin\omega_B t)\,\mathbf{i}_0 + (R\cos\omega_B t - R - a)\,\mathbf{j}_0$, $\mathbf{k}\times\mathbf{i}_0 = \mathbf{j}_0$, $\mathbf{k}\times\mathbf{j}_0 = -\mathbf{i}_0$ and $\omega_B R = v_B$, the three terms are

$$\mathbf{v}_A - \mathbf{v}_B = (v_A - v_B\cos\omega_B t)\,\mathbf{i}_0 - v_B\sin\omega_B t\,\mathbf{j}_0, \qquad -\boldsymbol{\omega}_{B0}\times(\mathbf{r}_A - \mathbf{r}_B) = -\omega_B\,(R + a - R\cos\omega_B t)\,\mathbf{i}_0 - \omega_B\,(v_A t - R\sin\omega_B t)\,\mathbf{j}_0$$

Adding them, $\omega_B R\cos\omega_B t = v_B\cos\omega_B t$ cancels the $-v_B\cos\omega_B t$ term and $\omega_B R\sin\omega_B t = v_B\sin\omega_B t$ cancels the $-v_B\sin\omega_B t$ term:

$$\mathbf{v}_{A/B} = \left[v_A - v_B\,\frac{R + a}{R}\right]\mathbf{i}_0 - \frac{v_A v_B t}{R}\,\mathbf{j}_0 \quad (\text{components in } \mathcal{B}_0)$$

Answers and Checks

Case 1

$\mathbf{v}_{B/A} = (v_B - v_A)\,\mathbf{i}_0$ and $\mathbf{v}_{A/B} = (v_A - v_B)\,\mathbf{i}_0$.

Case 2

$\mathbf{v}_{B/A} = [v_B\cos(v_B t/R) - v_A]\,\mathbf{i}_0 + v_B\sin(v_B t/R)\,\mathbf{j}_0$, with modulus $v_{\text{rel}} = \sqrt{v_A^2 + v_B^2 - 2v_A v_B\cos(v_B t/R)}$, for an observer that translates with A ($\mathbf{v}_{A/B} = -\mathbf{v}_{B/A}$ for an observer that translates with B). No integration is needed beyond the differentiation above.

For an observer attached to B (turning with it), $\mathbf{v}_{A/B} = [v_A - v_B(R+a)/R]\,\mathbf{i}_0 - (v_A v_B t/R)\,\mathbf{j}_0$. It differs from $-\mathbf{v}_{B/A}$ because the observer attached to B rotates: at $t = 0$, $v_{A/B} = v_A - v_B(1 + a/R)$, whereas the translating-observer value is $v_A - v_B$; for $R \to \infty$ both coincide.

Checks

- **Dimensional check:** $[\mathbf{v}_{B/A}] = \mathrm{m/s}$; in Case 2 the argument of the trigonometric functions is $\dfrac{v_B t}{R} = \dfrac{(\mathrm{m/s})\,\mathrm{s}}{\mathrm{m}}$, which is dimensionless (rad), and $[v_B^2 + v_A^2 - 2v_A v_B\cos(\cdot)] = \mathrm{m^2/s^2}$, so v_{rel} is in m/s.
- **Role of the separation a:** it does not appear in any relative velocity for translating observers, because $\mathbf{v}_{B/A}$ is the time derivative of $\mathbf{r}_B - \mathbf{r}_A$ and a only enters the constant offset of that vector (it does appear for the rotating observer, through $\boldsymbol{\omega}_{B0} \times (\mathbf{r}_A - \mathbf{r}_B)$).
- **Limiting cases of Case 2 (sympy):**
- $t = 0$: $v_{\text{rel}} = \sqrt{v_A^2 + v_B^2 - 2v_A v_B} = |v_B - v_A|$, because both velocities are parallel to $\mathbf{i}_0$ at that instant.
- $R \to \infty$ (the circular track straightens): $\cos(v_B t/R) \to 1$ and $v_{\text{rel}} \to |v_B - v_A|$, which is the Case 1 result.
- $v_A = v_B = v$: $v_{\text{rel}} = v\sqrt{2 - 2\cos(vt/R)} = 2v\left|\sin\dfrac{vt}{2R}\right|$, which starts at 0 and never exceeds $2v$.
- $v_A = 0$: $v_{\text{rel}} = v_B$ for all t (relative to a stationary train, only the speed of B matters).
- **Symbolic check:** $\|\mathbf{v}_B - \mathbf{v}_A\|^2$ computed from the components equals $v_B^2 + v_A^2 - 2v_A v_B\cos(v_B t/R)$ with zero residual. The translating-observer results use only $\mathcal{B}_0$; the rotating observer of Case 2 uses $[{}_0R_B]$. Dimensions of that result: $[v_A] = [v_B a/R] = [v_A v_B t/R] = \mathrm{m/s}$.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 8 in its solution keys, so the results are verified by independent derivation (the symbolic check above).

Source: Problems.pdf

PROBLEM 09

Artificial Gravity in a Rotating Cylinder

Original statement · Problems.pdf, p. 8

Centrifugal + Coriolis Forces

In "2001: A Space Odyssey" an idea to obtain artificial gravity in a spaceship was presented. It consists of a cylinder rotating at constant angular velocity around its axis. The astronauts would live in the inner surface of the cylinder and the centrifugal force would act as artificial gravity.

(a) If the cylinder has radius R, determine the angular velocity ω required to obtain a given level artificial gravity g. Consider $R = 10$ m and $g = 9.81$ m/s^2 for a particular case.

(b) An astronaut runs inside the surface of the cylinder with a constant speed u with respect to the cylinder. Determine the artificial gravity that she feels (neglect the height of the astronaut h with respect to the radius of the cylinder R, $h \ll R$).

(c) What is the condition for her to feel no artificial gravity at all?

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a cylindrical hull of radius R spins at constant rate ω about its axis in deep space. An astronaut P of mass m stands on, or runs along, the inner surface.
- **Why the hull frame.** The astronaut lives in the hull: the floor, her running speed u and what she "feels" are all defined with respect to the cylinder. In the frame S_1 attached to the hull the floor is fixed, and the only price is to add the inertia forces of Notes Eq. 4.62. The centrifugal force is exactly the "artificial gravity" of the statement; the Coriolis force is what changes when she runs.
- **Hypotheses:**
  1. No external gravity (deep space); the only actual force on the astronaut is the total force $\mathbf{N}$ of the floor on her feet (normal reaction plus any friction; part (b) shows that, for uniform running, $\mathbf{N}$ comes out purely radial).
  2. $h \ll R$: she is a point particle on the surface, at distance R from the axis.

3. ω constant; the axis point O is fixed in an inertial frame S_0.

- **Artificial gravity felt.** A body feels as "weight" the force it has to receive from the floor. We define the artificial gravity as $g_{\text{art}} = |\mathbf{N}|/m$, the reading per unit mass of a scale under her feet, directed opposite to $\mathbf{N}$ (i.e. away from the axis, "down" for her).
- **Fixed (inertial) frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, with O on the axis and $\mathbf{k}_0$ along it.
- **Moving frame:** $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ fixed to the hull, $\mathbf{k}_1 = \mathbf{k}_0$. The origin of S_1 is the fixed point O; S_1 coincides with S_0 at $t = 0$.
- **Local cylindrical basis of the astronaut (in S_1):** $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}_1\}$, radial (from the axis to the floor), circumferential (in the sense of rotation) and axial; deduced in the next section.
- **Coordinates and degrees of freedom:** on the cylinder surface, 2 coordinates: azimuth φ in S_1 (measured from $\mathbf{i}_1$ in the sense of rotation) and axial position z. Running at constant speed u means $u_\theta = R\dot{\varphi}$ and $u_z = \dot{z}$ constant with $u_\theta^2 + u_z^2 = u^2$.

Symbol	Meaning	SI unit
R	radius of the cylinder (10 m in the numerical case)	m
ω	spin rate of the cylinder	rad/s
g	target artificial gravity (9.81 m/s^2)	m/s^2
u	running speed relative to the cylinder	m/s
u_θ, u_z	circumferential (positive in the sense of rotation) and axial components of the running velocity	m/s
m	mass of the astronaut	kg

Change of Basis and Position Vector

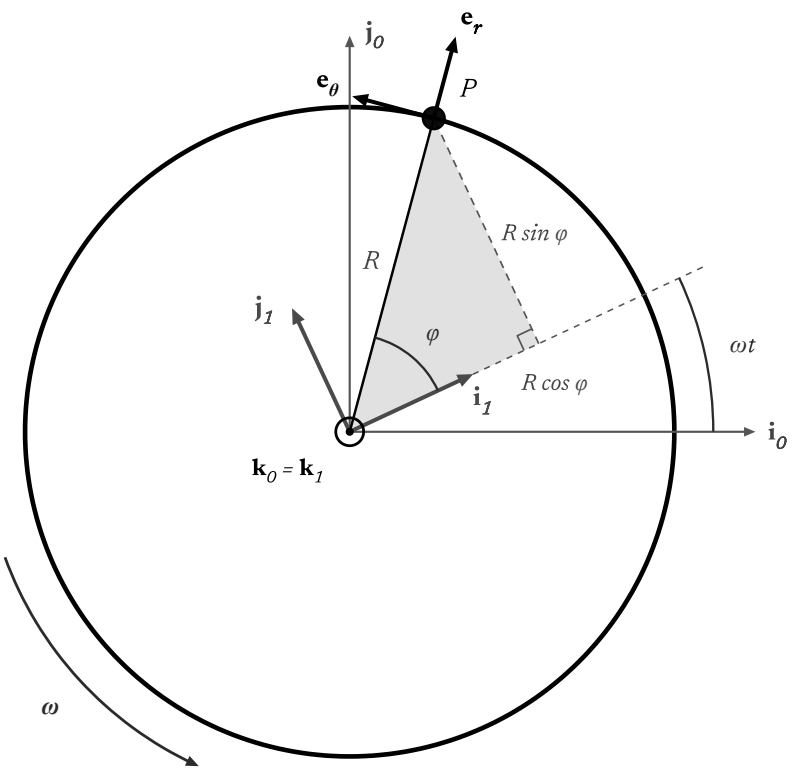


Figure 1. Cross-section along the axis ($\mathbf{k}_0 = \mathbf{k}_1$ out of the paper, $\odot$). The hull axes $\mathbf{i}_1, \mathbf{j}_1$ are $\mathbf{i}_0, \mathbf{j}_0$ turned counter-clockwise by ωt. In the shaded right triangle the hypotenuse is R, the leg along $\mathbf{i}_1$ is $R\cos\varphi$ and the leg along $\mathbf{j}_1$ is $R\sin\varphi$; $\mathbf{e}_r, \mathbf{e}_\theta$ in orange.

The basis $\mathcal{B}_1$. The hull turns about its axis $\mathbf{k}_0$, so the cross-section is drawn looking along the axis, with $\mathbf{k}_0 = \mathbf{k}_1$ pointing out of the paper, towards the reader ($\odot$). With $\mathbf{i}_0$ to the right, the right-hand rule $\mathbf{i}_0 \times \mathbf{j}_0 = \mathbf{k}_0$ puts $\mathbf{j}_0$ up the page, and a positive ω turns the hull counter-clockwise. After a time t, $\mathbf{i}_1$ has turned the angle ωt from $\mathbf{i}_0$:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of a right triangle with the leg $\cos\omega t$ along $\mathbf{i}_0$ (adjacent) and the leg $\sin\omega t$ along $+\mathbf{j}_0$ (opposite): $\mathbf{i}_1 = \cos\omega t\,\mathbf{i}_0 + \sin\omega t\,\mathbf{j}_0$.
- $\mathbf{j}_1$ is $\mathbf{j}_0$ turned the same angle counter-clockwise; turning from "up" towards the left, its opposite leg $\sin\omega t$ points along $-\mathbf{i}_0$, which gives the minus sign: $\mathbf{j}_1 = -\sin\omega t\,\mathbf{i}_0 + \cos\omega t\,\mathbf{j}_0$.
- $\mathbf{k}_1 = \mathbf{k}_0$, the axis of rotation.

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos\omega t\cos\omega t - (-\sin\omega t)\sin\omega t = \cos^2\omega t + \sin^2\omega t = 1$$

The columns are unit vectors ($\cos^2\omega t+\sin^2\omega t=1$) and mutually orthogonal ($-\cos\omega t\sin\omega t+\sin\omega t\cos\omega t=0$), so $[{}_0R_1][{}_0R_1]^T=I$.

Motion of S_1: $\boldsymbol{\omega}_{10}=\omega\mathbf{k}_1$, $\boldsymbol{\alpha}_{10}=\mathbf{0}$ (ω constant and $\mathbf{k}_1$ fixed), $\mathbf{r}_0^O=\mathbf{v}_0^O=\mathbf{a}_0^O=\mathbf{0}$. Poisson check: $\frac{d\mathbf{i}_1}{dt}\big|_0=-\omega\sin\omega t\,\mathbf{i}_0+\omega\cos\omega t\,\mathbf{j}_0=\omega\,\mathbf{j}_1=\omega\,\mathbf{k}_1\times\mathbf{i}_1$ (chain rule).

Position vector. To reach P we go along the axis to the cross-section that contains her (height z), and then radially out to the floor, a distance R:

$$\mathbf{r}_1^P=z\,\mathbf{k}_1+R\,\mathbf{e}_r$$

In the cross-section, $R\,\mathbf{e}_r$ is the hypotenuse (length R) of the shaded right triangle: the leg adjacent to φ lies along $\mathbf{i}_1$ and measures $R\cos\varphi$, the leg opposite to φ lies along $\mathbf{j}_1$ and measures $R\sin\varphi$. So, with φ the usual polar angle measured from $\mathbf{i}_1$:

$$\mathbf{e}_r=\cos\varphi\,\mathbf{i}_1+\sin\varphi\,\mathbf{j}_1,\qquad \mathbf{e}_\theta=\frac{d\mathbf{e}_r}{d\varphi}=-\sin\varphi\,\mathbf{i}_1+\cos\varphi\,\mathbf{j}_1$$

($\frac{d}{d\varphi}\cos\varphi=-\sin\varphi$, $\frac{d}{d\varphi}\sin\varphi=\cos\varphi$; $|\mathbf{e}_\theta|=\sqrt{\sin^2\varphi+\cos^2\varphi}=1$, and $\mathbf{e}_\theta$ points in the sense of increasing φ, the sense of rotation). The matrix with columns $\mathbf{e}_r,\mathbf{e}_\theta,\mathbf{k}_1$ in $\mathcal{B}_1$ is a simple rotation by φ (matrix $\begin{pmatrix}\cos\varphi&-\sin\varphi&0\\ \sin\varphi&\cos\varphi&0\\ 0&0&1\end{pmatrix}$, $\det=\cos^2\varphi+\sin^2\varphi=1$; RR^T has diagonal entries $\cos^2\varphi+\sin^2\varphi=1$ (and 1) and off-diagonal entries $\cos\varphi\sin\varphi-\sin\varphi\cos\varphi=0$, so $RR^T=I$).

Derivatives and cross products used later. In S_1, by the chain rule, $\frac{d\mathbf{e}_r}{dt}\big|_1=\frac{d\mathbf{e}_r}{d\varphi}\dot\varphi=\dot\varphi\,\mathbf{e}_\theta$ and $\frac{d\mathbf{e}_\theta}{dt}\big|_1=(-\cos\varphi\,\mathbf{i}_1-\sin\varphi\,\mathbf{j}_1)\,\dot\varphi=-\dot\varphi\,\mathbf{e}_r$. With $\mathbf{k}_1\times\mathbf{i}_1=\mathbf{j}_1$ and $\mathbf{k}_1\times\mathbf{j}_1=-\mathbf{i}_1$: $\mathbf{k}_1\times\mathbf{e}_r=\cos\varphi\,\mathbf{j}_1-\sin\varphi\,\mathbf{i}_1=\mathbf{e}_\theta$ and $\mathbf{k}_1\times\mathbf{e}_\theta=-\sin\varphi\,\mathbf{j}_1-\cos\varphi\,\mathbf{i}_1=-\mathbf{e}_r$. The part of $\mathbf{r}_1^P$ normal to $\boldsymbol{\omega}_{10}$ is $\mathbf{r}_\perp=R\,\mathbf{e}_r$ (the axial part $z\,\mathbf{k}_1$ is parallel to it).

Kinematics

In part (a) the astronaut is at rest on the floor, so $\mathbf{v}_1^P=\mathbf{0}$ and $\mathbf{a}_1^P=\mathbf{0}$. In part (b) she runs with a constant relative velocity:

Relative kinematics. $\mathbf{r}_1^P=R\,\mathbf{e}_r+z\,\mathbf{k}_1$ with $R\dot\varphi=u_\theta$, $\dot z=u_z$ (both constant). Differentiating in S_1 with the product and chain rules (R constant, $\mathbf{k}_1$ constant in S_1):

$$\mathbf{v}_1^P=R\dot\varphi\,\mathbf{e}_\theta+\dot z\,\mathbf{k}_1=u_\theta\,\mathbf{e}_\theta+u_z\,\mathbf{k}_1$$

$$\mathbf{a}_1^P = \dot{u}_\theta\,\mathbf{e}_\theta + u_\theta\,(-\dot{\varphi}\,\mathbf{e}_r) + \dot{u}_z\,\mathbf{k}_1 = -\frac{u_\theta^2}{R}\,\mathbf{e}_r$$

($\dot{u}_\theta = \dot{u}_z = 0$ and $\dot{\varphi} = u_\theta/R$; the relative path is a helix of radius R; in S_1 she needs a centripetal acceleration u_θ^2/R).

Forces and Equations of Motion

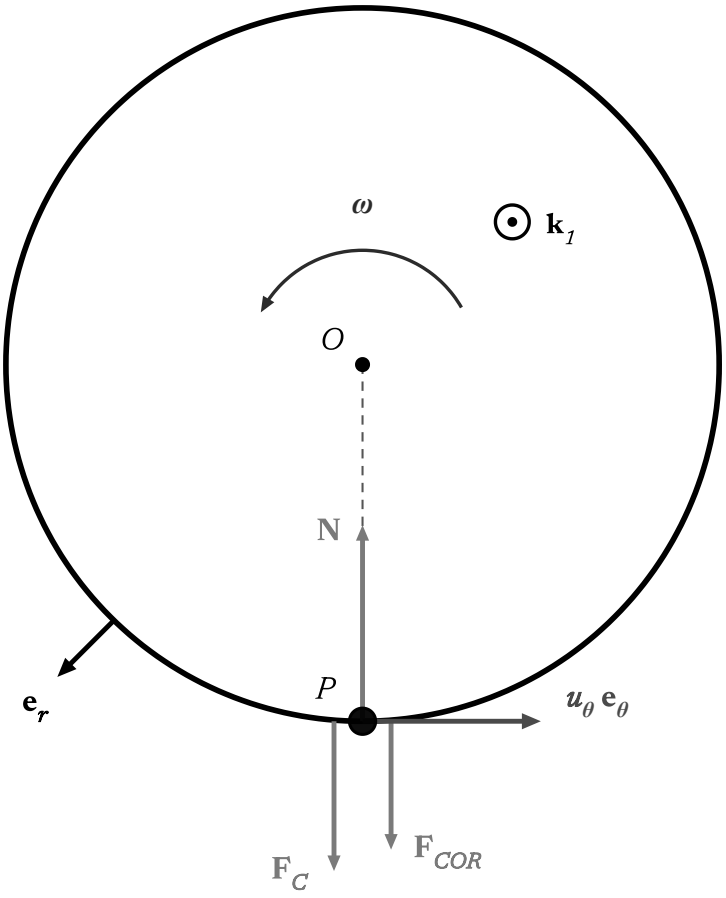


Figure 2. Forces in S_1 on the astronaut at P ($\mathbf{k}_1$ out of the paper). With $u_\theta > 0$ (running in the sense of rotation) both $\mathbf{F}_C = m\omega^2 R\,\mathbf{e}_r$ and $\mathbf{F}_{COR} = 2m\omega u_\theta\,\mathbf{e}_r$ point away from the axis; the floor force $\mathbf{N}$ points to the axis and balances them plus $m\mathbf{a}_1^P$.

The only actual force on the astronaut is the floor force $\mathbf{N}$ (no gravity in deep space). The inertia forces of Notes Eq. 4.62 are $\mathbf{F}_A = \mathbf{0}$ (fixed origin) and $\mathbf{F}_E = \mathbf{0}$ ($\boldsymbol{\alpha}_{10} = \mathbf{0}$); $\mathbf{F}_C = m\omega^2\mathbf{r}_\perp$ (Slide 30) and $\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$ (Slide 31).

Centrifugal force. With $\mathbf{r}_\perp = R\,\mathbf{e}_r$, $\mathbf{k}_1 \times \mathbf{e}_r = \mathbf{e}_\theta$ and $\mathbf{k}_1 \times \mathbf{e}_\theta = -\mathbf{e}_r$:

$$\mathbf{F}_C = -m\,\boldsymbol{\omega}\mathbf{k}_1 \times (\boldsymbol{\omega}\mathbf{k}_1 \times R\,\mathbf{e}_r) = -m\,\boldsymbol{\omega}\mathbf{k}_1 \times (R\omega\,\mathbf{e}_\theta) = m\omega^2 R\,\mathbf{e}_r$$

Coriolis force (running case), with $\mathbf{k}_1 \times \mathbf{e}_\theta = -\mathbf{e}_r$ and $\mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$:

$$\mathbf{F}_{COR} = -2m\,\boldsymbol{\omega}\mathbf{k}_1 \times (u_\theta\,\mathbf{e}_\theta + u_z\,\mathbf{k}_1) = 2m\omega u_\theta\,\mathbf{e}_r$$

Newton's law in S_1 (Notes Eq. 4.63), $m\mathbf{a}_1^P = \mathbf{N} + \mathbf{F}_C + \mathbf{F}_{COR}$. The unknown is the floor force $\mathbf{N}$ (the relative motion is prescribed, so no coordinate is unknown):

$$-m\frac{u_\theta^2}{R}\,\mathbf{e}_r = \mathbf{N} + m\omega^2 R\,\mathbf{e}_r + 2m\omega u_\theta\,\mathbf{e}_r$$

$$\mathbf{N} = -m\left(\omega^2 R + 2\omega u_\theta + \frac{u_\theta^2}{R}\right)\mathbf{e}_r = -\frac{m}{R}\left(\omega R + u_\theta\right)^2\mathbf{e}_r$$

(the bracket is the square $(\omega R + u_\theta)^2 = \omega^2 R^2 + 2\omega R u_\theta + u_\theta^2$ divided by R). At rest ($u_\theta = u_z = 0$) it reduces to the static balance $m\,\mathbf{0} = \mathbf{N} + m\omega^2 R\,\mathbf{e}_r$, $\mathbf{N} = -m\omega^2 R\,\mathbf{e}_r$.

Answers and Checks

(a) Spin rate for a given artificial gravity

> **Pedagogical Justification:** an astronaut at rest on the floor has $\mathbf{v}_1^P = \mathbf{0}$ and $\mathbf{a}_1^P = \mathbf{0}$, so Notes Eq. 4.63 becomes a static balance in S_1 between the floor reaction and the centrifugal force (the Coriolis force vanishes because $\mathbf{v}_1^P = \mathbf{0}$).

From the static balance, $\mathbf{N} = -m\omega^2 R\,\mathbf{e}_r$ and

$$g_{\text{art}} = \frac{|\mathbf{N}|}{m} = \omega^2 R$$

The floor pushes her toward the axis, and she feels a "gravity" $\omega^2 R$ pointing away from the axis. Setting $g_{\text{art}} = g$:

$$\boxed{\omega = \sqrt{\frac{g}{R}}}$$

Numbers ($R = 10$ m, $g = 9.81$ m/s^2):

$$\omega = \sqrt{\frac{9.81\ \text{m/s}^2}{10\ \text{m}}} = \sqrt{0.981}\ \text{s}^{-1} \approx 0.9905\ \text{rad/s}, \qquad T_{\text{rev}} = \frac{2\pi}{\omega} \approx 6.34\ \text{s}, \qquad \frac{60\,\omega}{2\pi} \approx 9.46\ \text{rpm}$$

and the floor moves at $\omega R \approx 9.90$ m/s in S_0.

(b) Artificial gravity felt while running

> **Pedagogical Justification:** now $\mathbf{v}_1^P \neq \mathbf{0}$, so the Coriolis force appears, and her relative path may be curved, so $\mathbf{a}_1^P \neq \mathbf{0}$. Both are computed in S_1 and inserted in Notes Eq. 4.63 to obtain $\mathbf{N}$. The statement does not fix the running direction, so we take a general constant relative velocity with circumferential part u_θ and axial part u_z (running "around" the hull is $u_z = 0$, $u_\theta = \pm u$).

The reaction $\mathbf{N} = -\frac{m}{R}(\omega R + u_\theta)^2\,\mathbf{e}_r$ is purely radial, so no friction is needed to keep the motion uniform. The artificial gravity she feels is

$$\boxed{g_{\text{art}} = \frac{(\omega R + u_\theta)^2}{R} = g\left(1 + \frac{u_\theta}{\omega R}\right)^2 = g + 2\omega u_\theta + \frac{u_\theta^2}{R}}$$

using $\omega^2 R = g$ from part (a). Particular cases:

- running around the hull **in the sense of rotation** ($u_\theta = +u$): $g_{\text{art}} = (\omega R + u)^2/R > g$ (she feels heavier);
- running **against the rotation** ($u_\theta = -u$): $g_{\text{art}} = (\omega R - u)^2/R < g$ for $u < 2\omega R$ (she feels lighter);
- running **along the axis** ($u_\theta = 0$, $u_z = u$): $g_{\text{art}} = g$ (the Coriolis force vanishes because $\mathbf{v}_1^P \parallel \boldsymbol{\omega}_{10}$).

Numbers ($R = 10$ m, $\omega = 0.9905$ rad/s): at $u = 2$ m/s, $g_{\text{art}} = 14.17$ m/s^2 ($1.44\,g$) with the rotation and 6.25 m/s^2 ($0.64\,g$) against it; at $u = 5$ m/s, 22.2 m/s^2 ($2.26\,g$) and 2.41 m/s^2 ($0.25\,g$).

(c) Condition for no artificial gravity

> **Pedagogical Justification:** $g_{\text{art}} = (\omega R + u_\theta)^2/R$ is a perfect square, so it vanishes only when its base vanishes.

$$g_{\text{art}} = 0 \iff \omega R + u_\theta = 0 \iff \boxed{u_\theta = -\omega R, \quad u_z = 0}$$

She must run around the hull, against the rotation, at the speed of the floor ωR (≈ 9.90 m/s for the numerical case, a sprinter's speed). Then, by Notes Eq. 3.26, her absolute velocity is $\mathbf{v}_0^P = \mathbf{v}_1^P + \boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = -\omega R\,\mathbf{e}_\theta + \omega R\,\mathbf{e}_\theta = \mathbf{0}$: she is at rest in inertial space, floating, while the floor slides under her feet. With an axial component ($u_z \neq 0$) and $u = \omega R$ the condition cannot be met, since then $|u_\theta| < \omega R$.

Checks

- **Inertial reading (cross-check).** In S_0 the astronaut moves on a circle (or helix) of radius R with tangential speed $\omega R + u_\theta$, so the floor must supply the centripetal force $m(\omega R + u_\theta)^2/R$: the same result. The rotating observer splits it into centrifugal $m\omega^2 R$, Coriolis $2m\omega u_\theta$ and relative centripetal $m u_\theta^2/R$ parts.
- **Limits:** $u_\theta \to 0$ gives $g_{\text{art}} = g$; for $|u_\theta| \ll \omega R$, $g_{\text{art}} \approx g \pm 2\omega u$, a correction due to the Coriolis force alone; a larger hull ($R \to \infty$ at fixed g) needs a smaller ω and makes the Coriolis correction $2\omega u = 2u\sqrt{g/R}$ smaller, which is why large habitats are more comfortable.
- **Orthogonality:** $\mathbf{e}_r \cdot \mathbf{e}_\theta = -\cos\varphi\sin\varphi + \sin\varphi\cos\varphi = 0$.
- **Dimensions:** $[\sqrt{g/R}] = \sqrt{\mathrm{m\,s^{-2}/m}} = \mathrm{s^{-1}}$; $[\omega^2 R] = [2\omega u_\theta] = [u_\theta^2/R] = \mathrm{m\,s^{-2}}$; $[N] = \mathrm{kg\,m\,s^{-2}} = \mathrm{N}$.
- **Verification (sympy):** $\mathbf{a}_1^P$, $\mathbf{F}_C$, $\mathbf{F}_{COR}$ and $\mathbf{N}$ are recomputed; the inertial double derivative of $[{}_0R_1]\{\mathbf{r}_1^P\}$ equals the Notes Eq. 3.27 decomposition and gives the same $\mathbf{N}$; the numbers above are reproduced.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 9 in its solution keys, so the results are verified by independent derivation (the inertial reading and the sympy check above).

Source: Problems.pdf (PDF page 8); no official key

PROBLEM 20

Heavy Particle on a Rotating Circumference

Original statement · Problems.pdf, p. 10

Energy in a Rotating Frame

A heavy particle is moving along a smooth circumference of radius R and center O. One of the diameters of the circumference is along the local vertical, and the circumference is rotating with constant angular velocity ω about it. The position of the particle with respect to the circumference is described by the angle θ between the particle and the lowest point of the circumference.

(a) Discuss the conservation of mechanical energy in the moving reference frame and compute its value.

(b) Compute the reaction of the circumference in the direction normal to its plane, as a function of θ.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a particle P of mass m slides on a smooth circular wire of radius R and centre O; the wire turns with constant angular velocity ω about its vertical diameter. The angle θ is measured from the lowest point.
- **Why the moving frame.** In the frame S_1 attached to the wire the constraint is a fixed circle, the motion has one coordinate θ, and the reaction has two clear components (in the plane of the wire, normal to it). The question itself asks for the energy "in the moving reference frame", which requires the inertia forces of Notes Eq. 4.62 and their potentials (Slides 29–31).
- **Hypotheses:**
  1. S_0 fixed to the ground is inertial; O is fixed; uniform gravity $-g\,\mathbf{k}$.
  2. The wire is smooth: the reaction has no component along the tangent of the wire.
  3. ω is constant.
- **Fixed (inertial) frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, $\mathbf{k}_0$ upward along the rotation axis.

- **Moving frame:** $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ attached to the wire: $\mathbf{k}_1 = \mathbf{k}_0$, $\mathbf{i}_1$ horizontal in the plane of the wire (on the side where $\theta > 0$), $\mathbf{j}_1 = \mathbf{k}_1 \times \mathbf{i}_1$ normal to the plane of the wire. The plane of the wire is at azimuth ωt from $\mathbf{i}_0$.
- **Wire basis:** $\{\mathbf{e}_r, \mathbf{e}_\theta, -\mathbf{j}_1\}$, radial (from O to P), tangent (increasing θ) and normal to the plane; deduced in the next section.
- **Coordinate and degrees of freedom:** the particle is constrained to a curve fixed in S_1: **1 DOF**, θ, the angle of OP measured at O from the downward vertical $-\mathbf{k}_1$, positive towards $+\mathbf{i}_1$.
- **Unknown reaction:** $\mathbf{N} = N_r\,\mathbf{e}_r + N_b\,\mathbf{j}_1$, with N_r in the plane of the wire (along the radius) and N_b normal to that plane.

Symbol	Meaning	SI unit
m	mass of the particle	kg
R	radius of the circumference	m
ω	spin rate of the circumference (constant)	rad/s
g	gravitational acceleration	m/s^2
θ	angle from the lowest point	rad
$N_r,\ N_b$	reaction components (radial in-plane, normal to the plane)	N
$\theta_0,\ \dot\theta_0$	initial angle and angular rate in S_1	rad, rad/s

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from above

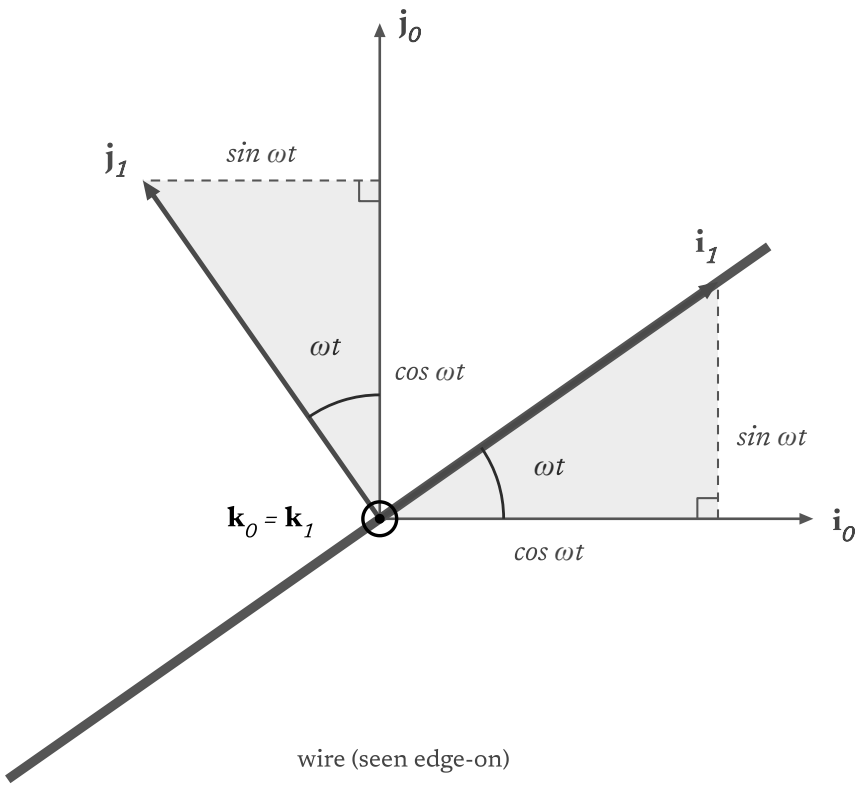


Figure 1. View from $+\mathbf{k}_0$ ($\mathbf{k}_0 = \mathbf{k}_1$ out of the paper, $\odot$). The wire is seen edge-on, along $\mathbf{i}_1$, turned counter-clockwise by ωt; $\mathbf{j}_1 = \mathbf{k}_1 \times \mathbf{i}_1$ is normal to it. The opposite leg of $\mathbf{j}_1$ points along $-\mathbf{i}_0$: hence $\mathbf{j}_1 = -\sin\omega t\,\mathbf{i}_0 + \cos\omega t\,\mathbf{j}_0$.

The rotation is about the vertical $\mathbf{k}_0$, so it is drawn looking down from $+\mathbf{k}_0$: $\mathbf{k}_0 = \mathbf{k}_1$ points out of the paper ($\odot$), $\mathbf{i}_0$ to the right and, by the right-hand rule $\mathbf{i}_0 \times \mathbf{j}_0 = \mathbf{k}_0$, $\mathbf{j}_0$ up the page. The plane of the wire is seen edge-on, as a line through O along $\mathbf{i}_1$; it turns counter-clockwise ($\omega > 0$) and makes the angle ωt with $\mathbf{i}_0$:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of a right triangle with the leg $\cos\omega t$ along $\mathbf{i}_0$ (adjacent) and the leg $\sin\omega t$ along $+\mathbf{j}_0$ (opposite): $\mathbf{i}_1 = \cos\omega t\,\mathbf{i}_0 + \sin\omega t\,\mathbf{j}_0$.
- $\mathbf{j}_1 = \mathbf{k}_1 \times \mathbf{i}_1$ is $\mathbf{i}_1$ turned $90°$ counter-clockwise, normal to the trace of the wire. It has turned ωt from $\mathbf{j}_0$, towards the left, so its opposite leg $\sin\omega t$ lies along $-\mathbf{i}_0$: $\mathbf{j}_1 = -\sin\omega t\,\mathbf{i}_0 + \cos\omega t\,\mathbf{j}_0$ (that is the minus sign of the matrix).
- $\mathbf{k}_1 = \mathbf{k}_0$.

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos\omega t\cos\omega t - (-\sin\omega t)\sin\omega t = \cos^2\omega t + \sin^2\omega t = 1$$

The columns are unit vectors and mutually orthogonal ($-\cos\omega t\sin\omega t + \sin\omega t\cos\omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.

Motion of S_1: $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1, \boldsymbol{\alpha}_{10} = \mathbf{0}, \mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$. Poisson check (chain rule): $\frac{d\mathbf{i}_1}{dt}\big|_0 = -\omega\sin\omega t\,\mathbf{i}_0 + \omega\cos\omega t\,\mathbf{j}_0 = \omega\,\mathbf{j}_1 = \omega\,\mathbf{k}_1 \times \mathbf{i}_1$.

The position vector in the plane of the wire

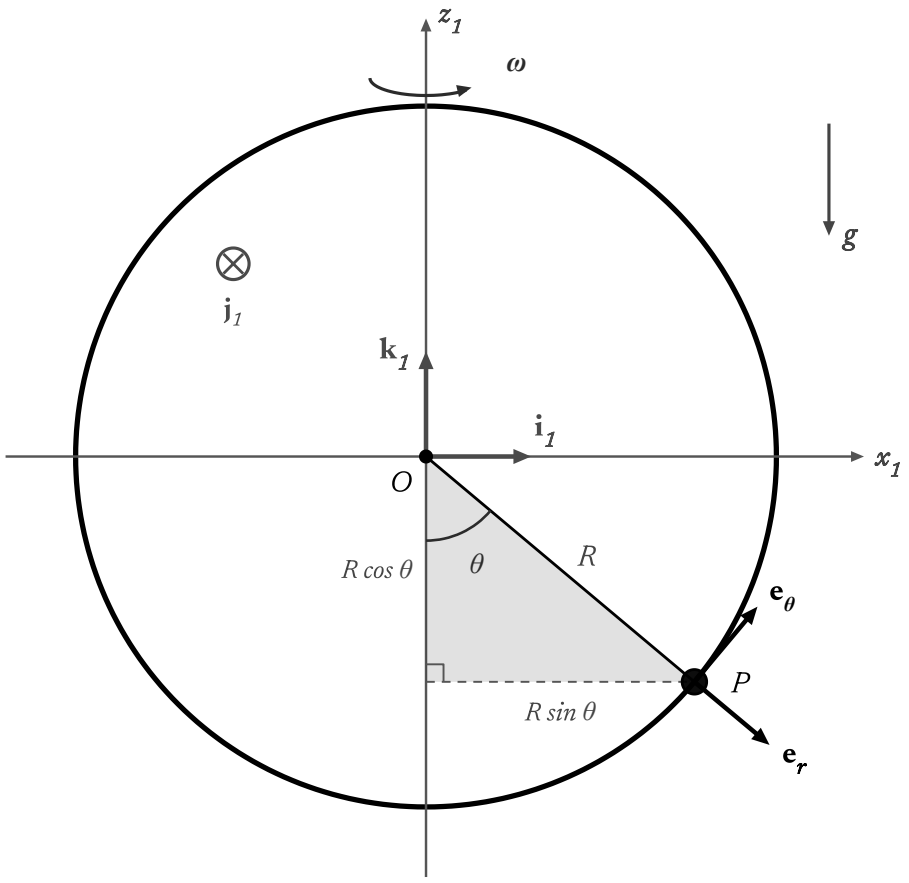


Figure 2. Plane of the wire, fixed in S_1 ($\mathbf{i}_1$ right, $\mathbf{k}_1$ up, so $\mathbf{j}_1$ points into the paper, $\otimes$). In the shaded right triangle OFP the hypotenuse is R, the leg opposite θ is $R\sin\theta$ along $\mathbf{i}_1$ and the adjacent leg is $R\cos\theta$ along $-\mathbf{k}_1$. $\mathbf{e}_r, \mathbf{e}_\theta$ in orange.

The plane of the wire is fixed in S_1 and is drawn with $\mathbf{i}_1$ to the right and $\mathbf{k}_1$ up. Then $\mathbf{j}_1 = \mathbf{k}_1 \times \mathbf{i}_1$ points into the paper ($\otimes$): fingers from $\mathbf{k}_1$ (up) to $\mathbf{i}_1$ (right), clockwise, so the thumb points away from the reader.

The centre O is the origin, so $\mathbf{r}_1^P = \overrightarrow{OP}$, of length R, making the angle θ with the downward vertical $-\mathbf{k}_1$. Let F be the point of the vertical axis at the height of P. In the shaded right triangle OFP (right angle at F) the hypotenuse is $|OP| = R$; the leg opposite to θ is $FP = R\sin\theta$, horizontal, along $+\mathbf{i}_1$; the leg adjacent to θ is $OF = R\cos\theta$, vertical and pointing down, along $-\mathbf{k}_1$. Hence

$$\mathbf{r}_1^P = R\,\mathbf{e}_r, \qquad \mathbf{e}_r = \sin\theta\,\mathbf{i}_1 - \cos\theta\,\mathbf{k}_1$$

Check: $\theta = 0$ gives $-R\,\mathbf{k}_1$, the lowest point; $\theta = \pi/2$ gives $R\,\mathbf{i}_1$, the end of the horizontal diameter. The part of $\mathbf{r}_1^P$ normal to the axis is the horizontal leg, $\mathbf{r}_\perp = R\sin\theta\,\mathbf{i}_1$.

Relation with the standard polar angle. Measured from $\mathbf{i}_1$ towards $\mathbf{k}_1$, the usual polar angle φ gives $\mathbf{e}_r = \cos\varphi\,\mathbf{i}_1 + \sin\varphi\,\mathbf{k}_1$. This matches $\sin\theta\,\mathbf{i}_1 - \cos\theta\,\mathbf{k}_1$ for $\varphi = \theta - \pi/2$: $\cos\left(\theta - \frac{\pi}{2}\right) = \cos\theta\cos\frac{\pi}{2} + \sin\theta\sin\frac{\pi}{2} = \sin\theta$ and $\sin\left(\theta - \frac{\pi}{2}\right) = \sin\theta\cos\frac{\pi}{2} - \cos\theta\sin\frac{\pi}{2} = -\cos\theta$. So θ is the polar angle measured from $-\mathbf{k}_1$, chosen so that $\theta = 0$ is the lowest point, as in the statement.

Wire basis. The tangent is the derivative with respect to θ ($\frac{d}{d\theta}\sin\theta = \cos\theta$, $\frac{d}{d\theta}(-\cos\theta) = \sin\theta$), already of unit length ($\sqrt{\cos^2\theta + \sin^2\theta} = 1$):

$$\mathbf{e}_\theta = \frac{d\mathbf{e}_r}{d\theta} = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{k}_1$$

Since $\mathbf{e}_r \times \mathbf{e}_\theta = (-\cos\theta)(\cos\theta)(\mathbf{k}_1 \times \mathbf{i}_1) + (\sin\theta)(\sin\theta)(\mathbf{i}_1 \times \mathbf{k}_1) = -\cos^2\theta\,\mathbf{j}_1 - \sin^2\theta\,\mathbf{j}_1 = -\mathbf{j}_1$ (with $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{i}_1 \times \mathbf{k}_1 = -\mathbf{j}_1$), the right-handed triad is $\{\mathbf{e}_r, \mathbf{e}_\theta, -\mathbf{j}_1\}$; $-\mathbf{j}_1$ points out of the paper in the plane view. Its columns in $\mathcal{B}_1$:

$$[{}_1R_w] = \begin{pmatrix} \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \\ -\cos\theta & \sin\theta & 0 \end{pmatrix}, \qquad \det[{}_1R_w] = \sin\theta(0\cdot 0 + \sin\theta) - \cos\theta(0\cdot 0 - \cos\theta) = 1$$

and the columns are orthonormal ($\sin^2\theta + \cos^2\theta = 1$, $\sin\theta\cos\theta - \cos\theta\sin\theta = 0$), so $[{}_1R_w][{}_1R_w]^T = I$.

Derivatives in S_1 (chain rule): $\frac{d\mathbf{e}_r}{dt}\big|_1 = \frac{d\mathbf{e}_r}{d\theta}\dot\theta = \dot\theta\,(\cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{k}_1) = \dot\theta\,\mathbf{e}_\theta$ and $\frac{d\mathbf{e}_\theta}{dt}\big|_1 = \dot\theta\,(-\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{k}_1) = -\dot\theta\,\mathbf{e}_r$.

Kinematics

$$\mathbf{r}_1^P = R\,\mathbf{e}_r, \qquad \mathbf{v}_1^P = R\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{a}_1^P = R\ddot\theta\,\mathbf{e}_\theta + R\dot\theta\left(-\dot\theta\,\mathbf{e}_r\right) = R\ddot\theta\,\mathbf{e}_\theta - R\dot\theta^2\,\mathbf{e}_r$$

(R is constant, so $\mathbf{v}_1^P = R\,\frac{d\mathbf{e}_r}{dt}\big|_1$; then the product rule on $R\dot\theta\,\mathbf{e}_\theta$: $\frac{d}{dt}(R\dot\theta) = R\ddot\theta$ and $\frac{d\mathbf{e}_\theta}{dt}\big|_1 = -\dot\theta\,\mathbf{e}_r$.)

The absolute velocity (Notes Eq. 3.26), used below, is $\mathbf{v}_0^P = \mathbf{v}_1^P + \boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = R\dot\theta\,\mathbf{e}_\theta + \omega R\sin\theta\,\mathbf{j}_1$ (from $\omega\mathbf{k}_1\times(R\sin\theta\,\mathbf{i}_1 - R\cos\theta\,\mathbf{k}_1) = R\omega\sin\theta\,\mathbf{j}_1$).

Forces and Equations of Motion

Forces seen from S_1 (Notes Eqs. 4.62–4.63):

- weight $-mg\,\mathbf{k}_1$ and reaction $\mathbf{N} = N_r\,\mathbf{e}_r + N_b\,\mathbf{j}_1$ (actual forces);
- $\mathbf{F}_A = \mathbf{0}$ (O fixed) and $\mathbf{F}_E = \mathbf{0}$ ($\boldsymbol{\alpha}_{10} = \mathbf{0}$);
- centrifugal force: $\mathbf{r}_\perp = R\sin\theta\,\mathbf{i}_1$ (the part of $\mathbf{r}_1^P$ normal to $\mathbf{k}_1$), so (Slide 30)

$$\mathbf{F}_C = -m\boldsymbol{\omega}_{10}\times\left(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P\right) = m\omega^2 R\sin\theta\,\mathbf{i}_1$$

Component check: $\omega\mathbf{k}_1\times(R\sin\theta\,\mathbf{i}_1 - R\cos\theta\,\mathbf{k}_1) = R\omega\sin\theta\,\mathbf{j}_1$, then $\omega\mathbf{k}_1\times R\omega\sin\theta\,\mathbf{j}_1 = -R\omega^2\sin\theta\,\mathbf{i}_1$, and the minus sign of $\mathbf{F}_C$ gives $+m\omega^2 R\sin\theta\,\mathbf{i}_1$;

- Coriolis force, with $\mathbf{k}_1\times\mathbf{i}_1 = \mathbf{j}_1$ and $\mathbf{k}_1\times\mathbf{k}_1 = \mathbf{0}$:

$$\mathbf{F}_{COR} = -2m\,\boldsymbol{\omega}\mathbf{k}_1\times R\dot\theta\,(\cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{k}_1) = -2m\omega R\dot\theta\cos\theta\,\mathbf{j}_1$$

Equation of motion (projection on $\mathbf{e}_\theta$, where $\mathbf{N}$ has no component). Using $\mathbf{k}_1\cdot\mathbf{e}_\theta = \sin\theta$, $\mathbf{i}_1\cdot\mathbf{e}_\theta = \cos\theta$, $\mathbf{j}_1\cdot\mathbf{e}_\theta = 0$:

$$mR\ddot\theta = -mg\sin\theta + m\omega^2 R\sin\theta\cos\theta \quad\Longrightarrow\quad \ddot\theta = -\frac{g}{R}\sin\theta + \omega^2\sin\theta\cos\theta$$

Projection on $\mathbf{j}_1$ (normal to the plane). The relative motion stays in the plane, so $\mathbf{a}_1^P\cdot\mathbf{j}_1 = 0$; the weight and the centrifugal force lie in the plane; only N_b and the Coriolis force remain:

$$m\,\mathbf{a}_1^P \cdot \mathbf{j}_1 = 0 = N_b - 2m\omega R\dot\theta\cos\theta \quad\Longrightarrow\quad N_b = 2m\omega R\dot\theta\cos\theta$$

Projection on $\mathbf{e}_r$ ($\mathbf{k}_1\cdot\mathbf{e}_r = -\cos\theta, \mathbf{i}_1\cdot\mathbf{e}_r = \sin\theta$):

$$-mR\dot\theta^2 = N_r + mg\cos\theta + m\omega^2 R\sin^2\theta \quad\Longrightarrow\quad N_r = -mR\dot\theta^2 - mg\cos\theta - m\omega^2 R\sin^2\theta$$

Answers and Checks

(a) Mechanical energy in the moving frame

- **Justification:** E_1^P is conserved when every force that does work in S_1 derives from a potential (Slide 29). We check the forces one by one using their power in S_1, $\mathbf{F}\cdot\mathbf{v}_1^P$.
- **Reaction:** $\mathbf{N}\cdot\mathbf{v}_1^P = R\dot\theta\,(N_r\,\mathbf{e}_r + N_b\,\mathbf{j}_1)\cdot\mathbf{e}_\theta = 0$: the wire is smooth and **fixed in** S_1, so it does no work in S_1.
- **Coriolis force:** $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P = 0$ always (Slide 31); here explicitly $\mathbf{j}_1\cdot\mathbf{e}_\theta = 0$.
- **Euler force and origin force:** both zero.
- **Weight:** conservative, $V_g = mgz_1 = -mgR\cos\theta$ (zero at the height of O).
- **Centrifugal force:** $\boldsymbol{\omega}_{10}$ is constant, so it has the potential (Slide 30) $V_1 = -\frac{1}{2}m\omega^2 r_\perp^2 = -\frac{1}{2}m\omega^2R^2\sin^2\theta$ (zero on the axis).

Therefore E_1^P **is conserved**:

$$\boxed{E_1^P = \frac{1}{2}mR^2\dot\theta^2 - mgR\cos\theta - \frac{1}{2}m\omega^2R^2\sin^2\theta = \frac{1}{2}mR^2\dot\theta_0^2 - mgR\cos\theta_0 - \frac{1}{2}m\omega^2R^2\sin^2\theta_0}$$

its value being fixed by the initial conditions $\theta(0)=\theta_0, \dot\theta(0)=\dot\theta_0$ (the statement gives none, so the constant is left in terms of them).

Check by differentiation (chain rule on each term, then the EOM):

$$\frac{dE_1^P}{dt} = mR^2\dot\theta\ddot\theta + mgR\sin\theta\,\dot\theta - m\omega^2R^2\sin\theta\cos\theta\,\dot\theta = mR^2\dot\theta\left[\ddot\theta + \frac{g}{R}\sin\theta - \omega^2\sin\theta\cos\theta\right] = 0$$

The energy in S_0 is not conserved. In S_0 the wire moves, and its reaction does work: with $\mathbf{v}_0^P = R\dot\theta\,\mathbf{e}_\theta + \omega R\sin\theta\,\mathbf{j}_1$, the power of $\mathbf{N}$ in S_0 is $\mathbf{N}\cdot\mathbf{v}_0^P = N_b\,\omega R\sin\theta \neq 0$. This is why the moving frame is the natural one for an energy integral here.

(b) Reaction normal to the plane of the circumference

> **Pedagogical Justification:** the projection of Notes Eq. 4.63 on $\mathbf{j}_1$ gave $N_b = 2m\omega R\dot\theta\cos\theta$; to write it as a function of θ we eliminate $\dot\theta$ with the energy integral of part (a).

Solving the energy integral for $R\dot\theta$:

$$R^2\dot\theta^2 = \frac{2E_1^P}{m} + 2gR\cos\theta + \omega^2R^2\sin^2\theta \quad\Longrightarrow\quad R\dot\theta = \pm\sqrt{\frac{2E_1^P}{m} + 2gR\cos\theta + \omega^2R^2\sin^2\theta}$$

$$\boxed{N_b(\theta) = \pm\,2m\omega\cos\theta\,\sqrt{\frac{2E_1^P}{m} + 2gR\cos\theta + \omega^2R^2\sin^2\theta}}$$

with the sign of $\dot\theta$ (positive when θ increases) and E_1^P from part (a). For completeness, the in-plane component is $N_r = -mR\dot\theta^2 - mg\cos\theta - m\omega^2R\sin^2\theta$.

Checks

- **Origin of N_b.** The normal-to-plane reaction is entirely due to the Coriolis force: the wire must push the particle sideways so that it keeps turning with the plane while its distance to the axis changes. It vanishes when $\dot\theta = 0$ and when $\cos\theta = 0$ (the particle crosses the horizontal diameter, where $\mathbf{v}_1^P \parallel \boldsymbol{\omega}_{10}$).
- **Relative equilibria** (extension, from $\ddot\theta = 0$ with $\dot\theta = 0$): $\sin\theta\,\left(\omega^2\cos\theta - g/R\right) = 0$, i.e. $\theta = 0, \theta = \pi$, and, if $\omega^2R > g$, $\cos\theta = g/(\omega^2R)$; at all of them $N_b = 0$.
- **Limit $\omega \to 0$:** $N_b \to 0$ and $E_1^P \to \tfrac{1}{2}mR^2\dot\theta^2 - mgR\cos\theta$, the energy of the simple pendulum.
- **Orthogonality:** $\mathbf{e}_r \cdot \mathbf{e}_\theta = \sin\theta\cos\theta - \cos\theta\sin\theta = 0$, and both are normal to $\mathbf{j}_1$.
- **Dimensions:** each term of E_1^P is $\mathrm{kg\,m^2\,s^{-2}} = \mathrm{J}$; $[N_b] = [m\omega R\dot\theta] = \mathrm{kg\,s^{-1}\,m\,s^{-1}} = \mathrm{N}$; inside the root, $[2E_1^P/m] = [gR] = [\omega^2R^2] = \mathrm{m^2\,s^{-2}}$.
- **Verification (sympy):** the inertial double derivative of $[{}_0R_1]\{\mathbf{r}_1^P\}$ equals the Notes Eq. 3.27 decomposition; solving $m\mathbf{a}_0^P = -mg\mathbf{k} + \mathbf{N}$ in S_0 returns the same EOM, N_b and N_r; $dE_1^P/dt = 0$ along the EOM, while dE_0^P/dt equals the power $N_b\,\omega R\sin\theta$ of the reaction in S_0.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 20 in its solution keys, so the results are verified by independent derivation (the sympy check above).

Source: Problems.pdf (PDF page 10); no official key

PROBLEM 21

Apparent Weight on a Spinning Planet

Original statement · Problems.pdf, p. 10

Centrifugal Force

On a spinning spherical planet of radius R and gravitational acceleration g_p, the bodies weight [sic] twice as much on the poles as on the equator. Determine the duration of the day on that planet.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a spherical planet of radius R spins with constant angular velocity Ω about its polar axis. At its surface the gravitational attraction is g_p per unit mass, directed toward the centre. A body of mass m rests on the ground.
- **Why the planet frame.** "Weight" is what a scale fixed to the ground reads: it is defined in the frame of the planet, where the body is at rest. In that frame the dynamics reduce to a static balance once the inertia forces (Notes Eq. 4.62) are added; the centrifugal force is what makes the weight depend on latitude.
- **Hypotheses:**
  1. The frame S_0 with origin at the centre O of the planet and axes fixed with respect to the stars is inertial (O unaccelerated).
  2. Ω is constant; the planet is a rigid sphere and g_p is the same everywhere on its surface (spherical mass distribution).
  3. The apparent weight is the magnitude of the contact force $\mathbf{N}$ the ground (or a scale) exerts on the body at rest.
- **Fixed (inertial) frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, $\mathbf{k}_0$ along the polar axis (towards the north pole).
- **Moving frame:** $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ fixed to the planet, $\mathbf{k}_1 = \mathbf{k}_0$, with $\mathbf{i}_1$ in the equatorial plane and in the meridian plane of the body; S_1 coincides with S_0 at $t = 0$.
- **Radial unit vector:** $\mathbf{e}_r$, from O to the body; deduced in the next section.
- **Coordinates and degrees of freedom:** the body is at rest in S_1 at latitude λ (angle of OP above the equatorial plane): there is no motion to determine; the unknown is Ω (and then the day $T_{\text{day}} = 2\pi/\Omega$).

Symbol	Meaning	SI unit
R	radius of the planet	m
g_p	gravitational acceleration at the surface	m/s^2
Ω	spin rate of the planet	rad/s
λ	latitude of the body	rad
m	mass of the body	kg
T_{day}	duration of the day (one rotation, sidereal)	s

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from the north pole

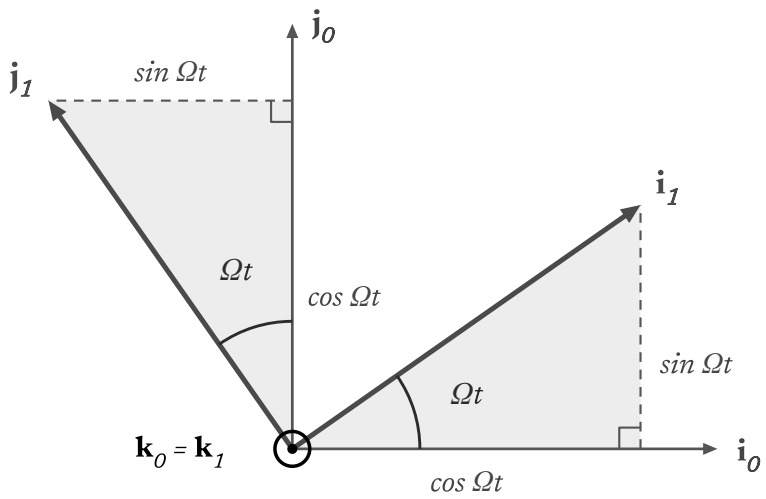


Figure 1. View from above the north pole ($\mathbf{k}_0 = \mathbf{k}_1$ out of the paper, $\odot$). $\mathbf{i}_1$ is $\mathbf{i}_0$ turned counter-clockwise by Ωt (legs $\cos\Omega t$, $\sin\Omega t$); $\mathbf{j}_1$ turns the same angle from $\mathbf{j}_0$, so its opposite leg points along $-\mathbf{i}_0$.

The planet turns about $\mathbf{k}_0$, so the equatorial plane is drawn looking down from $+\mathbf{k}_0$ (from above the north pole): $\mathbf{k}_0 = \mathbf{k}_1$ points out of the paper ($\odot$), $\mathbf{i}_0$ to the right and, by the right-hand rule $\mathbf{i}_0 \times \mathbf{j}_0 = \mathbf{k}_0$, $\mathbf{j}_0$ up the page. With $\Omega > 0$ the planet turns counter-clockwise, and after a time t the angle from $\mathbf{i}_0$ to $\mathbf{i}_1$ is Ωt:

- $\mathbf{i}_1$: hypotenuse 1, adjacent leg $\cos\Omega t$ along $\mathbf{i}_0$, opposite leg $\sin\Omega t$ along $+\mathbf{j}_0$, so $\mathbf{i}_1 = \cos\Omega t\,\mathbf{i}_0 + \sin\Omega t\,\mathbf{j}_0$.
- $\mathbf{j}_1$ turns the same angle from $\mathbf{j}_0$, towards the left: adjacent leg $\cos\Omega t$ along $\mathbf{j}_0$, opposite leg $\sin\Omega t$ along $-\mathbf{i}_0$, so $\mathbf{j}_1 = -\sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{j}_0$ (the minus sign).
- $\mathbf{k}_1 = \mathbf{k}_0$.

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & -\sin\Omega t & 0 \\ \sin\Omega t & \cos\Omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos\Omega t\cos\Omega t - (-\sin\Omega t)\sin\Omega t = \cos^2\Omega t + \sin^2\Omega t = 1$$

The columns have unit length and $-\cos\Omega t\sin\Omega t + \sin\Omega t\cos\Omega t = 0$, so $[{}_0R_1][{}_0R_1]^T = I$.

Motion of S_1: $\boldsymbol{\omega}_{10} = \Omega\mathbf{k}_1$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$, $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$. Poisson check (chain rule): $\frac{d\mathbf{i}_1}{dt}\big|_0 = -\Omega\sin\Omega t\,\mathbf{i}_0 + \Omega\cos\Omega t\,\mathbf{j}_0 = \Omega\,\mathbf{j}_1 = \Omega\,\mathbf{k}_1 \times \mathbf{i}_1$.

The position of the body in its meridian plane

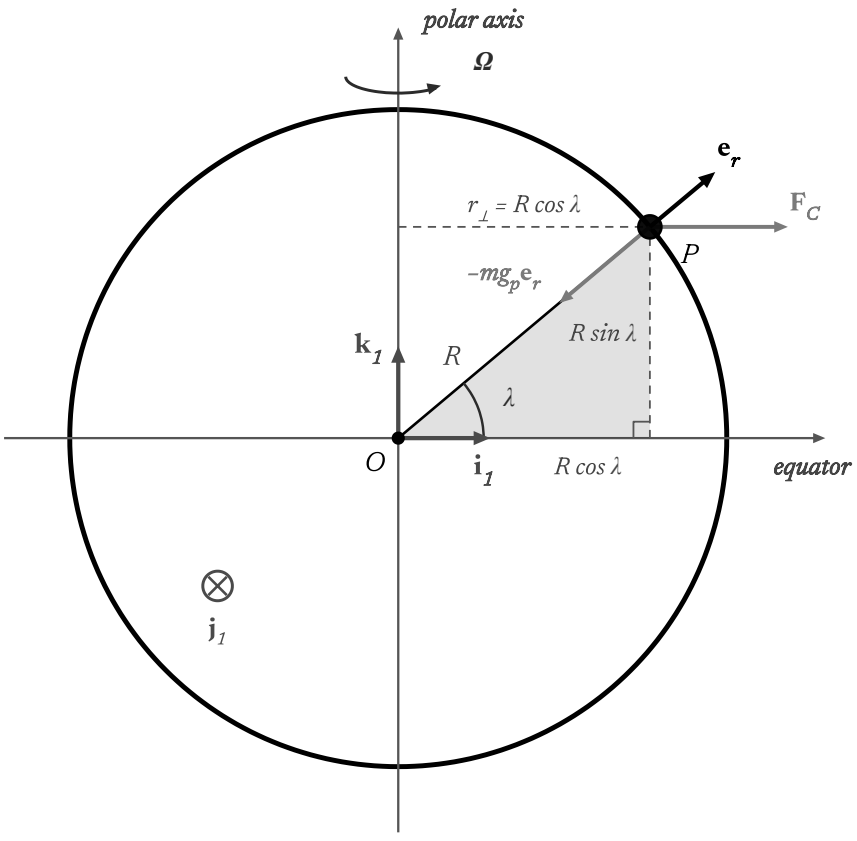


Figure 2. Meridian plane of the body ($\mathbf{i}_1$ right, $\mathbf{k}_1$ up, $\mathbf{j}_1$ into the paper, $\otimes$). In the shaded right triangle OFP the hypotenuse is R, the leg adjacent to λ is $R\cos\lambda$ along $\mathbf{i}_1$ and the opposite leg is $R\sin\lambda$ along $\mathbf{k}_1$. The centrifugal force $m\Omega^2\mathbf{r}_\perp$ is horizontal, away from the axis; the attraction $-mg_p\mathbf{e}_r$ points to O.

The meridian plane of the body contains $\mathbf{i}_1$ and $\mathbf{k}_1$; it is drawn with $\mathbf{i}_1$ to the right and $\mathbf{k}_1$ up, so $\mathbf{j}_1 = \mathbf{k}_1 \times \mathbf{i}_1$ points into the paper ($\otimes$; fingers from "up" to "right", clockwise, thumb away from the reader).

P is on the surface, at distance R from the centre: $\mathbf{r}_1^P = \overrightarrow{OP} = R\,\mathbf{e}_r$. Let F be the foot of the perpendicular from P to the equatorial line. In the shaded right triangle OFP (right angle at F) the hypotenuse is $|OP| = R$, the leg adjacent to λ is $OF = R\cos\lambda$ along $\mathbf{i}_1$ and the leg opposite to λ is $FP = R\sin\lambda$ along $\mathbf{k}_1$:

$$\mathbf{r}_1^P = R\,\mathbf{e}_r, \qquad \mathbf{e}_r = \cos\lambda\,\mathbf{i}_1 + \sin\lambda\,\mathbf{k}_1$$

(here the latitude is the usual polar angle of the meridian plane, measured from $\mathbf{i}_1$ towards $\mathbf{k}_1$, so no shift is needed). The part of $\mathbf{r}_1^P$ normal to the axis is the horizontal dashed segment from the axis to P, of length equal to the leg OF:

$$\mathbf{r}_\perp = R\cos\lambda\,\mathbf{i}_1$$

Checks: $\lambda = \pi/2$ gives $R\,\mathbf{k}_1$ (north pole, $\mathbf{r}_\perp = \mathbf{0}$); $\lambda = 0$ gives $R\,\mathbf{i}_1$ (equator, $r_\perp = R$).

Kinematics

The body is at rest in S_1: λ and R are constant and $\mathbf{i}_1, \mathbf{k}_1$ are fixed in S_1, so $\mathbf{v}_1^P = \mathbf{0}$ and $\mathbf{a}_1^P = \mathbf{0}$: there are no relative-motion terms and no Coriolis force. (Its absolute acceleration, by Notes Eq. 3.27, is only the centripetal term $\boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P) = -\Omega^2 R\cos\lambda\,\mathbf{i}_1$.)

Forces and Equations of Motion

1. Forces on the body in the planet frame

> **Pedagogical Justification:** the body is at rest in S_1, so $\mathbf{v}_1^P = \mathbf{0}$ and $\mathbf{a}_1^P = \mathbf{0}$. In Notes Eq. 4.63, $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$, the Coriolis force vanishes ($\mathbf{v}_1^P = \mathbf{0}$), $\mathbf{F}_A = \mathbf{0}$ and $\mathbf{F}_E = \mathbf{0}$; only the centrifugal force remains.

Actual forces: attraction $-mg_p\,\mathbf{e}_r$ and contact force $\mathbf{N}$. Centrifugal force (Slide 30), with $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$, $\mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$:

$$\mathbf{F}_C = -m\,\Omega\mathbf{k}_1 \times (\Omega\mathbf{k}_1 \times R\,\mathbf{e}_r) = -m\,\Omega\mathbf{k}_1 \times (R\Omega\cos\lambda\,\mathbf{j}_1) = m\Omega^2 R\cos\lambda\,\mathbf{i}_1 = m\Omega^2\,\mathbf{r}_\perp$$

It is horizontal and points away from the axis, as drawn.

Static balance in S_1

The equation of motion is the static balance of Notes Eq. 4.63 with $\mathbf{a}_1^P = \mathbf{0}$:

$$\mathbf{0} = -mg_p\,\mathbf{e}_r + \mathbf{N} + m\Omega^2 R\cos\lambda\,\mathbf{i}_1 \quad\Longrightarrow\quad \mathbf{N} = mg_p\,(\cos\lambda\,\mathbf{i}_1 + \sin\lambda\,\mathbf{k}_1) - m\Omega^2 R\cos\lambda\,\mathbf{i}_1$$

Answers and Checks

2. Weight at the pole and at the equator

- **Pole** ($\lambda = \pi/2$): $\cos\lambda = 0$, $\mathbf{r}_\perp = \mathbf{0}$, no centrifugal force:

$$\mathbf{N}_{\text{pole}} = mg_p\,\mathbf{k}_1, \qquad W_{\text{pole}} = mg_p$$

- **Equator** ($\lambda = 0$): $\mathbf{e}_r = \mathbf{i}_1$ and the centrifugal force is radial and outward:

$$\mathbf{N}_{\text{eq}} = m\left(g_p - \Omega^2 R\right)\mathbf{i}_1, \qquad W_{\text{eq}} = m\left(g_p - \Omega^2 R\right)$$

A body can rest on the equator only if $N_{\text{eq}} \geq 0$ (the ground can push but not pull), i.e. $\Omega^2 R \leq g_p$.

3. Condition $W_{\text{pole}} = 2W_{\text{eq}}$ and length of the day

$$mg_p = 2m\left(g_p - \Omega^2 R\right) \quad\Longrightarrow\quad 2\Omega^2 R = 2g_p - g_p = g_p \quad\Longrightarrow\quad \Omega = \sqrt{\frac{g_p}{2R}}$$

(this satisfies $\Omega^2 R = g_p/2 \leq g_p$. The other possibility, $mg_p = 2m(\Omega^2 R - g_p)$, would mean $\Omega^2 R > g_p$: equatorial bodies would fly off, so it is rejected.) The day is the time of one rotation:

$$\boxed{T_{\text{day}} = \frac{2\pi}{\Omega} = 2\pi\sqrt{\frac{2R}{g_p}}}$$

Checks

- **Meaning.** The centrifugal force at the equator cancels half of the attraction, so $\Omega^2 R = g_p/2$: the planet spins at $1/\sqrt{2} \approx 0.71$ of the break-up rate $\sqrt{g_p/R}$ at which equatorial bodies would float ($W_{\text{eq}} = 0$).
- **Any latitude** (extension, same balance): $|\mathbf{N}|^2 = m^2\left[g_p^2 - \left(2g_p\Omega^2 R - \Omega^4 R^2\right)\cos^2\lambda\right]$, which with $\Omega^2 R = g_p/2$ gives $W(\lambda) = mg_p\sqrt{1 - \tfrac{3}{4}\cos^2\lambda}$: from $mg_p/2$ at the equator to mg_p at the poles (e.g. $mg_p\sqrt{10}/4 \approx 0.79\,mg_p$ at 45°). Away from the poles and the equator $\mathbf{N}$ is not radial: the plumb line does not point to the centre.
- **Numbers** (illustration only, Earth-like $R = 6.371 \times 10^6$ m, $g_p = 9.81$ m/s^2): $T_{\text{day}} \approx 7.16 \times 10^3$ s ≈ 1.99 h.
- **Limit:** a slower spin ($\Omega \to 0$) gives $W_{\text{eq}} \to W_{\text{pole}}$; the ratio 2 requires a fast rotation.
- **Orthogonality:** $\mathbf{e}_r \cdot \mathbf{j}_1 = 0$ and $|\mathbf{e}_r| = \sqrt{\cos^2\lambda + \sin^2\lambda} = 1$.
- **Dimensions:** $[\Omega^2 R] = \mathrm{s}^{-2}\,\mathrm{m} = [g_p]$; $[T_{\text{day}}] = \sqrt{\mathrm{m}/(\mathrm{m\,s^{-2}})} = \mathrm{s}$.
- **Verification (sympy):** the inertial acceleration of a point at rest on the planet, obtained by differentiating $[{}_0R_1]\{\mathbf{r}_1^P\}$ twice, equals $\boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P)$; the balance gives $W_{\text{pole}} = mg_p$, $W_{\text{eq}} = m|g_p - \Omega^2 R|$, $\Omega = \sqrt{g_p/(2R)}$ and $T_{\text{day}} = 2\pi\sqrt{2R/g_p}$.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 21 in its solution keys, so the results are verified by independent derivation (the sympy check above).

Source: Problems.pdf (PDF page 10); no official key

PROBLEM 22

Particle on a Curve

Original statement · Problems.pdf, p. 10

1D Curve Constraint

A heavy point particle P of mass m is constrained to move without friction along the curve:

$$y = a\left(1 - \cos\frac{x}{a}\right) \tag{4}$$

in the range $-\pi a \leq x \leq \pi a$, where a is a positive constant. Initially, the particle is located at $x = 0$ with velocity v_0 along the positive x axis.

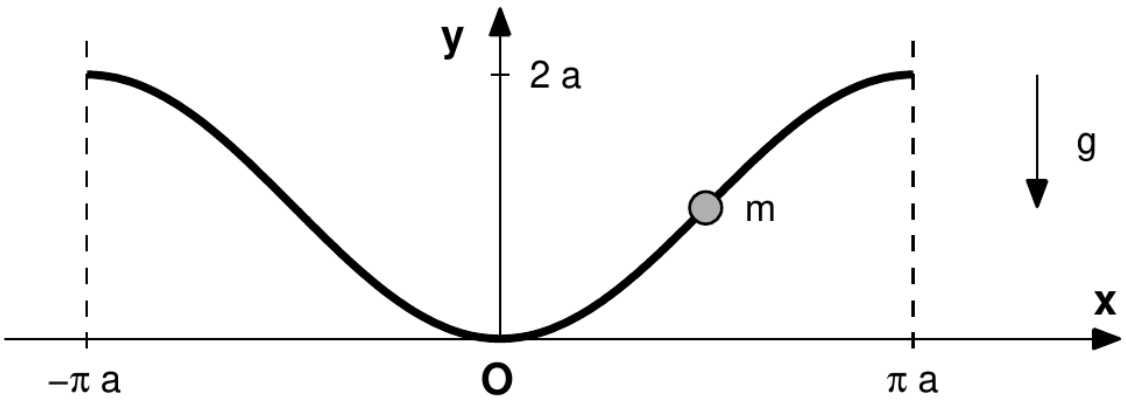


Figure: Heavy particle m on the curve y = a(1 - cos(x/a)) between x = -pi a and x = pi a, gravity g downward (Problems.pdf, p. 10).

(a) Reduce the problem to integrals.

(b) If $v_0 = \sqrt{ga}$, determine the minimum and maximum value of x along the motion.

(c) What should be the minimum value of v_0 in order to reach $x = \pi a$?

Let's assume now that the curve containing the point particle rotates around the OY axis with constant angular velocity $\boldsymbol{\omega} = \Omega\mathbf{j}$.

(d) Define a convenient moving reference frame with origin in O, $OX_1Y_1Z_1$. Determine the inertia force that acts on the particle in this moving reference frame as a function of m, Ω, x_1 and $\dot{x}_1$, where x_1 represents the position coordinate of P along OX_1.

(e) Reduce the problem to quadratures and determine the value of the reaction force.

(f) Find out a trascendental [sic] equation whose solutions are the equilibrium positions of the point particle in $OX_1Y_1Z_1$.

(g) If Ω were a function of time, would it be possible to use the energy equation in the same way as before? Justify your answer.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a particle P of mass m moves without friction along the smooth planar curve

$$y = a\left(1 - \cos\frac{x}{a}\right), \qquad x \in [-\pi a, \pi a]$$

in a vertical plane under uniform downward gravity $\mathbf{g} = -g\,\mathbf{j}$. In parts (a)–(c) the curve is fixed; in parts (d)–(g) it turns about the vertical axis OY with constant angular velocity $\boldsymbol{\omega} = \Omega\,\mathbf{j}$.

- **Hypotheses:** the frame $OXYZ$ with OY vertical upwards is inertial; the curve is smooth, so its reaction has no component along the curve; in (d)–(g) the rotation is imposed (whatever torque is needed is supplied).
- **Initial state ($t = 0$):** $x(0) = 0 \implies y(0) = a(1 - \cos 0) = 0$, with initial velocity $\mathbf{v}_0 = v_0\,\mathbf{i}$ ($v_0 > 0$).
- **Fixed (inertial) frame:** $S_0 = OXYZ = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$, basis $\mathcal{B}_0$, with $\mathbf{j}$ vertical upwards and the curve in the plane OXY at $t = 0$.
- **Moving frame (parts d to g):** $S_1 = OX_1Y_1Z_1 = \{O; \mathbf{i}_1, \mathbf{j}, \mathbf{k}_1\}$, basis $\mathcal{B}_1$, attached to the curve: OX_1 in the plane of the curve, $OY_1 \equiv OY$ (the axis of rotation, so $\mathbf{j}_1 = \mathbf{j}$) and OZ_1 normal to the plane of the curve. S_1 coincides with S_0 at $t = 0$.

- **Bases attached to the curve:** the unit tangent $\mathbf{e}_t$ (in the sense of increasing abscissa) and the in-plane unit normal $\mathbf{n}$ at P; the third vector is the normal to the plane of the curve ($\mathbf{k}$, or $\mathbf{k}_1$ when the curve turns). They are deduced in the next section.
- **Coordinates and degrees of freedom:** the particle has 3 coordinates; the curve imposes two holonomic constraints, $z = 0$ (plane of the curve) and $y = a(1 - \cos(x/a))$ (in S_1: $z_1 = 0$ and $y_1 = a(1 - \cos(x_1/a))$). CDOF $= 3 - 2 = 1$, parametrized by the abscissa $x(t)$ in (a)–(c) and by $x_1(t)$, the position of P along OX_1, in (d)–(g). In S_0 the constraints of the rotating curve are rheonomic (they turn with Ωt), but the count is the same.

Symbol	Meaning	SI unit
m	mass of P	kg
a	length scale of the curve	m
g	gravitational acceleration	m/s^2
v_0	initial speed at O	m/s
Ω	angular velocity of the curve (constant, parts d–f)	rad/s
x, x_1	abscissa of P (fixed curve, rotating curve)	m

Change of Basis and Position Vector

Position vector and the slope triangle

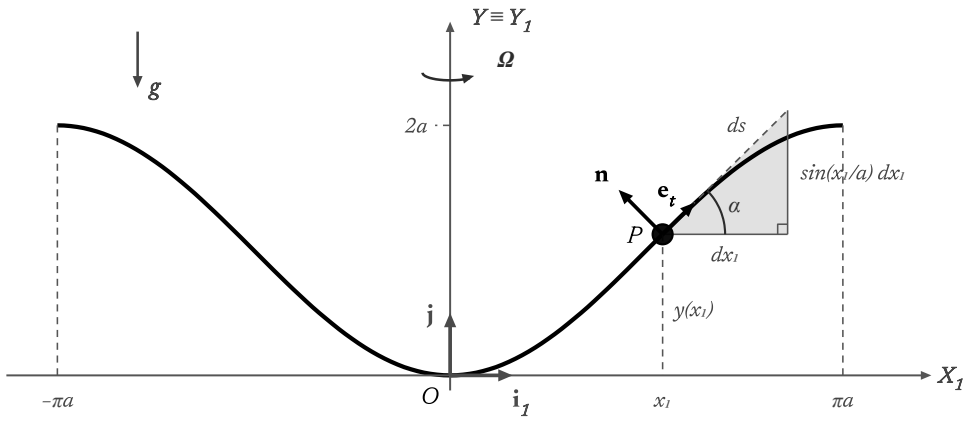


Figure 1. Plane of the curve (fixed in S_0 for (a)–(c), fixed in S_1 for (d)–(g)). The dashed lines are the components x_1 and $y(x_1) = a(1 - \cos(x_1/a))$ of the position. In the shaded slope triangle the leg adjacent to the tangent angle α is dx_1, the opposite leg is $dy = \sin(x_1/a)\,dx_1$ and the hypotenuse is the arc element ds. $\mathbf{i}_1, \mathbf{j}$ in blue, $\mathbf{e}_t, \mathbf{n}$ in orange.

The plane of the curve is drawn with the horizontal axis to the right and $\mathbf{j}$ up. For the fixed curve the axis is OX with $\mathbf{i}$; for the rotating curve the same drawing holds in S_1, with OX_1, $\mathbf{i}_1$ and x_1. The position of P is the sum of its two dashed components, the abscissa along the horizontal axis and the height given by the equation of the curve:

$$\mathbf{r}^P = x\,\mathbf{i} + a\left(1-\cos\frac{x}{a}\right)\mathbf{j} \quad \text{(fixed curve)}, \qquad \mathbf{r}_1^P = x_1\,\mathbf{i}_1 + a\left(1-\cos\frac{x_1}{a}\right)\mathbf{j} \quad (\text{in } S_1)$$

Checks: $x=0$ gives $\mathbf{r}^P=\mathbf{0}$ (the initial position O); $x=\pm\pi a$ gives the height $a(1-\cos(\pm\pi)) = a(1-(-1)) = 2a$, the two crests.

Slope of the curve. By the chain rule, $\frac{d}{dx}\cos\frac{x}{a} = -\sin\frac{x}{a}\cdot\frac{d}{dx}\left(\frac{x}{a}\right) = -\frac{1}{a}\sin\frac{x}{a}$, so

$$\frac{dy}{dx} = \frac{d}{dx}\left[a\left(1-\cos\frac{x}{a}\right)\right] = a\cdot\frac{1}{a}\sin\frac{x}{a} = \sin\left(\frac{x}{a}\right)$$

Slope triangle. Moving a distance dx along the curve, the height changes by $dy = \sin(x/a)\,dx$. In the shaded right triangle of the figure the leg adjacent to the angle α of the tangent is dx, the leg opposite to α is $dy=\sin(x/a)\,dx$ and the hypotenuse is the arc element ds. Pythagoras' theorem gives

$$ds^2 = dx^2 + dy^2 = dx^2 + \sin^2\left(\frac{x}{a}\right)dx^2 = \left[1+\sin^2\left(\frac{x}{a}\right)\right]dx^2 \;\Rightarrow\; ds = \sqrt{1+\sin^2\left(\frac{x}{a}\right)}\,dx$$

and, from the definitions of tangent, cosine and sine in the same triangle,

$$\tan\alpha = \frac{dy}{dx} = \sin\frac{x}{a}, \qquad \cos\alpha = \frac{dx}{ds} = \frac{1}{\sqrt{1+\sin^2(x/a)}}, \qquad \sin\alpha = \frac{dy}{ds} = \frac{\sin(x/a)}{\sqrt{1+\sin^2(x/a)}}$$

Unit tangent and in-plane normal. The unit tangent points along the hypotenuse, at the angle α from the horizontal axis: $\mathbf{e}_t = \cos\alpha\,\mathbf{i} + \sin\alpha\,\mathbf{j}$. The plane is seen with $\mathbf{k}=\mathbf{i}\times\mathbf{j}$ out of the paper ($\odot$), so a counter-clockwise turn of $+90^\circ$ in the drawing is a positive rotation about $\mathbf{k}$; it takes $(\cos\alpha,\sin\alpha)$ to $(-\sin\alpha,\cos\alpha)$ and gives the in-plane normal $\mathbf{n}$:

$$\mathbf{e}_t = \frac{\mathbf{i}+\sin(x/a)\,\mathbf{j}}{\sqrt{1+\sin^2(x/a)}}, \qquad \mathbf{n} = \frac{-\sin(x/a)\,\mathbf{i}+\mathbf{j}}{\sqrt{1+\sin^2(x/a)}}$$

Checks: $|\mathbf{e}_t|^2 = |\mathbf{n}|^2 = \dfrac{1+\sin^2(x/a)}{1+\sin^2(x/a)} = 1$; $\mathbf{e}_t \cdot \mathbf{n} = \dfrac{-\sin(x/a)+\sin(x/a)}{1+\sin^2(x/a)} = 0$; $\mathbf{e}_t \times \mathbf{n} = \dfrac{1\cdot 1\,(\mathbf{i}\times\mathbf{j}) - \sin^2(x/a)\,(\mathbf{j}\times\mathbf{i})}{1+\sin^2(x/a)} = \mathbf{k}$, so $\{\mathbf{e}_t, \mathbf{n}, \mathbf{k}\}$ is right-handed. In S_1 the same formulas hold with x_1, $\mathbf{i}_1$ and $\mathbf{k}_1$.

The basis $\mathcal{B}_1$ seen from above (parts d to g)

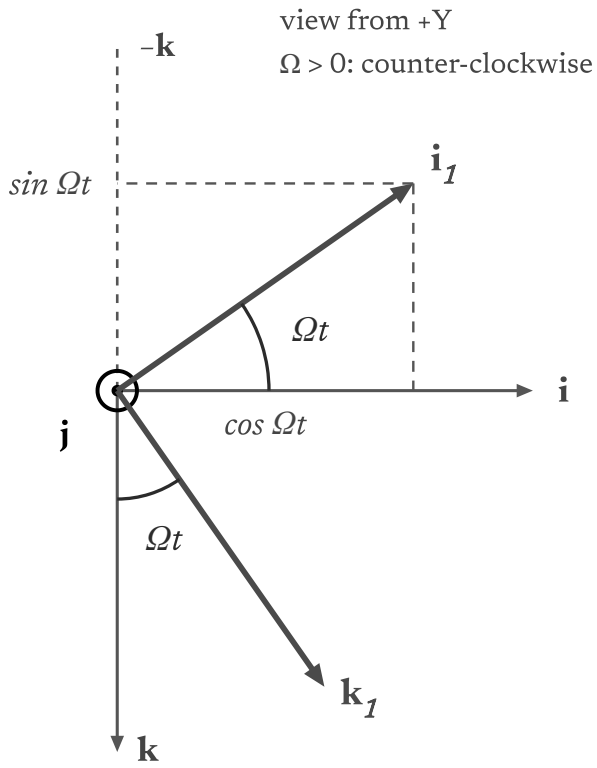


Figure 2. View from $+Y$, with $\mathbf{j}$ out of the paper ($\odot$): $\mathbf{k} = \mathbf{i}\times\mathbf{j}$ points down the page. The plane of the curve contains $\mathbf{i}_1$ and $\mathbf{j}$ and turns counter-clockwise by Ωt: the components of $\mathbf{i}_1$ are $\cos\Omega t$ along $\mathbf{i}$ and $\sin\Omega t$ along $-\mathbf{k}$ (up the page), hence the minus sign in $\mathbf{i}_1 = \cos\Omega t\,\mathbf{i} - \sin\Omega t\,\mathbf{k}$.

The curve turns about $\mathbf{j}$, so the rotation is drawn looking down the axis, from $+Y$. In that view $\mathbf{j}$ points out of the paper, towards the reader ($\odot$). With $\mathbf{i}$ to the right, the right-hand rule $\mathbf{k} = \mathbf{i}\times\mathbf{j}$ (fingers from $\mathbf{i}$, to the right, towards $\mathbf{j}$, towards the reader) puts the thumb, $\mathbf{k}$, down the page; $-\mathbf{k}$ points up the page.

With $\boldsymbol{\omega} = \Omega\,\mathbf{j}$ and $\Omega > 0$ the plane of the curve turns counter-clockwise in this view (thumb along $\mathbf{j}$, towards the reader). After a time t the axis $\mathbf{i}_1$ has turned the angle Ωt from $\mathbf{i}$:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of the dashed right triangle. The leg adjacent to Ωt lies along $\mathbf{i}$ and measures $\cos\Omega t$; the leg opposite to Ωt points up the page, the direction $-\mathbf{k}$, and measures $\sin\Omega t$. Hence $\mathbf{i}_1 = \cos\Omega t\,\mathbf{i} + \sin\Omega t\,(-\mathbf{k})$: **the minus sign comes from $\mathbf{k}$ pointing down the page in this view.**

- $\mathbf{k}_1$ starts on $\mathbf{k}$ (down the page) and turns the same angle counter-clockwise, so its tip moves towards $+\mathbf{i}$: its leg along $\mathbf{k}$ (adjacent) is $\cos \Omega t$ and its leg along $\mathbf{i}$ (opposite) is $+\sin \Omega t$.
- $\mathbf{j}_1 = \mathbf{j}$, the axis of the rotation.

$$\mathbf{i}_1 = \cos \Omega t\, \mathbf{i} - \sin \Omega t\, \mathbf{k}, \qquad \mathbf{j}_1 = \mathbf{j}, \qquad \mathbf{k}_1 = \sin \Omega t\, \mathbf{i} + \cos \Omega t\, \mathbf{k}$$

The columns of the change-of-basis matrix are these components:

$$[{}_0R_1] = \begin{pmatrix} \cos \Omega t & 0 & \sin \Omega t \\ 0 & 1 & 0 \\ -\sin \Omega t & 0 & \cos \Omega t \end{pmatrix}, \qquad \det [{}_0R_1] = \cos \Omega t \cos \Omega t - \sin \Omega t\,(-\sin \Omega t) = \cos^2 \Omega t + \sin^2 \Omega t = 1$$

Each column has unit length ($\cos^2 \Omega t + \sin^2 \Omega t = 1$) and the first and third columns are orthogonal ($\cos \Omega t \sin \Omega t - \sin \Omega t \cos \Omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.

Motion of S_1. The origin is fixed ($\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$), $\boldsymbol{\omega}_{10} = \Omega\, \mathbf{j}$ and $\boldsymbol{\alpha}_{10} = \dot{\Omega}\, \mathbf{j} = \mathbf{0}$ for constant Ω. Check with Poisson's formula: by the chain rule, $\frac{d\mathbf{i}_1}{dt}\big|_0 = -\Omega \sin \Omega t\, \mathbf{i} - \Omega \cos \Omega t\, \mathbf{k} = -\Omega\, \mathbf{k}_1$, and $\boldsymbol{\omega}_{10} \times \mathbf{i}_1 = \Omega\, \mathbf{j} \times \mathbf{i}_1 = -\Omega\, \mathbf{k}_1$.

Kinematics

Fixed curve (parts a to c). The basis $\mathcal{B}_0$ is constant. The height depends on time through $x(t)$, so by the chain rule $\dot{y} = \frac{dy}{dx}\, \dot{x} = \sin(x/a)\, \dot{x}$:

$$\mathbf{v}^P = \dot{x}\, \mathbf{i} + \sin \frac{x}{a}\, \dot{x}\, \mathbf{j}, \qquad v = \frac{ds}{dt} = \frac{ds}{dx}\, \dot{x} = \sqrt{1 + \sin^2\left(\frac{x}{a}\right)}\, \dot{x}, \qquad \mathbf{v}^P = v\, \mathbf{e}_t$$

Differentiating again, product rule on $\sin(x/a)\, \dot{x}$ with the chain rule $\frac{d}{dt} \sin \frac{x}{a} = \frac{1}{a} \cos \frac{x}{a}\, \dot{x}$:

$$\mathbf{a}^P = \ddot{x}\, \mathbf{i} + \left(\sin \frac{x}{a}\, \ddot{x} + \frac{1}{a} \cos \frac{x}{a}\, \dot{x}^2 \right) \mathbf{j}$$

Rotating curve (parts d to g). In S_1 the vectors $\mathbf{i}_1, \mathbf{j}$ are constant and only the components of $\mathbf{r}_1^P$ change; the same chain and product rules give

$$\mathbf{v}_1^P = \dot{x}_1\,\mathbf{i}_1 + \sin\frac{x_1}{a}\,\dot{x}_1\,\mathbf{j}, \qquad \mathbf{a}_1^P = \ddot{x}_1\,\mathbf{i}_1 + \left(\sin\frac{x_1}{a}\,\ddot{x}_1 + \frac{\dot{x}_1^2}{a}\cos\frac{x_1}{a}\right)\mathbf{j}$$

Projections used below, with $\sqrt{\cdot} \equiv \sqrt{1+\sin^2(x_1/a)}$: $\mathbf{a}_1^P \cdot \mathbf{e}_t = \dfrac{\left(1+\sin^2\frac{x_1}{a}\right)\ddot{x}_1 + \frac{1}{a}\sin\frac{x_1}{a}\cos\frac{x_1}{a}\,\dot{x}_1^2}{\sqrt{\cdot}}$ and $\mathbf{a}_1^P \cdot \mathbf{n} = \dfrac{-\sin\frac{x_1}{a}\,\ddot{x}_1 + \sin\frac{x_1}{a}\,\ddot{x}_1 + \frac{\dot{x}_1^2}{a}\cos\frac{x_1}{a}}{\sqrt{\cdot}} = \dfrac{\dot{x}_1^2\cos(x_1/a)}{a\sqrt{\cdot}}$ (the same with x for the fixed curve).

Forces and Equations of Motion

Real forces. The weight $\mathbf{W} = -mg\,\mathbf{j}$ (conservative, $V = mgy$) and the reaction $\mathbf{N}$ of the smooth curve, perpendicular to $\mathbf{e}_t$, so $\mathbf{N}\cdot\mathbf{v} = 0$. With the projections $\mathbf{j}\cdot\mathbf{e}_t = \sin\alpha = \dfrac{\sin(x/a)}{\sqrt{1+\sin^2(x/a)}}$ and $\mathbf{j}\cdot\mathbf{n} = \cos\alpha = \dfrac{1}{\sqrt{1+\sin^2(x/a)}}$.

Equation of motion, fixed curve.

> **Pedagogical Justification:** projecting Newton's second law $m\mathbf{a}^P = \mathbf{W} + \mathbf{N}$ on $\mathbf{e}_t$ removes $\mathbf{N}$.

$m\,\mathbf{a}^P\cdot\mathbf{e}_t = \mathbf{W}\cdot\mathbf{e}_t$; multiplying both sides by $\sqrt{1+\sin^2(x/a)}$ and dividing by m:

$$\left(1+\sin^2\frac{x}{a}\right)\ddot{x} + \frac{1}{a}\sin\frac{x}{a}\cos\frac{x}{a}\,\dot{x}^2 + g\sin\frac{x}{a} = 0$$

The energy integral of part (a) is the first integral of this equation (work-energy theorem). Check: differentiating $\frac{1}{2}m(1+\sin^2(x/a))\dot{x}^2 + mga(1-\cos(x/a)) = \text{const}$ with the product rule and the chain rule ($\frac{d}{dt}\sin^2\frac{x}{a} = \frac{2}{a}\sin\frac{x}{a}\cos\frac{x}{a}\,\dot{x}$, $\frac{d}{dt}\cos\frac{x}{a} = -\frac{1}{a}\sin\frac{x}{a}\,\dot{x}$) gives $m\left(1+\sin^2\frac{x}{a}\right)\dot{x}\ddot{x} + \frac{m}{a}\sin\frac{x}{a}\cos\frac{x}{a}\,\dot{x}^3 + mg\sin\frac{x}{a}\,\dot{x} = 0$, which is $m\dot{x}$ times the equation above.

(d) Inertia forces in the moving frame

> **Pedagogical Justification:** in the non-inertial frame S_1, $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63) with the inertia forces of Notes Eq. 4.62 (Slide 28). The origin is fixed and Ω is constant, so the force of the acceleration of the origin and the Euler force vanish ($\mathbf{F}_A = \mathbf{F}_E = \mathbf{0}$).

1. Centrifugal force. With $\mathbf{j}\times\mathbf{i}_1 = -\mathbf{k}_1$, $\mathbf{j}\times\mathbf{j} = \mathbf{0}$ and $\mathbf{j}\times\mathbf{k}_1 = \mathbf{i}_1$:

$$\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P=\Omega\mathbf{j}\times(x_1\mathbf{i}_1+y\mathbf{j})=-\Omega x_1\,\mathbf{k}_1,\qquad \mathbf{F}_{cf}=-m\boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P)=-m\,\Omega\mathbf{j}\times(-\Omega x_1\mathbf{k}_1)=m\Omega^2x_1\mathbf{i}_1$$

2. Coriolis force:

$$\mathbf{F}_{Cor}=-2m\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P=-2m(\Omega\mathbf{j})\times(\dot{x}_1\mathbf{i}_1+\dot{y}_1\mathbf{j})=-2m\Omega\dot{x}_1\,(\mathbf{j}\times\mathbf{i}_1)=2m\Omega\dot{x}_1\,\mathbf{k}_1$$

The total inertia force is $\mathbf{F}_i=m\Omega^2x_1\,\mathbf{i}_1+2m\Omega\dot{x}_1\,\mathbf{k}_1$. Its projections: $\mathbf{F}_{cf}\cdot\mathbf{e}_t=\dfrac{m\Omega^2x_1}{\sqrt{\cdot}}$, $\mathbf{F}_{cf}\cdot\mathbf{n}=-\dfrac{m\Omega^2x_1\sin(x_1/a)}{\sqrt{\cdot}}$, and $\mathbf{F}_{Cor}$ is along $\mathbf{k}_1$, normal to the plane of the curve.

Equations of motion in S_1. The reaction of the smooth curve is $\mathbf{N}=N_n\,\mathbf{n}+N_z\,\mathbf{k}_1$ (no tangential component). Projections of $m\mathbf{a}_1^P=\mathbf{W}+\mathbf{N}+\mathbf{F}_{cf}+\mathbf{F}_{Cor}$:

- on $\mathbf{e}_t$ (multiplied by $\sqrt{\cdot}/m$), the tangential projection of the centrifugal force adds Ω^2x_1 to the right-hand side:

$$\left(1+\sin^2\frac{x_1}{a}\right)\ddot{x}_1+\frac{1}{a}\sin\frac{x_1}{a}\cos\frac{x_1}{a}\,\dot{x}_1^2+g\sin\frac{x_1}{a}=\Omega^2x_1$$

- on $\mathbf{n}$: $\dfrac{m\dot{x}_1^2\cos(x_1/a)}{a\sqrt{\cdot}}=N_n-\dfrac{mg}{\sqrt{\cdot}}-\dfrac{m\Omega^2x_1\sin(x_1/a)}{\sqrt{\cdot}}$;
- on $\mathbf{k}_1$: $0=N_z+2m\Omega\dot{x}_1$ ($\mathbf{a}_1^P$ and $\mathbf{W}$ have no $\mathbf{k}_1$ component).

Answers and Checks

Part (a): Reduction of Motion to Integrals

> **Pedagogical Justification:** The frictionless constraint does zero work ($W_N=0$). Gravity is conservative with potential energy $V(y)=mgy=mga(1-\cos(x/a))$. Total mechanical energy is constant.

$$E=\frac{1}{2}mv^2+mgy=\frac{1}{2}mv_0^2+mg(0)=\frac{1}{2}mv_0^2$$

$$\frac{1}{2}m\left[1+\sin^2\left(\frac{x}{a}\right)\right]\dot{x}^2+mga\left(1-\cos\frac{x}{a}\right)=\frac{1}{2}mv_0^2$$

Multiply by $2/m$:

$$\left[1+\sin^2\left(\frac{x}{a}\right)\right]\dot{x}^2=v_0^2-2ga\left(1-\cos\frac{x}{a}\right)$$

Isolate $\dot{x}=\frac{dx}{dt}$:

$$\frac{dx}{dt}=\pm\sqrt{\frac{v_0^2-2ga(1-\cos(x/a))}{1+\sin^2(x/a)}}$$

Separation of variables and Barrow's rule from $x(0)=0$ yield the quadrature:

$$\int_0^{x(t)}\sqrt{\frac{1+\sin^2(x'/a)}{v_0^2-2ga(1-\cos(x'/a))}}\,dx'=\pm t$$

Part (b): Turning Points for $v_0=\sqrt{ga}$

> **Pedagogical Justification:** Turning points occur where the velocity vanishes ($v=0\iff\dot{x}=0$).

$$v_0^2-2ga\left(1-\cos\frac{x}{a}\right)=0$$

Substitute $v_0^2=ga$:

$$ga-2ga\left(1-\cos\frac{x}{a}\right)=0$$

Divide by ga:

$$1 - 2\left(1 - \cos\frac{x}{a}\right) = 0 \implies 2\cos\frac{x}{a} = 1 \implies \cos\frac{x}{a} = \frac{1}{2}$$

In the domain $x/a \in [-\pi, \pi]$:

$$\frac{x}{a} = \pm\frac{\pi}{3} \implies x_{min} = -\frac{\pi}{3}a, \qquad x_{max} = +\frac{\pi}{3}a$$

Part (c): Minimum Velocity v_0 to Reach $x = \pi a$

> **Pedagogical Justification:** At the crest $x = \pi a$, the height is $y(\pi a) = a(1 - \cos\pi) = a(1 - (-1)) = 2a$. For the particle to reach this crest, its kinetic energy at the peak must be non-negative ($T(\pi a) \geq 0$).

$$E \geq V(\pi a) \implies \frac{1}{2}mv_0^2 \geq mg(2a) = 2mga$$

$$v_0^2 \geq 4ga \implies v_{0,min} = \sqrt{4ga} = 2\sqrt{ga}$$

At exactly $v_0 = 2\sqrt{ga}$ the crest is a maximum of V (zero force there), so it is reached only asymptotically ($t \to \infty$); for $v_0 > 2\sqrt{ga}$ it is reached in finite time. $2\sqrt{ga}$ is therefore the infimum of the admissible values.

Part (d): Moving Reference Frame and Inertia Force

The moving frame is $S_1 = \{O; \mathbf{i}_1, \mathbf{j}, \mathbf{k}_1\}$, attached to the curve and turning with $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}$; the position of P in it is $\mathbf{r}_1^P = x_1\mathbf{i}_1 + y(x_1)\mathbf{j}$ and its relative velocity $\mathbf{v}_1^P = \dot{x}_1\mathbf{i}_1 + \sin(x_1/a)\dot{x}_1\mathbf{j}$. The inertia force is

$$\mathbf{F}_i = m\Omega^2 x_1\,\mathbf{i}_1 + 2m\Omega\dot{x}_1\,\mathbf{k}_1$$

(centrifugal plus Coriolis; $\mathbf{F}_A = \mathbf{F}_E = \mathbf{0}$).

Part (e): Quadratures and Reaction Force in the Moving Frame

- Effective potential in S_1:

$$V_{eff}(x_1) = mgy(x_1) - \frac{1}{2}m\Omega^2 x_1^2 = mga\left(1 - \cos\frac{x_1}{a}\right) - \frac{1}{2}m\Omega^2 x_1^2$$

- Energy conservation in S_1 (the curve is fixed and smooth in S_1, so $\mathbf{N}\cdot\mathbf{v}_1^P = 0$; $\mathbf{F}_{Cor}\cdot\mathbf{v}_1^P = 0$ always; gravity is conservative and $V_{cf} = -\frac{1}{2}m\Omega^2 x_1^2$ does not depend on time because Ω is constant):

$$E' = \frac{1}{2}m\left[1 + \sin^2(x_1/a)\right]\dot{x}_1^2 + mga\left(1 - \cos\frac{x_1}{a}\right) - \frac{1}{2}m\Omega^2 x_1^2 = \text{constant}$$

- **Initial value.** At $x_1 = 0$ the position is $\mathbf{0}$, so $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = \mathbf{0}$ and the relative speed equals v_0: $E' = \frac{1}{2}mv_0^2$. Solving for $\dot{x}_1$ and separating variables (Barrow's rule from $x_1(0) = 0$):

$$\dot{x}_1^2 = \frac{v_0^2 - 2ga\left(1 - \cos\frac{x_1}{a}\right) + \Omega^2 x_1^2}{1 + \sin^2(x_1/a)} \implies \pm t = \int_0^{x_1}\sqrt{\frac{1 + \sin^2(\xi/a)}{v_0^2 - 2ga\left(1 - \cos(\xi/a)\right) + \Omega^2\xi^2}}\,d\xi$$

- **Reaction.** $\mathbf{N} = N_n\,\mathbf{n} + N_z\,\mathbf{k}_1$ with the in-plane unit normal $\mathbf{n} = \left(-\sin\frac{x_1}{a}\,\mathbf{i}_1 + \mathbf{j}\right)/\sqrt{1 + \sin^2(x_1/a)}$. Along the normal to the plane, from the $\mathbf{k}_1$ projection: $N_z = -F_{Cor,z} = -2m\Omega\dot{x}_1$. Along $\mathbf{n}$, solving the $\mathbf{n}$ projection for N_n:

$$N_n = \frac{m}{\sqrt{1 + \sin^2(x_1/a)}}\left[g + \Omega^2 x_1\sin\frac{x_1}{a} + \frac{\dot{x}_1^2}{a}\cos\frac{x_1}{a}\right]$$

with $\dot{x}_1^2$ from the energy integral above.

Part (f): Transcendental Equilibrium Equation

Equilibrium occurs where $\frac{dV_{eff}}{dx_1} = 0$ (relative rest in the tangential equation, $\ddot{x}_1 = \dot{x}_1 = 0$):

$$\frac{dV_{eff}}{dx_1} = mg\sin\left(\frac{x_1}{a}\right) - m\Omega^2 x_1 = 0 \implies g\sin\left(\frac{x_1}{a}\right) - \Omega^2 x_1 = 0$$

Part (g): Case $\Omega = \Omega(t)$

If Ω varies with time, an Euler force $\mathbf{F}_{Euler} = -m\dot{\Omega}\,\mathbf{j} \times \mathbf{r}_1^P = m\dot{\Omega}x_1\,\mathbf{k}_1$ appears; it is normal to the plane of the curve, so $\mathbf{F}_{Euler} \cdot \mathbf{v}_1^P = 0$ and it does no work in S_1 (it is balanced by N_z). The energy is lost for another reason: the centrifugal potential depends explicitly on time, $\frac{\partial V_{cf}}{\partial t} = -m\Omega\dot{\Omega}x_1^2 \neq 0$, so $\frac{dE'}{dt} = -m\Omega\dot{\Omega}x_1^2 \neq 0$. Thus the mechanical energy in S_1 is **not** conserved and the energy equation cannot be used in the same way.

Checks

- **Shape of the potential.** The potential well has width $2\pi a$ and barrier height $2a$ (potential energy $2mga$). Rotation creates a centrifugal destabilizing potential $-\frac{1}{2}m\Omega^2x_1^2$ so that the origin loses stability for $\Omega^2 > g/a$. Off-centre equilibria satisfy $\sin u = \lambda u$ with $u = x_1/a$ and $\lambda = \Omega^2 a/g$; since $\sin u/u$ decreases monotonically from 1 to 0 on $(0, \pi]$, they exist (one pair $\pm u^*$ with $0 < u^* < \pi$) only for $\lambda < 1$, i.e. $\Omega^2 < g/a$, and there they are unstable ($V''_{eff} = (mg/a)(\cos u^* - \lambda) < 0$).
- **Limit $\Omega \to 0$:** $V_{eff} \to mga\,(1 - \cos(x_1/a))$, $N_z \to 0$, the quadrature of (e) reduces to that of (a) and the equilibrium equation $g\sin(x_1/a) = 0$ leaves only $x_1 = 0$ inside $(-\pi a, \pi a)$.
- **Orthogonality:** $\mathbf{e}_t \cdot \mathbf{n} = 0$, $|\mathbf{e}_t| = |\mathbf{n}| = 1$ and $\mathbf{e}_t \times \mathbf{n} = \mathbf{k}_1$ (worked out in the change-of-basis section); $[{}_0R_1]$ is orthogonal with $\det = 1$.
- **Numeric check:** a scan of $\sin u = \lambda u$ on $(0, \pi]$ finds one root for $\lambda = 0.5$ ($u^* = 1.8955$) and for 0.9, and none for $\lambda = 1, 1.1, 2$.
- **Dimensions:** $[dy/dx] = [\sin(x/a)] = 1$ and $[ds] = [dx] = \mathrm{m}$; $[g/a] = [\Omega^2] = \mathrm{s}^{-2}$, so $[\lambda] = [\Omega^2 a/g] = 1$; every term of E' is in J ($[mga] = [m\Omega^2x_1^2] = [mv_0^2] = \mathrm{kg\,m^2s^{-2}}$); $[N_z] = [m\Omega\dot{x}_1] = \mathrm{N}$; $[g\sin(x/a)] = [\Omega^2 x_1] = \mathrm{m\,s^{-2}}$.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 22 in its solution keys, so the results are verified by independent derivation (including the numeric scan of the equilibrium equation above).

Source: Problems.pdf

PROBLEM 25

Ijon Tichy's Space Walk

Original statement · Problems.pdf, p. 13

Orbiting Frame + Inertia Forces

A spacecraft S is orbiting a celestial body F. The trajectory of the spacecraft with respect to an inertial reference frame Fxy is given by

$$r = \frac{p}{1 + e\cos\nu},$$

where p and e are known constants and $\nu(t)$ is a known function of time. The space explorer Ijon Tichy needs to perform some extra-vehicular activities during his Seventh Voyage (footnote 1: *The Star Diaries* by Stanislaw Lem). He attaches himself with a massless cord of length L to the spacecraft and goes for a space walk. The cord remains in tension at all times. The astronaut is very light compared to the spacecraft, so his motion does not perturb the motion of the spacecraft. Furthermore, we assume that we can consider him as a point particle of mass m. We define a moving reference frame Sx_1y_1 as depicted in the figure.

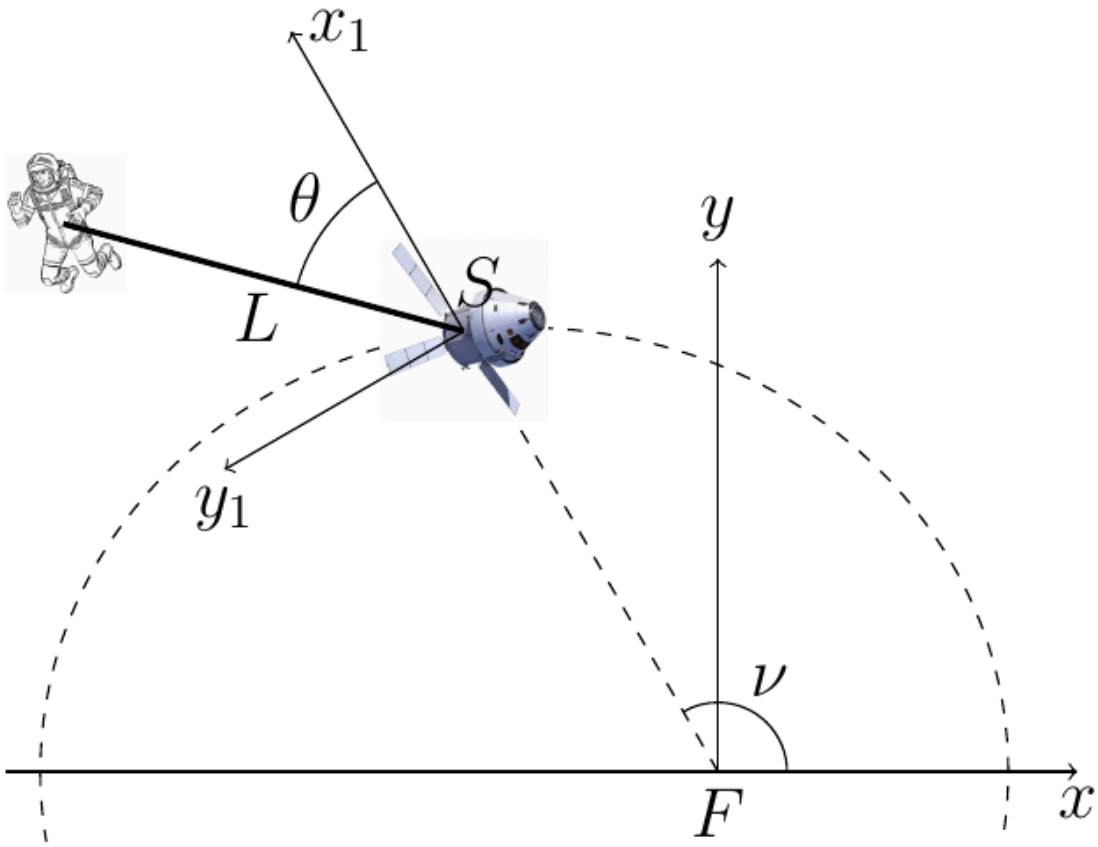


Figure: Spacecraft S on its orbit around F, true anomaly ν, moving axes Sx1 (radial) and Sy1, astronaut on a cord of length L at angle θ from Sx1 (Problems.pdf, p. 13).

(a) Determine the acceleration of the spacecraft with respect to Fxy as a function of time.

(b) Determine the tangent vector to the trajectory of S as a function of time.

(c) Determine the relative velocity and relative acceleration of Ijon Tichy with respect to Sx_1y_1 as a function of θ and its time derivatives.

(d) Determine the fictitious forces acting on Ijon Tichy if we analyze the problem with respect to the frame Sx_1y_1.

The length L is very small compared to the distance between F and S. This allows us to take an approximation: in the following, assume that the gravitational acceleration on the astronaut and on the spacecraft are exactly the same vector (magnitude and direction) as a function of time. (In

actual problems the small differences in gravity at the position of the astronaut and the spacecraft can give rise to small but important effects, and this approximation is usually not taken for that reason.)

(e) Write down one ordinary differential equation that once solved (with given initial conditions) would yield the angle θ as a function of time (you do not need to solve it).

(f) Assuming that you solved the previous equation, and therefore knew $\theta(t)$, determine the value of the tension on the cord as a function of time.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a spacecraft S moves on the conic $r = p/(1 + e\cos\nu)$ around the body F, with the true anomaly $\nu(t)$ a known function of time. The astronaut P (mass m) is tied to S by a massless cord of length L that is always taut; θ is the angle from the axis Sx_1 to the cord SP (figure of the statement).
- **Hypotheses:** Fxy is inertial. The motion of S is prescribed (the astronaut does not perturb it). The motion is planar, in the orbital plane Fxy. The cord is massless and taut, so it pulls P towards S with a tension $T > 0$. From part (e) on, the gravitational acceleration at P equals the one at S.
- **Why a moving frame:** the cord fixes the distance $SP = L$, so in a frame attached to S the astronaut moves on a circle of radius L and one angle, θ, describes it. The price is that Sx_1y_1 accelerates and rotates, so inertia forces must be added (Notes Eqs. 4.62–4.63).
- **Fixed (inertial) frame:** $S_0 \equiv Fxy = \{F; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$, basis $\mathcal{B}_0$, with $\mathbf{k} = \mathbf{i} \times \mathbf{j}$ normal to the orbital plane.
- **Moving frame:** $S_1 \equiv Sx_1y_1 = \{S; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, basis $\mathcal{B}_1$. From the figure of the statement, Sx_1 lies on the line FS and points away from F, and Sy_1 is obtained from Sx_1 by a $+90^\circ$ turn about $\mathbf{k}$ (it points in the direction of increasing ν); $\mathbf{k}_1 = \mathbf{k}$. Its origin moves on the orbit and its axes turn with the radius FS.
- **Basis of the cord:** $\{\mathbf{e}_c, \mathbf{e}_\theta, \mathbf{k}\}$, with $\mathbf{e}_c$ along SP (from S to P) and $\mathbf{e}_\theta$ in the direction of increasing θ; deduced in the next section.
- **Degrees of freedom:** the astronaut moves in the plane (2 coordinates) with one constraint, $|\mathbf{SP}| = L$, so CDOF $= 2 - 1 = 1$, with generalized coordinate θ. The spacecraft adds no unknown, because $\nu(t)$ is given.

Symbol	Meaning	SI unit
p	semi-latus rectum of the orbit of S	m
e	eccentricity of the orbit	dimensionless
$\nu(t)$	true anomaly of S (known)	rad
$r(t)$	distance FS	m
L	cord length	m
m	mass of the astronaut	kg
$\theta(t)$	angle from Sx_1 to the cord	rad
$T(t)$	cord tension	N
μ_F	gravitational parameter GM_F of F (used in the official key only)	m^3/s^2

Change of Basis and Position Vector

The polar basis $\mathcal{B}_1$ in the orbital plane

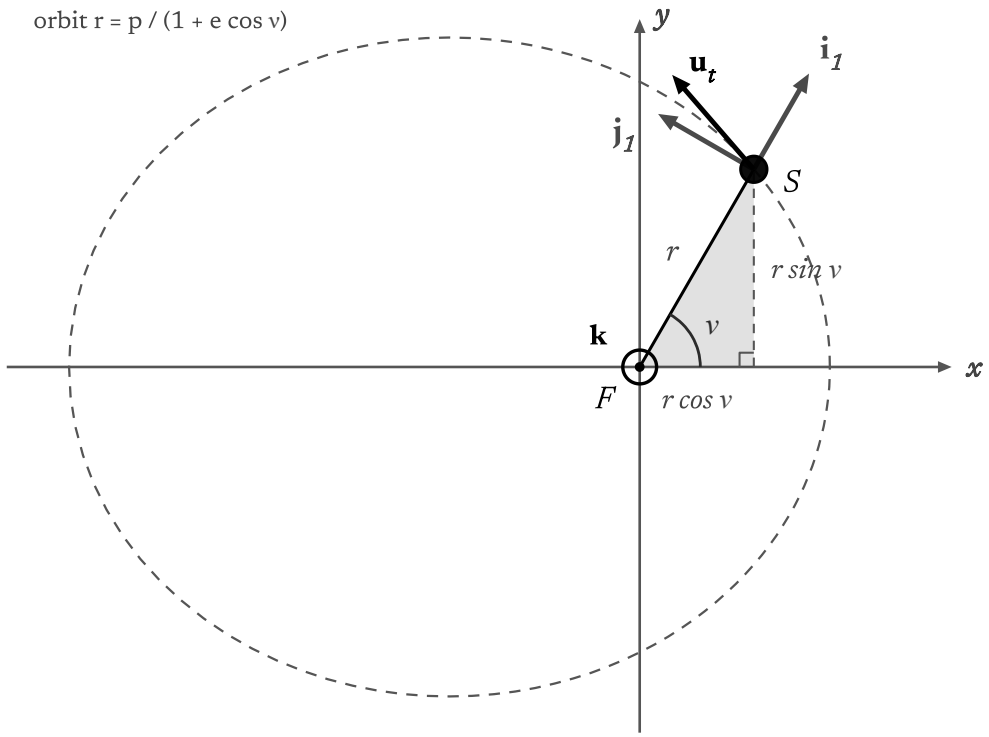


Figure 1. Orbital plane seen from $+\mathbf{k}$ ($\odot$). The shaded right triangle has hypotenuse $|FS| = r$, leg $r\cos\nu$ along $\mathbf{i}$ (adjacent to ν) and leg $r\sin\nu$ along $\mathbf{j}$ (opposite). $\mathbf{i}_1$ is the radial direction and $\mathbf{j}_1$ is $\mathbf{i}_1$ turned $+90°$ (blue); $\mathbf{u}_t$ is the unit tangent of the orbit (orange).

The orbital plane is drawn as seen from $+\mathbf{k}$: $\mathbf{k}$ points out of the paper, towards the reader ($\odot$), $\mathbf{i}$ to the right and $\mathbf{j}$ up (indeed $\mathbf{i}\times\mathbf{j}=\mathbf{k}$: fingers from right to up, thumb towards the reader). The true anomaly ν is measured at F from Fx, counter-clockwise, so it is already the standard polar angle of S and no change of origin of the angle is needed.

Position of the spacecraft. In the shaded right triangle (right angle at the foot of the perpendicular from S to Fx) the hypotenuse is $|FS| = r$, the leg adjacent to ν lies along $\mathbf{i}$ and measures $r\cos\nu$, and the leg opposite to ν lies along $\mathbf{j}$ and measures $r\sin\nu$:

$$\mathbf{r}_0^S = \overrightarrow{FS} = r\cos\nu\,\mathbf{i} + r\sin\nu\,\mathbf{j} = r\,\mathbf{i}_1, \qquad r = \frac{p}{1+e\cos\nu}$$

Radial unit vector. Dividing the same triangle by r gives a triangle with hypotenuse 1, adjacent leg $\cos\nu$ and opposite leg $\sin\nu$: $\mathbf{i}_1 = \cos\nu\,\mathbf{i} + \sin\nu\,\mathbf{j}$.

Transverse unit vector. $\mathbf{j}_1$ is $\mathbf{i}_1$ turned $+90^\circ$ about $\mathbf{k}$, which in this view (with $\mathbf{k}$ towards the reader) is a counter-clockwise quarter turn: $\mathbf{j}_1 = \mathbf{k}\times\mathbf{i}_1$. With $\mathbf{k}\times\mathbf{i} = \mathbf{j}$ and $\mathbf{k}\times\mathbf{j} = -\mathbf{i}$:

$$\mathbf{j}_1 = \cos\nu\,(\mathbf{k}\times\mathbf{i}) + \sin\nu\,(\mathbf{k}\times\mathbf{j}) = -\sin\nu\,\mathbf{i} + \cos\nu\,\mathbf{j}$$

The minus sign comes from $\mathbf{k}\times\mathbf{j} = -\mathbf{i}$: in the drawing, turning the vertical part $\sin\nu\,\mathbf{j}$ of $\mathbf{i}_1$ a quarter turn counter-clockwise makes it point to the left, along $-\mathbf{i}$. For $0 < \nu < \pi/2$ the tip of $\mathbf{j}_1$ is indeed to the left of the y axis, as drawn. Summary:

$$\mathbf{i}_1 = \cos\nu\,\mathbf{i} + \sin\nu\,\mathbf{j}, \qquad \mathbf{j}_1 = -\sin\nu\,\mathbf{i} + \cos\nu\,\mathbf{j}, \qquad \mathbf{k}_1 = \mathbf{k}$$

Change-of-basis matrix (columns are $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ written in $\mathcal{B}_0$):

$$[{}_0R_1] = \begin{pmatrix} \cos\nu & -\sin\nu & 0 \\ \sin\nu & \cos\nu & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1\cdot(\cos\nu\cos\nu - (-\sin\nu)\sin\nu) = \cos^2\nu + \sin^2\nu = 1$$

The product $[{}_0R_1][{}_0R_1]^T$ has diagonal entries $\cos^2\nu + \sin^2\nu = 1$ (and 1) and off-diagonal entries $\cos\nu\sin\nu - \sin\nu\cos\nu = 0$, so it is the identity.

Angular velocity and acceleration of S_1. S_1 performs a simple rotation of angle $\nu(t)$ about $\mathbf{k}$ (vault note *Topic 5 - Relative Motion*, Sec. 5.5; Slide 16). Check with Poisson's formula: by the chain rule, $\left.\frac{d\mathbf{i}_1}{dt}\right|_0 = \frac{d\cos\nu}{d\nu}\dot\nu\,\mathbf{i} + \frac{d\sin\nu}{d\nu}\dot\nu\,\mathbf{j} = \dot\nu\,(-\sin\nu\,\mathbf{i} + \cos\nu\,\mathbf{j}) = \dot\nu\,\mathbf{j}_1 = \dot\nu\,\mathbf{k}\times\mathbf{i}_1$. Hence

$$\boldsymbol{\omega}_{10} = \dot\nu\,\mathbf{k}, \qquad \boldsymbol{\alpha}_{10} = \left.\frac{d(\dot\nu\,\mathbf{k})}{dt}\right|_0 = \ddot\nu\,\mathbf{k}$$

Origin of S_1: $\mathbf{r}_0^S = r\,\mathbf{i}_1$, with $r = p/(1+e\cos\nu)$; $\mathbf{v}_0^S$ and $\mathbf{a}_0^S$ are computed in the kinematics.

The cord in the moving frame

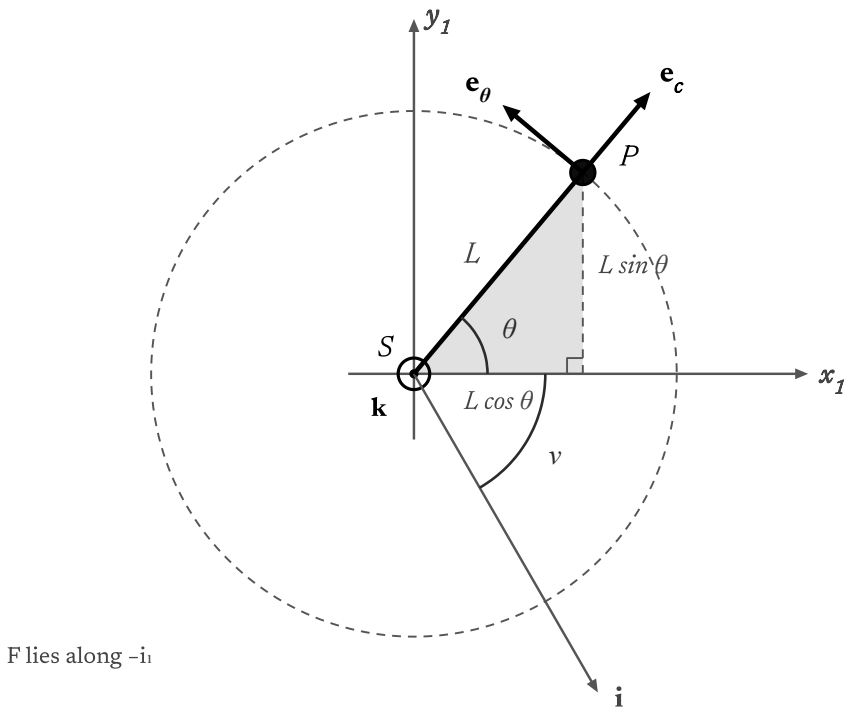


Figure 2. Frame Sx_1y_1 seen from $+\mathbf{k}$ ($\odot$), with $\mathbf{i}_1$ to the right. In the shaded triangle the hypotenuse is the cord $|SP| = L$, the leg adjacent to θ is $L\cos\theta$ (along $\mathbf{i}_1$) and the opposite leg is $L\sin\theta$ (along $\mathbf{j}_1$). $\mathbf{e}_c, \mathbf{e}_\theta$ in orange. The inertial axis $\mathbf{i}$ (grey) is at $-\nu$ from $\mathbf{i}_1$, so the cord makes $\theta + \nu$ with it.

The frame Sx_1y_1 is drawn with $\mathbf{i}_1$ to the right, $\mathbf{j}_1$ up and $\mathbf{k}$ out of the paper ($\odot$); F lies along $-\mathbf{i}_1$, at the distance r. The angle θ is measured from $\mathbf{i}_1$ towards $\mathbf{j}_1$, so it is the standard polar angle of P in $\mathcal{B}_1$. In the shaded right triangle the hypotenuse is the cord, $|SP| = L$; the leg adjacent to θ lies along $\mathbf{i}_1$ and measures $L\cos\theta$; the opposite leg lies along $\mathbf{j}_1$ and measures $L\sin\theta$. Dividing by L:

$$\mathbf{e}_c = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1, \qquad \mathbf{r}_1^P = \overrightarrow{SP} = L\,\mathbf{e}_c = L\cos\theta\,\mathbf{i}_1 + L\sin\theta\,\mathbf{j}_1$$

The transverse vector is the derivative of $\mathbf{e}_c$ with respect to θ ($\frac{d}{d\theta}\cos\theta = -\sin\theta$, $\frac{d}{d\theta}\sin\theta = \cos\theta$), of length $\sqrt{\sin^2\theta + \cos^2\theta} = 1$; it equals $\mathbf{k} \times \mathbf{e}_c$, the same quarter turn as before:

$$\mathbf{e}_\theta = \frac{d\mathbf{e}_c}{d\theta} = -\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1$$

$\{\mathbf{e}_c, \mathbf{e}_\theta, \mathbf{k}\}$ is again a rotation of $\mathcal{B}_1$ by θ about $\mathbf{k}$, with determinant $\cos^2\theta + \sin^2\theta = 1$. Products used below: $\mathbf{k}\times\mathbf{e}_c = \mathbf{e}_\theta$ and $\mathbf{k}\times\mathbf{e}_\theta = -\mathbf{e}_c$.

Absolute position of the astronaut. To go from F to P we pass through the spacecraft:

$$\mathbf{r}_0^P = \overrightarrow{FS} + \overrightarrow{SP} = r\,\mathbf{i}_1 + L\,\mathbf{e}_c$$

Angle of the cord in the inertial frame. Substituting $\mathbf{i}_1$ and $\mathbf{j}_1$ in $\mathbf{e}_c$ and using the addition formulas $\cos(\theta+\nu) = \cos\theta\cos\nu - \sin\theta\sin\nu$ and $\sin(\theta+\nu) = \sin\theta\cos\nu + \cos\theta\sin\nu$:

$$\mathbf{e}_c = \cos\theta\,(\cos\nu\,\mathbf{i} + \sin\nu\,\mathbf{j}) + \sin\theta\,(-\sin\nu\,\mathbf{i} + \cos\nu\,\mathbf{j}) = \cos(\theta+\nu)\,\mathbf{i} + \sin(\theta+\nu)\,\mathbf{j}$$

so the cord makes the angle $\theta+\nu$ with the inertial axis Fx (the grey axis $\mathbf{i}$ of the second figure is at $-\nu$ from $\mathbf{i}_1$). This is used in the checks.

Kinematics

(a) Acceleration of the spacecraft with respect to Fxy

> **Pedagogical Justification:** $\mathbf{r}_0^S = r\,\mathbf{i}_1$ is written in the rotating basis $\mathcal{B}_1$, so its derivative in S_0 follows from the Coriolis theorem (Notes Eq. 3.8, Slide 17): derivative of the components in S_1 plus $\boldsymbol{\omega}_{10}\times(\cdot)$.

Velocity. In S_1 only the component r changes, and $\dot\nu\,\mathbf{k}\times r\,\mathbf{i}_1 = r\dot\nu\,(\mathbf{k}\times\mathbf{i}_1) = r\dot\nu\,\mathbf{j}_1$:

$$\mathbf{v}_0^S = \left.\frac{d(r\,\mathbf{i}_1)}{dt}\right|_1 + \boldsymbol{\omega}_{10}\times r\,\mathbf{i}_1 = \dot r\,\mathbf{i}_1 + r\dot\nu\,\mathbf{j}_1$$

Acceleration. Apply the theorem again. The derivative of the components in S_1 uses the product rule on $r\dot\nu$, $\frac{d}{dt}(r\dot\nu) = \dot r\dot\nu + r\ddot\nu$:

$$\left.\frac{d\mathbf{v}_0^S}{dt}\right|_1 = \ddot{r}\,\mathbf{i}_1 + (\dot{r}\dot{\nu} + r\ddot{\nu})\,\mathbf{j}_1$$

The rotation term, with $\mathbf{k}\times\mathbf{i}_1 = \mathbf{j}_1$ and $\mathbf{k}\times\mathbf{j}_1 = -\mathbf{i}_1$:

$$\boldsymbol{\omega}_{10}\times\mathbf{v}_0^S = \dot{\nu}\,\mathbf{k}\times(\dot{r}\,\mathbf{i}_1 + r\dot{\nu}\,\mathbf{j}_1) = \dot{r}\dot{\nu}\,\mathbf{j}_1 - r\dot{\nu}^2\,\mathbf{i}_1$$

Adding both:

$$\mathbf{a}_0^S = \left(\ddot{r} - r\dot{\nu}^2\right)\mathbf{i}_1 + (2\dot{r}\dot{\nu} + r\ddot{\nu})\,\mathbf{j}_1$$

Derivatives of $r(\nu(t))$. Write $r = p\,D^{-1}$ with $D = 1 + e\cos\nu$. By the chain rule, $\dfrac{dD}{dt} = \dfrac{dD}{d\nu}\dot{\nu} = -e\sin\nu\,\dot{\nu}$, so

$$\dot{r} = \frac{dr}{dD}\frac{dD}{dt} = -p\,D^{-2}\,(-e\sin\nu\,\dot{\nu}) = \frac{p\,e\,\dot{\nu}\sin\nu}{(1+e\cos\nu)^2}$$

Differentiate again with the product rule on $N = pe\,\dot{\nu}\sin\nu$ and D^{-2}: $\dot{N} = pe\left(\ddot{\nu}\sin\nu + \dot{\nu}^2\cos\nu\right)$ (chain rule $\frac{d}{dt}\sin\nu = \cos\nu\,\dot{\nu}$) and $\frac{d}{dt}D^{-2} = -2D^{-3}\dot{D} = 2e\sin\nu\,\dot{\nu}\,D^{-3}$:

$$\ddot{r} = \frac{pe\left(\ddot{\nu}\sin\nu + \dot{\nu}^2\cos\nu\right)}{(1+e\cos\nu)^2} + \frac{2pe^2\dot{\nu}^2\sin^2\nu}{(1+e\cos\nu)^3}$$

Since $\nu(t)$ is known, $r, \dot{r}, \ddot{r}, \dot{\nu}, \ddot{\nu}$ are known functions of time, and so is $\mathbf{a}_0^S$. In $\mathcal{B}_0$ (substituting the columns of $[{}_0R_1]$), $\mathbf{a}_0^S = \left[(\ddot{r} - r\dot{\nu}^2)\cos\nu - (2\dot{r}\dot{\nu} + r\ddot{\nu})\sin\nu\right]\mathbf{i} + \left[(\ddot{r} - r\dot{\nu}^2)\sin\nu + (2\dot{r}\dot{\nu} + r\ddot{\nu})\cos\nu\right]\mathbf{j}$.

(b) Tangent vector to the trajectory of S

> **Pedagogical Justification:** the unit tangent is the velocity divided by the speed, $\mathbf{u}_t = \mathbf{v}_0^S/|\mathbf{v}_0^S|$ (intrinsic frame of Topic 1). Taking $\dot{\nu} > 0$ (prograde motion, as in the figure) the common factor $r\dot{\nu}$ can be extracted.

From (a), $\dot{r} = r\dot{\nu}\,\dfrac{e\sin\nu}{1+e\cos\nu}$ (because $\dfrac{p}{(1+e\cos\nu)^2} = \dfrac{r}{1+e\cos\nu}$). Then

$$\mathbf{v}_0^S = \frac{r\dot{\nu}}{1+e\cos\nu}\left[e\sin\nu\,\mathbf{i}_1 + (1+e\cos\nu)\,\mathbf{j}_1\right]$$

The bracket is the hypotenuse of a right triangle with legs $e\sin\nu$ (along $\mathbf{i}_1$) and $1+e\cos\nu$ (along $\mathbf{j}_1$); its length is $\sqrt{e^2\sin^2\nu + 1 + 2e\cos\nu + e^2\cos^2\nu} = \sqrt{1+2e\cos\nu+e^2}$, hence

$$\mathbf{u}_t(t) = \frac{e\sin\nu\,\mathbf{i}_1 + (1+e\cos\nu)\,\mathbf{j}_1}{\sqrt{1+2e\cos\nu+e^2}}, \qquad |\mathbf{v}_0^S| = \frac{r\dot{\nu}\sqrt{1+2e\cos\nu+e^2}}{1+e\cos\nu}$$

with $\nu = \nu(t)$. In $\mathcal{B}_0$, substituting $\mathbf{i}_1$ and $\mathbf{j}_1$, the $\mathbf{i}$ component of the bracket is $e\sin\nu\cos\nu - (1+e\cos\nu)\sin\nu = -\sin\nu$ and the $\mathbf{j}$ component is $e\sin^2\nu + (1+e\cos\nu)\cos\nu = e + \cos\nu$, so $\mathbf{u}_t = \left[-\sin\nu\,\mathbf{i} + (e+\cos\nu)\,\mathbf{j}\right]/\sqrt{1+2e\cos\nu+e^2}$ (the orange vector of the first figure). For a circular orbit ($e=0$) $\mathbf{u}_t = \mathbf{j}_1$, perpendicular to the radius, as expected.

(c) Relative velocity and acceleration of the astronaut in Sx_1y_1

> **Pedagogical Justification:** in S_1 the basis $\mathcal{B}_1$ is constant, so only the components of $\mathbf{r}_1^P = L\cos\theta\,\mathbf{i}_1 + L\sin\theta\,\mathbf{j}_1$ are differentiated (Notes Eq. 3.7); L is constant.

Chain rule: $\frac{d}{dt}\cos\theta = -\sin\theta\,\dot{\theta}$, $\frac{d}{dt}\sin\theta = \cos\theta\,\dot{\theta}$:

$$\mathbf{v}_1^P = \left.\frac{d\mathbf{r}_1^P}{dt}\right|_1 = -L\dot{\theta}\sin\theta\,\mathbf{i}_1 + L\dot{\theta}\cos\theta\,\mathbf{j}_1 = L\dot{\theta}\,\mathbf{e}_\theta$$

Differentiate again; product rule on $\dot{\theta}\sin\theta$: $\frac{d}{dt}(\dot{\theta}\sin\theta) = \ddot{\theta}\sin\theta + \dot{\theta}^2\cos\theta$, and on $\dot{\theta}\cos\theta$: $\ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta$:

$$\mathbf{a}_1^P = -L\left(\ddot{\theta}\sin\theta + \dot{\theta}^2\cos\theta\right)\mathbf{i}_1 + L\left(\ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta\right)\mathbf{j}_1 = L\ddot{\theta}\,\mathbf{e}_\theta - L\dot{\theta}^2\,\mathbf{e}_c$$

This is uniform-radius circular motion: a tangential part $L\ddot{\theta}$ and a centripetal part $L\dot{\theta}^2$ towards S.

Forces and Equations of Motion

(d) Fictitious forces in Sx_1y_1

Pedagogical Justification: in the non-inertial frame S_1, Newton's law reads $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63), with the four inertia forces of Notes Eq. 4.62 (Slide 28). Each one is computed with the motion of S_1 found in the change-of-basis section.

1. Force due to the acceleration of the origin ($\mathbf{F}_A = -m\mathbf{a}_0^S$), with $\mathbf{a}_0^S$ from (a):

$$\mathbf{F}_A = -m\left(\ddot{r} - r\dot{\nu}^2\right)\mathbf{i}_1 - m\left(2\dot{r}\dot{\nu} + r\ddot{\nu}\right)\mathbf{j}_1$$

2. Euler force ($\mathbf{F}_E = -m\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P$). With $\mathbf{k} \times \mathbf{i}_1 = \mathbf{j}_1, \mathbf{k} \times \mathbf{j}_1 = -\mathbf{i}_1$:

$$\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P = \ddot{\nu}\,\mathbf{k} \times \left(L\cos\theta\,\mathbf{i}_1 + L\sin\theta\,\mathbf{j}_1\right) = L\ddot{\nu}\left(\cos\theta\,\mathbf{j}_1 - \sin\theta\,\mathbf{i}_1\right) \;\Rightarrow\; \mathbf{F}_E = mL\ddot{\nu}\left(\sin\theta\,\mathbf{i}_1 - \cos\theta\,\mathbf{j}_1\right) = -mL\ddot{\nu}\,\mathbf{e}_\theta$$

3. Centrifugal force ($\mathbf{F}_C = -m\boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P)$). First $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = L\dot{\nu}\,\mathbf{e}_\theta$ (same computation as above with $\dot{\nu}$), then $\dot{\nu}\,\mathbf{k} \times L\dot{\nu}\,\mathbf{e}_\theta = L\dot{\nu}^2\left(-\sin\theta\,\mathbf{j}_1 - \cos\theta\,\mathbf{i}_1\right) = -L\dot{\nu}^2\,\mathbf{e}_c$:

$$\mathbf{F}_C = mL\dot{\nu}^2\,\mathbf{e}_c = mL\dot{\nu}^2\left(\cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1\right)$$

in agreement with $\mathbf{F}_C = m\omega_{10}^2\mathbf{r}_\perp$ (Slide 30), since $\mathbf{r}_1^P$ is perpendicular to $\mathbf{k}$.

4. Coriolis force ($\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$). With $\mathbf{k} \times \mathbf{e}_\theta = -\mathbf{e}_c$:

$$\mathbf{F}_{COR} = -2m\,\dot{\nu}\,\mathbf{k} \times L\dot{\theta}\,\mathbf{e}_\theta = 2mL\dot{\nu}\dot{\theta}\,\mathbf{e}_c = 2mL\dot{\nu}\dot{\theta}\left(\cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1\right)$$

The total inertia force is $\mathbf{F}_i = \mathbf{F}_A + \mathbf{F}_E + \mathbf{F}_C + \mathbf{F}_{COR}$. Because $\nu(t)$ is in general not uniform ($\ddot{\nu} \neq 0$ on an elliptic orbit), the Euler force is present.

(e) Equation of motion for $\theta(t)$

> **Pedagogical Justification:** the real forces on the astronaut are the cord tension $\mathbf{T} = -T\,\mathbf{e}_c$ (towards S) and gravity $m\mathbf{g}_P$. Projecting Notes Eq. 4.63 on $\mathbf{e}_\theta$ removes the unknown tension, because $\mathbf{e}_c \cdot \mathbf{e}_\theta = 0$.

Gravity hypothesis. S is in free flight around F (its conic orbit is a Keplerian orbit, Topic 4), so its acceleration is the gravitational acceleration at S: $\mathbf{a}_0^S = \mathbf{g}_S$. The statement assumes $\mathbf{g}_P = \mathbf{g}_S$. Hence the weight cancels the force due to the acceleration of the origin exactly:

$$m\mathbf{g}_P + \mathbf{F}_A = m\mathbf{a}_0^S - m\mathbf{a}_0^S = \mathbf{0}$$

Newton's law in S_1:

$$m\left(L\ddot{\theta}\,\mathbf{e}_\theta - L\dot{\theta}^2\,\mathbf{e}_c\right) = -T\,\mathbf{e}_c - mL\ddot{\nu}\,\mathbf{e}_\theta + mL\dot{\nu}^2\,\mathbf{e}_c + 2mL\dot{\nu}\dot{\theta}\,\mathbf{e}_c$$

Component along $\mathbf{e}_\theta$: $mL\ddot{\theta} = -mL\ddot{\nu}$. (Without the approximation, the same projection gives $L(\ddot{\theta} + \ddot{\nu}) = (\mathbf{g}_P - \mathbf{a}_0^S)\cdot\mathbf{e}_\theta$, the tidal term that the statement neglects.)

Component along $\mathbf{e}_c$:

$$-mL\dot{\theta}^2 = -T + mL\dot{\nu}^2 + 2mL\dot{\nu}\dot{\theta}$$

Answers and Checks

(a) Acceleration of the spacecraft

$\mathbf{a}_0^S = \left(\ddot{r} - r\dot{\nu}^2\right)\mathbf{i}_1 + \left(2\dot{r}\dot{\nu} + r\ddot{\nu}\right)\mathbf{j}_1$, with $r = p/(1+e\cos\nu)$, $\dot{r} = \dfrac{p\,e\,\dot{\nu}\sin\nu}{(1+e\cos\nu)^2}$ and $\ddot{r}$ as in the kinematics, all known functions of t through $\nu(t)$.

(b) Tangent vector

$$\mathbf{u}_t(t) = \frac{e\sin\nu\,\mathbf{i}_1 + (1+e\cos\nu)\,\mathbf{j}_1}{\sqrt{1+2e\cos\nu+e^2}} = \frac{-\sin\nu\,\mathbf{i} + (e+\cos\nu)\,\mathbf{j}}{\sqrt{1+2e\cos\nu+e^2}}.$$

(c) Relative velocity and acceleration

$\mathbf{v}_1^P = L\dot\theta\,\mathbf{e}_\theta = -L\dot\theta\sin\theta\,\mathbf{i}_1 + L\dot\theta\cos\theta\,\mathbf{j}_1$ and $\mathbf{a}_1^P = L\ddot\theta\,\mathbf{e}_\theta - L\dot\theta^2\,\mathbf{e}_c$.

(d) Fictitious forces

$\mathbf{F}_A = -m\mathbf{a}_0^S$, $\mathbf{F}_E = -mL\ddot\nu\,\mathbf{e}_\theta$, $\mathbf{F}_C = mL\dot\nu^2\,\mathbf{e}_c$, $\mathbf{F}_{COR} = 2mL\dot\nu\dot\theta\,\mathbf{e}_c$.

(e) Differential equation for $\theta(t)$

From the $\mathbf{e}_\theta$ component:

$$\ddot\theta + \ddot\nu(t) = 0$$

This is the requested ODE; $\ddot\nu(t)$ is a known function.

Its integral (not requested, but immediate). Integrate from 0 to t; the antiderivative of $\ddot\theta + \ddot\nu$ is $\dot\theta + \dot\nu$, and Barrow's rule gives

$$\int_0^t \left(\ddot\theta + \ddot\nu\right) dt' = \Big[\dot\theta + \dot\nu\Big]_0^t = \dot\theta(t) + \dot\nu(t) - \dot\theta(0) - \dot\nu(0) = 0 \;\Rightarrow\; \dot\theta + \dot\nu = \Omega_c \equiv \dot\theta(0) + \dot\nu(0)$$

Integrating once more, $[\theta + \nu]_0^t = \Omega_c t$, so $\theta(t) = \theta(0) + \nu(0) - \nu(t) + \Omega_c t$.

(f) Tension on the cord

From the $\mathbf{e}_c$ component:

$$T = mL\left(\dot\theta^2 + 2\dot\theta\dot\nu + \dot\nu^2\right) = mL\left(\dot\theta + \dot\nu\right)^2$$

With $\theta(t)$ known (and $\nu(t)$ given), $T(t) = mL\left(\dot\theta(t) + \dot\nu(t)\right)^2$; by the integral of (e) it is in fact constant, $T = mL\,\Omega_c^2$. The cord is taut ($T > 0$) as long as $\Omega_c \neq 0$.

General form (as in the official key). Keeping gravity and $\mathbf{a}_0^S$ separate, with $\mathbf{g}_P = \mathbf{g}_S = -\dfrac{\mu_F}{r^2}\mathbf{i}_1$, the $\mathbf{e}_c$ component reads $-mL\dot\theta^2 = -T + m(\mathbf{g}_S - \mathbf{a}_0^S)\cdot\mathbf{e}_c + mL\dot\nu^2 + 2mL\dot\nu\dot\theta$. With $\mathbf{i}_1\cdot\mathbf{e}_c = \cos\theta$, $\mathbf{j}_1\cdot\mathbf{e}_c = \sin\theta$ (the legs of the cord triangle divided by L):

$$T = mL\dot\theta^2 - \frac{\mu_F m\cos\theta}{r^2} + mL\dot\nu^2 + 2mL\dot\theta\dot\nu + m\left(r\dot\nu^2 - \ddot r\right)\cos\theta - m\left(2\dot r\dot\nu + r\ddot\nu\right)\sin\theta$$

For the Keplerian orbit of S, $\ddot{r} - r\dot{\nu}^2 = -\mu_F/r^2$ and $2\dot{r}\dot{\nu} + r\ddot{\nu} = \frac{1}{r}\frac{d}{dt}\left(r^2\dot{\nu}\right) = 0$ (Topic 4: conservation of $h = r^2\dot{\nu}$ and $p = h^2/\mu_F$), so the second and fifth terms cancel each other and the sixth vanishes, and this expression reduces to $T = mL(\dot{\theta} + \dot{\nu})^2$.

Checks

- **Meaning of $\dot{\theta} + \dot{\nu} = \text{const}$.** $\theta + \nu$ is the angle of the cord measured from the inertial axis Fx (shown in the change-of-basis section with the addition formulas). With equal gravity on S and P, both fall together and the astronaut sees no net gravity: relative to a non-rotating frame attached to S he simply turns around S at the constant rate Ω_c, and the cord supplies the centripetal force $mL\Omega_c^2$. In the rotating frame Sx_1y_1 this appears as the balance between the Euler force (which drives $\ddot{\theta} = -\ddot{\nu}$), and the centrifugal and Coriolis forces (which, together with $mL\dot{\theta}^2$, add up to the tension).
- **Limiting cases.** Circular orbit ($e = 0$): $\dot{\nu}$ is constant, $\ddot{\nu} = 0$, the Euler force vanishes and $\dot{\theta}$ is constant. If the astronaut does not rotate in inertial space ($\Omega_c = 0$), the tension vanishes and the taut-cord hypothesis is lost: he would drift freely at constant relative position in a non-rotating frame.
- **Orthogonality:** $\mathbf{i}_1 \cdot \mathbf{j}_1 = -\cos\nu\sin\nu + \sin\nu\cos\nu = 0$, $\mathbf{e}_c \cdot \mathbf{e}_\theta = -\cos\theta\sin\theta + \sin\theta\cos\theta = 0$, all of unit length; $|\mathbf{u}_t| = 1$ by construction.
- **Dimensions:** $[\dot{r}] = \mathrm{m}\cdot\mathrm{s}^{-1}$ (p in m times $\dot{\nu}$ in s^{-1}); $[\ddot{r}] = [r\dot{\nu}^2] = [r\ddot{\nu}] = \mathrm{m}\cdot\mathrm{s}^{-2}$; $[mL\ddot{\nu}] = [mL\dot{\nu}^2] = [mL\dot{\nu}\dot{\theta}] = \mathrm{kg}\cdot\mathrm{m}\cdot\mathrm{s}^{-2} = \mathrm{N}$; $[\mu_F m/r^2] = \mathrm{m}^3\mathrm{s}^{-2}\cdot\mathrm{kg}/\mathrm{m}^2 = \mathrm{N}$; $\mathbf{u}_t$ is dimensionless.
- **Symbolic check (sympy):** $\mathbf{a}_0^S$ from differentiating $r(\nu(t))\,\mathbf{i}_1$ twice in $\mathcal{B}_0$, the formulas for $\dot{r}$ and $\ddot{r}$, the unit tangent, $\mathbf{v}_1^P$, $\mathbf{a}_1^P$, the decomposition of Notes Eq. 3.27, the four inertia forces, $\ddot{\theta} = -\ddot{\nu}$, $T = mL(\dot{\theta} + \dot{\nu})^2$, the equality of the general formula with the key, and $\mathbf{a}_0^S = -\mu_F/r^2\,\mathbf{i}_1$ for $\dot{\nu} = h/r^2$, $p = h^2/\mu_F$: all verified.
- **Check against the official key:** (`Problems.pdf`, PDF pages 73–74) it gives $\mathbf{a}_0^S$, $\dot{r}$, $\ddot{r}$, the Euler force $mL\ddot{\nu}(\sin\theta\,\mathbf{i}_1 - \cos\theta\,\mathbf{j}_1)$ and the general tension formula written above; all of them coincide with the results of this solution, and the general formula reduces to $mL(\dot{\theta} + \dot{\nu})^2$ for the Keplerian orbit.

Source: Problems.pdf (PDF page 13); key on PDF pages 73-74

PROBLEM 26

Heavy Particle on a Rotating Half-Circumference (Moving-Frame Method)

Original statement · Problems.pdf, p. 14

Rotating Wire + Inertia Forces

A heavy point particle P of mass m is constrained to move without friction along the half-circumference wire of radius a shown in the figure. The wire containing the point particle rotates around the OY axis with constant angular velocity $\boldsymbol{\omega} = \Omega\mathbf{j}$. Initially, the particle is located at $x = 0$ with velocity v_0 along the positive x axis.

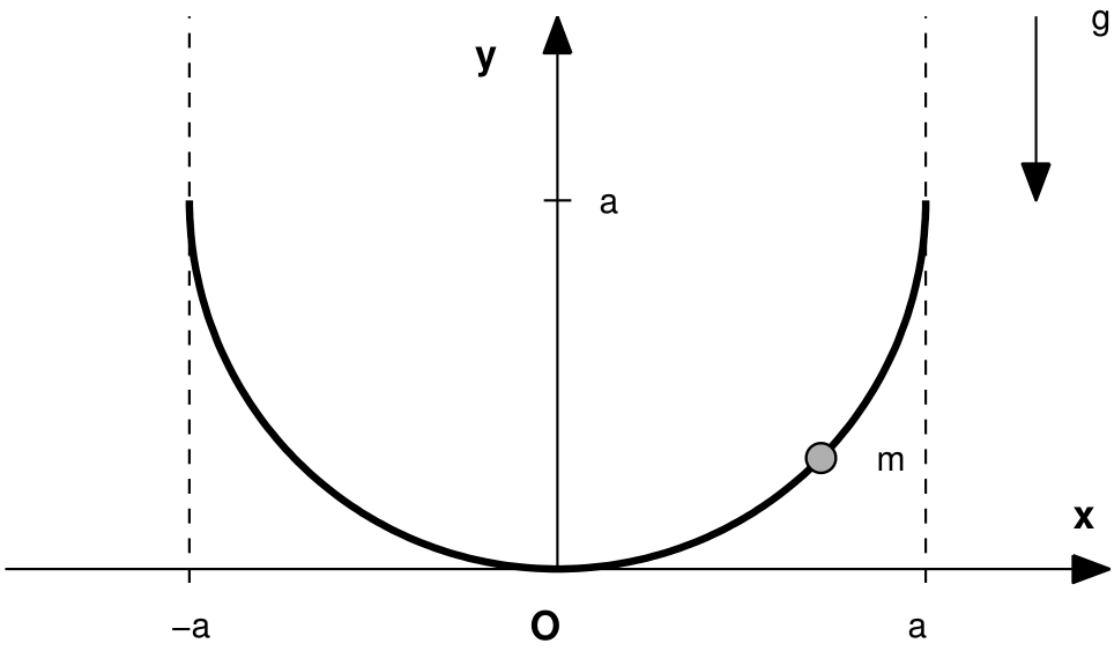


Figure: Half-circumference wire of radius a with lowest point O, ends at x = ±a, rotating about the vertical axis OY; particle of mass m on the wire (Problems.pdf, p. 14).

(a) Determine the number of degrees of freedom of the problem.

(b) Define a convenient moving reference frame with origin in O, $OX_1Y_1Z_1$. Express the inertia force that acts on the particle in this moving reference frame.

(c) Express also the components of the weight of the particle with respect to $OX_1Y_1Z_1$.

(d) Determine if there are any equilibrium points.

(e) Reduce the problem to integrals (i.e. quadratures) and determine the value of the reaction force.

(f) Is it possible to use the energy equation? If Ω were a function of time, would it be possible to use the energy equation? Justify your answer.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a heavy particle P (mass m) slides without friction on a half-circumference wire of radius a. In the plane of the wire the lowest point is O, the centre is $C = (0, a)$ and the ends are $(\pm a, a)$ (figure of the statement). The wire turns about the vertical axis OY with constant angular velocity $\boldsymbol{\omega} = \Omega\,\mathbf{j}$.
- **Hypotheses:** the frame $OXYZ$ with OY vertical upwards is inertial; gravity is $-g\,\mathbf{j}$; the wire is smooth, so its reaction has no component along the wire; the rotation is imposed (whatever torque is needed is supplied). It is solved with the relative-motion method of Topic 5 (moving frame, Notes Eq. 3.27 and inertia forces, Notes Eqs. 4.62–4.63).
- **Why a moving frame:** in a frame that turns with the wire the constraint is a fixed circle, so the motion of P is a one-dimensional motion on a fixed curve. The cost is to add the centrifugal and Coriolis forces; there is no Euler force and no force from the acceleration of the origin because Ω is constant and O is fixed.
- **Initial conditions ($t = 0$):** P at O ($x = 0$) with velocity v_0 along $+x$.
- **Fixed (inertial) frame:** $S_0 = OXYZ = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$, basis $\mathcal{B}_0 = \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$, with $\mathbf{j}$ vertical upwards.
- **Moving frame** (requested in (b)): $S_1 = OX_1Y_1Z_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, basis $\mathcal{B}_1$, attached to the wire: OX_1 in the plane of the wire (the x axis of the figure), $OY_1 \equiv OY$ (the axis of rotation) and OZ_1 normal to the plane of the wire. S_1 coincides with S_0 at $t = 0$.
- **Intrinsic basis of the wire:** $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$ at P: unit tangent in the sense of increasing θ, principal normal and binormal. They are deduced in the next section.
- **Coordinate:** θ, the angle of CP measured at the centre C from the downward vertical CO, positive towards $+X_1$, with $\theta \in [-\pi/2, \pi/2]$ on the wire. Ωt is prescribed, so it is a known function of time, not a coordinate.

Symbol	Meaning	SI unit
m	mass of P	kg
a	radius of the wire	m
Ω	angular velocity of the wire (constant)	rad/s
g	gravitational acceleration	m/s^2
v_0	initial speed at O	m/s
θ	angle of CP from the downward vertical	rad

(a) Degrees of freedom

> **Pedagogical Justification:** CDOF = (coordinates of the free particle) − (independent holonomic constraints).

P has 3 coordinates. In S_1 the wire imposes two fixed equations, $z_1 = 0$ (plane of the wire) and $x_1^2 + (y_1 - a)^2 = a^2$ (circle of centre C). In S_0 the same constraints are rheonomic (they turn with Ωt), but the count is the same: CDOF $= 3 - 2 = 1$, with generalized coordinate θ.

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from above

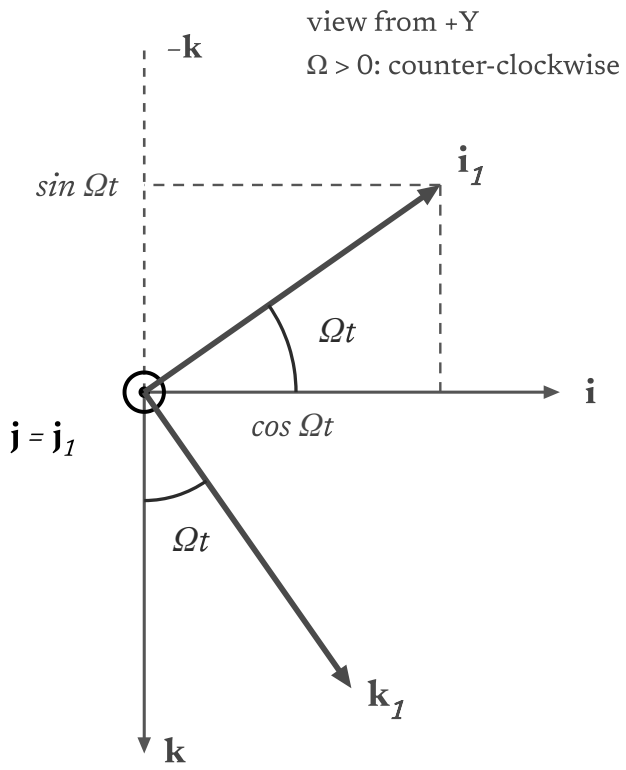


Figure 1. View from $+Y$, with $\mathbf{j} = \mathbf{j}_1$ out of the paper ($\odot$): then $\mathbf{k} = \mathbf{i} \times \mathbf{j}$ points down the page. $\mathbf{i}_1, \mathbf{k}_1$ are $\mathbf{i}, \mathbf{k}$ turned counter-clockwise by Ωt; the components of $\mathbf{i}_1$ are $\cos \Omega t$ along $\mathbf{i}$ and $\sin \Omega t$ along $-\mathbf{k}$ (up the page), hence the minus sign.

The rotation is about $\mathbf{j}$, so it is drawn looking down the axis, from $+Y$. In that view $\mathbf{j}$ points out of the paper, towards the reader ($\odot$). Take $\mathbf{i}$ to the right. The direction of $\mathbf{k}$ then follows from the right-hand rule, $\mathbf{k} = \mathbf{i} \times \mathbf{j}$: the fingers go from $\mathbf{i}$ (right) to $\mathbf{j}$ (towards the reader) and the thumb points down the page. So in this view $\mathbf{k}$ points down and $-\mathbf{k}$ points up.

With $\boldsymbol{\omega} = \Omega\,\mathbf{j}$ and $\Omega > 0$, the wire turns counter-clockwise in this view (right-hand rule again: thumb along $\mathbf{j}$, towards the reader). After a time t the axis $\mathbf{i}_1$ has turned the angle Ωt from $\mathbf{i}$:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of the right triangle drawn with dashed lines. The leg adjacent to Ωt lies along $\mathbf{i}$ and measures $\cos \Omega t$. The leg opposite to Ωt points up the page, which is the direction $-\mathbf{k}$, and measures $\sin \Omega t$. Hence $\mathbf{i}_1 = \cos \Omega t\, \mathbf{i} + \sin \Omega t\, (-\mathbf{k})$. **The minus sign comes from $\mathbf{k}$ pointing down the page in this view.**

- $\mathbf{k}_1$ starts on $\mathbf{k}$ (down the page) and turns the same angle Ωt counter-clockwise, which moves its tip towards $+\mathbf{i}$. Its leg along $\mathbf{k}$ (adjacent) is $\cos\Omega t$ and its leg along $\mathbf{i}$ (opposite) is $+\sin\Omega t$.
- $\mathbf{j}_1 = \mathbf{j}$, because it is the axis of the rotation.

$$\mathbf{i}_1 = \cos\Omega t\,\mathbf{i} - \sin\Omega t\,\mathbf{k}, \qquad \mathbf{j}_1 = \mathbf{j}, \qquad \mathbf{k}_1 = \sin\Omega t\,\mathbf{i} + \cos\Omega t\,\mathbf{k}$$

The columns of the change-of-basis matrix are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$ (vault note *Topic 5 - Relative Motion*, Sec. 4.1):

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & 0 & \sin\Omega t \\ 0 & 1 & 0 \\ -\sin\Omega t & 0 & \cos\Omega t \end{pmatrix}, \qquad \det[{}_0R_1] = \cos\Omega t\cos\Omega t - \sin\Omega t\,(-\sin\Omega t) = \cos^2\Omega t + \sin^2\Omega t = 1$$

Each column has unit length ($\cos^2\Omega t + \sin^2\Omega t = 1$) and the first and third are orthogonal ($\cos\Omega t\sin\Omega t - \sin\Omega t\cos\Omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$; also $\mathbf{i}_1 \times \mathbf{j}_1 = \mathbf{k}_1$, so $\mathcal{B}_1$ is right-handed.

Motion of S_1. $\mathbf{r}_0^O = \mathbf{0}$, $\mathbf{v}_0^O = \mathbf{0}$, $\mathbf{a}_0^O = \mathbf{0}$ (fixed origin); $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}_1$ and $\boldsymbol{\alpha}_{10} = \dot{\Omega}\,\mathbf{j}_1 = \mathbf{0}$. Check with Poisson's formula: by the chain rule, $\left.\frac{d\mathbf{i}_1}{dt}\right|_0 = \frac{d(\cos\Omega t)}{dt}\,\mathbf{i} - \frac{d(\sin\Omega t)}{dt}\,\mathbf{k} = -\Omega\sin\Omega t\,\mathbf{i} - \Omega\cos\Omega t\,\mathbf{k} = -\Omega\,\mathbf{k}_1$, and indeed $\boldsymbol{\omega}_{10}\times\mathbf{i}_1 = \Omega\,\mathbf{j}_1\times\mathbf{i}_1 = -\Omega\,\mathbf{k}_1$.

The position vector in the plane of the wire

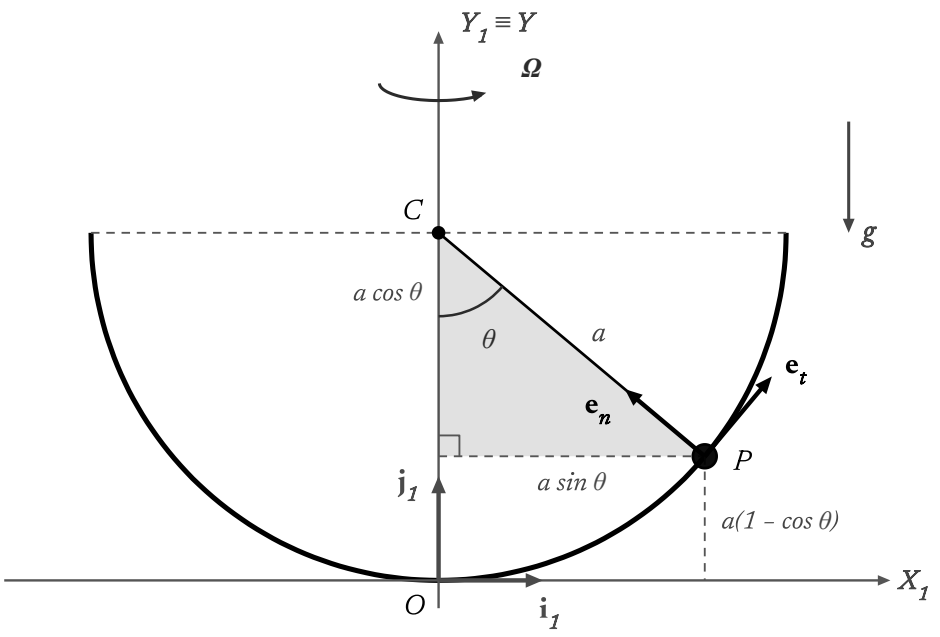


Figure 2. Plane of the wire, fixed in S_1: θ is measured at C from the downward vertical CO. In the shaded right triangle the hypotenuse is $|CP| = a$, the leg opposite θ is $a\sin\theta$ and the adjacent leg is $a\cos\theta$; the dashed lines are the components of $\mathbf{r}_1^P$. $\mathbf{i}_1, \mathbf{j}_1$ in blue, $\mathbf{e}_t, \mathbf{e}_n$ in orange.

The plane of the wire is fixed in S_1, so it is drawn with $\mathbf{i}_1$ to the right and $\mathbf{j}_1$ up. To go from O to P we pass through the centre, because the distance from C is the one that stays constant:

$$\mathbf{r}_1^P = \overrightarrow{OC} + \overrightarrow{CP}$$

- $\overrightarrow{OC} = a\,\mathbf{j}_1$: C is on the axis, at height a above O.
- $\overrightarrow{CP}$ has length a and makes the angle θ with the downward vertical $-\mathbf{j}_1$. Let F be the point of the axis at the height of P. In the shaded right triangle CFP (right angle at F) the hypotenuse is $|CP| = a$; the leg opposite to θ is $FP = a\sin\theta$, horizontal and towards $+\mathbf{i}_1$ for $\theta > 0$; the leg adjacent to θ is $CF = a\cos\theta$, vertical and pointing down, along $-\mathbf{j}_1$. So $\overrightarrow{CP} = a\sin\theta\,\mathbf{i}_1 - a\cos\theta\,\mathbf{j}_1$.

Adding the two vectors:

$$\mathbf{r}_1^P = a\sin\theta\,\mathbf{i}_1 + a\,(1-\cos\theta)\,\mathbf{j}_1$$

These are the two dashed components of the figure: the horizontal one $a\sin\theta$ and the height of P above O, $a - a\cos\theta$. Checks: $\theta = 0$ gives $\mathbf{r}_1^P = \mathbf{0}$ (the particle at O, the initial position); $\theta = \pi/2$ gives $a\,\mathbf{i}_1 + a\,\mathbf{j}_1$, the end (a, a) of the wire.

Relation with the standard polar angle. The usual polar angle φ about C is measured from $\mathbf{i}_1$, counter-clockwise, with $\overrightarrow{CP} = a\,(\cos\varphi\,\mathbf{i}_1 + \sin\varphi\,\mathbf{j}_1)$. Comparing with $\overrightarrow{CP} = a\sin\theta\,\mathbf{i}_1 - a\cos\theta\,\mathbf{j}_1$ requires $\cos\varphi = \sin\theta$ and $\sin\varphi = -\cos\theta$, which holds for

$$\varphi = \theta - \frac{\pi}{2}: \qquad \cos\left(\theta - \tfrac{\pi}{2}\right) = \cos\theta\cos\tfrac{\pi}{2} + \sin\theta\sin\tfrac{\pi}{2} = \sin\theta, \qquad \sin\left(\theta - \tfrac{\pi}{2}\right) = \sin\theta\cos\tfrac{\pi}{2} - \cos\theta\sin\tfrac{\pi}{2} = -\cos\theta$$

(with $\cos\frac{\pi}{2} = 0$ and $\sin\frac{\pi}{2} = 1$). So θ is the polar angle measured from $-\mathbf{j}_1$ instead of $\mathbf{i}_1$. It is chosen so that $\theta = 0$ is the lowest point O, where the particle starts and where the trivial equilibrium is.

Intrinsic basis of the wire. The tangent is the derivative of the position with respect to the coordinate. With $\frac{d}{d\theta}\sin\theta = \cos\theta$ and $\frac{d}{d\theta}(1-\cos\theta) = \sin\theta$:

$$\frac{d\mathbf{r}_1^P}{d\theta} = a\cos\theta\,\mathbf{i}_1 + a\sin\theta\,\mathbf{j}_1, \qquad \left|\frac{d\mathbf{r}_1^P}{d\theta}\right| = a\sqrt{\cos^2\theta + \sin^2\theta} = a \;\Rightarrow\; \mathbf{e}_t = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1$$

Differentiating $\mathbf{e}_t$ with respect to θ ($\frac{d}{d\theta}\cos\theta = -\sin\theta$, $\frac{d}{d\theta}\sin\theta = \cos\theta$) gives a vector of length $\sqrt{\sin^2\theta + \cos^2\theta} = 1$, the principal normal:

$$\mathbf{e}_n = \frac{d\mathbf{e}_t}{d\theta} = -\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1 = -\frac{\overrightarrow{CP}}{a}$$

so $\mathbf{e}_n$ points from P towards the centre C, as drawn. The binormal follows from $\mathbf{i}_1 \times \mathbf{j}_1 = \mathbf{k}_1$ and $\mathbf{j}_1 \times \mathbf{i}_1 = -\mathbf{k}_1$:

$$\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n = \cos^2\theta\,(\mathbf{i}_1 \times \mathbf{j}_1) - \sin^2\theta\,(\mathbf{j}_1 \times \mathbf{i}_1) = \left(\cos^2\theta + \sin^2\theta\right)\mathbf{k}_1 = \mathbf{k}_1$$

The matrix with columns $\mathbf{e}_t, \mathbf{e}_n, \mathbf{k}_1$ in $\mathcal{B}_1$ is a rotation by θ about $\mathbf{k}_1$, with determinant $\cos^2\theta + \sin^2\theta = 1$. In the polar language, $\mathbf{u}_\rho = -\mathbf{e}_n$ (radial, from C outwards) and $\mathbf{u}_\varphi = \mathbf{e}_t$.

Kinematics

(b) Relative velocity in $OX_1Y_1Z_1$

In S_1 the vectors $\mathbf{i}_1, \mathbf{j}_1$ are constant, so only the components of $\mathbf{r}_1^P$ change. By the chain rule, $\frac{d}{dt}\sin\theta = \cos\theta\,\dot\theta$ and $\frac{d}{dt}(1-\cos\theta) = \sin\theta\,\dot\theta$, that is $\mathbf{v}_1^P = \frac{d\mathbf{r}_1^P}{d\theta}\,\dot\theta$:

$$\mathbf{v}_1^P = a\dot\theta\cos\theta\,\mathbf{i}_1 + a\dot\theta\sin\theta\,\mathbf{j}_1 = a\dot\theta\,\mathbf{e}_t$$

Relative acceleration. Product rule on $a\dot\theta\,\mathbf{e}_t$ in S_1: $\frac{d}{dt}(a\dot\theta) = a\ddot\theta$ and, by the chain rule, $\frac{d\mathbf{e}_t}{dt}\big|_1 = \frac{d\mathbf{e}_t}{d\theta}\,\dot\theta = \dot\theta\,\mathbf{e}_n$:

$$\mathbf{a}_1^P = a\ddot\theta\,\mathbf{e}_t + a\dot\theta\,\dot\theta\,\mathbf{e}_n = a\ddot\theta\,\mathbf{e}_t + a\dot\theta^2\,\mathbf{e}_n$$

It is the circular motion of radius a: a tangential part and a centripetal part towards C.

Forces and Equations of Motion

(b, continued) Inertia forces

> **Pedagogical Justification:** Notes Eq. 4.62 (Slide 28) gives $\mathbf{F}_i = -m\mathbf{a}_0^O - m\boldsymbol{\alpha}_{10}\times\mathbf{r}_1^P - m\boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P) - 2m\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P$. The first two terms vanish ($\mathbf{a}_0^O = \mathbf{0}$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$).

Centrifugal force. With $\mathbf{j}_1\times\mathbf{i}_1 = -\mathbf{k}_1$ and $\mathbf{j}_1\times\mathbf{j}_1 = \mathbf{0}$, $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = \Omega\,\mathbf{j}_1\times a\sin\theta\,\mathbf{i}_1 = -\Omega a\sin\theta\,\mathbf{k}_1$; then, with $\mathbf{j}_1\times\mathbf{k}_1 = \mathbf{i}_1$, $\boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P) = \Omega\,\mathbf{j}_1\times(-\Omega a\sin\theta\,\mathbf{k}_1) = -\Omega^2 a\sin\theta\,\mathbf{i}_1$:

$$\mathbf{F}_C = m\Omega^2 a\sin\theta\,\mathbf{i}_1$$

It points horizontally away from the axis and equals $m\Omega^2\mathbf{r}_\perp$ (Slide 30), since $\mathbf{r}_\perp = a\sin\theta\,\mathbf{i}_1$ is the part of $\mathbf{r}_1^P$ perpendicular to $\mathbf{j}_1$ (the horizontal leg of the triangle).

Coriolis force. $\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P = \Omega\,\mathbf{j}_1\times\left(a\dot\theta\cos\theta\,\mathbf{i}_1 + a\dot\theta\sin\theta\,\mathbf{j}_1\right) = a\Omega\dot\theta\cos\theta\,(\mathbf{j}_1\times\mathbf{i}_1) = -\Omega a\dot\theta\cos\theta\,\mathbf{k}_1$, so

$$\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = 2m\Omega a\dot{\theta}\cos\theta\,\mathbf{k}_1$$

Total inertia force:

$$\mathbf{F}_i = m\Omega^2 a\sin\theta\,\mathbf{i}_1 + 2m\Omega a\dot{\theta}\cos\theta\,\mathbf{k}_1$$

Its projections on the wire basis use $\mathbf{i}_1\cdot\mathbf{e}_t = \cos\theta$, $\mathbf{i}_1\cdot\mathbf{e}_n = -\sin\theta$ and $\mathbf{k}_1\cdot\mathbf{e}_t = \mathbf{k}_1\cdot\mathbf{e}_n = 0$: $\mathbf{F}_C\cdot\mathbf{e}_t = m\Omega^2 a\sin\theta\cos\theta$ and $\mathbf{F}_C\cdot\mathbf{e}_n = -m\Omega^2 a\sin^2\theta$.

(c) Weight in $OX_1Y_1Z_1$

The axis of rotation is the vertical, so $\mathbf{j}_1 = \mathbf{j}$ at all times and

$$\mathbf{W} = -mg\,\mathbf{j}_1 \qquad (W_{X_1} = 0,\quad W_{Y_1} = -mg,\quad W_{Z_1} = 0)$$

In the wire basis, with $\mathbf{j}_1\cdot\mathbf{e}_t = \sin\theta$ and $\mathbf{j}_1\cdot\mathbf{e}_n = \cos\theta$: $\mathbf{W}\cdot\mathbf{e}_t = -mg\sin\theta$, $\mathbf{W}\cdot\mathbf{e}_n = -mg\cos\theta$, $\mathbf{W}\cdot\mathbf{k}_1 = 0$.

Reaction of the wire. The wire is smooth, so the reaction has no tangential component: $\mathbf{N} = N_n\,\mathbf{e}_n + N_z\,\mathbf{k}_1$.

Equations of motion.

> **Pedagogical Justification:** Newton's law in S_1, $m\mathbf{a}_1^P = \mathbf{W} + \mathbf{N} + \mathbf{F}_i$ (Notes Eq. 4.63), projected on the intrinsic basis. The tangent $\mathbf{e}_t$ eliminates the smooth reaction ($\mathbf{N}\cdot\mathbf{e}_t = 0$) and the Coriolis force ($\mathbf{k}_1\cdot\mathbf{e}_t = 0$); the other two projections give the reaction.

Projection on $\mathbf{e}_t$ (the equation of motion):

$$ma\ddot{\theta} = -mg\sin\theta + m\Omega^2 a\sin\theta\cos\theta \;\Rightarrow\; \ddot{\theta} = \sin\theta\left(\Omega^2\cos\theta - \frac{g}{a}\right)$$

Projection on $\mathbf{e}_n$:

$$ma\dot{\theta}^2 = N_n - mg\cos\theta - m\Omega^2 a\sin^2\theta$$

Projection on $\mathbf{k}_1$ ($\mathbf{a}_1^P\cdot\mathbf{k}_1 = 0$, $\mathbf{W}\cdot\mathbf{k}_1 = 0$):

$$0 = N_z + 2m\Omega a\dot\theta\cos\theta$$

Answers and Checks

(a) Degrees of freedom

CDOF $= 1$, with generalized coordinate θ; the rotation Ωt is prescribed.

(b) Velocity and inertia forces in $OX_1Y_1Z_1$

$\mathbf{v}_1^P = a\dot\theta\,\mathbf{e}_t = a\dot\theta\cos\theta\,\mathbf{i}_1 + a\dot\theta\sin\theta\,\mathbf{j}_1$ and $\mathbf{F}_i = m\Omega^2 a\sin\theta\,\mathbf{i}_1 + 2m\Omega a\dot\theta\cos\theta\,\mathbf{k}_1$.

(c) Weight

$\mathbf{W} = -mg\,\mathbf{j}_1$.

(d) Equilibrium points

Equilibria (relative rest, $\ddot\theta = \dot\theta = 0$, in the tangential equation):

1. $\sin\theta = 0$: $\theta = 0$, the lowest point O. It exists for every Ω.
2. $\cos\theta_* = \dfrac{g}{a\Omega^2}$: two symmetric points $\theta = \pm\theta_*$, which exist only if $\Omega^2 > g/a$ (then $0 < \theta_* < \pi/2$, inside the wire).

Stability with the potential of (f), $V_1(\theta) = mga(1-\cos\theta) - \frac{1}{2}m\Omega^2a^2\sin^2\theta$ (the tangential equation is $ma^2\ddot\theta = -dV_1/d\theta$). Its derivatives are $V_1' = ma\sin\theta\left(g - \Omega^2 a\cos\theta\right)$ and $V_1'' = mga\cos\theta - m\Omega^2a^2\cos 2\theta$:

- $V_1''(0) = ma\left(g - \Omega^2 a\right)$: $\theta = 0$ is stable if $\Omega^2 < g/a$ and unstable if $\Omega^2 > g/a$.
- $V_1''(\pm\theta_*) = mga\cos\theta_* - m\Omega^2a^2\left(2\cos^2\theta_* - 1\right) = m\Omega^2a^2\left(1 - \cos^2\theta_*\right) = m\Omega^2a^2\sin^2\theta_* > 0$ (using $ga = \Omega^2a^2\cos\theta_*$): the off-axis equilibria are stable whenever they exist.

(e) Quadratures and reaction force

> **Pedagogical Justification:** multiplying the tangential equation by $\dot\theta$ makes both sides exact derivatives; the result is the energy integral of (f). The projections on $\mathbf{e}_n$ and $\mathbf{k}_1$ then give the reaction.

First integral. Multiply $ma^2\ddot\theta = -mga\sin\theta + m\Omega^2a^2\sin\theta\cos\theta$ by $\dot\theta$. By the chain rule, $\frac{d}{dt}\left(\frac{1}{2}\dot\theta^2\right) = \dot\theta\ddot\theta$, $\frac{d}{dt}\cos\theta = -\sin\theta\,\dot\theta$ and $\frac{d}{dt}\left(\frac{1}{2}\sin^2\theta\right) = \sin\theta\cos\theta\,\dot\theta$, so

$$\frac{d}{dt}\left[\tfrac{1}{2}ma^2\dot\theta^2 - mga\cos\theta - \tfrac{1}{2}m\Omega^2a^2\sin^2\theta\right] = 0$$

Integrate from 0 to t and apply Barrow's rule, with $\theta(0)=0$ and $a\dot\theta(0)=v_0$ (at O, $\mathbf{r}_1^P=\mathbf{0}$, so $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P=\mathbf{0}$ and the given absolute velocity equals the relative one, $v_0\,\mathbf{i}_1=a\dot\theta(0)\,\mathbf{e}_t(0)$):

$$\left[\tfrac{1}{2}ma^2\dot\theta^2-mga\cos\theta-\tfrac{1}{2}m\Omega^2a^2\sin^2\theta\right]_0^t=0\;\Rightarrow\;\dot\theta^2=\frac{v_0^2}{a^2}-\frac{2g}{a}\left(1-\cos\theta\right)+\Omega^2\sin^2\theta\equiv\Phi(\theta)$$

Quadrature. Separate variables ($d\theta=\sqrt{\Phi}\,dt$, positive root since $\dot\theta(0)=v_0/a>0$, valid until the first turning point $\Phi=0$) and integrate from 0 to t:

$$t=\int_0^{\theta(t)}\frac{d\theta'}{\sqrt{\dfrac{v_0^2}{a^2}-\dfrac{2g}{a}\left(1-\cos\theta'\right)+\Omega^2\sin^2\theta'}}$$

Reaction force. From the projection on $\mathbf{e}_n$, $N_n=ma\dot\theta^2+mg\cos\theta+m\Omega^2a\sin^2\theta$; substituting $\dot\theta^2=\Phi(\theta)$:

$$N_n(\theta)=\frac{mv_0^2}{a}+mg\left(3\cos\theta-2\right)+2m\Omega^2a\sin^2\theta$$

From the projection on $\mathbf{k}_1$:

$$N_z=-2m\Omega a\dot\theta\cos\theta=\mp2m\Omega a\cos\theta\sqrt{\Phi(\theta)}$$

(upper sign while $\dot\theta>0$). The total reaction is $\mathbf{N}=N_n\left(-\sin\theta\,\mathbf{i}_1+\cos\theta\,\mathbf{j}_1\right)+N_z\,\mathbf{k}_1$.

Staying on the wire. The particle reaches the end $\theta=\pi/2$ only if $\Phi(\pi/2)=v_0^2/a^2-2g/a+\Omega^2\geq0$; otherwise it oscillates between the two roots of $\Phi(\theta)=0$. The condition is also sufficient: Φ is a concave quadratic in $\cos\theta$, so on $[0,\pi/2]$ its minimum is at an endpoint, and $\Phi(0)=v_0^2/a^2>0$.

(f) Energy equation

> **Pedagogical Justification:** E_1^P is conserved when every force that does work in S_1 derives from a time-independent potential (Slide 29; vault note *Topic 5 - Relative Motion*, Sec. 12.6).

Constant Ω: yes. In S_1: the wire is fixed and smooth, so $\mathbf{N}\cdot\mathbf{v}_1^P=0$; the Coriolis force never does work, $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P=0$ (Slide 31); the weight has the potential $mgy_1=mga(1-\cos\theta)$; the centrifugal force has the potential $-\tfrac{1}{2}m\Omega^2r_\perp^2=-\tfrac{1}{2}m\Omega^2a^2\sin^2\theta$ (Slide 30), time-independent because Ω is constant. Hence

$$E_1^P = \tfrac{1}{2}ma^2\dot\theta^2 + mga\,(1-\cos\theta) - \tfrac{1}{2}m\Omega^2a^2\sin^2\theta = \tfrac{1}{2}mv_0^2$$

which is the first integral used in (e). (In S_0 the mechanical energy is not conserved: the reaction N_z is perpendicular to the moving wire but not to the absolute velocity, so the motor that keeps Ω constant does work.)

Variable $\Omega(t)$: no energy integral. Two things change. The Euler force appears, $\mathbf{F}_E = -m\dot\Omega\,\mathbf{j}_1 \times a\sin\theta\,\mathbf{i}_1 = m\dot\Omega a\sin\theta\,\mathbf{k}_1$; it is normal to the plane of the wire, so $\mathbf{F}_E\cdot\mathbf{v}_1^P = 0$ and it does no work in S_1 (it is balanced by N_z). But the centrifugal potential now depends explicitly on time, $V_C = -\tfrac{1}{2}m\Omega(t)^2a^2\sin^2\theta$, and differentiating E_1^P along the motion (tangential equation with $\Omega(t)$) gives

$$\frac{dE_1^P}{dt} = \frac{\partial V_C}{\partial t} = -m\Omega\dot\Omega a^2\sin^2\theta \neq 0$$

so E_1^P is not a constant of motion and the problem cannot be reduced to a quadrature with the energy equation. The work-energy relation $dE_1^P/dt = \partial V_C/\partial t$ is still valid, but it is not a first integral.

Checks

- **Bifurcation.** For $\Omega^2 < g/a$ gravity dominates and the bottom is the only (stable) equilibrium. At $\Omega_c = \sqrt{g/a}$ the bottom becomes unstable and two stable equilibria $\pm\theta_*$ appear, where the tangential components of weight and centrifugal force balance: $mg\sin\theta_* = m\Omega^2a\sin\theta_*\cos\theta_*$.
- **Limits.** $\Omega\to 0$: $\Phi = v_0^2/a^2 - 2g(1-\cos\theta)/a$ and $N_n = mv_0^2/a + mg(3\cos\theta - 2)$, the simple-pendulum results; $N_z\to 0$. $\Omega\to\infty$: $\cos\theta_*\to 0$, the equilibria move to the ends of the wire.
- **Coriolis and the reaction.** N_z is exactly the force the wire must exert to cancel the Coriolis force; it changes sign with $\dot\theta$, pushing the particle forward or backward with the rotation.
- **Orthogonality:** $\mathbf{e}_t\cdot\mathbf{e}_n = -\cos\theta\sin\theta + \sin\theta\cos\theta = 0$ and $|\mathbf{e}_t| = |\mathbf{e}_n| = 1$.
- **Dimensions:** $[a\sin\theta] = [a(1-\cos\theta)] = \mathrm{m}$, with θ in rad (dimensionless); $[\Omega^2] = [g/a] = \mathrm{s}^{-2}$, so every term of Φ is in s^{-2} and $[t] = \mathrm{rad}/\mathrm{s}^{-1} = \mathrm{s}$; $[mv_0^2/a] = [mg] = [m\Omega^2a] = \mathrm{kg}\cdot\mathrm{m}\cdot\mathrm{s}^{-2} = \mathrm{N}$; $[m\Omega a\dot\theta] = \mathrm{N}$; $[E_1^P] = [ma^2\dot\theta^2] = \mathrm{J}$.
- **Symbolic check (sympy):** $[{}_0R_1]$ and $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}$; Notes Eq. 3.27 against the double derivative of $\mathbf{r}_1^P$ in $\mathcal{B}_0$; $\mathbf{F}_C$, $\mathbf{F}_{COR}$; the equation of motion and both reaction components from $m\mathbf{a}_0^P = \mathbf{W} + \mathbf{N}$ in S_0; $dE_1^P/dt = 0$; $\Phi(\theta)$ and $N_n(\theta)$; the stability derivatives; and, for $\Omega(t)$, the Euler force, its zero power and $dE_1^P/dt = -m\Omega\dot\Omega a^2\sin^2\theta$. A numerical integration ($g = 9.81$, $a = 1$, $\Omega = 4$, $v_0 = 2$, SI) keeps $\dot\theta^2 - \Phi(\theta)$ below 3×10^{-10}.

- **Check against the official key:** `Problems.pdf` has no entry for Problem 26 in its solution keys, so the results are verified by independent derivation (the symbolic and numerical checks above).

Source: Problems.pdf

PROBLEM 27

Bead on a Rotating Sine-Shaped Wire with a Spring

Original statement · Problems.pdf, p. 15

Rotating Wire + Spring

A heavy bead P of mass m slides without friction along a thin wire with shape given by the following equation:

$$y_1(x_1) = \sin x_1.$$

Moreover, the bead is joined to a straight rail located at $y_1 = 2$ by a spring of stiffness k. The connection of the spring with the rail can move freely along the rail, so that the spring is always vertical. The whole setup rotates with respect to the y_1 axis with a constant angular velocity ω. A sketch of the setup and the definition of the right-handed, rotating $Ox_1y_1z_1$ reference frame is given below.

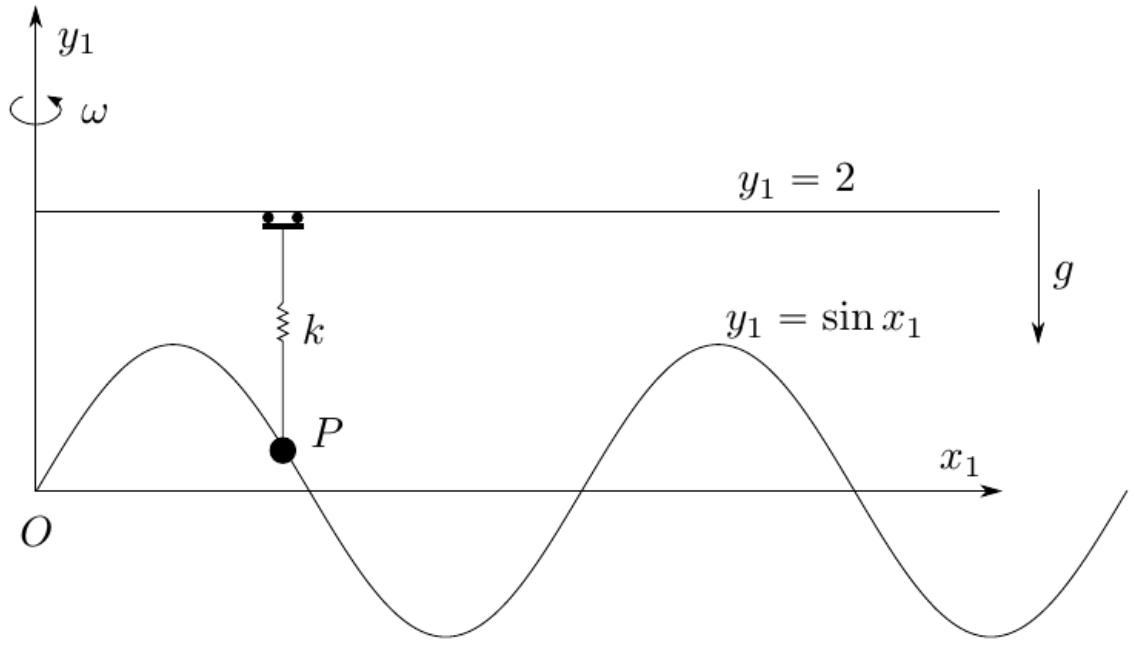


Figure: Sine-shaped wire y1 = sin x1 rotating about the vertical axis Oy1 with angular velocity ω; bead P joined by a vertical spring k to the rail y1 = 2 (Problems.pdf, p. 15).

(a) Write down the general expression for the components of the relative velocity and acceleration of P in the reference frame $Ox_1y_1z_1$ as functions of the axial position of the bead, x_1, and its time derivatives $\dot{x}_1$ and $\ddot{x}_1$.

(b) Identify all forces acting on the bead and then write down the three components of Newton's second law in the $Ox_1y_1z_1$ reference frame. From these equations, obtain an explicit expression of the components of the reaction force on the wire as a function of x_1, $\dot{x}_1$ and the parameters of the problem only, and a second order differential equation for x_1 that does not depend on the reaction force.

(c) Is it possible to use the equation of conservation of mechanical energy of the relative motion of P? If so, write it down in terms of x_1, $\dot{x}_1$ and the parameters of the problem. What is the relation of this equation with the differential equation for x_1 in the previous point?

(d) For $\omega = 0$ and $x_1 \in [0, 2\pi]$, discuss the existence of equilibrium positions as a function of the parameter $\lambda = mg/k$. Are these equilibrium positions stable?

(e) Consider now that $\omega \neq 0$, and $k = 0$. We drop the bead P at $x_1 = \pi$ with no initial velocity. Sketch the shape of the potential and discuss qualitatively the motion of P if the parameter $\beta = \omega^2/g$ is small or large. Write down the system of equations that would allow to solve for the critical value of β that divides between the two behaviors.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a heavy bead P (mass m) slides on the smooth plane wire $y_1 = \sin x_1$. A vertical spring of stiffness k joins it to the rail $y_1 = 2$; the upper end slides freely on the rail, so the spring stays vertical. Wire, rail and spring rotate about the vertical axis Oy_1 with constant angular velocity ω.
- **Hypotheses:** the non-rotating frame with origin O and vertical axis Oy_1 is inertial; gravity is $-g\,\mathbf{j}_1$; the wire is smooth (no reaction along the tangent). The statement does not give the natural length of the spring; it is taken as zero (the form of the official energy integral implies this), so the spring force is k times its length $2 - \sin x_1$. Lengths in the wire equation are measured in metres (the equation $y_1 = \sin x_1$ hides a unit length of 1 m).

- **Why the rotating frame:** in $Ox_1y_1z_1$ the wire, the rail and the spring are fixed, so the bead moves on a fixed curve with one coordinate x_1; the rotation enters only through the centrifugal and Coriolis forces (Notes Eq. 4.62).
- **Fixed (inertial) frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, basis $\mathcal{B}_0$, with $\mathbf{j}_0$ vertical upwards.
- **Rotating frame:** $S_1 = Ox_1y_1z_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, basis $\mathcal{B}_1$, attached to the wire: Ox_1 in the plane of the wire, Oy_1 the vertical axis of rotation ($\mathbf{j}_1 = \mathbf{j}_0$), Oz_1 normal to the plane of the wire; coincident with S_0 at $t = 0$.
- **Directions attached to the wire:** the tangent $\mathbf{t}$ (in the sense of increasing x_1), the in-plane normal $\mathbf{n}$ and the normal to the plane $\mathbf{k}_1$; deduced in the next section.
- **Degrees of freedom:** 3 coordinates minus 2 constraints ($z_1 = 0$ and $y_1 = \sin x_1$): CDOF $= 1$, generalized coordinate x_1.

Symbol	Meaning	SI unit
m	mass of the bead	kg
k	spring stiffness	N/m
ω	angular velocity of the setup (constant)	rad/s
g	gravitational acceleration	m/s^2
x_1	axial position of the bead	m
$\lambda = mg/k$	weight-to-stiffness ratio	m
$\beta = \omega^2/g$	centrifugal-to-gravity ratio	m^{-1}

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from above

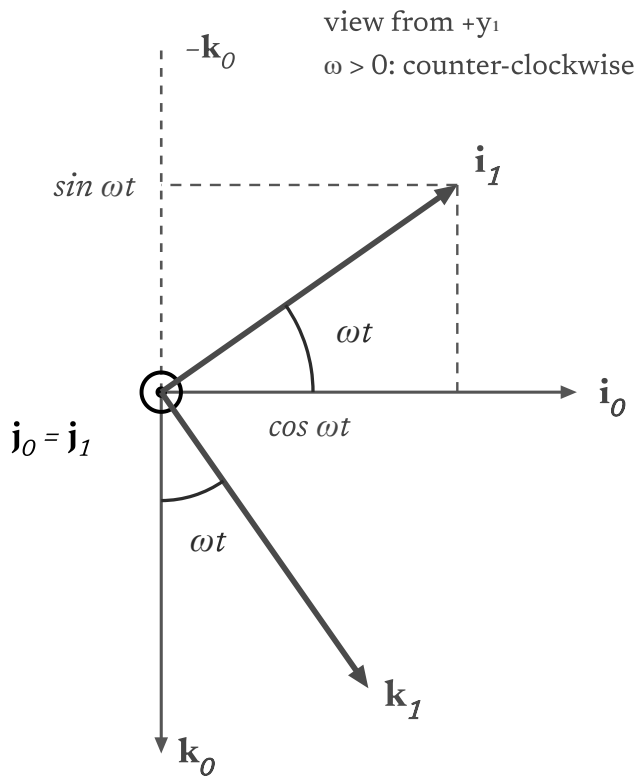


Figure 1. View from $+y_1$, with $\mathbf{j}_0 = \mathbf{j}_1$ out of the paper ($\odot$): $\mathbf{k}_0 = \mathbf{i}_0 \times \mathbf{j}_0$ points down the page. The plane of the wire turns counter-clockwise by ωt: the components of $\mathbf{i}_1$ are $\cos\omega t$ along $\mathbf{i}_0$ and $\sin\omega t$ along $-\mathbf{k}_0$ (up the page), hence $\mathbf{i}_1 = \cos\omega t\,\mathbf{i}_0 - \sin\omega t\,\mathbf{k}_0$.

The setup turns about $\mathbf{j}_0 = \mathbf{j}_1$, so the rotation is drawn looking down the axis, from $+y_1$: $\mathbf{j}_0$ points out of the paper, towards the reader ($\odot$). With $\mathbf{i}_0$ to the right, the right-hand rule $\mathbf{k}_0 = \mathbf{i}_0 \times \mathbf{j}_0$ (fingers from $\mathbf{i}_0$, to the right, towards $\mathbf{j}_0$, towards the reader) puts $\mathbf{k}_0$ down the page; $-\mathbf{k}_0$ points up the page. With $\omega > 0$ the wire turns counter-clockwise in this view (thumb along $\mathbf{j}_0$, towards the reader), and after a time t the axis $\mathbf{i}_1$ has turned ωt from $\mathbf{i}_0$:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of the dashed right triangle: the leg adjacent to ωt lies along $\mathbf{i}_0$ and measures $\cos\omega t$; the opposite leg points up the page, the direction $-\mathbf{k}_0$, and measures $\sin\omega t$. Hence $\mathbf{i}_1 = \cos\omega t\,\mathbf{i}_0 + \sin\omega t\,(-\mathbf{k}_0)$: **the minus sign comes from $\mathbf{k}_0$ pointing down the page in this view.**
- $\mathbf{k}_1$ starts on $\mathbf{k}_0$ (down the page) and turns the same angle counter-clockwise, so its tip moves towards $+\mathbf{i}_0$: adjacent leg $\cos\omega t$ along $\mathbf{k}_0$, opposite leg $+\sin\omega t$ along $\mathbf{i}_0$.
- $\mathbf{j}_1 = \mathbf{j}_0$, the axis of the rotation.

$$\mathbf{i}_1 = \cos\omega t\,\mathbf{i}_0 - \sin\omega t\,\mathbf{k}_0, \qquad \mathbf{j}_1 = \mathbf{j}_0, \qquad \mathbf{k}_1 = \sin\omega t\,\mathbf{i}_0 + \cos\omega t\,\mathbf{k}_0$$

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & 0 & \sin\omega t \\ 0 & 1 & 0 \\ -\sin\omega t & 0 & \cos\omega t \end{pmatrix}, \qquad \det[{}_0R_1] = \cos\omega t\cos\omega t - \sin\omega t\,(-\sin\omega t) = \cos^2\omega t + \sin^2\omega t = 1$$

Multiplying out, $[{}_0R_1][{}_0R_1]^T$ has diagonal entries $\cos^2\omega t + \sin^2\omega t = 1$ (and 1) and off-diagonal entries $\pm(\cos\omega t\sin\omega t - \sin\omega t\cos\omega t) = 0$, so it is the identity.

Motion of S_1. $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$; $\boldsymbol{\omega}_{10} = \omega\,\mathbf{j}_1$ (constant), $\boldsymbol{\alpha}_{10} = \mathbf{0}$. Check with Poisson's formula: by the chain rule, $\frac{d\mathbf{i}_1}{dt}\big|_0 = -\omega\sin\omega t\,\mathbf{i}_0 - \omega\cos\omega t\,\mathbf{k}_0 = -\omega\,\mathbf{k}_1$, and $\omega\,\mathbf{j}_1 \times \mathbf{i}_1 = -\omega\,\mathbf{k}_1$.

Position vector, tangent and normal in the plane of the wire

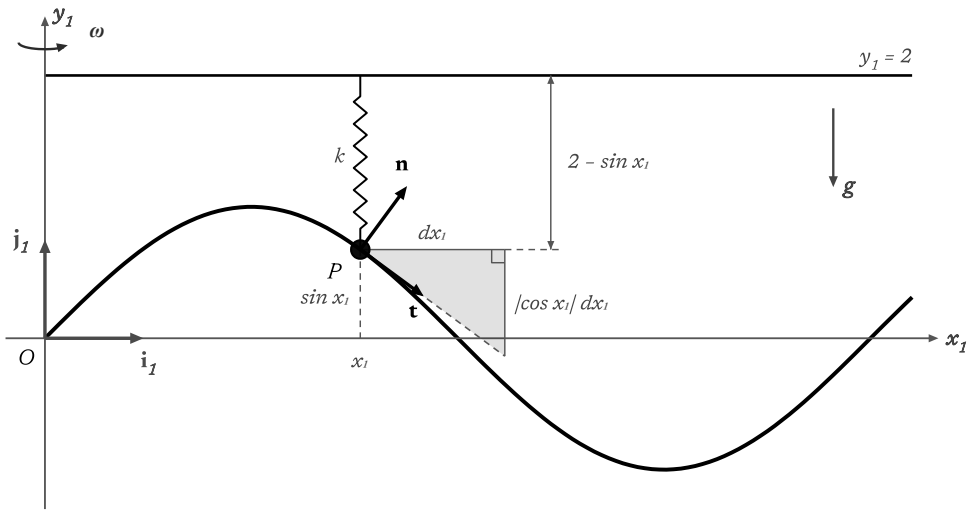


Figure 2. Plane of the wire, fixed in $Ox_1y_1z_1$. The dashed lines are the components x_1 and $\sin x_1$ of $\mathbf{r}_1^P$; the spring is vertical, of length $2 - \sin x_1$. In the shaded slope triangle the horizontal leg is dx_1 and the vertical leg is $dy_1 = \cos x_1\,dx_1$ (negative here: the wire goes down), so the tangent direction is $\mathbf{t} = \mathbf{i}_1 + \cos x_1\,\mathbf{j}_1$ and the in-plane normal is $\mathbf{n} = -\cos x_1\,\mathbf{i}_1 + \mathbf{j}_1$ (orange, drawn as unit vectors).

The plane of the wire is fixed in S_1 and is drawn with $\mathbf{i}_1$ to the right and $\mathbf{j}_1$ up ($\mathbf{k}_1 = \mathbf{i}_1 \times \mathbf{j}_1$ out of the paper). The position of the bead is the sum of its two dashed components, the abscissa x_1 and the height given by the wire:

$$\mathbf{r}_1^P = x_1\,\mathbf{i}_1 + \sin x_1\,\mathbf{j}_1$$

The spring goes vertically from P to the rail $y_1 = 2$, so its length is the difference of heights, $2 - \sin x_1 > 0$ (always positive because $\sin x_1 \le 1$).

Slope triangle. By the chain rule, $dy_1 = \frac{d}{dx_1}(\sin x_1)\, dx_1 = \cos x_1\, dx_1$. In the shaded triangle the horizontal leg is dx_1 (along $\mathbf{i}_1$) and the vertical leg is $\cos x_1\, dx_1$ (along $\mathbf{j}_1$; drawn downwards because $\cos x_1 < 0$ at the position shown). The hypotenuse points along the wire, so a (not normalised) tangent vector is the vector sum of the two legs divided by dx_1, and its length follows from Pythagoras' theorem:

$$\mathbf{t} = \mathbf{i}_1 + \cos x_1\, \mathbf{j}_1, \qquad |\mathbf{t}|^2 = 1 + \cos^2 x_1$$

In-plane normal. Turning $\mathbf{t}$ by $+90^\circ$ about $\mathbf{k}_1$ (counter-clockwise in the drawing, since $\mathbf{k}_1$ points towards the reader) takes the components $(1, \cos x_1)$ to $(-\cos x_1, 1)$:

$$\mathbf{n} = \mathbf{k}_1 \times \mathbf{t} = -\cos x_1\, \mathbf{i}_1 + \mathbf{j}_1, \qquad \mathbf{t} \cdot \mathbf{n} = -\cos x_1 + \cos x_1 = 0$$

(with $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$ and $\mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$). $\mathbf{t}$, $\mathbf{n}$ and $\mathbf{k}_1$ are mutually orthogonal; $\mathbf{n}$ has the same length as $\mathbf{t}$.

Kinematics

(a) Relative velocity and acceleration

> **Pedagogical Justification:** $\mathcal{B}_1$ is constant in S_1, so the relative velocity and acceleration are the time derivatives of the components of $\mathbf{r}_1^P$ (Notes Eq. 3.7).

Chain rule: $\frac{d}{dt}\sin x_1(t) = \cos x_1\, \dot{x}_1$. Product and chain rule: $\frac{d}{dt}\left(\cos x_1\, \dot{x}_1\right) = -\sin x_1\, \dot{x}_1 \cdot \dot{x}_1 + \cos x_1\, \ddot{x}_1 = -\sin x_1\, \dot{x}_1^2 + \cos x_1\, \ddot{x}_1$. Hence

$$\mathbf{v}_1^P = \dot{x}_1\, \mathbf{i}_1 + \cos x_1\, \dot{x}_1\, \mathbf{j}_1 + 0\, \mathbf{k}_1 = \dot{x}_1\, \mathbf{t}$$

$$\mathbf{a}_1^P = \ddot{x}_1\, \mathbf{i}_1 + \left(\cos x_1\, \ddot{x}_1 - \sin x_1\, \dot{x}_1^2\right) \mathbf{j}_1 + 0\, \mathbf{k}_1$$

Forces and Equations of Motion

(b) Forces on the bead, Newton's law, reaction and equation of motion

Pedagogical Justification: in the non-inertial frame S_1, $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63). Of the four inertia forces (Notes Eq. 4.62, Slide 28), only the centrifugal and the Coriolis forces survive, because $\mathbf{a}_0^O = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$.

Real forces.

1. Weight: $\mathbf{W} = -mg\,\mathbf{j}_1$.
2. Spring: its length is $2 - \sin x_1 > 0$ and it pulls the bead up towards the rail: $\mathbf{F}_s = k\,(2 - \sin x_1)\,\mathbf{j}_1$.
3. Wire reaction: $\mathbf{N} = N_x\,\mathbf{i}_1 + N_y\,\mathbf{j}_1 + N_z\,\mathbf{k}_1$ with $\mathbf{N}\cdot\mathbf{t} = 0$, that is $N_x + N_y\cos x_1 = 0$ (smooth wire).

Inertia forces. With $\mathbf{j}_1 \times \mathbf{i}_1 = -\mathbf{k}_1, \mathbf{j}_1 \times \mathbf{j}_1 = \mathbf{0}, \mathbf{j}_1 \times \mathbf{k}_1 = \mathbf{i}_1$:

$$\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = \omega\,\mathbf{j}_1 \times (x_1\,\mathbf{i}_1 + \sin x_1\,\mathbf{j}_1) = -\omega x_1\,\mathbf{k}_1, \qquad \boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) = \omega\,\mathbf{j}_1 \times (-\omega x_1\,\mathbf{k}_1) = -\omega^2 x_1\,\mathbf{i}_1$$

$$\mathbf{F}_C = m\omega^2 x_1\,\mathbf{i}_1, \qquad \mathbf{F}_{COR} = -2m\,\omega\,\mathbf{j}_1 \times (\dot{x}_1\,\mathbf{i}_1 + \cos x_1\,\dot{x}_1\,\mathbf{j}_1) = 2m\omega\dot{x}_1\,\mathbf{k}_1$$

Three components of Newton's law in S_1:

$$\mathbf{i}_1: \quad m\ddot{x}_1 = N_x + m\omega^2 x_1$$

$$\mathbf{j}_1: \quad m\left(\cos x_1\,\ddot{x}_1 - \sin x_1\,\dot{x}_1^2\right) = N_y - mg + k\,(2 - \sin x_1)$$

$$\mathbf{k}_1: \quad 0 = N_z + 2m\omega\dot{x}_1$$

Out-of-plane reaction. From the $\mathbf{k}_1$ equation, $N_z = -2m\omega\dot{x}_1$: the wire cancels the Coriolis force.

In-plane reaction. Substitute $N_x = -N_y \cos x_1$ in the $\mathbf{i}_1$ equation, $m\ddot{x}_1 = -N_y \cos x_1 + m\omega^2 x_1$. Multiply it by $\cos x_1$ and substitute $m \cos x_1\, \ddot{x}_1$ in the $\mathbf{j}_1$ equation:

$$-N_y \cos^2 x_1 + m\omega^2 x_1 \cos x_1 - m \sin x_1\, \dot{x}_1^2 = N_y - mg + k\,(2 - \sin x_1)$$

Collect N_y:

$$N_y = \frac{m\left(\omega^2 x_1 \cos x_1 - \dot{x}_1^2 \sin x_1\right) + mg - k\,(2 - \sin x_1)}{1 + \cos^2 x_1}, \qquad N_x = -N_y \cos x_1$$

so the reaction is

$$\mathbf{N} = \frac{m\left(\omega^2 x_1 \cos x_1 - \dot{x}_1^2 \sin x_1\right) + mg - k\,(2 - \sin x_1)}{1 + \cos^2 x_1}\,(-\cos x_1\,\mathbf{i}_1 + \mathbf{j}_1) - 2m\omega\dot{x}_1\,\mathbf{k}_1$$

(the in-plane part is along $\mathbf{n}$, as it must be for a smooth wire).

Equation of motion. Project Newton's law on the tangent $\mathbf{t} = \mathbf{i}_1 + \cos x_1\,\mathbf{j}_1$: add the $\mathbf{i}_1$ equation and $\cos x_1$ times the $\mathbf{j}_1$ equation; $N_x + N_y \cos x_1 = 0$ removes the reaction:

$$m\ddot{x}_1 + m\cos x_1\left(\cos x_1\,\ddot{x}_1 - \sin x_1\,\dot{x}_1^2\right) = m\omega^2 x_1 + \cos x_1\,[-mg + k\,(2 - \sin x_1)]$$

$$m\left(1 + \cos^2 x_1\right)\ddot{x}_1 - m \sin x_1 \cos x_1\,\dot{x}_1^2 = m\omega^2 x_1 - mg\cos x_1 + k\cos x_1\,(2 - \sin x_1)$$

Answers and Checks

(a) Relative velocity and acceleration

$\mathbf{v}_1^P = (\dot{x}_1,\ \cos x_1\,\dot{x}_1,\ 0)$ and $\mathbf{a}_1^P = \left(\ddot{x}_1,\ \cos x_1\,\ddot{x}_1 - \sin x_1\,\dot{x}_1^2,\ 0\right)$ in $\mathcal{B}_1$.

(b) Reaction and equation of motion

$N_x = -N_y \cos x_1$, $N_y = \dfrac{m\left(\omega^2 x_1 \cos x_1 - \dot{x}_1^2 \sin x_1\right) + mg - k\,(2 - \sin x_1)}{1 + \cos^2 x_1}$, $N_z = -2m\omega\dot{x}_1$; equation of motion $m\left(1 + \cos^2 x_1\right)\ddot{x}_1 - m \sin x_1 \cos x_1\,\dot{x}_1^2 = m\omega^2 x_1 - mg\cos x_1 + k\cos x_1\,(2 - \sin x_1)$.

(c) Conservation of the mechanical energy of the relative motion

> **Pedagogical Justification:** E_1^P is conserved if all the forces that do work in S_1 derive from a time-independent potential (Slide 29). Here $\boldsymbol{\omega}_{10}$ is constant and the wire is fixed in S_1.

Yes, it can be used. Check of each force: $\mathbf{N} \cdot \mathbf{v}_1^P = \dot{x}_1 \left(N_x + N_y \cos x_1\right) = 0$; $\mathbf{F}_{COR} \cdot \mathbf{v}_1^P = 0$ (Slide 31); the weight has potential $mgy_1 = mg \sin x_1$; the spring has potential $\frac{1}{2}k \left(2 - y_1\right)^2 = \frac{1}{2}k \left(2 - \sin x_1\right)^2$ (its free end slides without friction, so only its length matters); the centrifugal force has potential $-\frac{1}{2}m\omega^2 r_\perp^2 = -\frac{1}{2}m\omega^2 x_1^2$ (Slide 30). With $T_1^P = \frac{1}{2}m|\mathbf{v}_1^P|^2 = \frac{1}{2}m\dot{x}_1^2 \left(1 + \cos^2 x_1\right)$:

$$E_1^P = \tfrac{1}{2}m\dot{x}_1^2 \left(1 + \cos^2 x_1\right) + mg \sin x_1 + \tfrac{1}{2}k \left(2 - \sin x_1\right)^2 - \tfrac{1}{2}m\omega^2 x_1^2 = \text{const}$$

Expanding $\frac{1}{2}k(2 - \sin x_1)^2 = 2k - 2k \sin x_1 + \frac{1}{2}k \sin^2 x_1$ and dropping the constant $2k$ gives the form of the official key, $\frac{1}{2}m\dot{x}_1^2(1 + \cos^2 x_1) + \sin x_1 \left(mg - 2k\right) + \frac{1}{2}k \sin^2 x_1 - \frac{1}{2}m\omega^2 x_1^2 = C$.

Relation with the equation of motion. Differentiate E_1^P in time. Product and chain rules: $\frac{d}{dt}\left[\frac{1}{2}m\dot{x}_1^2(1 + \cos^2 x_1)\right] = m\dot{x}_1\ddot{x}_1(1 + \cos^2 x_1) - m\dot{x}_1^3 \sin x_1 \cos x_1$, $\frac{d}{dt}(mg \sin x_1) = mg \cos x_1 \, \dot{x}_1$, $\frac{d}{dt}\left[\frac{1}{2}k(2 - \sin x_1)^2\right] = -k(2 - \sin x_1) \cos x_1 \, \dot{x}_1$, $\frac{d}{dt}\left(-\frac{1}{2}m\omega^2 x_1^2\right) = -m\omega^2 x_1 \dot{x}_1$. Therefore

$$\frac{dE_1^P}{dt} = \dot{x}_1 \left[m \left(1 + \cos^2 x_1\right) \ddot{x}_1 - m \sin x_1 \cos x_1 \, \dot{x}_1^2 - m\omega^2 x_1 + mg \cos x_1 - k \cos x_1 \left(2 - \sin x_1\right)\right]$$

The bracket is exactly the equation of motion of (b), so $dE_1^P/dt = 0$ along every motion: the energy equation is a first integral of the equation of motion (obtained by multiplying it by $\dot{x}_1$ and integrating once), and the equation of motion is recovered by differentiating it (for $\dot{x}_1 \neq 0$).

(d) Equilibria for $\omega = 0$, $x_1 \in [0, 2\pi]$

> **Pedagogical Justification:** with $\omega = 0$ the equation of motion is $m(1 + \cos^2 x_1)\ddot{x}_1 - m \sin x_1 \cos x_1 \dot{x}_1^2 = -dV/dx_1$, with $V(x_1) = mg \sin x_1 + \frac{1}{2}k(2 - \sin x_1)^2$. Equilibria are the critical points of V; minima are stable, maxima unstable.

First derivative (chain rule):

$$V'(x_1) = mg \cos x_1 - k \left(2 - \sin x_1\right) \cos x_1 = k \cos x_1 \left(\lambda - 2 + \sin x_1\right)$$

Second derivative (product rule):

$$V''(x_1) = -k \sin x_1 \, (\lambda - 2 + \sin x_1) + k \cos^2 x_1$$

Equilibria.

1. $\cos x_1 = 0$: $x_1 = \pi/2$ (crest of the wire) and $x_1 = 3\pi/2$ (trough). They exist for every λ.
2. $\sin x_1 = 2 - \lambda$: it has solutions only if $-1 \le 2 - \lambda \le 1$, that is $1 \le \lambda \le 3$. For $1 < \lambda < 2$ they are $x_1 = \arcsin(2 - \lambda)$ and $\pi - \arcsin(2 - \lambda)$ (on both sides of the crest); for $\lambda = 2$, $x_1 = 0, \pi, 2\pi$; for $2 < \lambda < 3$, $x_1 = \pi + \arcsin(\lambda - 2)$ and $2\pi - \arcsin(\lambda - 2)$ (on both sides of the trough). At $\lambda = 1$ and $\lambda = 3$ they merge with $\pi/2$ and $3\pi/2$.

Stability.

- $x_1 = \pi/2$: $V''(\pi/2) = -k(\lambda - 1) = k(1 - \lambda)$: stable for $\lambda < 1$ (a stiff spring holds the bead on the crest), unstable for $\lambda > 1$.
- $x_1 = 3\pi/2$: $V''(3\pi/2) = +k(\lambda - 3)$: stable for $\lambda > 3$ (a heavy bead sits in the trough), unstable for $\lambda < 3$.
- $\sin x_1 = 2 - \lambda$ ($1 < \lambda < 3$): the bracket vanishes, so $V'' = k \cos^2 x_1 > 0$: stable.
- Boundary cases: for $\lambda = 1$ near $\pi/2$ ($u = x_1 - \pi/2$) and for $\lambda = 3$ near $3\pi/2$ ($u = x_1 - 3\pi/2$), the Taylor expansion gives $V' \approx \frac{1}{2} k u^3$, so V has a quartic minimum: stable, but with zero linear frequency.

> **Discrepancy with the official solution**
> The official key (Problems.pdf, PDF page 74) states the results in terms of a parameter α: "$\pi/2$ stable if $\alpha > 1$; $3\pi/2$ stable if $\alpha < 1/3$; $\sin x_1 = 2 - 1/\alpha$, which exists and is stable if $1/3 < \alpha < 1$". These coincide with the results above only if $\alpha = 1/\lambda = k/(mg)$, the reciprocal of the parameter $\lambda = mg/k$ defined in the statement. Check: at $\pi/2$, $V''(\pi/2) = k(1 - \lambda) > 0 \iff \lambda < 1 \iff 1/\lambda > 1$. The key also writes the in-plane reaction direction as $(-\cos x \, \mathbf{i}_1 + \mathbf{j}_1)$, a typo for $(-\cos x_1 \, \mathbf{i}_1 + \mathbf{j}_1)$; its reaction formulas and energy integral otherwise agree with (b) and (c).

(e) $\omega \neq 0$, $k = 0$: bead dropped at $x_1 = \pi$

> **Pedagogical Justification:** the energy integral of (c) with $k = 0$ is $\frac{1}{2} m \dot{x}_1^2 (1 + \cos^2 x_1) + V(x_1) = V(\pi)$, so the bead can only be where $V(x_1) \le V(\pi)$; the motion is read from the shape of V (turning points where $V = V(\pi)$).

Dimensionless potential. With $k = 0$, $V = mg \sin x_1 - \frac{1}{2} m \omega^2 x_1^2$. Define $\tilde{V} = 2V/(mg)$:

$$\tilde{V}(x_1) = 2\sin x_1 - \beta x_1^2, \qquad \beta = \frac{\omega^2}{g}, \qquad \tilde{V}(\pi) = -\beta\pi^2$$

and the energy equation becomes $\dfrac{\dot{x}_1^2}{g}\left(1+\cos^2 x_1\right) = \tilde{V}(\pi) - \tilde{V}(x_1)$.

Shape of the potential (sketch). $\tilde{V}$ is a sine wave of amplitude 2 riding on the downward parabola $-\beta x_1^2$: wells near the troughs $x_1 \approx 3\pi/2, 7\pi/2, \ldots$, humps near the crests $x_1 \approx 5\pi/2, 9\pi/2, \ldots$, and $\tilde{V} \to -\infty$ as $|x_1| \to \infty$. The humps flatten and disappear as β grows, because $\tilde{V}' = 2\cos x_1 - 2\beta x_1$ can vanish only where $|\beta x_1| \le 1$. For example, $\beta = 0.02\ \mathrm{m}^{-1}$: $\tilde{V}(\pi) = -0.197$ m, $\tilde{V}(3\pi/2) = -2.44$ m, $\tilde{V}(5\pi/2) = +0.77$ m; $\beta = 0.1\ \mathrm{m}^{-1}$: $\tilde{V}(\pi) = -0.99$ m, $\tilde{V}(5\pi/2) = -4.17$ m.

Start of the motion. $\tilde{V}'(\pi) = 2\cos\pi - 2\beta\pi = -2(1+\beta\pi) < 0$, so the generalized force $-V'$ is positive: the bead starts moving towards larger x_1 (down the wire into the trough near $3\pi/2$), and $x_1 = \pi$ is a turning point.

Small β (slow rotation). The hump near $5\pi/2$ is higher than the starting level, $\tilde{V} > -\beta\pi^2$ there, so the bead stops at the turning point x_t with $\tilde{V}(x_t) = \tilde{V}(\pi)$ and returns: it oscillates periodically between $x_1 = \pi$ and x_t (in the example $\beta = 0.02\ \mathrm{m}^{-1}$, $0.77 > -0.197$).

Large β (fast rotation). The centrifugal potential dominates, no hump to the right of π reaches the starting level, $\tilde{V}(x_1) < \tilde{V}(\pi)$ for all $x_1 > \pi$, and the bead moves outwards without bound ($x_1 \to \infty$) (in the example $\beta = 0.1\ \mathrm{m}^{-1}$, $-4.17 < -0.99$).

Critical value. The two behaviours are separated when the first hump to the right of π has its maximum exactly at the starting level: at that x_c the potential is stationary and equal to $\tilde{V}(\pi)$. The system of equations for (x_c, β_c) is

$$\begin{cases} \tilde{V}(x_c) = 2\sin x_c - \beta_c x_c^2 = -\beta_c\pi^2 \\ \tilde{V}'(x_c) = 2\cos x_c - 2\beta_c x_c = 0 \end{cases}$$

with the condition $\tilde{V}''(x_c) = -2\sin x_c - 2\beta_c < 0$ (a maximum). Solving it numerically (extension: the statement asks only for the system) gives $x_c \approx 2.401\pi \approx 7.544$ m and $\beta_c \approx 0.0405\ \mathrm{m}^{-1}$. Hence the bead oscillates for $\beta < \beta_c$ and escapes for $\beta > \beta_c$.

Discrepancy with the official solution

The official key writes the same system of equations for x_c and β_c, but states "$\beta > \beta_c$, the bead oscillates; $\beta < \beta_c$, the bead moves towards infinity". This is reversed. Proof: the centrifugal term $-\beta x_1^2$ is the only one that pushes the bead outwards, so a larger β lowers the humps; in the extreme $\beta \to \infty$, $\tilde{V}' = 2\cos x_1 - 2\beta x_1 < 0$ for all $x_1 \ge \pi$

(no hump at all), while for $\beta \to 0$ the hump at $5\pi/2$ has $\tilde{V} \to 2 > \tilde{V}(\pi) \to 0$. Numerical integration of the equation of motion (sympy) gives a bounded oscillation $x_1 \in [\pi, 2.31\pi]$ for $\beta = 0.95\beta_c$ and escape for $\beta = 1.05\beta_c$.

Checks

- **Role of each force.** The spring pulls the bead towards the rail, which on this wire favours the crest $\pi/2$; the weight favours the trough $3\pi/2$; the ratio $\lambda = mg/k$ decides, with the intermediate equilibria $\sin x_1 = 2 - \lambda$ where the vertical forces mg and $k(2 - \sin x_1)$ balance exactly. The centrifugal force pushes the bead away from the axis and, for fast rotation, throws it out; the Coriolis force only loads the wire (N_z) and never changes the energy.
- **Limits.** $\omega \to 0$: $N_z \to 0$ and the equation of motion is that of a bead on a fixed wire. $k \to 0$ in (d): $\lambda \to \infty$, only $3\pi/2$ is stable (bead in the trough). $k \to \infty$: $\lambda \to 0$, only $\pi/2$ is stable.
- **Orthogonality:** $\mathbf{t} \cdot \mathbf{n} = 0$ and $\mathbf{t} \cdot \mathbf{k}_1 = \mathbf{n} \cdot \mathbf{k}_1 = 0$; $[{}_0R_1]$ is orthogonal with $\det = 1$.
- **Dimensions** (with the implicit unit length of the wire): $[m\omega^2 x_1] = [mg] = [k(2 - \sin x_1)] = \mathrm{N}$; $[m\omega\dot{x}_1] = \mathrm{kg \cdot s^{-1} \cdot m \cdot s^{-1}} = \mathrm{N}$; every term of E_1^P is in J ($[k(2 - \sin x_1)^2] = \mathrm{N/m \cdot m^2}$); $[\lambda] = \mathrm{m}$ is compared with lengths ($2 - \sin x_1$); $[\beta x_1^2] = \mathrm{m^{-1} \cdot m^2} = \mathrm{m} = [2 \sin x_1]$.
- **Symbolic and numerical check (sympy):** $[{}_0R_1]$; $\mathbf{v}_1^P$, $\mathbf{a}_1^P$; Notes Eq. 3.27 against differentiation in $\mathcal{B}_0$; $\mathbf{F}_C$, $\mathbf{F}_{COR}$; N_x, N_y, N_z and the equation of motion solved from the three components (and again from $m\mathbf{a}_0^P = \mathbf{F}$ in S_0); $dE_1^P/dt = \dot{x}_1 \times$(equation of motion); V', V'' and the boundary cases; the critical point (x_c, β_c) and the two behaviours by numerical integration: all verified.
- **Check against the official key:** the key of `Problems.pdf` (PDF page 74) gives the reaction, the energy integral and the equilibria, which agree with (b), (c) and (d) up to the two points documented in the discrepancy notes above (the parameter $\alpha = 1/\lambda$ of the stability statements and the reversed inequality for β_c); both are proved there.

Source: Problems.pdf (PDF page 15); key on PDF page 74

PROBLEM 28

Spherical Pendulum with a Vertically Oscillating Suspension Point

Original statement · Problems.pdf, p. 16

Translating Frame + Inertia Force

A point particle P of mass m is attached with a massless cord of length L to the point A, as shown in the figure. The point A is oscillating vertically with respect to the fixed reference frame $Oxyz$ according to the law $z = a\sin\omega t$, where a and ω are known constants. The position of the particle P with respect to the moving reference frame $Ax'y'z'$ is completely specified by the angles φ and θ shown in the figure.

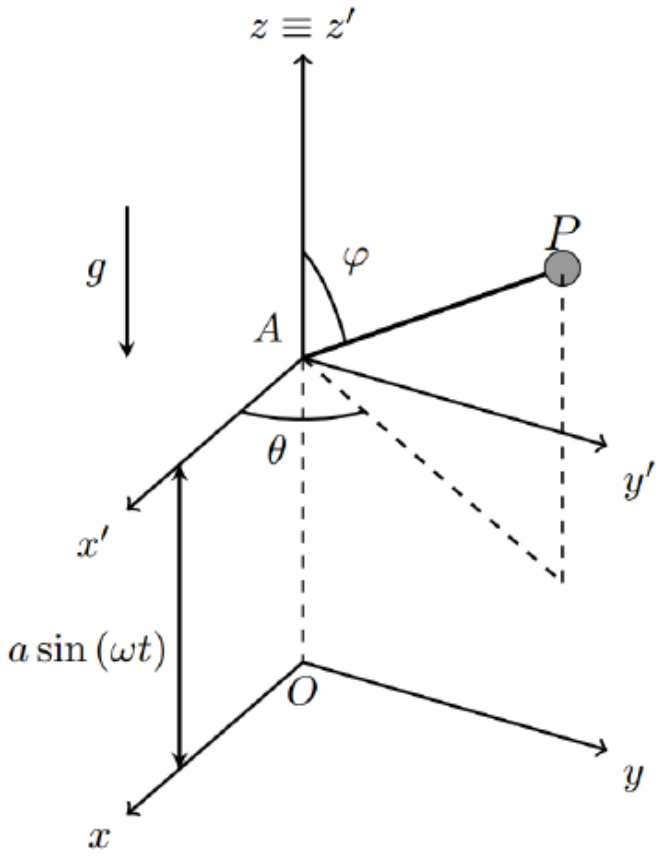


Figure: Point A oscillating vertically as a sin(ωt) above O; translating axes Ax'y'z'; particle P on a cord from A, located by the angle φ from Az' and the azimuth θ (Problems.pdf, p. 16).

At $t = 0$ the relative position and relative velocity of the particle are defined by the values

$$\theta_0 = 0, \qquad \varphi_0 = 30^\circ, \qquad \dot{\theta}_0 = \sqrt{\frac{2g}{\sqrt{3}L}}, \qquad \dot{\varphi}_0 = 0,$$

where g is the gravitational acceleration. We define the following unit vectors that define a spherical coordinate system

$$\mathbf{e}_r = \sin\varphi\cos\theta\,\mathbf{i} + \sin\varphi\sin\theta\,\mathbf{j} + \cos\varphi\,\mathbf{k}, \qquad \mathbf{e}_\varphi = \cos\varphi\cos\theta\,\mathbf{i} + \cos\varphi\sin\theta\,\mathbf{j} - \sin\varphi\,\mathbf{k}, \qquad \mathbf{e}_\theta = -\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}.$$

The partial derivatives of these vectors with respect to φ and θ are

$$\frac{\partial \mathbf{e}_r}{\partial \varphi} = \mathbf{e}_\varphi, \quad \frac{\partial \mathbf{e}_\varphi}{\partial \varphi} = -\mathbf{e}_r, \quad \frac{\partial \mathbf{e}_\theta}{\partial \varphi} = \mathbf{0}, \quad \frac{\partial \mathbf{e}_r}{\partial \theta} = \sin\varphi\,\mathbf{e}_\theta, \quad \frac{\partial \mathbf{e}_\varphi}{\partial \theta} = \cos\varphi\,\mathbf{e}_\theta, \quad \frac{\partial \mathbf{e}_\theta}{\partial \theta} = -\sin\varphi\,\mathbf{e}_r - \cos\varphi\,\mathbf{e}_\varphi.$$

(a) Show that the relative acceleration of P with respect to $Ax'y'z'$ can be expressed as $\mathbf{a}'_P = -L\left(\dot{\varphi}^2 + \dot{\theta}^2\sin^2\varphi\right)\mathbf{e}_r + L\left(\ddot{\varphi} - \dot{\theta}^2\sin\varphi\cos\varphi\right)\mathbf{e}_\varphi + L\left(\ddot{\theta}\sin\varphi + 2\dot{\theta}\dot{\varphi}\cos\varphi\right)\mathbf{e}_\theta$.

(b) Write down a system of three ordinary differential equations that once solved would yield the angles θ and φ, and the tension on the cord as a function of time (you do not need to solve it).

(c) At $t = 0$, determine the relative acceleration of P with respect to $Ax'y'z'$.

(d) At $t = 0$, determine the value of the tension on the cord.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a particle P (mass m) hangs from a massless cord of length L fixed at A ; A moves on the vertical axis with $z_A = a\sin\omega t$. In the frame $Ax'y'z'$ the particle is located by the spherical angles φ (from the upward axis Az' to AP, figure of the

statement) and θ (azimuth from Ax').

- **Hypotheses:** $Oxyz$ is inertial with Oz upwards and gravity $-g\,\mathbf{k}$; the cord is massless and taut, so it acts on P with a tension $\mathbf{T} = -T\,\mathbf{e}_r$, $T \geq 0$, directed from P to A; no friction or drag. Here ω is the frequency of the oscillation of A, not an angular velocity of the frame.
- **Why a moving frame:** in $Ax'y'z'$ the cord has fixed length, so P moves on a sphere of radius L described by two angles. The frame does not rotate, so the only inertia force is the one due to the acceleration of its origin.
- **Fixed (inertial) frame:** $S_0 = Oxyz = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$, basis $\mathcal{B}_0$.
- **Moving frame:** $S_1 = Ax'y'z' = \{A; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$: its axes stay parallel to those of $Oxyz$ (figure of the statement), so it uses the same basis $\mathcal{B}_1 = \mathcal{B}_0$ and only translates. In the notation of the theory, $\mathbf{a}'_P \equiv \mathbf{a}_1^P$.
- **Spherical basis** $\{\mathbf{e}_r, \mathbf{e}_\varphi, \mathbf{e}_\theta\}$ of the statement, attached to P; it is deduced in the next section.
- **Degrees of freedom:** 3 coordinates minus the constraint $|\mathbf{AP}| = L$: CDOF $= 2$, generalized coordinates (φ, θ). The motion of A is prescribed.
- **Initial conditions:** $\theta_0 = 0$, $\varphi_0 = 30^\circ = \pi/6$, $\dot\theta_0 = \sqrt{2g/(\sqrt{3}L)}$, $\dot\varphi_0 = 0$ (relative values).

Symbol	Meaning	SI unit
m	mass of P	kg
L	cord length	m
a	amplitude of the motion of A	m
ω	frequency of the motion of A	rad/s
g	gravitational acceleration	m/s^2
φ, θ	polar and azimuthal angles of AP in $Ax'y'z'$	rad
T	cord tension	N

Change of Basis and Position Vector

Translation of the frame and the meridian plane

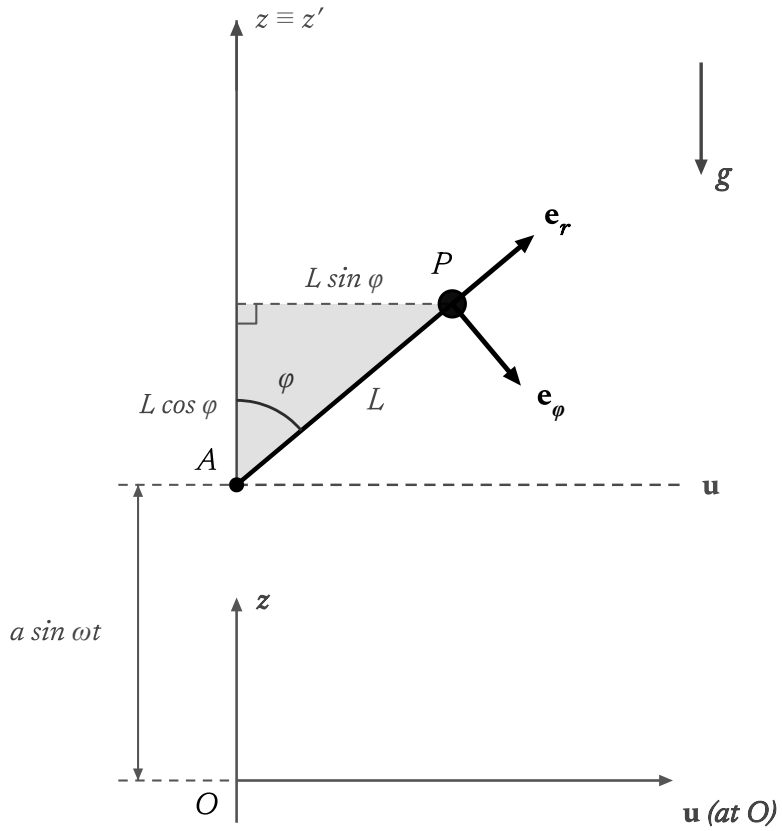


Figure 1. Vertical plane that contains Az' and the cord. A moves on the fixed axis Oz ($z \equiv z'$), at height $a \sin \omega t$; the axes of $Ax'y'z'$ stay parallel to those of $Oxyz$. In the shaded right triangle the hypotenuse is the cord $|AP| = L$, the leg adjacent to φ is vertical, $L \cos \varphi$ along $\mathbf{k}$, and the opposite leg is horizontal, $L \sin \varphi$ along $\mathbf{u} = \cos \theta \, \mathbf{i} + \sin \theta \, \mathbf{j}$. $\mathbf{e}_r, \mathbf{e}_\varphi$ in orange.

Change of basis $S_0 \to S_1$. The axes of $Ax'y'z'$ stay parallel to those of $Oxyz$, so $\mathbf{i}' = \mathbf{i}, \mathbf{j}' = \mathbf{j}$, $\mathbf{k}' = \mathbf{k}$ and the change-of-basis matrix is the identity: $[{}_0R_1] = I$, with $\det I = 1$ and $II^T = I$. The frame does not rotate: $\boldsymbol{\omega}_{10} = \mathbf{0}, \boldsymbol{\alpha}_{10} = \mathbf{0}$.

Motion of the origin. A is on the fixed axis Oz at the height $a \sin \omega t$: $\mathbf{r}_0^A = \overrightarrow{OA} = a \sin \omega t \, \mathbf{k}$. By the chain rule, $\frac{d}{dt} \sin \omega t = \omega \cos \omega t$ and $\frac{d}{dt} \cos \omega t = -\omega \sin \omega t$:

$$\mathbf{v}_0^A = a\omega \cos \omega t \, \mathbf{k}, \qquad \mathbf{a}_0^A = -a\omega^2 \sin \omega t \, \mathbf{k}$$

The meridian triangle. The first figure is the vertical plane that contains the axis Az' and the cord. Let $\mathbf{u}$ be the horizontal unit vector of that plane, pointing from the axis towards P. In the shaded right triangle the hypotenuse is the cord, $|AP| = L$; the angle φ is measured at A from the upward vertical, so the leg adjacent to φ is vertical and measures $L\cos\varphi$ (along $+\mathbf{k}$ while $\varphi < \pi/2$), and the leg opposite to φ is horizontal and measures $L\sin\varphi$ (along $\mathbf{u}$):

$$\overrightarrow{AP} = L\sin\varphi\,\mathbf{u} + L\cos\varphi\,\mathbf{k}$$

Since φ is measured from the vertical (not from the horizontal $\mathbf{u}$, as a polar angle in that plane would be), the usual polar angle of P in the meridian plane is $\psi = \pi/2 - \varphi$: indeed $\cos(\pi/2-\varphi) = \cos\frac{\pi}{2}\cos\varphi + \sin\frac{\pi}{2}\sin\varphi = \sin\varphi$ and $\sin(\pi/2-\varphi) = \sin\frac{\pi}{2}\cos\varphi - \cos\frac{\pi}{2}\sin\varphi = \cos\varphi$, which are the horizontal and vertical components above.

The azimuth seen from above

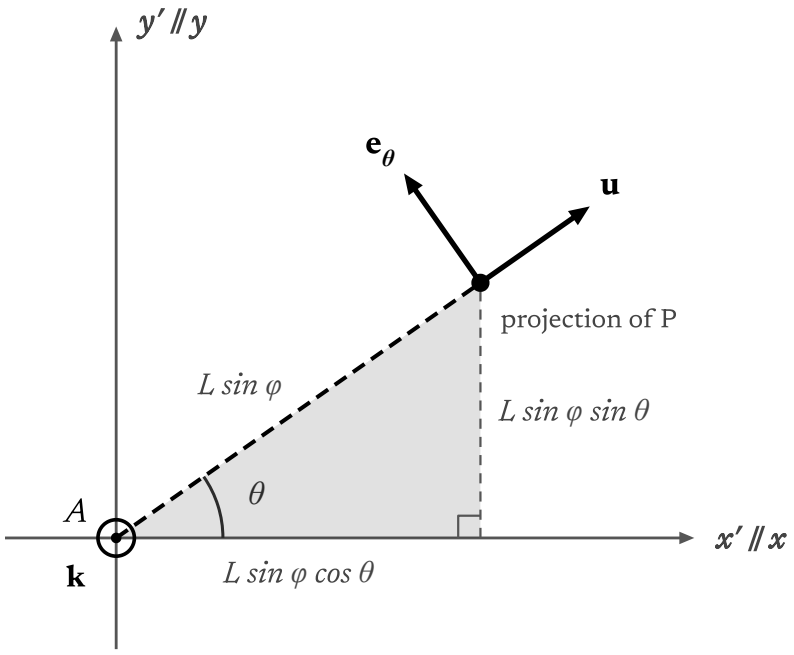


Figure 2. View from $+z$, with $\mathbf{k}$ out of the paper ($\odot$). The horizontal projection of AP has length $L\sin\varphi$ (the horizontal leg of the meridian triangle) and makes the azimuth θ with Ax'; its components are $L\sin\varphi\cos\theta$ (adjacent) and $L\sin\varphi\sin\theta$ (opposite). $\mathbf{u}$ is the unit vector of that direction and $\mathbf{e}_\theta$ is $\mathbf{u}$ turned $+90^\circ$ about $\mathbf{k}$.

The second figure looks down the vertical axis, from $+z$: $\mathbf{k}$ points out of the paper, towards the reader ($\odot$), with Ax' to the right and Ay' up ($\mathbf{i}\times\mathbf{j}=\mathbf{k}$, thumb towards the reader). The horizontal projection of $\overrightarrow{AP}$ is the horizontal leg of the meridian triangle, of length $L\sin\varphi$, and makes the azimuth θ with Ax' (counter-clockwise). In this second right triangle the hypotenuse is $L\sin\varphi$, the leg adjacent to θ is $L\sin\varphi\cos\theta$ along $\mathbf{i}$ and the opposite leg is $L\sin\varphi\sin\theta$ along $\mathbf{j}$; dividing by $L\sin\varphi$,

$$\mathbf{u}=\cos\theta\,\mathbf{i}+\sin\theta\,\mathbf{j}$$

Position vector. Substituting $\mathbf{u}$ in $\overrightarrow{AP}$:

$$\mathbf{r}_1^P=\overrightarrow{AP}=L\left(\sin\varphi\cos\theta\,\mathbf{i}+\sin\varphi\sin\theta\,\mathbf{j}+\cos\varphi\,\mathbf{k}\right)=L\,\mathbf{e}_r,\qquad \mathbf{r}_0^P=\overrightarrow{OA}+\overrightarrow{AP}=a\sin\omega t\,\mathbf{k}+L\,\mathbf{e}_r$$

which is the $\mathbf{e}_r$ of the statement. Since $\varphi<\pi/2$ means $\mathbf{e}_r\cdot\mathbf{k}=\cos\varphi>0$, at $t=0$ ($\varphi_0=30^\circ$) the particle is **above** A, as drawn in the figure of the statement.

The other two vectors by differentiation. $\mathbf{e}_\varphi$ is the unit vector in the direction in which P moves when φ grows; differentiating $\mathbf{e}_r$ with respect to φ ($\frac{\partial}{\partial\varphi}\sin\varphi=\cos\varphi$, $\frac{\partial}{\partial\varphi}\cos\varphi=-\sin\varphi$) gives a vector of length $\sqrt{\cos^2\varphi(\cos^2\theta+\sin^2\theta)+\sin^2\varphi}=1$:

$$\mathbf{e}_\varphi=\frac{\partial\mathbf{e}_r}{\partial\varphi}=\cos\varphi\cos\theta\,\mathbf{i}+\cos\varphi\sin\theta\,\mathbf{j}-\sin\varphi\,\mathbf{k}=\cos\varphi\,\mathbf{u}-\sin\varphi\,\mathbf{k}$$

In the meridian figure it is perpendicular to the cord and points down and away from the axis (the minus sign: when φ grows the particle goes down, because φ is measured from the upward vertical). Differentiating $\mathbf{e}_r$ with respect to θ ($\frac{\partial}{\partial\theta}\cos\theta=-\sin\theta$, $\frac{\partial}{\partial\theta}\sin\theta=\cos\theta$) gives $\sin\varphi\,(-\sin\theta\,\mathbf{i}+\cos\theta\,\mathbf{j})$, of length $\sin\varphi$; dividing by it,

$$\mathbf{e}_\theta=\frac{1}{\sin\varphi}\frac{\partial\mathbf{e}_r}{\partial\theta}=-\sin\theta\,\mathbf{i}+\cos\theta\,\mathbf{j}=\mathbf{k}\times\mathbf{u}$$

that is, $\mathbf{u}$ turned $+90^\circ$ about $\mathbf{k}$ (counter-clockwise in the view from above). These are the vectors of the statement, and the partial derivatives given there follow from the same rules.

Change-of-basis matrix of the spherical basis. The matrix with columns $\mathbf{e}_r,\mathbf{e}_\varphi,\mathbf{e}_\theta$ (components in $\{\mathbf{i},\mathbf{j},\mathbf{k}\}$)

$$[{}_1R_s] = \begin{pmatrix} \sin\varphi\cos\theta & \cos\varphi\cos\theta & -\sin\theta \\ \sin\varphi\sin\theta & \cos\varphi\sin\theta & \cos\theta \\ \cos\varphi & -\sin\varphi & 0 \end{pmatrix}$$

has orthonormal columns ($|\mathbf{e}_r|^2 = \sin^2\varphi(\cos^2\theta + \sin^2\theta) + \cos^2\varphi = 1$, $\mathbf{e}_r \cdot \mathbf{e}_\varphi = \sin\varphi\cos\varphi(\cos^2\theta + \sin^2\theta) - \cos\varphi\sin\varphi = 0$, $\mathbf{e}_r \cdot \mathbf{e}_\theta = \sin\varphi(-\cos\theta\sin\theta + \sin\theta\cos\theta) = 0$, $\mathbf{e}_\varphi \cdot \mathbf{e}_\theta = 0$ likewise), so $[{}_1R_s]^T[{}_1R_s] = I$; and $\mathbf{e}_r \times \mathbf{e}_\varphi = \mathbf{e}_\theta$, so $\det[{}_1R_s] = 1$ (right-handed). The last row gives the vertical: $\mathbf{k} = \cos\varphi\,\mathbf{e}_r - \sin\varphi\,\mathbf{e}_\varphi$ (the two legs of the meridian triangle, now read along the cord and across it).

Kinematics

(a) Relative acceleration

Pedagogical Justification: $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are constant in S_1; the spherical vectors depend on time only through $\varphi(t)$ and $\theta(t)$, so their derivative in S_1 follows from the multivariable chain rule $\dfrac{d\mathbf{e}}{dt} = \dfrac{\partial \mathbf{e}}{\partial\varphi}\dot\varphi + \dfrac{\partial \mathbf{e}}{\partial\theta}\dot\theta$ with the partial derivatives of the statement.

Time derivatives of the basis (in S_1):

$$\dot{\mathbf{e}}_r = \dot\varphi\,\mathbf{e}_\varphi + \dot\theta\sin\varphi\,\mathbf{e}_\theta, \qquad \dot{\mathbf{e}}_\varphi = -\dot\varphi\,\mathbf{e}_r + \dot\theta\cos\varphi\,\mathbf{e}_\theta, \qquad \dot{\mathbf{e}}_\theta = -\dot\theta\,(\sin\varphi\,\mathbf{e}_r + \cos\varphi\,\mathbf{e}_\varphi)$$

Relative velocity (L constant):

$$\mathbf{v}_1^P = L\,\dot{\mathbf{e}}_r = L\dot\varphi\,\mathbf{e}_\varphi + L\dot\theta\sin\varphi\,\mathbf{e}_\theta$$

Relative acceleration. Product rule on each term: $\frac{d}{dt}\left(L\dot\varphi\,\mathbf{e}_\varphi\right) = L\ddot\varphi\,\mathbf{e}_\varphi + L\dot\varphi\,\dot{\mathbf{e}}_\varphi$ and $\frac{d}{dt}\left(L\dot\theta\sin\varphi\,\mathbf{e}_\theta\right) = L\ddot\theta\sin\varphi\,\mathbf{e}_\theta + L\dot\theta\cos\varphi\,\dot\varphi\,\mathbf{e}_\theta + L\dot\theta\sin\varphi\,\dot{\mathbf{e}}_\theta$ (chain rule $\frac{d}{dt}\sin\varphi = \cos\varphi\,\dot\varphi$). Substituting the basis derivatives:

$$\mathbf{a}_1^P = L\ddot\varphi\,\mathbf{e}_\varphi + L\dot\varphi\left(-\dot\varphi\,\mathbf{e}_r + \dot\theta\cos\varphi\,\mathbf{e}_\theta\right) + L\ddot\theta\sin\varphi\,\mathbf{e}_\theta + L\dot\theta\dot\varphi\cos\varphi\,\mathbf{e}_\theta - L\dot\theta^2\sin\varphi\,(\sin\varphi\,\mathbf{e}_r + \cos\varphi\,\mathbf{e}_\varphi)$$

Collecting along $\mathbf{e}_r$ ($-L\dot\varphi^2 - L\dot\theta^2\sin^2\varphi$), $\mathbf{e}_\varphi$ ($L\ddot\varphi - L\dot\theta^2\sin\varphi\cos\varphi$) and $\mathbf{e}_\theta$ ($L\dot\varphi\dot\theta\cos\varphi + L\ddot\theta\sin\varphi + L\dot\theta\dot\varphi\cos\varphi$):

$$\mathbf{a}'_P \equiv \mathbf{a}_1^P = -L\left(\dot\varphi^2 + \dot\theta^2\sin^2\varphi\right)\mathbf{e}_r + L\left(\ddot\varphi - \dot\theta^2\sin\varphi\cos\varphi\right)\mathbf{e}_\varphi + L\left(\ddot\theta\sin\varphi + 2\dot\theta\dot\varphi\cos\varphi\right)\mathbf{e}_\theta$$

which is the expression of the statement. The absolute acceleration is $\mathbf{a}_0^P = \mathbf{a}_0^A + \mathbf{a}_1^P$ (Notes Eq. 3.29, translating frame).

Forces and Equations of Motion

(b) Forces, equations of motion and tension

Pedagogical Justification: in the translating frame, Notes Eq. 4.63 with $\boldsymbol{\omega}_{10} = \boldsymbol{\alpha}_{10} = \mathbf{0}$ reduces to $m\mathbf{a}_1^P = \mathbf{F} - m\mathbf{a}_0^A$ (equivalently Notes Eq. 3.29, $\mathbf{a}_0^P = \mathbf{a}_0^A + \mathbf{a}_1^P$, inserted in $m\mathbf{a}_0^P = \mathbf{F}$). Projecting on the spherical basis separates the tension (only along $\mathbf{e}_r$) from the two equations of motion.

Forces in S_1. Tension $-T\,\mathbf{e}_r$; weight $-mg\,\mathbf{k}$; inertia force of the origin $\mathbf{F}_A = -m\mathbf{a}_0^A = ma\omega^2\sin\omega t\,\mathbf{k}$. Weight and $\mathbf{F}_A$ combine into an effective, time-dependent gravity:

$$-mg\,\mathbf{k} + ma\omega^2\sin\omega t\,\mathbf{k} = -m\,g_e(t)\,\mathbf{k}, \qquad g_e(t) = g - a\omega^2\sin\omega t$$

With $\mathbf{k} = \cos\varphi\,\mathbf{e}_r - \sin\varphi\,\mathbf{e}_\varphi$, this force is $-mg_e\cos\varphi\,\mathbf{e}_r + mg_e\sin\varphi\,\mathbf{e}_\varphi$.

Three components of $m\mathbf{a}_1^P = -T\,\mathbf{e}_r - mg_e\,\mathbf{k}$:

$$\mathbf{e}_r: \quad -mL\left(\dot\varphi^2 + \dot\theta^2\sin^2\varphi\right) = -T - mg_e(t)\cos\varphi$$

$$\mathbf{e}_\varphi: \quad mL\left(\ddot\varphi - \dot\theta^2\sin\varphi\cos\varphi\right) = mg_e(t)\sin\varphi$$

$$\mathbf{e}_\theta: \quad mL\left(\ddot\theta\sin\varphi + 2\dot\theta\dot\varphi\cos\varphi\right) = 0$$

Answers and Checks

(a) Relative acceleration

$\mathbf{a}'_P = -L\left(\dot\varphi^2 + \dot\theta^2\sin^2\varphi\right)\mathbf{e}_r + L\left(\ddot\varphi - \dot\theta^2\sin\varphi\cos\varphi\right)\mathbf{e}_\varphi + L\left(\ddot\theta\sin\varphi + 2\dot\theta\dot\varphi\cos\varphi\right)\mathbf{e}_\theta$, as shown in the kinematics.

(b) System of equations

Solving the three components for the unknowns:

$$\begin{cases} \ddot\varphi = \dot\theta^2\sin\varphi\cos\varphi + \dfrac{g - a\omega^2\sin\omega t}{L}\sin\varphi \\ \ddot\theta\sin\varphi + 2\dot\theta\dot\varphi\cos\varphi = 0 \\ T = mL\left(\dot\varphi^2 + \dot\theta^2\sin^2\varphi\right) - m\left(g - a\omega^2\sin\omega t\right)\cos\varphi \end{cases}$$

The first two equations, with the initial conditions, give $\varphi(t)$ and $\theta(t)$; the third gives $T(t)$. Multiplying the second by $\sin\varphi$, product and chain rules show it is $\frac{d}{dt}\left(\dot\theta\sin^2\varphi\right) = \ddot\theta\sin^2\varphi + 2\dot\theta\sin\varphi\cos\varphi\,\dot\varphi = 0$: the vertical component of the angular momentum about A, $mL^2\dot\theta\sin^2\varphi$, is conserved, because tension, weight and $\mathbf{F}_A$ have no moment about the vertical axis through A (Topic 4).

(c) Relative acceleration at $t = 0$

Data: $\sin\omega\cdot 0 = 0$, so $g_e(0) = g$; $\sin\varphi_0 = \frac{1}{2}$, $\cos\varphi_0 = \frac{\sqrt3}{2}$, $\dot\theta_0^2 = \dfrac{2g}{\sqrt3 L}$, $\dot\varphi_0 = 0$.

Angular accelerations. From the first equation of (b):

$$\ddot\varphi_0 = \frac{2g}{\sqrt3 L}\cdot\frac{1}{2}\cdot\frac{\sqrt3}{2} + \frac{g}{L}\cdot\frac{1}{2} = \frac{g}{2L} + \frac{g}{2L} = \frac{g}{L}$$

From the second, $\ddot\theta_0\sin\varphi_0 = -2\dot\theta_0\cdot 0\cdot\cos\varphi_0 = 0$, so $\ddot\theta_0 = 0$.

Components of $\mathbf{a}_1^P(0)$. $\dot\theta_0^2\sin^2\varphi_0 = \dfrac{2g}{\sqrt3 L}\cdot\dfrac{1}{4} = \dfrac{g}{2\sqrt3 L}$ and $\dot\theta_0^2\sin\varphi_0\cos\varphi_0 = \dfrac{2g}{\sqrt3 L}\cdot\dfrac{\sqrt3}{4} = \dfrac{g}{2L}$:

$$\mathbf{a}_1^P(0) = -\frac{g}{2\sqrt3}\,\mathbf{e}_r + L\left(\frac{g}{L} - \frac{g}{2L}\right)\mathbf{e}_\varphi + 0\,\mathbf{e}_\theta = -\frac{g}{2\sqrt3}\,\mathbf{e}_r + \frac{g}{2}\,\mathbf{e}_\varphi$$

At $\theta = 0$, $\varphi = 30^\circ$: $\mathbf{e}_r = \frac{1}{2}\mathbf{i} + \frac{\sqrt3}{2}\mathbf{k}$ and $\mathbf{e}_\varphi = \frac{\sqrt3}{2}\mathbf{i} - \frac{1}{2}\mathbf{k}$, so the $\mathbf{i}$ component is $-\frac{g}{4\sqrt3} + \frac{\sqrt3 g}{4} = \frac{(3-1)g}{4\sqrt3} = \frac{g}{2\sqrt3}$ and the $\mathbf{k}$ component is $-\frac{g}{4} - \frac{g}{4} = -\frac{g}{2}$:

$$\mathbf{a}_1^P(0) = \frac{g}{2\sqrt{3}}\,\mathbf{i} - \frac{g}{2}\,\mathbf{k}, \qquad \left|\mathbf{a}_1^P(0)\right| = g\sqrt{\frac{1}{12} + \frac{1}{4}} = \frac{g}{\sqrt{3}}$$

(d) Tension at $t = 0$

From the third equation of (b):

$$T(0) = mL\left(0 + \frac{g}{2\sqrt{3}L}\right) - mg\,\frac{\sqrt{3}}{2} = mg\left(\frac{1}{2\sqrt{3}} - \frac{3}{2\sqrt{3}}\right) = -\frac{mg}{\sqrt{3}}$$

Interpretation. The value is negative: at $t = 0$ the cord would have to push the particle outwards. A cord can only pull, so with the data as stated the cord is not taut at $t = 0$ and the taut-cord model of (a)–(c) does not apply. The actual motion starts as a free motion (the cord goes slack), with $m\mathbf{a}_1^P = -mg_e\,\mathbf{k}$, i.e. $\mathbf{a}_1^P(0) = -g\,\mathbf{k}$ instead of the value of (c). The result $T(0) = -mg/\sqrt{3}$ would be meaningful as a compression if P were held by a massless rigid rod instead of a cord.

> **Data inconsistency in the statement (no official key)**
> With φ measured from the upward axis Az' (figure and $\mathbf{e}_r \cdot \mathbf{k} = \cos\varphi$), the given $\dot\theta_0$ cannot keep the cord taut: $T(0) = -mg/\sqrt{3} < 0$. The given value $\dot\theta_0^2 = 2g/(\sqrt{3}L) = g/(L\cos 30^\circ)$ is exactly the rate of a steady **conical pendulum hanging below** A at 30° from the downward vertical. Under that alternative reading (which contradicts the figure and the given $\mathbf{e}_r$, so it is only a remark), the same method gives $\ddot\varphi_0 = 0$, $\mathbf{a}_1^P(0) = -\dfrac{g}{\sqrt{3}}\,\mathbf{i}$ (purely centripetal, towards the axis) and $T(0) = mL\dot\theta_0^2\sin^2\varphi_0 + mg\cos\varphi_0 = \dfrac{mg}{2\sqrt{3}} + \dfrac{\sqrt{3}mg}{2} = \dfrac{2mg}{\sqrt{3}}$. The answers above follow the statement as written.

Checks

- **Effective gravity.** In the translating frame of A the oscillation of the support only modulates gravity, $g_e(t) = g - a\omega^2\sin\omega t$ (Slide 28: the force due to the acceleration of the origin is indistinguishable from a gravitational field). When $a\omega^2 > g$, g_e changes sign periodically; a fast, small-amplitude vibration can then stabilise the upper position (Kapitza pendulum, which needs roughly $a^2\omega^2 > 2gL$ with $a \ll L$; extension, not course material).
- **Limits.** $a \to 0$ or $\omega \to 0$: $g_e = g$ and the system is the classical spherical pendulum written with φ from the upward vertical (equilibrium $\varphi = 0$ unstable, $\varphi = \pi$ stable). $\dot\theta = 0$: the θ equation is satisfied identically and the motion is planar. $\varphi = 0$: $\mathbf{e}_r = \mathbf{k}$, the

particle is on the axis above A and the azimuth is undefined, as the factor $\sin\varphi$ of $\partial\mathbf{e}_r/\partial\theta$ shows.

- **Signs at $t=0$.** $\ddot{\varphi}_0 = g/L > 0$: in the taut-cord model both the weight and the centrifugal effect of $\dot{\theta}_0$ push φ to grow, but part (d) shows that the cord would be slack at $t=0$, so the real motion starts as a free fall.
- **Orthogonality:** the spherical basis is orthonormal and right-handed ($\det[{}_1R_s]=1$), and $[{}_0R_1]=I$, as worked out in the change-of-basis section.
- **Dimensions:** $[a\omega^2]=\mathrm{m}\cdot\mathrm{s}^{-2}=[g]$; $[\dot{\theta}^2]=[g/L]=\mathrm{s}^{-2}$, so each term of the $\ddot{\varphi}$ equation is in s^{-2}; $[mL\dot{\theta}^2]=[mg]=\mathrm{N}$; $[L\dot{\varphi}^2]=[L\ddot{\theta}]=\mathrm{m}\cdot\mathrm{s}^{-2}$.
- **Symbolic check (sympy):** orthonormality and handedness of the spherical basis and the six partial derivatives of the statement; $\mathbf{v}_1^P$ and $\mathbf{a}_1^P$ by direct differentiation; $\mathbf{a}_0^P=\mathbf{a}_0^A+\mathbf{a}_1^P$; the three equations of (b) solved from the vector equation; $\ddot{\varphi}_0=g/L, \ddot{\theta}_0=0, \mathbf{a}_1^P(0)$, $|\mathbf{a}_1^P(0)|=g/\sqrt{3}, T(0)=-mg/\sqrt{3}$; and the alternative reading ($\ddot{\varphi}_0=0, T(0)=2mg/\sqrt{3}$): all verified.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 28 in its solution keys, so the results are verified by independent derivation (the symbolic check above); the data inconsistency of the statement is documented in the warning above.

Source: Problems.pdf

PROBLEM 30

Particle on a Rotating Quarter Circumference with a Spring

Original statement · Problems.pdf, p. 18

Rotating Frame + Path Variables

A point particle P of mass m is constrained to move on a smooth quarter of circumference as shown in the figure below. We define a moving reference frame $OX_1Y_1Z_1$ that rotates with respect to an inertial reference frame $OXYZ$. The rotation of the moving reference frame is performed along a vertical axis defined as $x_1 = -R$ and $y_1 = 0$; the angular velocity is $\boldsymbol{\omega} = \omega\mathbf{k} = \omega\mathbf{k}_1$. The particle is attached to a linear spring with the other end attached to the point B of the reference frame $OX_1Y_1Z_1$ with coordinates $(-2R, 0, 0)$. The constant of the spring is $K = \alpha m\omega^2$. The angle between $\overline{OP}$ and the negative side of the z_1 axis, ϕ, is used to define the position of the particle. Neglect the action of gravity on the particle.

Initially the particle is located at $\phi = 0$ and it is set to motion with a relative velocity $-\beta\omega R\,\mathbf{i}_1$. Both α and β are positive nondimensional parameters.

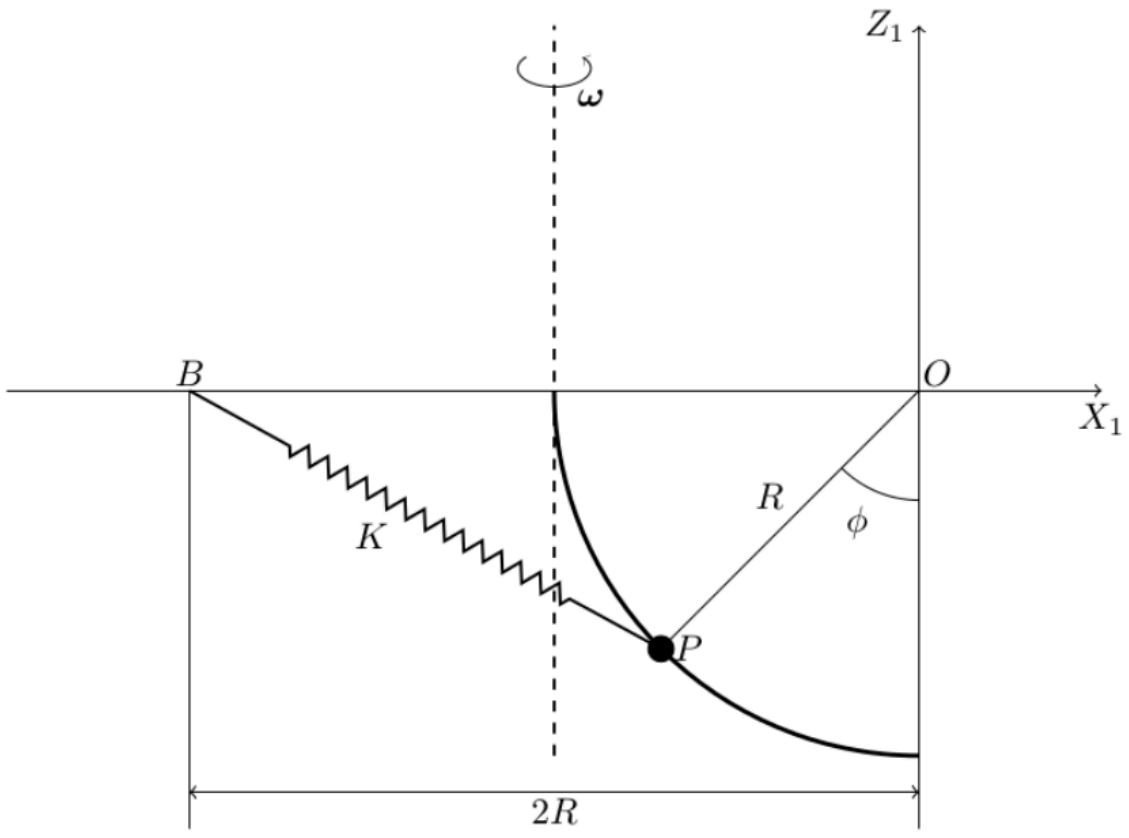


Figure: Quarter circumference of radius R centred at O in the plane X1Z1, rotation axis x1 = −R (dashed), spring of constant K from B = (−2R, 0, 0) to P, angle φ measured from the negative z1 axis (Problems.pdf, p. 18).

(a) Compute the relative velocity and the relative acceleration of the particle in the path variable reference frame $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$ as functions of the coordinate ϕ and their derivatives.

(b) Write down the three components of Newton's second law expressed in the path variables reference frame $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$.

(c) Express the energy conservation equation in the relative reference frame. Discuss if this equation is equivalent to one of the above ones.

Let us now consider a different problem. Depending on the values of the parameters, we may find the equilibrium of the forces at different positions.

(d) Obtain the value of α to have an equilibrium position when the particle is positioned at ϕ equal to $\pi/6$.

For this value of α, the particle is again located at $\phi = 0$ and is set to motion with the relative velocity defined before.

(e) Describe the possible types of motion for the different values of β. In particular, obtain the critical value of β that makes the particle reach the position $\phi = \pi/2$ and the reaction force of the curve to the particle at that instant.

(f) For a value of β higher than the critical one, the particle leaves the arc of circumference and its motion is not anymore constrained. Under these conditions compute the maximum height reached by the particle, z_{max}, as a function of β. Note that the particle is still under the action of the spring.

Worked Solution

Reference Frames, Bases and Coordinates

- **System:** a particle P of mass m slides on a smooth quarter circumference of radius R and centre O, drawn in the vertical plane X_1Z_1 of a frame that rotates with constant angular velocity $\omega\mathbf{k}_1$ about the vertical axis $x_1 = -R$, $y_1 = 0$. A spring of constant $K = \alpha m\omega^2$ joins P to the point $B = (-2R, 0, 0)$ of the rotating frame. Gravity is neglected.

- **Why a rotating frame.** The wire and the spring anchor B are fixed in $OX_1Y_1Z_1$, so the constraint is a fixed circle there and the position of P is described by one angle. The price is the inertia force of Notes Eqs. 4.62–4.63, which is easy to compute because $\boldsymbol{\omega}_{10}$ is constant. In the inertial frame the wire itself moves, and the constraint would be harder to write.
- **Hypotheses:**
  1. $OXYZ$ is inertial; the frame rotates uniformly, $\boldsymbol{\omega}_{10} = \omega \mathbf{k}_1$ with ω constant.
  2. The arc is smooth (no friction) and acts as a bilateral constraint (a bead on a wire): the reaction has no tangential component but its normal and binormal components can take either sign.
  3. The spring is linear and, since no natural length is given, of **zero natural length**: $\mathbf{F}_K = -K\,\overrightarrow{BP}$.
  4. Gravity is neglected, as the statement says.
- **Fixed (inertial) frame.** The only points that do not move are those of the rotation axis. We take $S_0 = \{Q; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, basis $\mathcal{B}_0$, with origin at the point Q of the axis at the height of O (the point $x_1 = -R$, $y_1 = z_1 = 0$, the upper end of the arc). It plays the role of the statement's $OXYZ$; moving the origin of an inertial frame to another fixed point does not change any velocity or acceleration. $\mathbf{k}_0 = \mathbf{k}$ is vertical.
- **Moving frame.** $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, basis $\mathcal{B}_1$, is the statement's $OX_1Y_1Z_1$, attached to the arc, with $\mathbf{k}_1 = \mathbf{k}_0$. Its origin O is at distance R from the axis, so it moves on a horizontal circle. We choose $\mathbf{i}_1 = \mathbf{i}_0$ at $t = 0$.
- **Path (intrinsic) basis** $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$ at P: tangent in the direction of increasing ϕ, principal normal towards the centre of curvature and binormal; it is deduced in the next section.
- **Degrees of freedom and coordinate:** on the arc, P has one degree of freedom, the angle $\phi \in [0, \pi/2]$ between $\overline{OP}$ and the negative z_1 axis. After leaving the arc in part (f), P is free: three degrees of freedom (x_1, y_1, z_1).
- **Initial conditions:** $\phi(0) = 0$, $\mathbf{v}_1^P(0) = -\beta\omega R\,\mathbf{i}_1$.

Symbol	Meaning	SI unit
m	mass of P	kg
R	radius of the arc and distance from O to the axis	m
ω	constant angular velocity of the frame	rad/s
$K = \alpha m \omega^2$	spring constant	N/m = kg/s^2
α, β	positive nondimensional parameters	1
ϕ	angle between $\overline{OP}$ and the $-z_1$ axis	rad
N_n, N_b	components of the reaction of the arc	N

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from above

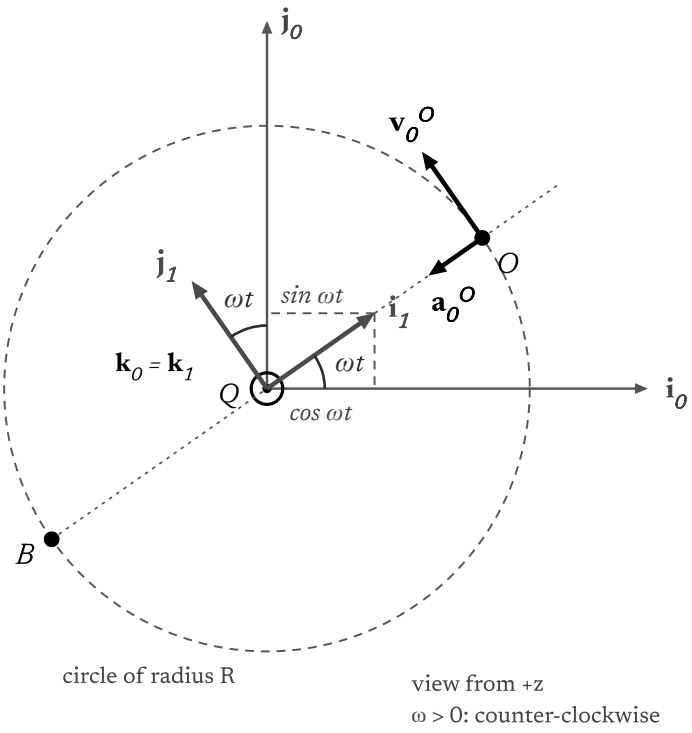


Figure 1. View from $+z$, with $\mathbf{k}_0 = \mathbf{k}_1$ out of the paper ($\odot$) at the point Q of the axis: then $\mathbf{j}_0 = \mathbf{k}_0 \times \mathbf{i}_0$ points up the page. $\mathbf{i}_1, \mathbf{j}_1$ are $\mathbf{i}_0, \mathbf{j}_0$ turned counter-clockwise by ωt; the dashed legs of $\mathbf{i}_1$ are $\cos \omega t$ along $\mathbf{i}_0$ and $\sin \omega t$ along $\mathbf{j}_0$. The origin O of S_1 runs on the dashed circle of radius R with $\mathbf{v}_0^O = \omega R \, \mathbf{j}_1$ and $\mathbf{a}_0^O = -\omega^2 R \, \mathbf{i}_1$.

The rotation is about the vertical $\mathbf{k}_0$, so it is drawn looking down the axis, from $+z$, with the axis at Q. In that view $\mathbf{k}_0$ points out of the paper, towards the reader ($\odot$). Take $\mathbf{i}_0$ to the right. Then $\mathbf{j}_0 = \mathbf{k}_0 \times \mathbf{i}_0$ (right-hand rule: thumb along $\mathbf{k}_0$, towards the reader, and the fingers turn $\mathbf{i}_0$ counter-clockwise by $90°$) points up the page.

With $\boldsymbol{\omega}_{10} = \omega \mathbf{k}_0$ and $\omega > 0$, the frame turns counter-clockwise in this view (thumb towards the reader). After a time t it has turned the angle ωt:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of the dashed right triangle. The leg adjacent to ωt lies along $\mathbf{i}_0$ and measures $\cos \omega t$; the leg opposite to ωt points up the page, along $+\mathbf{j}_0$, and measures $\sin \omega t$. Hence $\mathbf{i}_1 = \cos \omega t \, \mathbf{i}_0 + \sin \omega t \, \mathbf{j}_0$, with no minus sign.
- $\mathbf{j}_1$ starts on $\mathbf{j}_0$ (up the page) and turns the same angle counter-clockwise, so its tip moves to the left, towards $-\mathbf{i}_0$. Its leg along $\mathbf{j}_0$ (adjacent) is $\cos \omega t$ and its leg along $-\mathbf{i}_0$ (opposite) is $\sin \omega t$: $\mathbf{j}_1 = -\sin \omega t \, \mathbf{i}_0 + \cos \omega t \, \mathbf{j}_0$. **The minus sign comes from the counter-clockwise turn carrying the upward axis to the left.**

- $\mathbf{k}_1 = \mathbf{k}_0$, the axis of the rotation.

The columns of $[{}_0R_1]$ are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- **Checks.** Expanding along the third row, $\det[{}_0R_1] = 1 \cdot (\cos\omega t\cos\omega t - (-\sin\omega t)\sin\omega t) = \cos^2\omega t + \sin^2\omega t = 1$. The columns have unit length ($\cos^2\omega t + \sin^2\omega t = 1$) and the first two are orthogonal ($-\cos\omega t\sin\omega t + \sin\omega t\cos\omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.
- **Angular motion of S_1:** $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$ and, ω being constant, $\boldsymbol{\alpha}_{10} = \left.\dfrac{d\boldsymbol{\omega}_{10}}{dt}\right|_0 = \mathbf{0}$.
- **Motion of the origin O.** O lies on the line of $\mathbf{i}_1$ through Q, at distance R from the axis (the figure shows its dashed circle), so $\mathbf{r}_0^O = \overrightarrow{QO} = R\,\mathbf{i}_1$. By the Coriolis theorem (Notes Eq. 3.8), since $R\,\mathbf{i}_1$ is constant in S_1:

$$\mathbf{v}_0^O = \left.\frac{d(R\mathbf{i}_1)}{dt}\right|_1 + \boldsymbol{\omega}_{10}\times R\,\mathbf{i}_1 = \mathbf{0} + \omega R\,(\mathbf{k}_1\times\mathbf{i}_1) = \omega R\,\mathbf{j}_1$$

$$\mathbf{a}_0^O = \left.\frac{d(\omega R\mathbf{j}_1)}{dt}\right|_1 + \boldsymbol{\omega}_{10}\times\omega R\,\mathbf{j}_1 = \omega^2 R\,(\mathbf{k}_1\times\mathbf{j}_1) = -\omega^2 R\,\mathbf{i}_1$$

- **Direct check in $\mathcal{B}_0$.** $\mathbf{r}_0^O = R\cos\omega t\,\mathbf{i}_0 + R\sin\omega t\,\mathbf{j}_0$. By the chain rule, $\frac{d}{dt}\cos\omega t = -\omega\sin\omega t$ and $\frac{d}{dt}\sin\omega t = \omega\cos\omega t$, so $\mathbf{v}_0^O = \omega R\,(-\sin\omega t\,\mathbf{i}_0 + \cos\omega t\,\mathbf{j}_0) = \omega R\,\mathbf{j}_1$ and, differentiating again, $\mathbf{a}_0^O = -\omega^2 R\,(\cos\omega t\,\mathbf{i}_0 + \sin\omega t\,\mathbf{j}_0) = -\omega^2 R\,\mathbf{i}_1$, the vectors drawn at O in orange.

The position vector in the plane of the arc

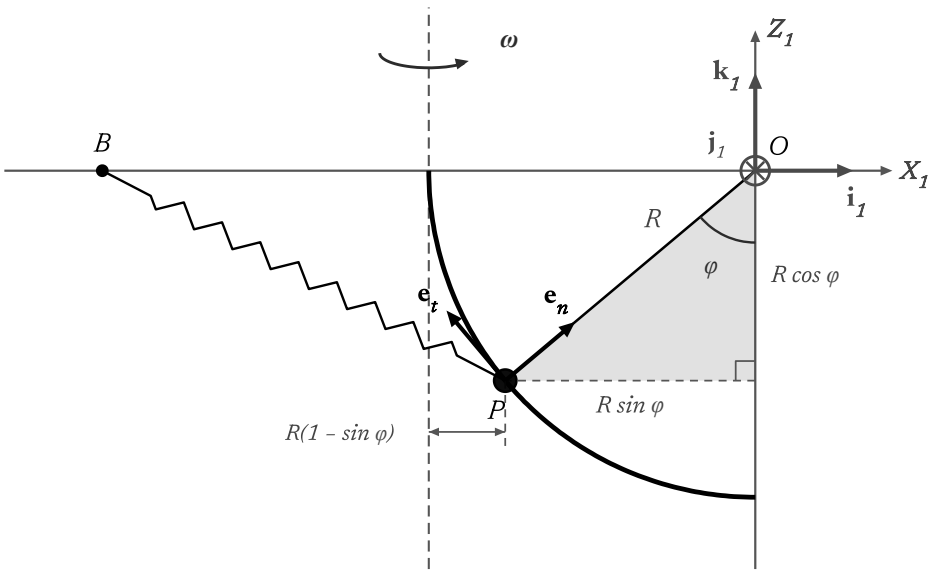


Figure 2. Plane X_1Z_1, fixed in S_1 ($\mathbf{j}_1$ points into the paper, $\otimes$). ϕ is measured at O from the downward axis $-z_1$. In the shaded right triangle the hypotenuse is $|OP| = R$, the leg adjacent to ϕ is $R\cos\phi$ (downwards) and the opposite leg is $R\sin\phi$ (to the left); the distance from P to the rotation axis is $R(1-\sin\phi)$. $\mathbf{i}_1, \mathbf{k}_1$ in blue, $\mathbf{e}_t, \mathbf{e}_n$ in orange.

The plane X_1Z_1 is fixed in S_1; it is drawn with $\mathbf{i}_1$ to the right and $\mathbf{k}_1$ up. The direction out of the paper is then $\mathbf{i}_1 \times \mathbf{k}_1 = -\mathbf{j}_1$, so $\mathbf{j}_1$ points **into** the paper ($\otimes$).

Let F be the point of the z_1 axis at the height of P. In the shaded right triangle OFP (right angle at F) the hypotenuse is $|OP| = R$ and the angle at O is ϕ:

- the leg adjacent to ϕ is $OF = R\cos\phi$, along the z_1 axis and pointing down: $\overrightarrow{OF} = -R\cos\phi\,\mathbf{k}_1$;
- the leg opposite to ϕ is $FP = R\sin\phi$, horizontal and pointing to the left, towards B: $\overrightarrow{FP} = -R\sin\phi\,\mathbf{i}_1$.

So $\mathbf{r}_1^P = \overrightarrow{OF} + \overrightarrow{FP}$:

$$\mathbf{r}_1^P = -R\sin\phi\,\mathbf{i}_1 - R\cos\phi\,\mathbf{k}_1$$

Checks: $\phi = 0$ gives $-R\,\mathbf{k}_1$, the lowest point of the arc on the $-z_1$ axis (the initial position); $\phi = \pi/2$ gives $-R\,\mathbf{i}_1$, the upper end of the arc, which lies on the rotation axis $x_1 = -R$. The horizontal distance from P to the rotation axis is $x_1 - (-R) = R - R\sin\phi = R(1 - \sin\phi)$, the dimension drawn in the figure; it is the lever arm of the centrifugal force.

Relation with the standard polar angle. In the plane X_1Z_1 the usual polar angle ϑ about O is measured from $\mathbf{i}_1$ towards $\mathbf{k}_1$, with $\mathbf{r}_1^P = R\,(\cos\vartheta\,\mathbf{i}_1 + \sin\vartheta\,\mathbf{k}_1)$. Comparing with the expression above requires $\cos\vartheta = -\sin\phi$ and $\sin\vartheta = -\cos\phi$, which holds for

$$\vartheta = -\frac{\pi}{2} - \phi: \qquad \cos\left(\tfrac{\pi}{2} + \phi\right) = \cos\tfrac{\pi}{2}\cos\phi - \sin\tfrac{\pi}{2}\sin\phi = -\sin\phi, \qquad \sin\left(\tfrac{\pi}{2} + \phi\right) = \sin\tfrac{\pi}{2}\cos\phi + \cos\tfrac{\pi}{2}\sin\phi = \cos\phi$$

so $\cos\vartheta = \cos\left(\frac{\pi}{2} + \phi\right) = -\sin\phi$ (the cosine is even) and $\sin\vartheta = -\sin\left(\frac{\pi}{2} + \phi\right) = -\cos\phi$ (the sine is odd), with $\cos\frac{\pi}{2} = 0$, $\sin\frac{\pi}{2} = 1$. Thus ϕ is measured from $-\mathbf{k}_1$ instead of $\mathbf{i}_1$, and it grows clockwise in the drawing (from the downward axis towards B), while ϑ grows counter-clockwise.

Path basis. On the arc we take the arc length $s = R\phi$ (increasing with ϕ). The tangent is the derivative of the position with respect to ϕ, divided by its length. With $\frac{d}{d\phi}(-R\sin\phi) = -R\cos\phi$ and $\frac{d}{d\phi}(-R\cos\phi) = R\sin\phi$:

$$\frac{d\mathbf{r}_1^P}{d\phi} = -R\cos\phi\,\mathbf{i}_1 + R\sin\phi\,\mathbf{k}_1, \qquad \left|\frac{d\mathbf{r}_1^P}{d\phi}\right| = R\sqrt{\cos^2\phi + \sin^2\phi} = R \;\Rightarrow\; \mathbf{e}_t = -\cos\phi\,\mathbf{i}_1 + \sin\phi\,\mathbf{k}_1$$

Differentiating $\mathbf{e}_t$ with respect to ϕ ($\frac{d}{d\phi}(-\cos\phi) = \sin\phi$, $\frac{d}{d\phi}\sin\phi = \cos\phi$) gives a vector of length $\sqrt{\sin^2\phi + \cos^2\phi} = 1$, the principal normal:

$$\mathbf{e}_n = \frac{d\mathbf{e}_t}{d\phi} = \sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{k}_1 = -\frac{\mathbf{r}_1^P}{R}$$

so $\mathbf{e}_n$ points from P to the centre of curvature O, along the hypotenuse of the triangle, as drawn. The binormal:

$$\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n = \begin{vmatrix} \mathbf{i}_1 & \mathbf{j}_1 & \mathbf{k}_1 \\ -\cos\phi & 0 & \sin\phi \\ \sin\phi & 0 & \cos\phi \end{vmatrix} = (0-0)\,\mathbf{i}_1 - (-\cos^2\phi - \sin^2\phi)\,\mathbf{j}_1 + (0-0)\,\mathbf{k}_1 = \mathbf{j}_1$$

the direction into the paper, normal to the plane of the arc. The matrix whose columns are $\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b$ in $\mathcal{B}_1$ is

$$[{}_1R_t] = \begin{pmatrix} -\cos\phi & \sin\phi & 0 \\ 0 & 0 & 1 \\ \sin\phi & \cos\phi & 0 \end{pmatrix}, \qquad \det[{}_1R_t] = -\cos\phi\,(0-\cos\phi) - \sin\phi\,(0-\sin\phi) = 1$$

and its columns are unit and mutually orthogonal ($\mathbf{e}_t \cdot \mathbf{e}_n = -\cos\phi\sin\phi + \sin\phi\cos\phi = 0$), so $[{}_1R_t][{}_1R_t]^T = I$.

- **Useful projections** (read from the columns): $\mathbf{i}_1 \cdot \mathbf{e}_t = -\cos\phi$, $\mathbf{i}_1 \cdot \mathbf{e}_n = \sin\phi$, $\mathbf{k}_1 \cdot \mathbf{e}_t = \sin\phi$, $\mathbf{k}_1 \cdot \mathbf{e}_n = \cos\phi$, $\mathbf{j}_1 = \mathbf{e}_b$.

Kinematics

(a) Relative velocity and acceleration in the path basis

Pedagogical Justification: The relative velocity and acceleration are the first and second derivatives of $\mathbf{r}_1^P$ taken in S_1, where $\mathbf{i}_1, \mathbf{k}_1$ are constant. Only ϕ depends on time, so each derivative is a chain rule.

Differentiate $\mathbf{r}_1^P = -R\sin\phi\,\mathbf{i}_1 - R\cos\phi\,\mathbf{k}_1$ with $\dfrac{d}{dt}\sin\phi = \cos\phi\,\dot\phi$ and $\dfrac{d}{dt}\cos\phi = -\sin\phi\,\dot\phi$:

$$\mathbf{v}_1^P = \left.\frac{d\mathbf{r}_1^P}{dt}\right|_1 = -R\cos\phi\,\dot\phi\,\mathbf{i}_1 + R\sin\phi\,\dot\phi\,\mathbf{k}_1 = R\dot\phi\,(-\cos\phi\,\mathbf{i}_1 + \sin\phi\,\mathbf{k}_1) = R\dot\phi\,\mathbf{e}_t$$

which is $\dot s\,\mathbf{e}_t$ with $s = R\phi$. For the acceleration use the product rule on $R\dot\phi\,\mathbf{e}_t$, and the chain rule on $\mathbf{e}_t(\phi(t))$:

$$\left.\frac{d\mathbf{e}_t}{dt}\right|_1 = \frac{d\mathbf{e}_t}{d\phi}\,\dot\phi = (\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{k}_1)\,\dot\phi = \dot\phi\,\mathbf{e}_n$$

$$\mathbf{a}_1^P = \left.\frac{d\mathbf{v}_1^P}{dt}\right|_1 = R\ddot\phi\,\mathbf{e}_t + R\dot\phi\,\left.\frac{d\mathbf{e}_t}{dt}\right|_1 = R\ddot\phi\,\mathbf{e}_t + R\dot\phi^2\,\mathbf{e}_n$$

The normal part is $\dot s^2/\rho$ with radius of curvature $\rho = R$, as expected for a circle. There is no binormal component.

Forces and Equations of Motion

(b) Newton's second law in the path basis

Pedagogical Justification: S_1 is not inertial, so we use Newton's law with inertia forces, $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eqs. 4.62–4.63, Slides 27–28). This keeps the simple relative acceleration of part (a) and puts the frame's motion into four forces.

Actual forces.

- Spring (zero natural length): $\overrightarrow{BP} = \mathbf{r}_1^P - \mathbf{r}_1^B = (-R\sin\phi + 2R)\,\mathbf{i}_1 - R\cos\phi\,\mathbf{k}_1 = (2R - R\sin\phi)\,\mathbf{i}_1 - R\cos\phi\,\mathbf{k}_1$, with $\mathbf{r}_1^B = -2R\,\mathbf{i}_1$, so

$$\mathbf{F}_K = -\alpha m\omega^2 R\,[(2 - \sin\phi)\,\mathbf{i}_1 - \cos\phi\,\mathbf{k}_1]$$

- Reaction of the smooth arc: $\mathbf{N} = N_n\,\mathbf{e}_n + N_b\,\mathbf{e}_b$ (no $\mathbf{e}_t$ component, because the arc is smooth).

Inertia forces (Slide 28), with $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$, $\mathbf{a}_0^O = -\omega^2 R\,\mathbf{i}_1$:

- Force due to the acceleration of the origin: $\mathbf{F}_A = -m\mathbf{a}_0^O = m\omega^2 R\,\mathbf{i}_1$.
- Euler force: $\mathbf{F}_E = -m\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P = \mathbf{0}$.
- Centrifugal force. First $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = \omega\mathbf{k}_1 \times (-R\sin\phi\,\mathbf{i}_1 - R\cos\phi\,\mathbf{k}_1) = -\omega R\sin\phi\,\mathbf{j}_1$ (using $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$); then $\boldsymbol{\omega}_{10} \times (-\omega R\sin\phi\,\mathbf{j}_1) = -\omega^2 R\sin\phi\,(\mathbf{k}_1 \times \mathbf{j}_1) = \omega^2 R\sin\phi\,\mathbf{i}_1$ (using $\mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$), so $\mathbf{F}_C = -m\omega^2 R\sin\phi\,\mathbf{i}_1$.
- Coriolis force: $\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = \omega R\dot{\phi}\,\mathbf{k}_1 \times (-\cos\phi\,\mathbf{i}_1 + \sin\phi\,\mathbf{k}_1) = -\omega R\dot{\phi}\cos\phi\,\mathbf{j}_1$, so $\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = 2m\omega R\dot{\phi}\cos\phi\,\mathbf{j}_1$.

The first and third combine into the centrifugal force about the **true** rotation axis, whose distance to P is $x_1 + R = R(1 - \sin\phi)$ (Slide 30, and the dimension of the figure of the arc):

$$\mathbf{F}_A + \mathbf{F}_C = m\omega^2 R\,(1 - \sin\phi)\,\mathbf{i}_1$$

Projections. With the projections listed at the end of the change-of-basis section:

$$\mathbf{F}_K \cdot \mathbf{e}_t = -\alpha m\omega^2 R\,[(2 - \sin\phi)(-\cos\phi) + (-\cos\phi)(\sin\phi)] = -\alpha m\omega^2 R\,[-2\cos\phi + \sin\phi\cos\phi - \sin\phi\cos\phi] = 2\alpha m\omega^2 R\cos\phi$$

$$\mathbf{F}_K \cdot \mathbf{e}_n = -\alpha m\omega^2 R\left[(2-\sin\phi)\sin\phi - \cos^2\phi\right] = -\alpha m\omega^2 R\left[2\sin\phi - 1\right] = \alpha m\omega^2 R\,(1-2\sin\phi)$$

$$(\mathbf{F}_A + \mathbf{F}_C)\cdot\mathbf{e}_t = -m\omega^2 R\,(1-\sin\phi)\cos\phi, \qquad (\mathbf{F}_A + \mathbf{F}_C)\cdot\mathbf{e}_n = m\omega^2 R\,(1-\sin\phi)\sin\phi$$

(in the second line $\sin^2\phi + \cos^2\phi = 1$ was used), and $\mathbf{F}_K$, $\mathbf{F}_A + \mathbf{F}_C$ have no $\mathbf{e}_b = \mathbf{j}_1$ component, while $\mathbf{F}_{COR}$ lies entirely along $\mathbf{e}_b$.

(b, continued) Equation of motion and reaction

The three components of $m\mathbf{a}_1^P = \mathbf{F}_K + \mathbf{N} + \mathbf{F}_A + \mathbf{F}_E + \mathbf{F}_C + \mathbf{F}_{COR}$:

$$\mathbf{e}_t: \quad mR\ddot{\phi} = 2\alpha m\omega^2 R\cos\phi - m\omega^2 R\,(1-\sin\phi)\cos\phi$$

$$\mathbf{e}_n: \quad mR\dot{\phi}^2 = N_n + \alpha m\omega^2 R\,(1-2\sin\phi) + m\omega^2 R\,(1-\sin\phi)\sin\phi$$

$$\mathbf{e}_b: \quad 0 = N_b + 2m\omega R\dot{\phi}\cos\phi$$

The tangential component contains no reaction and is the **equation of motion**; the other two give the **reaction**:

$$\boxed{\ddot{\phi} = \omega^2\cos\phi\,(2\alpha - 1 + \sin\phi)}$$

$$\boxed{N_n = mR\dot{\phi}^2 - \alpha m\omega^2 R\,(1-2\sin\phi) - m\omega^2 R\,\sin\phi\,(1-\sin\phi), \qquad N_b = -2m\omega R\dot{\phi}\cos\phi}$$

The binormal reaction balances the Coriolis force: the arc has to push P sideways to keep it in the rotating plane.

Answers and Checks

(a) Relative velocity and acceleration

$$\boxed{\mathbf{v}_1^P = R\dot{\phi}\,\mathbf{e}_t, \qquad \mathbf{a}_1^P = R\ddot{\phi}\,\mathbf{e}_t + R\dot{\phi}^2\,\mathbf{e}_n}$$

(b) Newton's second law in the path basis

The three components are the $\mathbf{e}_t$, $\mathbf{e}_n$ and $\mathbf{e}_b$ equations of the previous section: $\ddot{\phi} = \omega^2\cos\phi\,(2\alpha - 1 + \sin\phi)$, $N_n = mR\dot{\phi}^2 - \alpha m\omega^2 R\,(1 - 2\sin\phi) - m\omega^2 R\sin\phi\,(1 - \sin\phi)$ and $N_b = -2m\omega R\dot{\phi}\cos\phi$.

(c) Energy conservation in the relative frame

> **Pedagogical Justification:** In S_1 (Slide 29) the mechanical energy E_1^P is conserved if every force that does work in S_1 has a potential. Here: $\mathbf{N}$ does no work in S_1 because the arc is fixed in S_1 and $\mathbf{N} \perp \mathbf{v}_1^P = R\dot{\phi}\,\mathbf{e}_t$; $\mathbf{F}_{COR} \perp \mathbf{v}_1^P$ never does work (Slide 31); $\mathbf{F}_E = \mathbf{0}$; the spring is conservative; and $\mathbf{F}_A + \mathbf{F}_C$ is the centrifugal force about an axis fixed in S_1 with constant ω, which has the potential of Slide 30.

Potentials. Spring: $V_K = \tfrac{1}{2}K\,|\overrightarrow{BP}|^2$ with

$$|\overrightarrow{BP}|^2 = R^2\left[(2 - \sin\phi)^2 + \cos^2\phi\right] = R^2\left[4 - 4\sin\phi + \sin^2\phi + \cos^2\phi\right] = R^2\,(5 - 4\sin\phi)$$

so $V_K = \tfrac{1}{2}\alpha m\omega^2 R^2\,(5 - 4\sin\phi)$. Centrifugal (Slide 30), with $r_\perp = x_1 + R = R(1 - \sin\phi)$:

$$V_C = -\tfrac{1}{2}m\omega^2 r_\perp^2 = -\tfrac{1}{2}m\omega^2 R^2\,(1 - \sin\phi)^2$$

(Check: in Cartesian form $V_C = -\tfrac{1}{2}m\omega^2(x_1 + R)^2$ and $-\partial V_C/\partial x_1 = m\omega^2(x_1 + R)$, the $\mathbf{i}_1$ component of $\mathbf{F}_A + \mathbf{F}_C$.)

Energy equation:

$$\boxed{E_1^P = \tfrac{1}{2}mR^2\dot{\phi}^2 + \tfrac{1}{2}\alpha m\omega^2 R^2\,(5 - 4\sin\phi) - \tfrac{1}{2}m\omega^2 R^2\,(1 - \sin\phi)^2 = \text{const}}$$

Initial value. At $t = 0$, $\mathbf{e}_t(0) = -\mathbf{i}_1$, so $\mathbf{v}_1^P(0) = R\dot{\phi}(0)\,(-\mathbf{i}_1) = -\beta\omega R\,\mathbf{i}_1$ gives $\dot{\phi}(0) = \beta\omega$, and $E_1^P(0) = \tfrac{1}{2}m\omega^2 R^2\,(\beta^2 + 5\alpha - 1)$. Equating and dividing by $\tfrac{1}{2}mR^2$:

$$\dot\phi^2 = \omega^2\left[\beta^2 + 5\alpha - 1 - \alpha(5 - 4\sin\phi) + (1-\sin\phi)^2\right] \implies \boxed{\dot\phi^2 = \omega^2\left[\beta^2 + (4\alpha - 2)\sin\phi + \sin^2\phi\right]}$$

Equivalence. Differentiate E_1^P in time with the chain rule ($\frac{d}{dt}\sin\phi = \cos\phi\,\dot\phi$, $\frac{d}{dt}(1-\sin\phi)^2 = 2(1-\sin\phi)(-\cos\phi\,\dot\phi)$):

$$\frac{dE_1^P}{dt} = mR^2\dot\phi\ddot\phi - 2\alpha m\omega^2R^2\cos\phi\,\dot\phi + m\omega^2R^2(1-\sin\phi)\cos\phi\,\dot\phi = mR^2\dot\phi\left[\ddot\phi - 2\alpha\omega^2\cos\phi + \omega^2(1-\sin\phi)\cos\phi\right]$$

The bracket is zero exactly when the $\mathbf{e}_t$ equation of part (b) holds. So the energy equation is **not new information**: it is the first integral of the tangential component (the $\mathbf{e}_t$ equation multiplied by $R\dot\phi$ and integrated once). The $\mathbf{e}_n$ and $\mathbf{e}_b$ components are independent of it and are needed to compute the reaction.

(d) Value of α for equilibrium at $\phi = \pi/6$

> **Pedagogical Justification:** Relative equilibrium in S_1 means $\dot\phi = 0$ and $\ddot\phi = 0$ for all time. The $\mathbf{e}_t$ equation then requires that the tangential force vanish.

$$\ddot\phi = 0 \iff \cos\phi\,(2\alpha - 1 + \sin\phi) = 0$$

The root $\cos\phi = 0$ ($\phi = \pi/2$, the particle on the axis) is an equilibrium for every α. Any other equilibrium must satisfy $\sin\phi_e = 1 - 2\alpha$. For $\phi_e = \pi/6$, $\sin\phi_e = \frac{1}{2}$:

$$2\alpha - 1 + \tfrac{1}{2} = 0 \implies \boxed{\alpha = \tfrac{1}{4}}$$

At that equilibrium, from part (b) with $\dot\phi = 0$: $N_b = 0$ and $N_n = -\frac{1}{4}m\omega^2R\,(1-1) - m\omega^2R\cdot\frac{1}{2}\cdot\frac{1}{2} = -\frac{1}{4}m\omega^2R$, i.e. the arc pulls P away from O with force $\frac{1}{4}m\omega^2R$.

(e) Types of motion for $\alpha = 1/4$, critical β and reaction at $\phi = \pi/2$

> **Pedagogical Justification:** The energy integral of part (c) is a first-order equation $\dot\phi^2 = \omega^2[\beta^2 - W(\phi)]$; motion is possible only where $W(\phi) \le \beta^2$, so the shape of W classifies the motions, as with a one-dimensional potential.

With $\alpha = \frac{1}{4}$, $4\alpha - 2 = -1$:

$$\dot\phi^2 = \omega^2\left[\beta^2 - W(\phi)\right], \qquad W(\phi) = \sin\phi\,(1-\sin\phi)$$

On $[0,\pi/2]$: $W(0)=0$, $W(\pi/2)=0$, and $\dfrac{dW}{d\phi} = \cos\phi\,(1-2\sin\phi)$ vanishes inside the interval only at $\sin\phi = \tfrac12$, i.e. $\phi=\pi/6$, where W has its maximum $W(\pi/6) = \tfrac12\cdot\tfrac12 = \tfrac14$. W plays the role of a potential barrier, so the equilibrium $\phi=\pi/6$ is **unstable**. Three types of motion follow:

- $\beta > \tfrac12$**:** $\dot\phi^2 \geq \omega^2(\beta^2-\tfrac14) > 0$ on the whole arc, so ϕ increases monotonically, crosses the barrier at $\pi/6$ and reaches $\phi=\pi/2$ with $\dot\phi^2 = \omega^2[\beta^2-0]$, i.e. $\dot\phi = \beta\omega$: the relative speed there equals the initial one, $\beta\omega R$. The particle then leaves the arc at its upper end (part f).
- $\beta = \tfrac12$**:** $\beta^2 - W = \tfrac14 - \sin\phi + \sin^2\phi = (\tfrac12-\sin\phi)^2$, so $\dot\phi = \omega(\tfrac12-\sin\phi)$. The particle approaches the unstable equilibrium $\phi=\pi/6$ asymptotically and never reaches it: near it, $\sin\phi \approx \tfrac12 + \tfrac{\sqrt3}{2}(\phi-\tfrac{\pi}{6})$ and $\dot\phi \approx -\tfrac{\sqrt3}{2}\omega\,(\phi-\tfrac{\pi}{6})$, an exponential approach with rate $\tfrac{\sqrt3}{2}\omega$.
- $\beta < \tfrac12$**:** $\dot\phi$ vanishes at the turning point ϕ^* with $W(\phi^*)=\beta^2$, i.e. $\sin^2\phi^* - \sin\phi^* + \beta^2 = 0$; the root reached first is

$$\sin\phi^* = \frac{1-\sqrt{1-4\beta^2}}{2} < \frac12$$

There $\ddot\phi = \omega^2\cos\phi^*\,(\sin\phi^*-\tfrac12) < 0$, so the particle turns back, returns to $\phi=0$ with $\dot\phi = -\beta\omega$ (relative velocity $+\beta\omega R\,\mathbf{i}_1$, same speed by energy conservation) and leaves the quarter circumference at its lower end.

Critical value. The particle reaches $\phi=\pi/2$ if and only if β exceeds

$$\boxed{\beta_c = \tfrac12}$$

Reaction at $\phi=\pi/2$. Here $\cos\phi=0$, $\sin\phi=1$, $\dot\phi=\beta\omega$, and $\mathbf{e}_n(\pi/2)=\mathbf{i}_1$. From part (b) with $\alpha=\tfrac14$:

$$N_b = -2m\omega R\,\beta\omega\cdot 0 = 0, \qquad N_n = mR\beta^2\omega^2 - \tfrac14 m\omega^2 R\,(1-2) - m\omega^2 R\cdot 1\cdot 0 = m\omega^2 R\left(\beta^2+\tfrac14\right)$$

$$\boxed{\mathbf{N}(\phi=\tfrac{\pi}{2}) = m\omega^2 R\left(\beta^2+\tfrac{1}{4}\right)\mathbf{i}_1 \xrightarrow{\beta\to\beta_c} \tfrac{1}{2}m\omega^2 R\,\mathbf{i}_1}$$

It points towards O: it supplies the centripetal force $mR\dot{\phi}^2$ and also balances the spring, which pulls P towards B with $\frac{1}{4}m\omega^2 R$. The centrifugal and Coriolis forces vanish there because P is on the axis and moves parallel to it.

(f) Maximum height after leaving the arc ($\beta > \beta_c$)

> **Pedagogical Justification:** Once free, P has three degrees of freedom, and the horizontal motion is coupled through the Coriolis and centrifugal forces. But every inertia force of a frame rotating about the vertical is horizontal, so the vertical component of $m\mathbf{a}_1^P = \mathbf{F}+\mathbf{F}_i$ involves only the spring and decouples.

Initial state of the free motion (time $\tau=0$ at separation): $\phi=\pi/2$ gives $\mathbf{r}_1^P = -R\,\mathbf{i}_1$ (on the axis, $z_1=0$), and $\mathbf{v}_1^P = R\dot{\phi}\,\mathbf{e}_t(\pi/2) = \beta\omega R\,\mathbf{k}_1$ (vertical, upward).

Vertical equation. For a free position $\mathbf{r}_1^P = x_1\mathbf{i}_1 + y_1\mathbf{j}_1 + z_1\mathbf{k}_1$:

- $\mathbf{k}_1\cdot\mathbf{F}_K = -K\,\mathbf{k}_1\cdot[(x_1+2R)\mathbf{i}_1 + y_1\mathbf{j}_1 + z_1\mathbf{k}_1] = -\alpha m\omega^2 z_1$;
- $\mathbf{k}_1\cdot(\mathbf{F}_A+\mathbf{F}_C) = 0$, because $\mathbf{F}_A+\mathbf{F}_C = m\omega^2\,[(x_1+R)\mathbf{i}_1 + y_1\mathbf{j}_1]$ is horizontal;
- $\mathbf{k}_1\cdot\mathbf{F}_{COR} = -2m\,\mathbf{k}_1\cdot(\omega\mathbf{k}_1\times\mathbf{v}_1^P) = 0$, because $\mathbf{k}_1\times\mathbf{v}_1^P \perp \mathbf{k}_1$.

Hence, with $\alpha=\frac{1}{4}$,

$$m\ddot{z}_1 = -\alpha m\omega^2 z_1 \implies \ddot{z}_1 + \frac{\omega^2}{4}z_1 = 0$$

a harmonic oscillator of angular frequency $\omega/2$. Its general solution and derivative (chain rule $\frac{d}{d\tau}\sin\frac{\omega\tau}{2} = \frac{\omega}{2}\cos\frac{\omega\tau}{2}$) are

$$z_1 = C_1\cos\frac{\omega\tau}{2} + C_2\sin\frac{\omega\tau}{2}, \qquad \dot{z}_1 = -\frac{\omega}{2}C_1\sin\frac{\omega\tau}{2} + \frac{\omega}{2}C_2\cos\frac{\omega\tau}{2}$$

Initial conditions: $z_1(0)=C_1=0$ and $\dot{z}_1(0)=\frac{\omega}{2}C_2=\beta\omega R$, so $C_2=2\beta R$:

$$z_1(\tau) = 2\beta R\,\sin\frac{\omega\tau}{2} \implies \boxed{z_{max}=2\beta R} \quad \text{reached at } \tau=\pi/\omega$$

Since $\mathbf{k}_1 = \mathbf{k}$, this is also the height in the inertial frame, measured from the horizontal plane through O and B. For a general α the same steps give $z_{max} = \beta R/\sqrt{\alpha}$. (Equivalently, the vertical motion conserves $\frac{1}{2}m\dot{z}_1^2 + \frac{1}{2}Kz_1^2$, and $\frac{1}{2}m(\beta\omega R)^2 = \frac{1}{2}\cdot\frac{1}{4}m\omega^2 z_{max}^2$ gives the same result.) The result assumes that the free particle does not hit the wire again.

Checks

- **Competition of two forces.** Along the arc the spring pulls P towards B (towards larger ϕ) with tangential force $2\alpha m\omega^2 R\cos\phi$, while the centrifugal force about the axis pushes it away from the axis (towards smaller ϕ) with $m\omega^2 R(1-\sin\phi)\cos\phi$. Both scale with ω^2 because $K = \alpha m\omega^2$, so ω only sets the time scale $1/\omega$ and the dynamics depends on α and β alone.
- **Limiting cases.** For $\alpha > \frac{1}{2}$ the factor $2\alpha - 1 + \sin\phi$ is positive on the whole arc and there is no interior equilibrium. For $\beta \to \beta_c^+$ the time to cross the barrier grows without bound (logarithmically), consistent with the asymptotic motion at $\beta = \beta_c$. As $\phi \to \pi/2$, both the Coriolis force and N_b vanish, as they must for motion parallel to $\boldsymbol{\omega}_{10}$.
- **Orthogonality:** $\mathbf{e}_t \cdot \mathbf{e}_n = 0$, $\mathbf{e}_t \times \mathbf{e}_n = \mathbf{j}_1$ and $\det[{}_1R_t] = \det[{}_0R_1] = 1$ (change-of-basis section).
- **Inertial check.** Computing $\mathbf{a}_0^P$ by differentiating $\mathbf{r}_0^P = \mathbf{r}_0^O + [{}_0R_1]\mathbf{r}_1^P$ twice gives exactly $\mathbf{a}_0^O + \mathbf{a}_1^P + \boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P) + 2\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P$ (Notes Eq. 3.27), and $m\mathbf{a}_0^P = \mathbf{F}_K + \mathbf{N}$ holds with the reactions of part (b) (checked symbolically with sympy).
- **Dimensions.** $[R\sin\phi] = [R\cos\phi] = \mathrm{m}$; $[K] = [\alpha m\omega^2] = \mathrm{kg\cdot s^{-2}} = \mathrm{N/m}$; $[\ddot{\phi}] = [\omega^2] = \mathrm{s^{-2}}$; $[N_n] = [m\omega^2 R] = \mathrm{kg\cdot m\cdot s^{-2}} = \mathrm{N}$; $[E_1^P] = [m\omega^2R^2] = \mathrm{J}$; $[z_{max}] = [\beta R] = \mathrm{m}$.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 30 in its solution keys, so the results are verified by independent derivation (energy equivalence of part (c) and the symbolic inertial check above).

Source: Problems.pdf

PROBLEM 33

Two Repelling Particles on a Rotating Parabola

Original statement · Problems.pdf, p. 21

Rotating Wire + Two Particles

Two heavy point particles, P_1 and P_2, of mass m each, slide without friction along a rigid wire parabola given by the equation $z = x^2/(2L)$, where L is a known length. The wire rotates about its axis with a fixed, known angular velocity ω. The two particles repel each other electrically with a force of magnitude $F_E = k/d^2$, where k is a known constant and d is the distance between the particles. We define the rotating reference frame $Oxyz$ attached to the parabola as shown in the figure below.

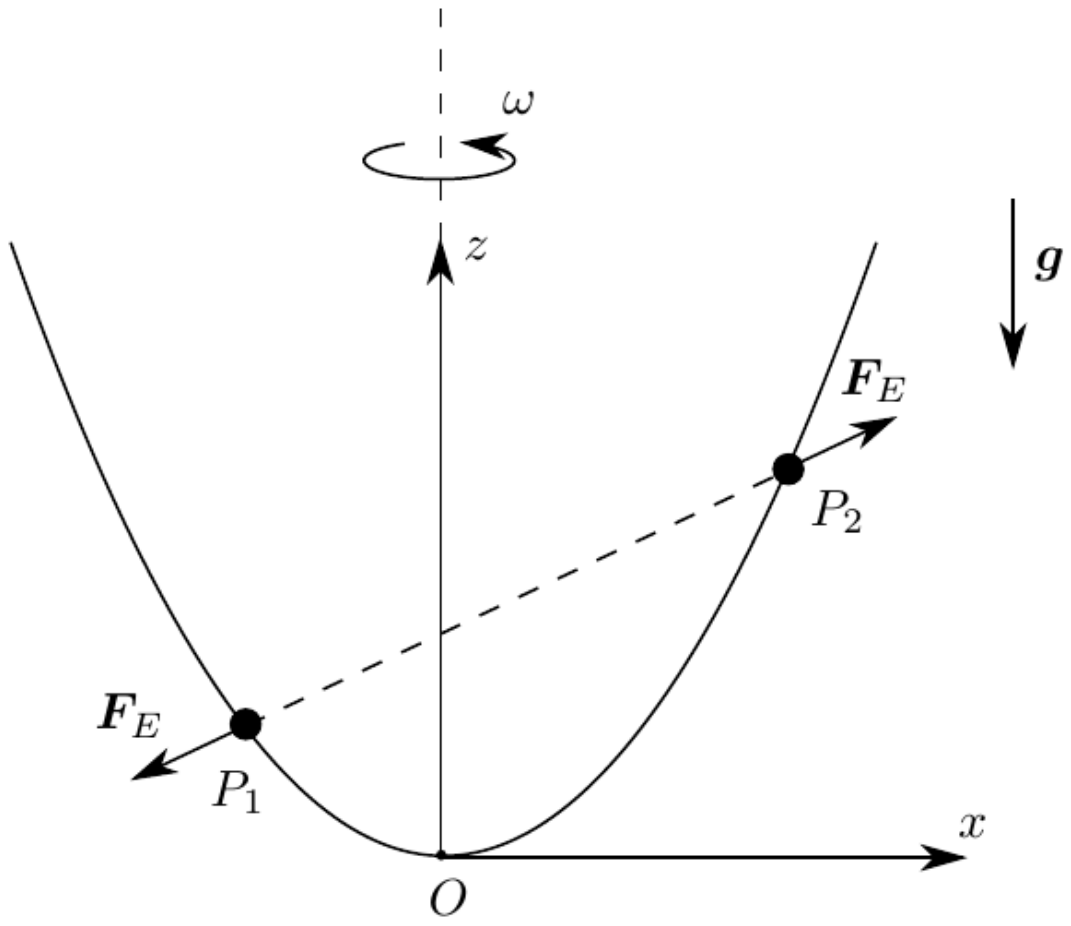


Figure: Parabola z = x²/(2L) in the plane Oxz rotating with angular velocity ω about Oz, particles P1 (left branch) and P2 (right branch) with the repulsive forces F_E along the line P1P2, gravity g downward (Problems.pdf, p. 21).

(a)

Write down the position, velocity, and acceleration vectors of each particle relative to $Oxyz$ in a generic instant of time, as a function of x_1, x_2 (the position of P_1 and P_2 along the Ox axis) and their derivatives.

(b) Write down Newton's second law for each particle in the $Oxyz$ reference frame.

(c) If the rotation of the wire is maintained by a motor, how much torque must it produce in the Oz direction to keep ω constant as a function of the variables of the problem and their derivatives?

(d) Find out the equilibrium configurations of the system in the reference frame $Oxyz$ as a function of the parameters $\gamma = g/(\omega^2 L)$ and $\kappa = k/(m\omega^2 L^3)$.

Consider now that P_1 and P_2 are constrained to move symmetrically on each branch of the parabola, and that $\gamma = \kappa = 2$. Initially, $x_1(0) = L$ and $v_1(0) = L\omega$ (with $\dot{x}_1(0) > 0$).

(e) Write down the equation that gives the minimum and maximum values of x_1 occurring in this case. *Hint: this equation is a third-order polynomial in x_1.*

Worked Solution

Reference Frames, Bases and Coordinates

- **System:** two particles P_1, P_2 of mass m slide on a smooth parabolic wire $z = x^2/(2L)$ that spins about its vertical axis Oz at constant ω. They repel each other with $F_E = k/d^2$ along the line that joins them, and both are heavy.
- **Why the rotating frame $Oxyz$.** In $Oxyz$ the wire is fixed, so each particle is located by a single coordinate x_i and its relative kinematics is that of a point on a fixed plane curve. The frame rotates uniformly about a fixed axis through O, so the inertia forces reduce to the centrifugal and Coriolis forces (Slide 28), and the centrifugal force has a potential (Slide 30).
- **Hypotheses:**
  1. The frame $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, with O the vertex of the parabola and $\mathbf{k}_0$ vertical upward, is inertial.
  2. The wire is smooth and bilateral (each particle is a bead): its reaction on P_i is perpendicular to the wire, with components in the plane of the wire and along $\mathbf{j}_1$.

3. ω is kept constant by the motor; the axis bearings exert no torque about Oz.
4. Electric repulsion is a central internal force: equal and opposite on the two particles, along $\overrightarrow{P_2P_1}$.

- **Fixed (inertial) frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, basis $\mathcal{B}_0$, origin at the vertex of the parabola, which lies on the axis and is fixed.
- **Moving frame:** $S_1 = Oxyz = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, basis $\mathcal{B}_1$, attached to the wire, with the wire in the plane Oxz and $\mathbf{k}_1 = \mathbf{k}_0$ along the axis. The two frames share the origin O. We choose $\mathbf{i}_1 = \mathbf{i}_0$ at $t = 0$.
- **Local basis of the wire at P_i:** unit tangent $\mathbf{t}_i$, in-plane unit normal $\mathbf{n}_i$ and $-\mathbf{j}_1$; deduced in the next section.
- **Degrees of freedom:** two, the abscissas x_1 and x_2 in $Oxyz$ (each bead has 3 coordinates and 2 constraints: $y = 0$ and $z = x^2/(2L)$). In part (e) the symmetry constraint $x_2 = -x_1$ leaves one.

Symbol	Meaning	SI unit
m	mass of each particle	kg
L	parameter of the parabola $z = x^2/(2L)$ (the radius of curvature at the vertex)	m
ω	constant angular velocity of the wire	rad/s
g	gravitational acceleration	m/s^2
k	electric repulsion constant, $F_E = k/d^2$	N·m^2
$\gamma = g/(\omega^2 L)$	weight versus centrifugal force	1
$\kappa = k/(m\omega^2 L^3)$	repulsion versus centrifugal force	1

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from above

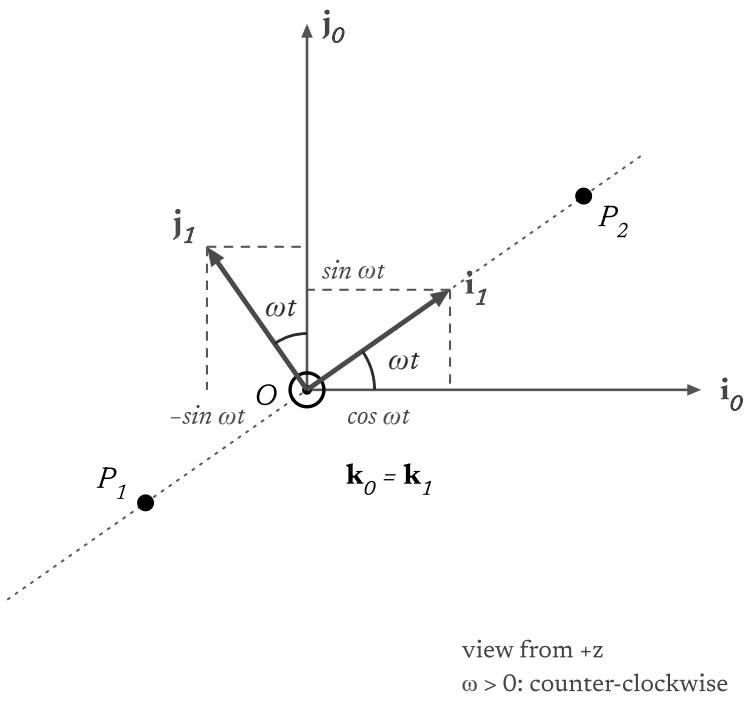


Figure 1. View from $+z$, with $\mathbf{k}_0 = \mathbf{k}_1$ out of the paper ($\odot$) at the vertex O: then $\mathbf{j}_0 = \mathbf{k}_0 \times \mathbf{i}_0$ points up the page. $\mathbf{i}_1, \mathbf{j}_1$ are $\mathbf{i}_0, \mathbf{j}_0$ turned counter-clockwise by ωt; the dashed legs give $\mathbf{i}_1 = \cos\omega t\,\mathbf{i}_0 + \sin\omega t\,\mathbf{j}_0$ and $\mathbf{j}_1 = -\sin\omega t\,\mathbf{i}_0 + \cos\omega t\,\mathbf{j}_0$. The plane of the wire is seen edge-on, along $\mathbf{i}_1$; P_1 ($x_1 < 0$) and P_2 ($x_2 > 0$) project on it.

The rotation is about the vertical $\mathbf{k}_0$, so it is drawn looking down the axis, from $+z$. In that view $\mathbf{k}_0$ points out of the paper, towards the reader ($\odot$). Take $\mathbf{i}_0$ to the right; then $\mathbf{j}_0 = \mathbf{k}_0 \times \mathbf{i}_0$ (right-hand rule: thumb along $\mathbf{k}_0$, towards the reader, fingers turning $\mathbf{i}_0$ counter-clockwise by $90°$) points up the page. With $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_0, \omega > 0$, the wire turns counter-clockwise in this view, and after a time t it has turned the angle ωt:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of a dashed right triangle: its leg adjacent to ωt lies along $\mathbf{i}_0$ and measures $\cos\omega t$; its opposite leg lies along $+\mathbf{j}_0$ and measures $\sin\omega t$. So $\mathbf{i}_1 = \cos\omega t\,\mathbf{i}_0 + \sin\omega t\,\mathbf{j}_0$.
- $\mathbf{j}_1$ starts on $\mathbf{j}_0$ (up the page) and turns counter-clockwise, so its tip moves to the left: its leg along $\mathbf{j}_0$ (adjacent) is $\cos\omega t$ and its leg along $-\mathbf{i}_0$ (opposite) is $\sin\omega t$. So $\mathbf{j}_1 = -\sin\omega t\,\mathbf{i}_0 + \cos\omega t\,\mathbf{j}_0$; **the minus sign comes from the counter-clockwise turn carrying the upward axis to the left.**
- $\mathbf{k}_1 = \mathbf{k}_0$, the axis of the rotation.

The plane of the wire contains $\mathbf{i}_1$ and $\mathbf{k}_1$, so in this view it is seen edge-on as the dotted line along $\mathbf{i}_1$; P_1 ($x_1 < 0$) and P_2 ($x_2 > 0$) lie on it, on either side of the axis. The columns of $[{}_0R_1]$ are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1 \cdot (\cos\omega t\cos\omega t - (-\sin\omega t)\sin\omega t) = \cos^2\omega t + \sin^2\omega t = 1$$

The columns have unit length ($\cos^2\omega t + \sin^2\omega t = 1$) and are mutually orthogonal (first and second: $-\cos\omega t\sin\omega t + \sin\omega t\cos\omega t = 0$; the third is orthogonal to both because their third components are zero), so $[{}_0R_1][{}_0R_1]^T = I$.

- **Motion of S_1:** $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$; $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$; $\boldsymbol{\alpha}_{10} = \mathbf{0}$ because ω is constant. Poisson check with the chain rule: $\frac{d\mathbf{i}_1}{dt}\big|_0 = -\omega\sin\omega t\,\mathbf{i}_0 + \omega\cos\omega t\,\mathbf{j}_0 = \omega\,\mathbf{j}_1$, and indeed $\omega\mathbf{k}_1 \times \mathbf{i}_1 = \omega\,\mathbf{j}_1$.

The position vectors and the local basis of the wire

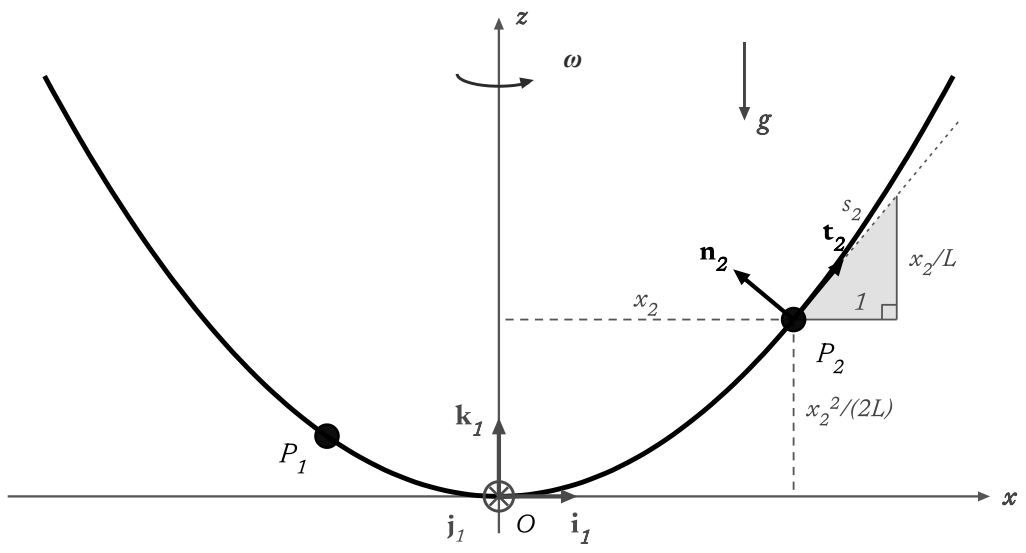


Figure 2. Plane Oxz, fixed in S_1 ($\mathbf{j}_1$ points into the paper, $\otimes$). The position of P_2 has the dashed components x_2 and $z_2 = x_2^2/(2L)$. The shaded slope triangle at P_2 has a horizontal leg proportional to 1, a vertical leg proportional to $dz/dx = x_2/L$ and the hypotenuse $s_2 = \sqrt{1 + x_2^2/L^2}$ along the tangent; dividing by s_2 gives $\mathbf{t}_2$, and turning it 90° towards the concave side gives $\mathbf{n}_2$ (orange).

The plane Oxz is fixed in S_1; it is drawn with $\mathbf{i}_1$ to the right and $\mathbf{k}_1$ up. The direction out of the paper is $\mathbf{i}_1 \times \mathbf{k}_1 = -\mathbf{j}_1$, so $\mathbf{j}_1$ points **into** the paper ($\otimes$).

Position. A bead at abscissa x_i is reached from O by moving x_i along $\mathbf{i}_1$ and then the height of the wire at that abscissa, $z_i = x_i^2/(2L)$, along $\mathbf{k}_1$ (the two dashed components drawn for P_2):

$$\mathbf{r}_1^{P_i} = x_i\,\mathbf{i}_1 + \frac{x_i^2}{2L}\,\mathbf{k}_1, \qquad i = 1, 2$$

Tangent. A tangent vector is the derivative of the position with respect to the coordinate; with $\frac{d}{dx}\left(\frac{x^2}{2L}\right) = \frac{2x}{2L} = \frac{x}{L}$:

$$\frac{d}{dx}\left(x\,\mathbf{i}_1 + \frac{x^2}{2L}\mathbf{k}_1\right) = \mathbf{i}_1 + \frac{x}{L}\,\mathbf{k}_1$$

It is the hypotenuse of the shaded slope triangle at P_2: horizontal leg 1, vertical leg $dz/dx = x_i/L$, hence length $s_i = \sqrt{1 + x_i^2/L^2}$ (Pythagoras). Dividing by s_i:

$$\mathbf{t}_i = \frac{1}{s_i}\left(\mathbf{i}_1 + \frac{x_i}{L}\mathbf{k}_1\right), \qquad \mathbf{n}_i = \frac{1}{s_i}\left(-\frac{x_i}{L}\mathbf{i}_1 + \mathbf{k}_1\right)$$

$\mathbf{n}_i$ is $\mathbf{t}_i$ turned 90° counter-clockwise in the drawing (the components (a, b) become $(-b, a)$), which points towards the concave side of the parabola, upwards and towards the axis. They are orthogonal, $\mathbf{t}_i \cdot \mathbf{n}_i = \frac{1}{s_i^2}\left(-\frac{x_i}{L} + \frac{x_i}{L}\right) = 0$, both have length $\sqrt{1 + x_i^2/L^2}/s_i = 1$, and, using $\mathbf{i}_1 \times \mathbf{k}_1 = -\mathbf{j}_1$ and $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$,

$$\mathbf{t}_i \times \mathbf{n}_i = \frac{1}{s_i^2}\left[\mathbf{i}_1 \times \mathbf{k}_1 - \frac{x_i^2}{L^2}\,\mathbf{k}_1 \times \mathbf{i}_1\right] = \frac{1}{s_i^2}\left[-\mathbf{j}_1 - \frac{x_i^2}{L^2}\mathbf{j}_1\right] = -\mathbf{j}_1$$

(the direction out of the paper), so $\{\mathbf{t}_i, \mathbf{n}_i, -\mathbf{j}_1\}$ is a right-handed orthonormal triad attached to the wire at P_i. The reaction of the smooth wire lies in the plane spanned by $\mathbf{n}_i$ and $\mathbf{j}_1$.

The chord between the beads

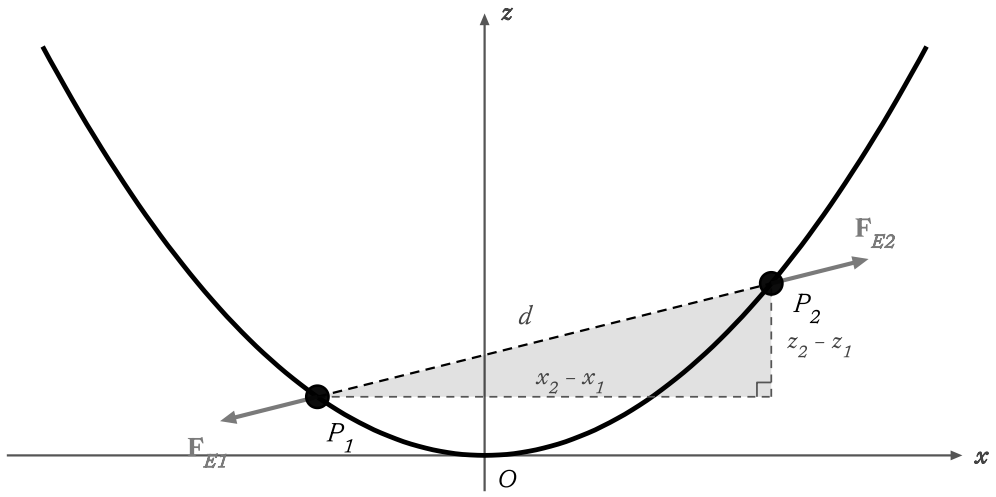


Figure 3. Chord P_1P_2: in the shaded right triangle the horizontal leg is $x_2 - x_1$, the vertical leg is $z_2 - z_1 = (x_2^2 - x_1^2)/(2L) = (x_2 - x_1)(x_1 + x_2)/(2L)$ and the hypotenuse is the distance d; the slope of the chord is $(x_1 + x_2)/(2L)$. The repulsions $\mathbf{F}_{E1}$ and $\mathbf{F}_{E2} = -\mathbf{F}_{E1}$ (pink) act along the chord, pushing the beads apart.

The repulsion acts along the chord P_1P_2. With $\Delta x = x_1 - x_2$, the shaded right triangle has the horizontal leg $|\Delta x|$, the vertical leg $\Delta z = z_1 - z_2 = \dfrac{x_1^2 - x_2^2}{2L} = \Delta x\,\dfrac{x_1 + x_2}{2L}$ (difference of squares) and the hypotenuse d. Hence

$$\overrightarrow{P_2P_1} = \mathbf{r}_1^{P_1} - \mathbf{r}_1^{P_2} = \Delta x\left[\mathbf{i}_1 + \frac{x_1 + x_2}{2L}\mathbf{k}_1\right], \qquad d^2 = |\overrightarrow{P_2P_1}|^2 = \Delta x^2\left[1 + \frac{(x_1 + x_2)^2}{4L^2}\right]$$

so the slope of the chord is $(x_1 + x_2)/(2L)$, the mean of the slopes x_i/L at its ends.

Kinematics

(a) Relative position, velocity and acceleration

Pedagogical Justification: Relative kinematics is computed in S_1, where $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ are constant, so only the scalar components are differentiated. Each $z_i = x_i^2/(2L)$ depends on time through $x_i(t)$, so the chain rule is needed.

Chain rule: $\dfrac{d}{dt}\left(\dfrac{x_i^2}{2L}\right) = \dfrac{2x_i}{2L}\,\dot{x}_i = \dfrac{x_i\dot{x}_i}{L}$. Hence

$$\mathbf{v}_1^{P_i} = \dot{x}_i\,\mathbf{i}_1 + \frac{x_i\dot{x}_i}{L}\,\mathbf{k}_1$$

Product rule on $x_i\dot{x}_i$: $\dfrac{d}{dt}(x_i\dot{x}_i) = \dot{x}_i\dot{x}_i + x_i\ddot{x}_i$. Hence

$$\mathbf{a}_1^{P_i} = \ddot{x}_i\,\mathbf{i}_1 + \frac{\dot{x}_i^2 + x_i\ddot{x}_i}{L}\,\mathbf{k}_1$$

Note $\mathbf{v}_1^{P_i} = s_i\dot{x}_i\,\mathbf{t}_i$, tangent to the wire, as it must be.

Forces and Equations of Motion

(b) Newton's second law in $Oxyz$

> **Pedagogical Justification:** $Oxyz$ is not inertial; Newton's law holds there if the inertia forces are added, $m\mathbf{a}_1^{P_i} = \mathbf{F} + \mathbf{F}_i$ (Notes Eqs. 4.62–4.63). Projecting on the tangent $\mathbf{t}_i$ eliminates the unknown reaction and gives the equation of motion; the other two projections give the reaction.

Actual forces on P_i:

- Weight: $-mg\,\mathbf{k}_1$.
- Reaction of the smooth wire: $\mathbf{N}_i = N_{in}\,\mathbf{n}_i + N_{iy}\,\mathbf{j}_1$ (so $\mathbf{N}_i \cdot \mathbf{t}_i = 0$).
- Electric repulsion. With $\overrightarrow{P_2P_1}$ and d of the chord triangle, the force on P_1 has magnitude k/d^2 and the direction of $\overrightarrow{P_2P_1}/d$, and the force on P_2 is opposite:

$$\mathbf{F}_{E1} = \frac{k}{d^2}\,\frac{\overrightarrow{P_2P_1}}{d} = \frac{k}{d^3}\,\Delta x\left[\mathbf{i}_1 + \frac{x_1 + x_2}{2L}\mathbf{k}_1\right] = -\mathbf{F}_{E2}$$

Inertia forces ($\mathbf{a}_0^O = \mathbf{0}$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$, so $\mathbf{F}_A = \mathbf{F}_E^{\text{Euler}} = \mathbf{0}$):

- Centrifugal: $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^{P_i} = \omega\mathbf{k}_1 \times \left(x_i\mathbf{i}_1 + \frac{x_i^2}{2L}\mathbf{k}_1\right) = \omega x_i\,\mathbf{j}_1$, then $\boldsymbol{\omega}_{10} \times \omega x_i\mathbf{j}_1 = \omega^2 x_i\,(\mathbf{k}_1 \times \mathbf{j}_1) = -\omega^2 x_i\,\mathbf{i}_1$, so $\mathbf{F}_{C,i} = m\omega^2 x_i\,\mathbf{i}_1$ (outward from the axis, Slide 30).
- Coriolis: $\boldsymbol{\omega}_{10} \times \mathbf{v}_1^{P_i} = \omega\mathbf{k}_1 \times \left(\dot{x}_i\mathbf{i}_1 + \frac{x_i\dot{x}_i}{L}\mathbf{k}_1\right) = \omega\dot{x}_i\,\mathbf{j}_1$, so $\mathbf{F}_{COR,i} = -2m\omega\dot{x}_i\,\mathbf{j}_1$.

(b, continued) Newton's equations

Newton's second law for each particle (vector form):

$$m\left[\ddot{x}_i\,\mathbf{i}_1+\frac{\dot{x}_i^2+x_i\ddot{x}_i}{L}\,\mathbf{k}_1\right]=-mg\,\mathbf{k}_1+N_{in}\,\mathbf{n}_i+N_{iy}\,\mathbf{j}_1+\mathbf{F}_{Ei}+m\omega^2 x_i\,\mathbf{i}_1-2m\omega\dot{x}_i\,\mathbf{j}_1,\qquad i=1,2$$

Projection on $s_i\mathbf{t}_i=\mathbf{i}_1+\dfrac{x_i}{L}\mathbf{k}_1$ (no reaction). Left side: $m\left[\ddot{x}_i+\dfrac{x_i}{L}\cdot\dfrac{\dot{x}_i^2+x_i\ddot{x}_i}{L}\right]=m\left[\left(1+\dfrac{x_i^2}{L^2}\right)\ddot{x}_i+\dfrac{x_i\dot{x}_i^2}{L^2}\right]$. Right side: weight $-mg\,x_i/L$; centrifugal $m\omega^2 x_i$; Coriolis 0; electric, for P_1,

$$\mathbf{F}_{E1}\cdot\left(\mathbf{i}_1+\frac{x_1}{L}\mathbf{k}_1\right)=\frac{k}{d^3}\,(x_1-x_2)\left[1+\frac{x_1(x_1+x_2)}{2L^2}\right]$$

and for P_2 the same with the indices exchanged (because $\mathbf{F}_{E2}=-\mathbf{F}_{E1}$ and $-\Delta x=x_2-x_1$). The **equations of motion** are

$$\boxed{m\left[\left(1+\frac{x_1^2}{L^2}\right)\ddot{x}_1+\frac{x_1\dot{x}_1^2}{L^2}\right]=m\left(\omega^2-\frac{g}{L}\right)x_1+\frac{k}{d^3}\,(x_1-x_2)\left[1+\frac{x_1(x_1+x_2)}{2L^2}\right]}$$

$$\boxed{m\left[\left(1+\frac{x_2^2}{L^2}\right)\ddot{x}_2+\frac{x_2\dot{x}_2^2}{L^2}\right]=m\left(\omega^2-\frac{g}{L}\right)x_2+\frac{k}{d^3}\,(x_2-x_1)\left[1+\frac{x_2(x_1+x_2)}{2L^2}\right]}$$

with $d^3=|x_1-x_2|^3\left[1+\dfrac{(x_1+x_2)^2}{4L^2}\right]^{3/2}$.

Projection on $\mathbf{j}_1$: $0=N_{iy}-2m\omega\dot{x}_i$, so $\boxed{N_{iy}=2m\omega\dot{x}_i}$: the wire pushes each bead sideways to cancel the Coriolis force.

Projection on $s_i\mathbf{n}_i=-\dfrac{x_i}{L}\mathbf{i}_1+\mathbf{k}_1$. Left side: $m\left[-\dfrac{x_i\ddot{x}_i}{L}+\dfrac{\dot{x}_i^2+x_i\ddot{x}_i}{L}\right]=\dfrac{m\dot{x}_i^2}{L}$. Right side: $N_{in}s_i-mg-\dfrac{m\omega^2x_i^2}{L}+\mathbf{F}_{Ei}\cdot(s_i\mathbf{n}_i)$, where for P_1

$$\mathbf{F}_{E1}\cdot\left(-\frac{x_1}{L}\mathbf{i}_1+\mathbf{k}_1\right)=\frac{k\,\Delta x}{d^3}\left[-\frac{x_1}{L}+\frac{x_1+x_2}{2L}\right]=\frac{k\,\Delta x}{d^3}\cdot\frac{x_2-x_1}{2L}=-\frac{k\,\Delta x^2}{2Ld^3}$$

and the same value is found for P_2. Therefore

$$\boxed{N_{in} = \frac{m}{s_i}\left[\frac{\dot{x}_i^2}{L} + g + \frac{\omega^2 x_i^2}{L}\right] + \frac{k\,(x_1 - x_2)^2}{2L\,s_i\,d^3}}, \qquad s_i = \sqrt{1 + \frac{x_i^2}{L^2}}$$

Answers and Checks

(a) Relative position, velocity and acceleration

$$\mathbf{r}_1^{P_i} = x_i\,\mathbf{i}_1 + \frac{x_i^2}{2L}\,\mathbf{k}_1, \qquad \mathbf{v}_1^{P_i} = \dot{x}_i\,\mathbf{i}_1 + \frac{x_i\dot{x}_i}{L}\,\mathbf{k}_1, \qquad \mathbf{a}_1^{P_i} = \ddot{x}_i\,\mathbf{i}_1 + \frac{\dot{x}_i^2 + x_i\ddot{x}_i}{L}\,\mathbf{k}_1 \qquad (i = 1, 2)$$

(b) Newton's second law in $Oxyz$

The vector equation of each bead, with its tangential projections (the two boxed equations of motion), $N_{iy} = 2m\omega\dot{x}_i$ and the boxed N_{in}, all derived in the previous section.

(c) Motor torque that keeps ω constant

> **Pedagogical Justification:** The motor torque acts on the wire; the wire acts on the particles through the reactions $\mathbf{N}_i$, and the particles act back with $-\mathbf{N}_i$. The wire is a massless guide (if it had mass and were symmetric about Oz, rotating at constant ω, its own angular momentum about Oz would be constant; this remark goes beyond the particle units and does not change the result). Hence the moment about Oz of all the actions on it must vanish.

Balance of moments about Oz on the wire: $M + \sum_i \left[\mathbf{r}^{P_i} \times (-\mathbf{N}_i)\right] \cdot \mathbf{k}_1 = 0$. Expand the cross product with $\mathbf{r}^{P_i} = x_i\mathbf{i}_1 + z_i\mathbf{k}_1$ and $\mathbf{N}_i = N_{ix}\mathbf{i}_1 + N_{iy}\mathbf{j}_1 + N_{iz}\mathbf{k}_1$; its $\mathbf{k}_1$ component is $x_i N_{iy} - 0 \cdot N_{ix} = x_i N_{iy}$. With $N_{iy} = 2m\omega\dot{x}_i$ from part (b):

$$\boxed{M = x_1 N_{1y} + x_2 N_{2y} = 2m\omega\,(x_1\dot{x}_1 + x_2\dot{x}_2)}$$

Check with the angular momentum of the particles in S_0. By Notes Eq. 3.26, $\mathbf{v}_0^{P_i} = \mathbf{v}_1^{P_i} + \boldsymbol{\omega}_{10} \times \mathbf{r}_1^{P_i} = \dot{x}_i\mathbf{i}_1 + \omega x_i\mathbf{j}_1 + \frac{x_i\dot{x}_i}{L}\mathbf{k}_1$, whose $\mathbf{k}_1$-moment about O is $m\left[x_i(\omega x_i) - 0 \cdot \dot{x}_i\right] = m\omega x_i^2$. So $H_{Oz} = m\omega(x_1^2 + x_2^2)$ and, by the product and chain rules, $\dot{H}_{Oz} = 2m\omega(x_1\dot{x}_1 + x_2\dot{x}_2)$. The weights are parallel to Oz and the electric pair is central ($(\mathbf{r}^{P_1} - \mathbf{r}^{P_2}) \times \mathbf{F}_{E1} = \mathbf{0}$), so only the reactions have a moment about Oz, and that moment equals M. Both routes agree with the official key (Problems.pdf, p. 75).

(d) Equilibrium configurations in $Oxyz$

> **Pedagogical Justification:** A relative equilibrium has $\dot{x}_i = \ddot{x}_i = 0$; then the two equations of motion become algebraic. Writing them with $u_i = x_i/L$ and dividing by $m\omega^2 L$ shows that only γ and κ matter.

With $\dot{x}_i = \ddot{x}_i = 0$, $x_i = Lu_i$ and $d = LD$, $D = |u_1 - u_2|\sqrt{1 + (u_1+u_2)^2/4}$, dividing by $m\omega^2 L$:

$$f_1 \equiv (1-\gamma)\,u_1 + \frac{\kappa}{D^3}(u_1 - u_2)\left[1 + \frac{u_1(u_1+u_2)}{2}\right] = 0, \qquad f_2 \equiv (1-\gamma)\,u_2 + \frac{\kappa}{D^3}(u_2 - u_1)\left[1 + \frac{u_2(u_1+u_2)}{2}\right] = 0$$

Sum. With $S = u_1 + u_2$, the electric terms add to $\dfrac{\kappa}{D^3}(u_1 - u_2)\left[\dfrac{u_1 S}{2} - \dfrac{u_2 S}{2}\right] = \dfrac{\kappa\,(u_1-u_2)^2 S}{2D^3}$, so

$$f_1 + f_2 = S\left[(1-\gamma) + \frac{\kappa\,(u_1-u_2)^2}{2D^3}\right] = 0$$

Either $S = 0$ (symmetric configuration) or $\gamma - 1 = \dfrac{\kappa(u_1-u_2)^2}{2D^3} > 0$.

Case $\gamma < 1$. If $S = 0$, $u_2 = -u_1$, $D = 2|u_1|$ and $f_1 = u_1\left[(1-\gamma) + \dfrac{2\kappa}{8|u_1|^3}\right]$: both terms of the bracket are positive and $u_1 \neq 0$ (the particles cannot coincide), so $f_1 \neq 0$. If $S \neq 0$ the bracket of the sum is positive. **No equilibrium exists.** (Centrifugal force beats gravity everywhere and the repulsion only pushes the particles further out.)

Case $\gamma = 1$. Both alternatives require $\kappa = 0$. For $\kappa > 0$ there is no equilibrium; only without repulsion ($\kappa = 0$) is every configuration an equilibrium (gravity and centrifugal force balance along the wire at every point: $(\omega^2 - g/L)x_i = 0$).

Case $\gamma > 1$, symmetric ($x_2 = -x_1$). $f_1 = u_1\left[(1-\gamma) + \dfrac{\kappa}{4|u_1|^3}\right] = 0$ gives $|u_1|^3 = \dfrac{\kappa}{4(\gamma-1)}$:

$$\boxed{x_1 = -x_2, \qquad x_1^2 = \left[\frac{\kappa L^3}{4(\gamma-1)}\right]^{2/3}}$$

Case $\gamma > 1$, asymmetric ($S \neq 0$). Insert $\dfrac{\kappa}{D^3} = \dfrac{2(\gamma-1)}{(u_1-u_2)^2}$ from the sum into $f_1 = 0$ and divide by $(\gamma - 1)$:

$$-u_1+\frac{2}{u_1-u_2}\left[1+\frac{u_1(u_1+u_2)}{2}\right]=0 \implies u_1(u_1-u_2)=2+u_1(u_1+u_2) \implies \boxed{x_1x_2=-L^2}$$

The particles are on opposite branches and the chord P_1P_2 passes through the focus $(0,L/2)$ of the parabola. Then $u_2=-1/u_1$, $S=u_1-1/u_1$, $u_1-u_2=u_1+1/u_1$, and $1+S^2/4=(u_1+1/u_1)^2/4$, so $D^3=|u_1+1/u_1|^6/8$ and the sum condition becomes

$$\gamma-1=\frac{4\kappa}{(u_1+1/u_1)^4} \implies \left|u_1+\frac{1}{u_1}\right|=\left(\frac{4\kappa}{\gamma-1}\right)^{1/4}$$

Since $|u_1+1/u_1|\geq 2$, with equality only for $|u_1|=1$ (which is the symmetric equilibrium), **asymmetric equilibria exist if and only if** $\kappa>4(\gamma-1)$, i.e. exactly when the symmetric equilibrium lies at $|x_1|>L$. They come in mirror pairs, $u_1=\frac{1}{2}\left[q\pm\sqrt{q^2-4}\right]$ with $q=(4\kappa/(\gamma-1))^{1/4}$ and $u_2=-1/u_1$ (plus the configurations with the labels exchanged). No equilibrium has both particles on the same branch.

(e) Symmetric motion with $\gamma=\kappa=2$: turning points

> **Pedagogical Justification:** In $Oxyz$ the wire (and the symmetry constraint) is fixed, $\boldsymbol{\omega}_{10}$ is constant and the origin is fixed. The reactions and the Coriolis forces do no work in S_1 (Slide 31), and the weights, the electric pair and the centrifugal forces have potentials (Slides 29–30). The mechanical energy of the pair in S_1 is therefore conserved, and its turning points ($\dot{x}_1=0$) give the extreme values of x_1.

Put $x_1=x>0$, $x_2=-x$ (the repulsion becomes infinite at $x=0$, so x_1 stays positive). Then $d=2x$ and, for each particle, $|\mathbf{v}_1^{P_i}|^2=\dot{x}^2\left(1+x^2/L^2\right)$.

- Kinetic energy: $T_1=2\cdot\frac{1}{2}m\dot{x}^2\left(1+\dfrac{x^2}{L^2}\right)=m\dot{x}^2\left(1+\dfrac{x^2}{L^2}\right)$.
- Gravity: $2\,mg\,\dfrac{x^2}{2L}=\dfrac{mgx^2}{L}$. Centrifugal: $2\left(-\frac{1}{2}m\omega^2x^2\right)=-m\omega^2x^2$. Electric: $\dfrac{k}{d}=\dfrac{k}{2x}$ (its gradient gives the repulsion k/d^2).

$$E_1=m\dot{x}^2\left(1+\frac{x^2}{L^2}\right)+m\left(\frac{g}{L}-\omega^2\right)x^2+\frac{k}{2x}=\text{const}$$

Initial value. $|\mathbf{v}_1^{P_1}(0)|=L\omega$ at $x=L$ gives $2\dot{x}(0)^2=L^2\omega^2$, so $T_1(0)=mL^2\omega^2$ and $E_1=mL^2\omega^2+m(g/L-\omega^2)L^2+k/(2L)$. Dividing by $m\omega^2L^2$, with $u=x/L$:

$$\frac{E_1}{m\omega^2 L^2} = \gamma + \frac{\kappa}{2} = 3 \qquad (\gamma = \kappa = 2)$$

Turning points ($\dot{x} = 0$): $(\gamma - 1)u^2 + \dfrac{\kappa}{2u} = \gamma + \dfrac{\kappa}{2}$, i.e. $u^2 + \dfrac{1}{u} = 3$. Multiply by $u > 0$:

$$\boxed{\frac{x_1^3}{L^3} - 3\,\frac{x_1}{L} + 1 = 0}$$

Roots. With $u = 2\cos\theta$ and $4\cos^3\theta - 3\cos\theta = \cos 3\theta$: $8\cos^3\theta - 6\cos\theta + 1 = 2\cos 3\theta + 1 = 0$, so $\cos 3\theta = -\frac{1}{2}$ and $\theta = 2\pi/9,\ 4\pi/9,\ 8\pi/9$. The roots are $u = 2\cos(2\pi/9) = 1.532$, $2\cos(4\pi/9) = 0.347$ and $2\cos(8\pi/9) = -1.879$ (rejected, $x_1 > 0$). Since the starting point $u = 1$ lies between the two positive roots (there $u^2 + 1/u = 2 < 3$),

$$x_{1,\min} = 2L\cos\frac{4\pi}{9} \approx 0.347\,L, \qquad x_{1,\max} = 2L\cos\frac{2\pi}{9} \approx 1.532\,L$$

and x_1 oscillates between them. The key also lists $x_1^3/L^3 - 3x_1/L - 1 = 0$ for $x_1 < 0$: that is the mirror image (a start on the negative branch) and does not apply to $x_1(0) = L > 0$.

Checks

- **Effective spring along the wire.** Gravity and the centrifugal force combine into $m(\omega^2 - g/L)x_i$ in the tangential equation: a restoring force if $\gamma > 1$ (gravity wins and pulls the beads to the vertex) and a repelling one if $\gamma < 1$. Equilibrium requires the repulsion to be balanced by this restoring force, hence the condition $\gamma > 1$.
- **Limits.** $\kappa \to 0$: the symmetric equilibrium collapses to $x_1 = x_2 = 0$ (no repulsion to separate the beads). $\gamma \to 1^+$: $x_1^2 \to \infty$, the restoring force vanishes and the beads are pushed to infinity. At the vertex, $x_i = 0$ gives $\mathbf{t}_i = \mathbf{i}_1$ and $\mathbf{n}_i = \mathbf{k}_1$, as the slope triangle degenerates to its horizontal leg.
- **Torque sign.** $M = \frac{d}{dt}\left[m\omega(x_1^2 + x_2^2)\right]$: when the beads move outward the moment of inertia of the system grows and the motor has to supply angular momentum ($M > 0$); when they move inward it must brake ($M < 0$).
- **Orthogonality:** $\mathbf{t}_i \cdot \mathbf{n}_i = 0$, $\mathbf{t}_i \times \mathbf{n}_i = -\mathbf{j}_1$ and $\det[{}_0R_1] = 1$ (change-of-basis section).
- **Inertial check.** Differentiating $[{}_0R_1]\mathbf{r}_1^{P_i}$ twice reproduces $\mathbf{a}_1^{P_i} + \boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^{P_i}) + 2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^{P_i}$ (Notes Eq. 3.27), and $dE_1/dt = 0$ along the equations of motion of part (b) (both checked symbolically with sympy, together with the equilibria of part (d)).

- **Dimensions.** $[x_i^2/(2L)] = \mathrm{m}$, $[x_i/L] = 1$; $[k/d^2] = \mathrm{N}\cdot\mathrm{m}^2/\mathrm{m}^2 = \mathrm{N}$; $[m\omega^2 x] = \mathrm{kg}\cdot\mathrm{s}^{-2}\cdot\mathrm{m} = \mathrm{N}$; $[M] = [m\omega x\dot{x}] = \mathrm{kg}\cdot\mathrm{s}^{-1}\cdot\mathrm{m}\cdot\mathrm{m/s} = \mathrm{N}\cdot\mathrm{m}$; $[\gamma] = \dfrac{\mathrm{m/s^2}}{\mathrm{s}^{-2}\,\mathrm{m}} = 1$; $[\kappa] = \dfrac{\mathrm{N}\cdot\mathrm{m}^2}{\mathrm{kg}\cdot\mathrm{s}^{-2}\cdot\mathrm{m}^3} = 1$.
- **Check against the official key:** the key of `Problems.pdf` gives the motor torque $2m\omega\left(\dot{x}_1 x_1 + \dot{x}_2 x_2\right)$, the equilibrium cases ($\gamma < 1$: none; $\gamma = 1$: only for $\kappa = 0$; $\gamma > 1$: symmetric with $x_1 = -x_2$ and $x_1^2 = \left[\kappa L^3/\left(4(\gamma-1)\right)\right]^{2/3}$, and asymmetric ones) and the turning-point equations $x_1^3/L^3 - 3x_1/L \pm 1 = 0$. They coincide with the results of (c), (d) and (e) (for $\gamma = \kappa = 2$ the turning-point condition $u^3 - 3u + 1 = 0$ is the positive-x_1 case).

Source: Problems.pdf (PDF page 21); key on PDF page 75

PROBLEM 34

Particle on a Circular Wire in an Oscillating Vertical Plane

Original statement · Problems.pdf, p. 22

Oscillating Plane + Euler Force

A heavy point particle P of mass m slides without friction on a circular wire of center C and radius R. The wire is located on a vertical plane Oxy that rotates about the vertical axis Oy with a prescribed angular frequency $\boldsymbol{\Omega}(t) = A\omega\sin(\omega t)\,\mathbf{j}$, where A and ω are known positive constants, as shown in the figure. The angle that the vector $\overrightarrow{CP}$ makes with the Ox axis is denoted φ.

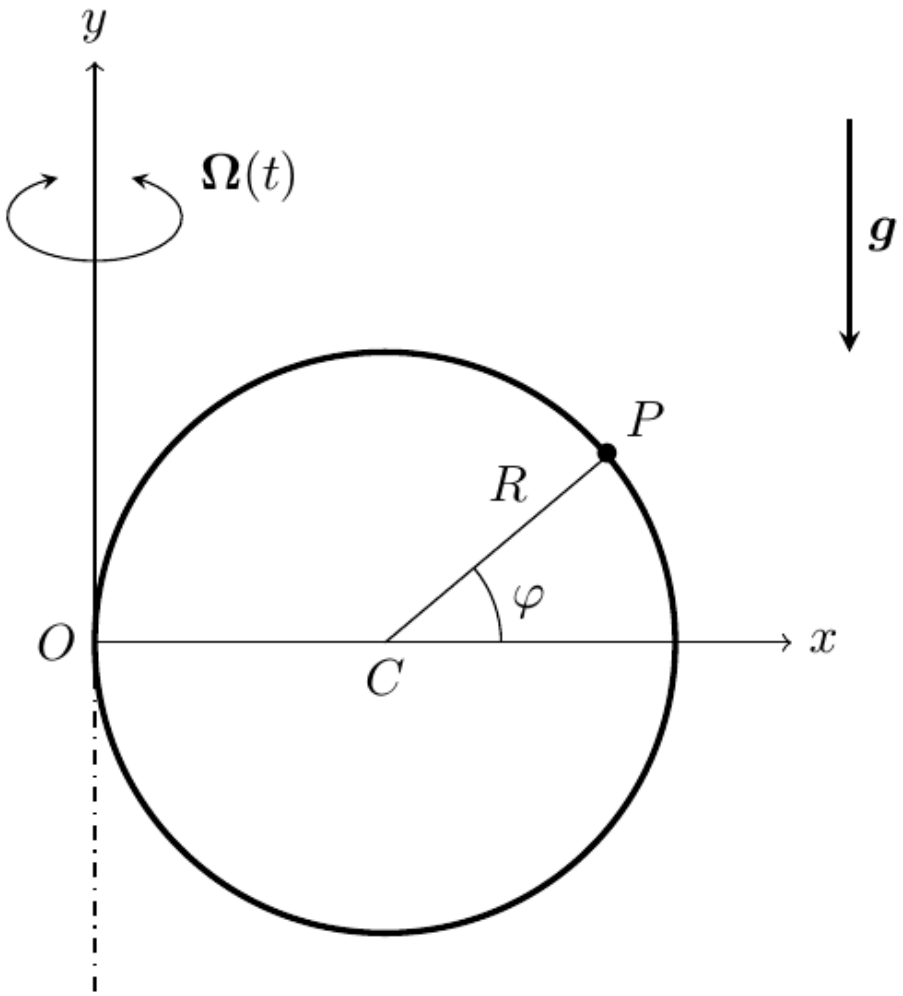


Figure: Circular wire of centre C and radius R in the vertical plane Oxy, passing through O, with C on the Ox axis; P on the wire at angle φ from Ox; the plane rotates about the vertical axis Oy with Ω(t); gravity g downward (Problems.pdf, p. 22).

Answer the following questions in a reasoned way:

(a) Write down the position, velocity and acceleration vectors of P relative to the reference frame $Oxyz$, as a function of φ and its derivatives.

(b) Identify the forces that act on P, and write down their mathematical expression for a generic instant of time t.

(c) Is the mechanical energy of P conserved as seen from the Oxy reference frame? Why?

(d) Write down a single second order differential equation to calculate φ as a function of time. Note that you do not need to solve this equation.

(e) Compute the reaction of the wire on the particle P as a function of φ, $\dot{\varphi}$ and t.

(f) Assume now that $A^2 \ll g/(R\omega^2)$ and that the motion of P is such that $\varphi = \delta - \pi/2$ with $\delta \ll 1$. Around what value of δ does the particle oscillate? Are there any values of ω for which the approximation $\delta \ll 1$ will not hold?

Worked Solution

Reference Frames, Bases and Coordinates

- **System:** a heavy bead P (mass m) slides on a smooth circular wire of radius R whose centre C lies on the Ox axis at distance R from O (the circle passes through O). The vertical plane Oxy of the wire swings about the vertical axis Oy with the prescribed angular velocity $\mathbf{\Omega}(t) = A\omega\sin(\omega t)\,\mathbf{j}$.
- **Why the frame $Oxyz$.** The wire is fixed in $Oxyz$, so P is described by the single angle φ and its relative kinematics is that of circular motion. Because Ω depends on time, the inertia forces now include the **Euler force** $-m\boldsymbol{\alpha}_{10}\times\mathbf{r}_1^P$ (Slide 28), which has no potential (Slide 31), and the centrifugal potential depends explicitly on time; the energy discussion of part (c) depends on both facts.
- **Hypotheses:**
  1. A frame S_0 with origin O on the fixed vertical axis is inertial; Oy is vertical upward and gravity is $-g\,\mathbf{j}$.
  2. The wire is smooth and bilateral: the reaction is perpendicular to the tangent of the wire, with a radial component in the plane of the wire and a component along $\mathbf{k}_1$, normal

to the plane.

3. The rotation of the plane is prescribed (imposed by an external drive whatever the motion of P).

- **Fixed (inertial) frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, basis $\mathcal{B}_0$, with $\mathbf{j}_0$ along the vertical rotation axis Oy.
- **Moving frame:** $S_1 = Oxyz = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, basis $\mathcal{B}_1$, attached to the plane of the wire ($\mathbf{i}_1$ along Ox, $\mathbf{j}_1$ along Oy upward, $\mathbf{k}_1 = \mathbf{i}_1 \times \mathbf{j}_1$ normal to the plane), with $\mathbf{j}_1 = \mathbf{j}_0$ (the rotation axis) and S_1 coinciding with S_0 at $t = 0$.
- **Polar basis of the wire** $\{\mathbf{e}_r, \mathbf{e}_\varphi, \mathbf{k}_1\}$ at P, attached to the centre C; deduced in the next section.
- **Degrees of freedom:** one, the angle φ of $\overrightarrow{CP}$ measured from Ox (3 coordinates minus 2 constraints: $z = 0$ and the circle of centre C).

Symbol	Meaning	SI unit
m	mass of P	kg
R	radius of the wire ($OC = R$)	m
A	amplitude parameter of $\Omega(t)$	1 (rad)
ω	frequency of the oscillation of the plane	rad/s
g	gravitational acceleration	m/s^2
φ	angle of $\overrightarrow{CP}$ from Ox	rad

Change of Basis and Position Vector

Angle turned by the plane

With $\psi(t)$ the angle of rotation about $\mathbf{j}_0$, $\dot{\psi} = \Omega(t)$ and $\psi(0) = 0$. Integrating with the substitution $u = \omega\tau$, $du = \omega\, d\tau$ (so $A\omega \sin(\omega\tau)\, d\tau = A \sin u\, du$, and the limits $\tau = 0, t$ become $u = 0, \omega t$), and Barrow's rule:

$$\psi(t) = \int_0^t A\omega \sin(\omega\tau)\, d\tau = A \int_0^{\omega t} \sin u\, du = A\,[-\cos u]_0^{\omega t} = A\,(-\cos\omega t + \cos 0) = A\,(1 - \cos\omega t)$$

Since $0 \le 1 - \cos \omega t \le 2$, the plane swings between $\psi = 0$ and $\psi = 2A$: it oscillates, it does not spin.

The basis $\mathcal{B}_1$ seen from above

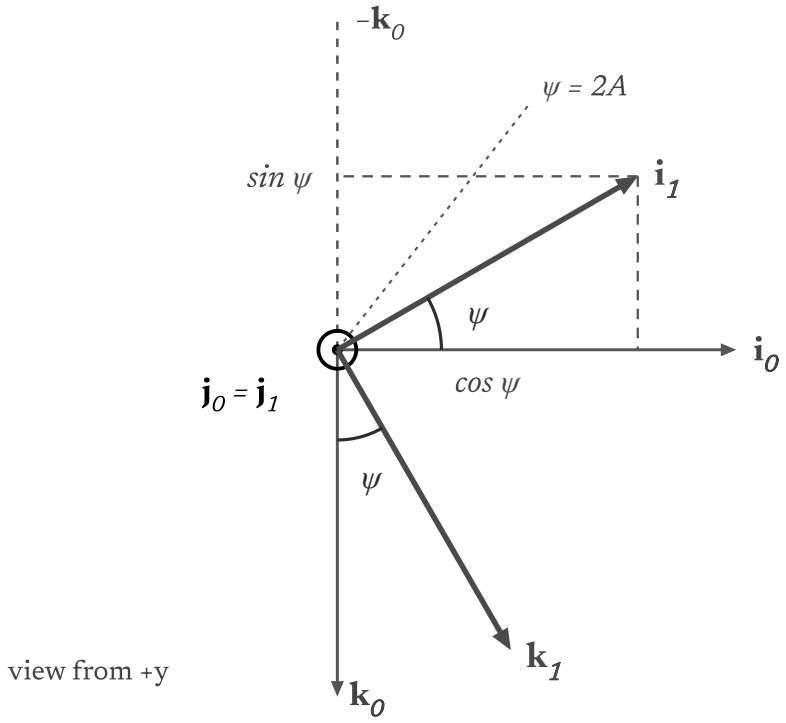


Figure 1. View from $+y$, with $\mathbf{j}_0 = \mathbf{j}_1$ out of the paper ($\odot$): then $\mathbf{k}_0 = \mathbf{i}_0 \times \mathbf{j}_0$ points down the page. $\mathbf{i}_1, \mathbf{k}_1$ are $\mathbf{i}_0, \mathbf{k}_0$ turned counter-clockwise by $\psi(t) = A(1 - \cos \omega t)$; the legs of $\mathbf{i}_1$ are $\cos \psi$ along $\mathbf{i}_0$ and $\sin \psi$ along $-\mathbf{k}_0$ (up the page), hence $\mathbf{i}_1 = \cos \psi \, \mathbf{i}_0 - \sin \psi \, \mathbf{k}_0$. The plane swings between $\psi = 0$ and $\psi = 2A$ (dotted).

The rotation is about $\mathbf{j}_0$, so it is drawn looking down the vertical axis, from $+y$. In that view $\mathbf{j}_0$ points out of the paper, towards the reader ($\odot$). Take $\mathbf{i}_0$ to the right. By the right-hand rule $\mathbf{k}_0 = \mathbf{i}_0 \times \mathbf{j}_0$: the fingers go from $\mathbf{i}_0$ (right) to $\mathbf{j}_0$ (towards the reader) and the thumb points down the page. So $\mathbf{k}_0$ points down and $-\mathbf{k}_0$ up. A positive rotation ψ about $\mathbf{j}_0$ (thumb towards the reader) is counter-clockwise in this view:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of the dashed right triangle. Its leg adjacent to ψ lies along $\mathbf{i}_0$ and measures $\cos \psi$; its leg opposite to ψ points up the page, along $-\mathbf{k}_0$, and measures $\sin \psi$. So $\mathbf{i}_1 = \cos \psi \, \mathbf{i}_0 - \sin \psi \, \mathbf{k}_0$; **the minus sign comes from $\mathbf{k}_0$ pointing down the page in this view.**
- $\mathbf{k}_1$ starts on $\mathbf{k}_0$ (down the page) and turns ψ counter-clockwise, so its tip moves towards $+\mathbf{i}_0$: its leg along $\mathbf{k}_0$ (adjacent) is $\cos \psi$ and its leg along $\mathbf{i}_0$ (opposite) is $+ \sin \psi$. So $\mathbf{k}_1 = \sin \psi \, \mathbf{i}_0 + \cos \psi \, \mathbf{k}_0$.
- $\mathbf{j}_1 = \mathbf{j}_0$, the axis of the rotation.

This is the simple rotation of angle ψ about $\mathbf{j}_0$ (Notes Eqs. 1.18–1.20); the columns of $[{}_0R_1]$ are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\psi & 0 & \sin\psi \\ 0 & 1 & 0 \\ -\sin\psi & 0 & \cos\psi \end{pmatrix}, \qquad \det[{}_0R_1] = \cos\psi\cdot\cos\psi - 0 + \sin\psi\cdot\sin\psi = 1$$

(expansion along the first row: $\cos\psi\,(1\cdot\cos\psi - 0) - 0 + \sin\psi\,(0 - 1\cdot(-\sin\psi))$). The columns are unit vectors and pairwise orthogonal ($\cos\psi\sin\psi - \sin\psi\cos\psi = 0$), so $[{}_0R_1][{}_0R_1]^T = I$. Poisson check, with the chain rule $\frac{d}{dt}\cos\psi = -\sin\psi\,\dot\psi$ and $\frac{d}{dt}\sin\psi = \cos\psi\,\dot\psi$: $\left.\dfrac{d\mathbf{i}_1}{dt}\right|_0 = \dot\psi\,(-\sin\psi\,\mathbf{i}_0 - \cos\psi\,\mathbf{k}_0) = -\dot\psi\,\mathbf{k}_1$, and indeed $\Omega\,\mathbf{j}_1\times\mathbf{i}_1 = -\Omega\,\mathbf{k}_1$.

- **Motion of S_1:** origin fixed on the axis, $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$;

$$\boldsymbol{\omega}_{10} = \Omega(t)\,\mathbf{j}_1 = A\omega\sin(\omega t)\,\mathbf{j}_1, \qquad \boldsymbol{\alpha}_{10} = \left.\frac{d\boldsymbol{\omega}_{10}}{dt}\right|_0 = \dot\Omega\,\mathbf{j}_1 = A\omega^2\cos(\omega t)\,\mathbf{j}_1$$

($\mathbf{j}_1 = \mathbf{j}_0$ is fixed, and the chain rule gives $\frac{d}{dt}\sin\omega t = \omega\cos\omega t$.)

The position vector in the plane of the wire

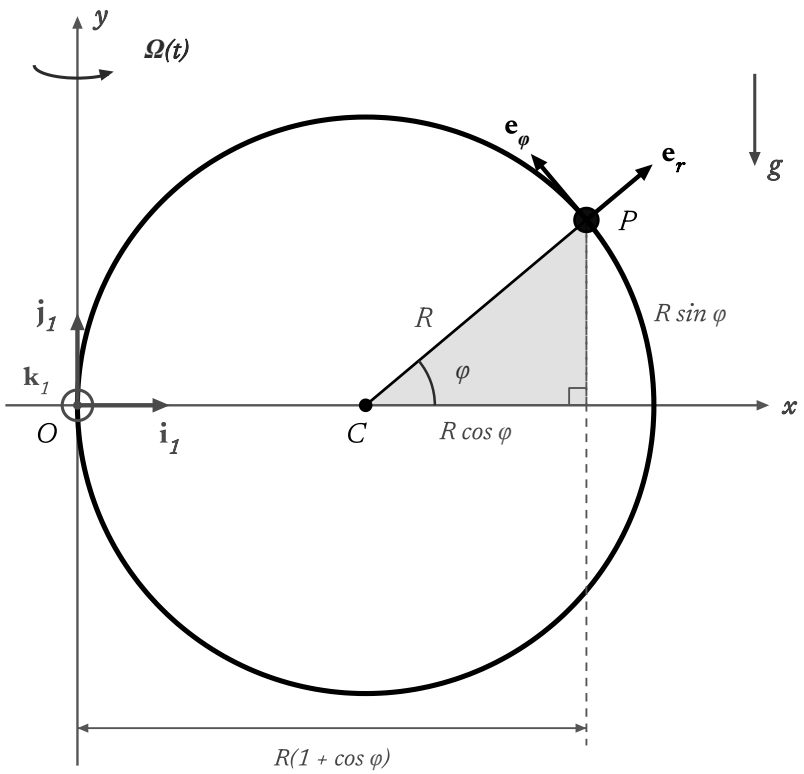


Figure 2. Plane Oxy, fixed in S_1 ($\mathbf{k}_1 = \mathbf{i}_1 \times \mathbf{j}_1$ points out of the paper, $\odot$). $OC = R\,\mathbf{i}_1$; in the shaded right triangle CFP the hypotenuse is $|CP| = R$, the leg adjacent to φ is $R\cos\varphi$ (along $\mathbf{i}_1$) and the opposite leg is $R\sin\varphi$ (along $\mathbf{j}_1$). The distance from P to the rotation axis Oy is $R(1+\cos\varphi)$. $\mathbf{i}_1, \mathbf{j}_1$ in blue, $\mathbf{e}_r, \mathbf{e}_\varphi$ in orange.

The plane Oxy is fixed in S_1 and is drawn with $\mathbf{i}_1$ to the right and $\mathbf{j}_1$ up, so $\mathbf{k}_1 = \mathbf{i}_1 \times \mathbf{j}_1$ points out of the paper ($\odot$). To go from O to P we pass through the centre, because the distance from C is the one that stays constant:

$$\mathbf{r}_1^P = \overrightarrow{OC} + \overrightarrow{CP}$$

- $\overrightarrow{OC} = R\,\mathbf{i}_1$: the circle passes through O and its centre lies on Ox at distance R.
- $\overrightarrow{CP}$ has length R and makes the angle φ with Ox. Let F be the foot of the perpendicular from P to the x axis. In the shaded right triangle CFP (right angle at F) the hypotenuse is $|CP| = R$, the leg adjacent to φ is $CF = R\cos\varphi$, along $\mathbf{i}_1$, and the leg opposite to φ is $FP = R\sin\varphi$, along $\mathbf{j}_1$. So $\overrightarrow{CP} = R\cos\varphi\,\mathbf{i}_1 + R\sin\varphi\,\mathbf{j}_1$.

Adding the two vectors:

$$\mathbf{r}_1^P = R\,(1+\cos\varphi)\,\mathbf{i}_1 + R\sin\varphi\,\mathbf{j}_1$$

Checks: $\varphi=\pi$ gives $\mathbf{r}_1^P=\mathbf{0}$ (the point O of the wire); $\varphi=-\pi/2$ gives $R\,\mathbf{i}_1 - R\,\mathbf{j}_1$, the lowest point of the wire. The $\mathbf{i}_1$ component $x = R(1+\cos\varphi)$ is the distance from P to the rotation axis Oy (the dimension at the bottom of the figure); it is the lever arm of the centrifugal and Euler forces.

Polar angle. Here φ is measured at C from the direction $+\mathbf{i}_1$, counter-clockwise towards $+\mathbf{j}_1$, so it is already the standard polar angle about C; no shift of origin of the angle is needed (unlike an angle measured from a vertical).

Polar basis. The radial unit vector is $\overrightarrow{CP}/R$, and the transverse one is its derivative with respect to φ ($\frac{d}{d\varphi}\cos\varphi = -\sin\varphi$, $\frac{d}{d\varphi}\sin\varphi = \cos\varphi$):

$$\mathbf{e}_r = \cos\varphi\,\mathbf{i}_1 + \sin\varphi\,\mathbf{j}_1, \qquad \mathbf{e}_\varphi = \frac{d\mathbf{e}_r}{d\varphi} = -\sin\varphi\,\mathbf{i}_1 + \cos\varphi\,\mathbf{j}_1, \qquad \frac{d\mathbf{e}_\varphi}{d\varphi} = -\cos\varphi\,\mathbf{i}_1 - \sin\varphi\,\mathbf{j}_1 = -\mathbf{e}_r$$

$\mathbf{e}_\varphi$ has length $\sqrt{\sin^2\varphi+\cos^2\varphi}=1$ and is the tangent to the wire in the direction of increasing φ. With $\mathbf{i}_1\times\mathbf{j}_1=\mathbf{k}_1$ and $\mathbf{j}_1\times\mathbf{i}_1=-\mathbf{k}_1$, $\mathbf{e}_r\times\mathbf{e}_\varphi = \cos^2\varphi\,\mathbf{k}_1 - \sin^2\varphi\,(-\mathbf{k}_1) = (\cos^2\varphi+\sin^2\varphi)\,\mathbf{k}_1 = \mathbf{k}_1$. Its matrix in $\mathcal{B}_1$ (columns $\mathbf{e}_r, \mathbf{e}_\varphi, \mathbf{k}_1$) is a rotation by φ about $\mathbf{k}_1$, with determinant $\cos^2\varphi+\sin^2\varphi = 1$. Useful projections: $\mathbf{i}_1\cdot\mathbf{e}_r = \cos\varphi$, $\mathbf{i}_1\cdot\mathbf{e}_\varphi = -\sin\varphi$, $\mathbf{j}_1\cdot\mathbf{e}_r = \sin\varphi$, $\mathbf{j}_1\cdot\mathbf{e}_\varphi = \cos\varphi$.

Kinematics

(a) Relative position, velocity and acceleration

> **Pedagogical Justification:** Relative quantities are derivatives in S_1, where $\mathbf{i}_1, \mathbf{j}_1$ are constant; only φ depends on time, so each component is differentiated with the chain rule.

With $\frac{d}{dt}\cos\varphi = -\sin\varphi\,\dot\varphi$ and $\frac{d}{dt}\sin\varphi = \cos\varphi\,\dot\varphi$ applied to $\mathbf{r}_1^P = R\,(1+\cos\varphi)\,\mathbf{i}_1 + R\sin\varphi\,\mathbf{j}_1$:

$$\mathbf{v}_1^P = -R\dot\varphi\sin\varphi\,\mathbf{i}_1 + R\dot\varphi\cos\varphi\,\mathbf{j}_1 = R\dot\varphi\,\mathbf{e}_\varphi$$

Product rule on each component, e.g. $\frac{d}{dt}\left(-R\dot\varphi\sin\varphi\right) = -R\ddot\varphi\sin\varphi - R\dot\varphi^2\cos\varphi$ and $\frac{d}{dt}\left(R\dot\varphi\cos\varphi\right) = R\ddot\varphi\cos\varphi - R\dot\varphi^2\sin\varphi$:

$$\mathbf{a}_1^P = -R\left(\ddot{\varphi}\sin\varphi + \dot{\varphi}^2\cos\varphi\right)\mathbf{i}_1 + R\left(\ddot{\varphi}\cos\varphi - \dot{\varphi}^2\sin\varphi\right)\mathbf{j}_1 = R\ddot{\varphi}\,\mathbf{e}_\varphi - R\dot{\varphi}^2\,\mathbf{e}_r$$

Forces and Equations of Motion

(b) Forces acting on P

> **Pedagogical Justification:** In the non-inertial frame $Oxyz$ we use $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eqs. 4.62–4.63). The actual forces are the weight and the reaction of the wire; the inertia forces follow from Slide 28 with $\mathbf{a}_0^O = \mathbf{0}, \boldsymbol{\omega}_{10} = \Omega\mathbf{j}_1, \boldsymbol{\alpha}_{10} = \dot{\Omega}\mathbf{j}_1$.

Actual forces:

- Weight: $\mathbf{W} = -mg\,\mathbf{j}_1$.
- Reaction of the smooth wire (no component along the tangent $\mathbf{e}_\varphi$): $\mathbf{N} = N_1\,\mathbf{e}_r + N_2\,\mathbf{k}_1 = N_1\cos\varphi\,\mathbf{i}_1 + N_1\sin\varphi\,\mathbf{j}_1 + N_2\,\mathbf{k}_1$.

Inertia forces (cross products with $\mathbf{j}_1\times\mathbf{i}_1 = -\mathbf{k}_1, \mathbf{j}_1\times\mathbf{j}_1 = \mathbf{0}, \mathbf{j}_1\times\mathbf{k}_1 = \mathbf{i}_1$):

- Origin: $\mathbf{F}_A = -m\mathbf{a}_0^O = \mathbf{0}$.
- Euler: $\boldsymbol{\alpha}_{10}\times\mathbf{r}_1^P = \dot{\Omega}\,\mathbf{j}_1\times[R(1+\cos\varphi)\mathbf{i}_1 + R\sin\varphi\,\mathbf{j}_1] = -\dot{\Omega}R(1+\cos\varphi)\,\mathbf{k}_1$, so

$$\mathbf{F}_E = -m\boldsymbol{\alpha}_{10}\times\mathbf{r}_1^P = mA\omega^2 R\cos(\omega t)\,(1+\cos\varphi)\,\mathbf{k}_1$$

- Centrifugal: $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = -\Omega R(1+\cos\varphi)\,\mathbf{k}_1$ and $\boldsymbol{\omega}_{10}\times(-\Omega R(1+\cos\varphi)\mathbf{k}_1) = -\Omega^2 R(1+\cos\varphi)\,\mathbf{i}_1$, so (Slide 30, distance to the axis $x = R(1+\cos\varphi)$)

$$\mathbf{F}_C = m\Omega^2 R\,(1+\cos\varphi)\,\mathbf{i}_1 = mA^2\omega^2 R\sin^2(\omega t)\,(1+\cos\varphi)\,\mathbf{i}_1$$

- Coriolis: $\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P = \Omega R\dot{\varphi}\,\mathbf{j}_1\times(-\sin\varphi\,\mathbf{i}_1 + \cos\varphi\,\mathbf{j}_1) = \Omega R\dot{\varphi}\sin\varphi\,\mathbf{k}_1$, so

$$\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P = -2mA\omega R\dot{\varphi}\sin(\omega t)\sin\varphi\,\mathbf{k}_1$$

(d) Equation of motion

> **Pedagogical Justification:** The tangential direction $\mathbf{e}_\varphi$ is the only one free of the unknown reaction ($\mathbf{N} \perp \mathbf{e}_\varphi$), and the Euler and Coriolis forces (along $\mathbf{k}_1$) also drop out.

Projections on $\mathbf{e}_\varphi = -\sin\varphi\,\mathbf{i}_1 + \cos\varphi\,\mathbf{j}_1$: $m\mathbf{a}_1^P \cdot \mathbf{e}_\varphi = mR\ddot{\varphi}$; $\mathbf{W}\cdot\mathbf{e}_\varphi = -mg\cos\varphi$; $\mathbf{F}_C \cdot \mathbf{e}_\varphi = -m\Omega^2 R(1+\cos\varphi)\sin\varphi$. Hence $mR\ddot{\varphi} = -mg\cos\varphi - m\Omega^2 R(1+\cos\varphi)\sin\varphi$, i.e.

$$\boxed{\ddot{\varphi} + \frac{g}{R}\cos\varphi + A^2\omega^2\sin^2(\omega t)\,(1+\cos\varphi)\sin\varphi = 0}$$

a nonlinear equation with a time-periodic coefficient (parametric forcing). It agrees with the official key.

(e) Reaction of the wire

> **Pedagogical Justification:** The remaining two components of $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$, along $\mathbf{e}_r$ and $\mathbf{k}_1$, each contain one reaction component.

Along $\mathbf{e}_r$: $m\mathbf{a}_1^P \cdot \mathbf{e}_r = -mR\dot{\varphi}^2$; $\mathbf{W}\cdot\mathbf{e}_r = -mg\sin\varphi$; $\mathbf{F}_C\cdot\mathbf{e}_r = m\Omega^2 R(1+\cos\varphi)\cos\varphi$. So $-mR\dot{\varphi}^2 = N_1 - mg\sin\varphi + m\Omega^2 R(1+\cos\varphi)\cos\varphi$:

$$\boxed{N_1 = mg\sin\varphi - mA^2R\omega^2\,(1+\cos\varphi)\sin^2(\omega t)\cos\varphi - mR\dot{\varphi}^2}$$

Along $\mathbf{k}_1$: $m\mathbf{a}_1^P\cdot\mathbf{k}_1 = 0 = N_2 + mA\omega^2 R\cos(\omega t)(1+\cos\varphi) - 2mA\omega R\dot{\varphi}\sin(\omega t)\sin\varphi$:

$$\boxed{N_2 = 2mAR\omega\,\dot{\varphi}\sin\varphi\sin(\omega t) - mAR\omega^2\,(1+\cos\varphi)\cos(\omega t)}$$

and $\mathbf{N} = N_1\cos\varphi\,\mathbf{i}_1 + N_1\sin\varphi\,\mathbf{j}_1 + N_2\,\mathbf{k}_1$. Both components agree with the official key. The normal component N_2 is what the wire must supply to drag P along with the swinging plane (it balances the Euler and Coriolis forces).

Answers and Checks

(a) Relative position, velocity and acceleration

$$\mathbf{r}_1^P = R\,(1+\cos\varphi)\,\mathbf{i}_1 + R\sin\varphi\,\mathbf{j}_1, \qquad \mathbf{v}_1^P = R\dot\varphi\,\mathbf{e}_\varphi, \qquad \mathbf{a}_1^P = R\ddot\varphi\,\mathbf{e}_\varphi - R\dot\varphi^2\,\mathbf{e}_r$$

(b) Forces acting on P

Weight $-mg\,\mathbf{j}_1$, reaction $N_1\,\mathbf{e}_r + N_2\,\mathbf{k}_1$, and the inertia forces $\mathbf{F}_A = \mathbf{0}$, $\mathbf{F}_E = mA\omega^2 R\cos(\omega t)(1+\cos\varphi)\,\mathbf{k}_1$, $\mathbf{F}_C = mA^2\omega^2 R\sin^2(\omega t)(1+\cos\varphi)\,\mathbf{i}_1$ and $\mathbf{F}_{COR} = -2mA\omega R\dot\varphi\sin(\omega t)\sin\varphi\,\mathbf{k}_1$.

(c) Is the mechanical energy conserved in Oxy?

> **Pedagogical Justification:** By Slide 29, E_1^P is conserved only if every force that does work in S_1 derives from a potential that does not depend explicitly on time. Compute the power of each force in S_1.

- $\mathbf{N}\cdot\mathbf{v}_1^P = R\dot\varphi\,(N_1\mathbf{e}_r + N_2\mathbf{k}_1)\cdot\mathbf{e}_\varphi = 0$ (the wire is fixed in S_1).
- $\mathbf{F}_E$ and $\mathbf{F}_{COR}$ are along $\mathbf{k}_1$, normal to the plane in which $\mathbf{v}_1^P$ lies: they do no work. (In general the Euler force has no potential, Slide 31; here it happens to be powerless.)
- The weight has the potential $mgy = mgR\sin\varphi$.
- The centrifugal force has the potential $V_C = -\tfrac{1}{2}m\Omega(t)^2x^2$, but **it depends explicitly on time** because Ω does.

Define $E_1^P = \tfrac{1}{2}mR^2\dot\varphi^2 + mgR\sin\varphi - \tfrac{1}{2}m\Omega^2R^2(1+\cos\varphi)^2$. Differentiating with the chain and product rules,

$$\frac{dE_1^P}{dt} = mR^2\dot\varphi\ddot\varphi + mgR\cos\varphi\,\dot\varphi + m\Omega^2R^2(1+\cos\varphi)\sin\varphi\,\dot\varphi - m\Omega\dot\Omega R^2(1+\cos\varphi)^2$$

The first three terms are $mR^2\dot\varphi$ times the equation of motion of part (d), so they cancel, leaving

$$\frac{dE_1^P}{dt} = -m\Omega\dot\Omega\,R^2(1+\cos\varphi)^2 = -mA^2\omega^3R^2\sin(\omega t)\cos(\omega t)\,(1+\cos\varphi)^2 \neq 0$$

No: the mechanical energy of P in Oxy is not conserved, because the centrifugal potential changes with time (the frame does not rotate at a constant rate). It would be conserved if Ω were constant. Nor is the energy conserved in S_0: there the wire moves and its reaction $N_2\mathbf{k}_1$

does work.

(d) Equation of motion

$$\ddot{\varphi} + \frac{g}{R}\cos\varphi + A^2\omega^2 \sin^2(\omega t)\,(1+\cos\varphi)\sin\varphi = 0$$

(e) Reaction of the wire

$\mathbf{N} = N_1\,\mathbf{e}_r + N_2\,\mathbf{k}_1$ with the boxed $N_1(\varphi,\dot\varphi,t)$ and $N_2(\varphi,\dot\varphi,t)$ of the previous section.

(f) Small oscillations about the lowest point

> **Pedagogical Justification:** For $\delta \ll 1$ the trigonometric functions are replaced by their first-order Taylor expansions; the condition $A^2 \ll g/(R\omega^2)$, i.e. $A^2\omega^2 \ll g/R$, lets us drop the parametric term $A^2\omega^2\sin^2(\omega t)\,\delta$ against $(g/R)\,\delta$. The result is a linear forced oscillator.

With $\varphi = \delta - \pi/2$, the subtraction formulas give $\cos\varphi = \cos\delta\cos\frac{\pi}{2} + \sin\delta\sin\frac{\pi}{2} = \sin\delta \approx \delta$ and $\sin\varphi = \sin\delta\cos\frac{\pi}{2} - \cos\delta\sin\frac{\pi}{2} = -\cos\delta \approx -1$; also $\ddot\varphi = \ddot\delta$. Substituting into the equation of part (d) and keeping first-order terms:

$$\ddot\delta + \frac{g}{R}\,\delta - A^2\omega^2\sin^2(\omega t)\,(1+\delta) = 0$$

Dropping $A^2\omega^2\sin^2(\omega t)\,\delta$ (small compared with $(g/R)\,\delta$) and using $\sin^2(\omega t) = \frac{1}{2}\,[1-\cos(2\omega t)]$:

$$\boxed{\ddot\delta + \frac{g}{R}\,\delta = \frac{A^2\omega^2}{2}\,[1-\cos(2\omega t)]}$$

Solution. With $\omega_n = \sqrt{g/R}$, a particular solution is a constant plus a term at 2ω. The constant δ_0 satisfies $\omega_n^2\delta_0 = A^2\omega^2/2$. For $\delta_p = C\cos(2\omega t)$, $\ddot\delta_p = -4\omega^2 C\cos(2\omega t)$ and $(\omega_n^2 - 4\omega^2)\,C = -A^2\omega^2/2$. Hence

$$\delta(t) = \frac{A^2 R\omega^2}{2g} - \frac{A^2\omega^2}{2\,(\omega_n^2 - 4\omega^2)}\cos(2\omega t) + a\cos\omega_n t + b\sin\omega_n t$$

(a, b set by the initial conditions). The particle oscillates, with natural frequency $\omega_n = \sqrt{g/R}$, about

$$\boxed{\delta_0 = \frac{A^2 R\omega^2}{2g}}$$

which is small, consistent with $A^2 \ll g/(R\omega^2)$. It is positive: on average the bead is displaced slightly outward, away from the axis, by the mean centrifugal force.

Breakdown. The forced amplitude $|C| = \dfrac{A^2\omega^2}{2\,|\omega_n^2 - 4\omega^2|}$ grows without bound when $2\omega \to \omega_n$, i.e. for

$$\boxed{\omega \approx \frac{1}{2}\sqrt{\frac{g}{R}}}$$

At exact resonance the particular solution is $\delta_p = -\dfrac{A^2\omega^2}{4\omega_n}\, t\sin(\omega_n t)$ (check: $\ddot\delta_p + \omega_n^2\delta_p = -\frac{A^2\omega^2}{2}\cos\omega_n t$), whose amplitude grows linearly in time. The assumption $\delta \ll 1$ then fails, as stated in the key.

Checks

- **Role of each inertia force.** The centrifugal force (along $\mathbf{i}_1$, in the plane) is the only inertia force that enters the equation of motion. The Euler and Coriolis forces are normal to the plane: they do not change φ, but they load the wire through N_2.
- **Limits.** $A \to 0$: $\ddot\varphi + (g/R)\cos\varphi = 0$, the simple pendulum about $\varphi = -\pi/2$ in a fixed plane, and $N_2 = 0$. $\omega \to 0$ with A fixed: $\Omega \to 0$ and the same pendulum is recovered.
- **Energy.** dE_1^P/dt averages to zero over a period of the forcing ($\sin\omega t\cos\omega t$ has zero mean) but is not zero instant by instant; near resonance the drive pumps energy into the oscillation.
- **Orthogonality:** $\mathbf{e}_r \cdot \mathbf{e}_\varphi = -\cos\varphi\sin\varphi + \sin\varphi\cos\varphi = 0$, $\mathbf{e}_r \times \mathbf{e}_\varphi = \mathbf{k}_1$ and $\det[{}_0R_1] = 1$ (change-of-basis section).
- **Inertial check.** Differentiating $[{}_0R_1]\mathbf{r}_1^P$ twice reproduces $\mathbf{a}_1^P + \boldsymbol{\alpha}_{10}\times\mathbf{r}_1^P + \boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P) + 2\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P$ (Notes Eq. 3.27), and $m\mathbf{a}_0^P = \mathbf{W} + \mathbf{N}$ holds with the reaction of part (e) (checked symbolically with sympy).
- **Dimensions.** $[R(1+\cos\varphi)] = [R\sin\varphi] = \mathrm{m}$; $[\psi] = [A] = 1$; $[g/R] = [A^2\omega^2] = \mathrm{s}^{-2}$ (same units as $\ddot\varphi$); $[mA^2R\omega^2] = [mR\dot\varphi^2] = [mg] = \mathrm{N}$; $[mAR\omega\dot\varphi] = \mathrm{kg\cdot m\cdot s^{-2}} = \mathrm{N}$; $[A^2R\omega^2/g] = \dfrac{\mathrm{m\cdot s^{-2}}}{\mathrm{m\cdot s^{-2}}} = 1$, as an angle must be.

- **Check against the official key:** the key of `Problems.pdf` gives the equation of motion $\ddot{\varphi} + (g/R)\cos\varphi + A^2\omega^2(1+\cos\varphi)\sin^2(\omega t)\sin\varphi = 0$, the reaction components N_1 and N_2 and the small-oscillation equation; all coincide with (d), (e) and (f) (no discrepancy).

Source: Problems.pdf (PDF page 22); key on PDF page 75

PROBLEM 36

Pendulum String Dragged by a Rotating Bar

Original statement · Problems.pdf, p. 24

Rotating Frame + String Constraint

A heavy point particle P with mass m hanging from a fixed point A on the ceiling with a massless string of constant length L. The particle is initially at rest. A bar b rotates about its center O in a horizontal plane at a distance h from the ceiling, with a known, constant angular velocity ω. The horizontal distance between O and A is R. At time $t = 0$, the bar collides with the string and drags it along its motion. The string slides over the bar without friction. We call C the instantaneous point of contact of the string with the bar. We define a reference frame S_1 that rotates with the bar as shown in the figure. We call ξ the distance between O and C; ℓ the length of the string segment between C and P; θ the angle between the vertical plane Ox_1z_1 and the plane defined by the bar and P; and φ the angle formed between the string segment CP and the bar. You are asked to study only the part of the motion in which the string remains tense and in contact with the bar.

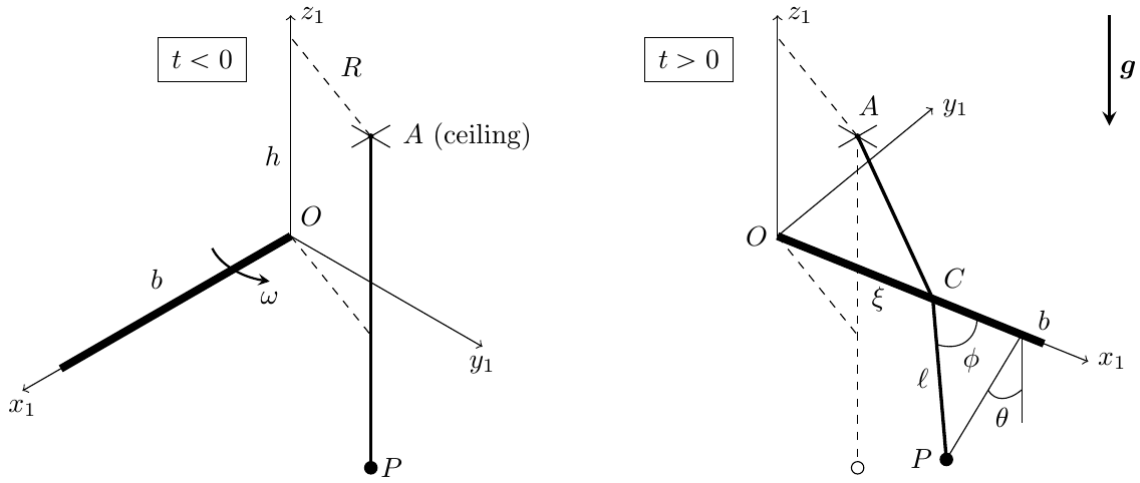


Figure: Left, before the impact (t negative): bar b rotating with ω about O in the horizontal plane x1y1, ceiling point A at height h above that plane and horizontal distance R from O, P hanging below A. Right, after the impact (t positive): frame S1 = Ox1y1z1 attached to the bar, contact point C at distance ξ from O, string A–C–P with segment CP of length ℓ at angle φ from the bar, θ the angle of the plane bar–P from the vertical; gravity g downward (Problems.pdf, p. 24).

(a) Write down the position and velocity of point A in S_1 as a function of time.

(b) Obtain a general expression for the position, velocity, and acceleration of P in S_1 as a function of $\xi, \ell, \theta, \varphi$, their derivatives, and time.

(c) Express AC, ℓ, and φ as a function of ξ and time.

(d) Identify and express all the forces acting on P.

(e) Write down a set of second order differential equations for the evolution of the unknowns of the problem.

(f) Compute the initial conditions for the previous equations at $t = 0$.

(g) Is the mechanical energy of the particle conserved in time? Justify your answer.

(h) If/when the string separates from the bar, express the geometric condition as a relation between θ and and [sic] time.

Worked Solution

Reference Frames, Bases and Coordinates

- **System:** a heavy particle P (mass m) hangs from the fixed ceiling point A by a string of length L. A bar turning at constant rate ω about the vertical axis through its centre O, in a horizontal plane a distance h below the ceiling, hits the string at $t = 0$ and drags it. The string then runs $A \to C \to P$, bent over the bar at the contact point C.
- **Why the frame S_1 attached to the bar.** In S_1 the bar is the fixed axis Ox_1, so the contact point is simply $C = (\xi, 0, 0)$ and the string geometry is described by ξ, ℓ, φ, θ. The price is that A moves in S_1 and that the centrifugal and Coriolis forces (Slides 27–28) have to be added. Since $\boldsymbol{\omega}_{10}$ is constant and O is fixed, the Euler force and the force of the origin's acceleration vanish, and the centrifugal force has a potential (Slide 30).
- **Hypotheses:**
  1. A frame S_0 with origin O and vertical axis $\mathbf{k}_0$ (upward) is inertial; gravity is $-g\,\mathbf{k}_0$.
  2. The string is massless and inextensible, and stays tense ($T > 0$) and in contact with the bar (the only stage of the motion studied, as the statement asks).
  3. The bar is thin (a straight line) and long enough for C to stay on it; the contact is frictionless.
  4. The rotation of the bar is prescribed: ω is kept constant whatever the string does.
  5. The bar reaches the string at $t = 0$ while P is at rest below A, so $L > h$.

- **Fixed (inertial) frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, basis $\mathcal{B}_0$, with $\mathbf{k}_0$ vertical upward and $\mathbf{i}_0$ pointing horizontally from O towards the vertical of A, so that $\mathbf{r}_0^A = R\,\mathbf{i}_0 + h\,\mathbf{k}_0$.
- **Moving frame:** $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, basis $\mathcal{B}_1$, attached to the bar: $\mathbf{i}_1$ along the bar and $\mathbf{k}_1 = \mathbf{k}_0$. At $t = 0$ the bar is below A, so S_1 coincides with S_0 at that instant.
- **Sense of rotation.** In the left drawing of the statement ($t < 0$) the vertical projection of A lies at positive x_1 and positive y_1, so the bar reaches it turning from $\mathbf{i}_1$ towards $\mathbf{j}_1$: $\boldsymbol{\omega}_{10} = +\omega\,\mathbf{k}_1$. (The same sign is implied by the Coriolis force of the official key.) The angle turned by the bar is ωt.
- **String basis:** $\mathcal{B}_2 = \{\mathbf{e}_{CP}, \mathbf{e}_\theta, \mathbf{e}_\varphi\}$, with $\mathbf{e}_{CP}$ the unit vector from C to P; it is deduced, with two drawings, in the next section.
- **Degrees of freedom:** P moves in space with the single constraint "string tense", so it has **two** degrees of freedom. We take ξ and θ; ℓ and φ are fixed by ξ and t through the constraints of part (c). The tension T is the additional unknown.

Symbol	Meaning	SI unit
m	mass of P	kg
L	total length of the string	m
h	height of the ceiling above the plane of the bar	m
R	horizontal distance between O and A	m
ω	angular velocity of the bar (constant)	rad/s
g	gravitational acceleration	m/s^2
ξ	distance OC along the bar	m
ℓ	length of the segment CP	m
φ	angle between $\overrightarrow{CP}$ and the bar ($\mathbf{i}_1$)	rad
θ	angle between the plane Ox_1z_1 and the plane bar–P	rad
T	tension of the string	N

Change of Basis and Position Vector

There are two rotations: the bar turns ωt about the vertical $\mathbf{k}_0$ ($\mathcal{B}_0 \to \mathcal{B}_1$), and the plane that contains the bar and P is turned θ about the bar from the vertical plane Ox_1z_1, inside which the segment CP makes the angle φ with the bar ($\mathcal{B}_1 \to \mathcal{B}_2$). Each one is drawn looking along its own axis.

The basis $\mathcal{B}_1$ seen from above

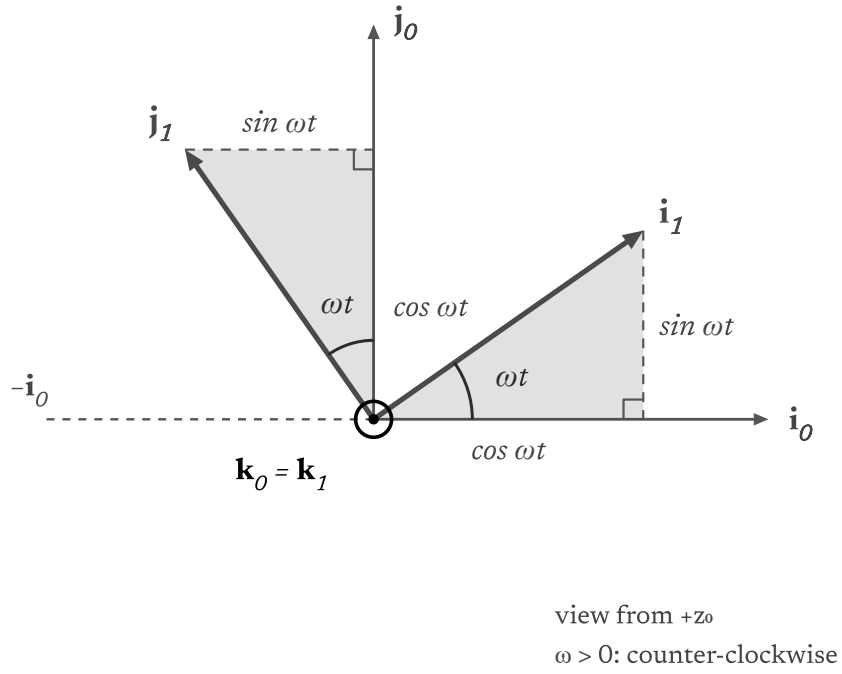


Figure 1. View from $+z_0$, with $\mathbf{k}_0 = \mathbf{k}_1$ out of the paper ($\odot$), so $\mathbf{j}_0 = \mathbf{k}_0 \times \mathbf{i}_0$ points up. $\mathbf{i}_1, \mathbf{j}_1$ are $\mathbf{i}_0, \mathbf{j}_0$ turned counter-clockwise by ωt. $\mathbf{i}_1$ has legs $\cos\omega t$ along $\mathbf{i}_0$ and $\sin\omega t$ along $\mathbf{j}_0$; $\mathbf{j}_1$ has legs $\cos\omega t$ along $\mathbf{j}_0$ and $\sin\omega t$ along $-\mathbf{i}_0$, hence its minus sign. The bar lies along $\mathbf{i}_1$.

The first rotation is about $\mathbf{k}_0$, so we look down the vertical from $+z_0$: $\mathbf{k}_0 = \mathbf{k}_1$ points out of the paper, towards the reader ($\odot$). Take $\mathbf{i}_0$ to the right, towards the point A' of the plane of the bar directly below A. The direction of $\mathbf{j}_0$ follows from the right-hand rule, $\mathbf{j}_0 = \mathbf{k}_0 \times \mathbf{i}_0$: the fingers go from $\mathbf{k}_0$ (towards the reader) to $\mathbf{i}_0$ (right) and the thumb points up the page. With $\boldsymbol{\omega}_{10} = \omega\,\mathbf{k}_0$ and $\omega > 0$, the bar turns counter-clockwise in this view (thumb along $\mathbf{k}_0$, towards the reader), and at time t it has turned the angle ωt from $\mathbf{i}_0$:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of a right triangle whose leg adjacent to ωt lies along $\mathbf{i}_0$ and measures $\cos\omega t$, and whose opposite leg points up the page, along $+\mathbf{j}_0$, and measures $\sin\omega t$. Hence $\mathbf{i}_1 = \cos\omega t\,\mathbf{i}_0 + \sin\omega t\,\mathbf{j}_0$.

- $\mathbf{j}_1$ starts on $\mathbf{j}_0$ (up the page) and turns the same angle counter-clockwise, so its tip leans to the left, towards $-\mathbf{i}_0$. Its leg along $\mathbf{j}_0$ (adjacent) is $\cos\omega t$ and its leg along $-\mathbf{i}_0$ (opposite) is $\sin\omega t$: $\mathbf{j}_1 = -\sin\omega t\,\mathbf{i}_0 + \cos\omega t\,\mathbf{j}_0$. **The minus sign comes from the counter-clockwise turn carrying the upward axis to the left, towards $-\mathbf{i}_0$.**
- $\mathbf{k}_1 = \mathbf{k}_0$, the axis of the rotation.

The columns of the change-of-basis matrix are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos\omega t\cos\omega t - (-\sin\omega t)\sin\omega t = \cos^2\omega t + \sin^2\omega t = 1$$

The product $[{}_0R_1][{}_0R_1]^T$ has diagonal entries $\cos^2\omega t + \sin^2\omega t = 1$ (first two rows) and 1 (third row), and off-diagonal entries $\cos\omega t\sin\omega t - \sin\omega t\cos\omega t = 0$ (first row times second row) and 0 (any row times the third), so $[{}_0R_1][{}_0R_1]^T = I$.

Motion of S_1: $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$ (the centre of the bar is fixed), $\boldsymbol{\omega}_{10} = \omega\,\mathbf{k}_1$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$ (ω constant and $\mathbf{k}_1 = \mathbf{k}_0$ fixed). Check with the chain rule: $\frac{d\mathbf{i}_1}{dt}\big|_0 = -\omega\sin\omega t\,\mathbf{i}_0 + \omega\cos\omega t\,\mathbf{j}_0 = \omega\,\mathbf{j}_1$, and indeed $\boldsymbol{\omega}_{10}\times\mathbf{i}_1 = \omega\,\mathbf{k}_1\times\mathbf{i}_1 = \omega\,\mathbf{j}_1$.

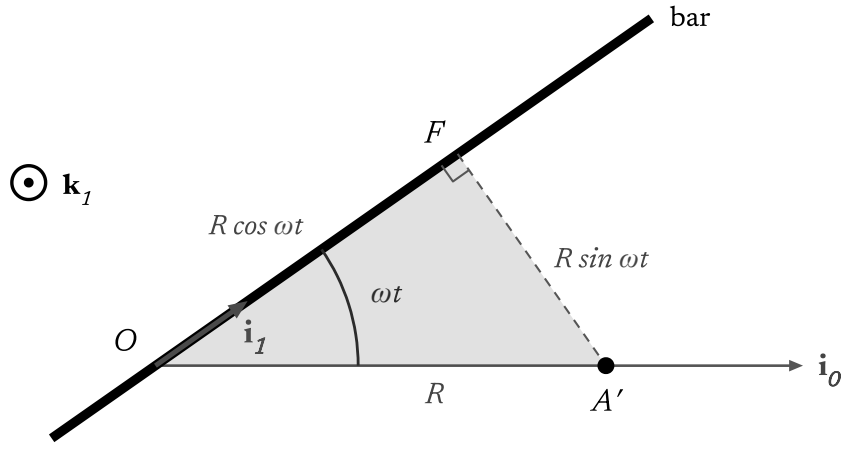


Figure 2. Same view from $+z_0$. A' is the point of the plane of the bar directly below the anchor A, at distance R from O along $\mathbf{i}_0$, that is ωt behind the bar. In the shaded right triangle OFA' the hypotenuse is R, the leg along the bar is $R\cos\omega t$ and the leg FA', parallel to $-\mathbf{j}_1$, is $R\sin\omega t$.

Position of A in S_1. In the same view, A' is at distance R from O along $\mathbf{i}_0$, that is, the angle ωt behind the bar (clockwise from $\mathbf{i}_1$). Drop the perpendicular from A' to the line of the bar, with foot F. In the right triangle OFA' (right angle at F) the hypotenuse is $OA' = R$, the leg adjacent to ωt is $OF = R\cos\omega t$, along $+\mathbf{i}_1$, and the opposite leg is $FA' = R\sin\omega t$, along $-\mathbf{j}_1$ because A' lies on the clockwise side of the bar. A is a height h above A', so

$$\mathbf{r}_1^A = R\cos\omega t\,\mathbf{i}_1 - R\sin\omega t\,\mathbf{j}_1 + h\,\mathbf{k}_1$$

The matrix gives the same result: the origins coincide, so $\mathbf{r}_1^A = [{}_0R_1]^T\mathbf{r}_0^A$,

$$[{}_0R_1]^T\begin{pmatrix} R \\ 0 \\ h \end{pmatrix} = \begin{pmatrix} R\cos\omega t \\ -R\sin\omega t \\ h \end{pmatrix}$$

The plane bar–P seen along the bar

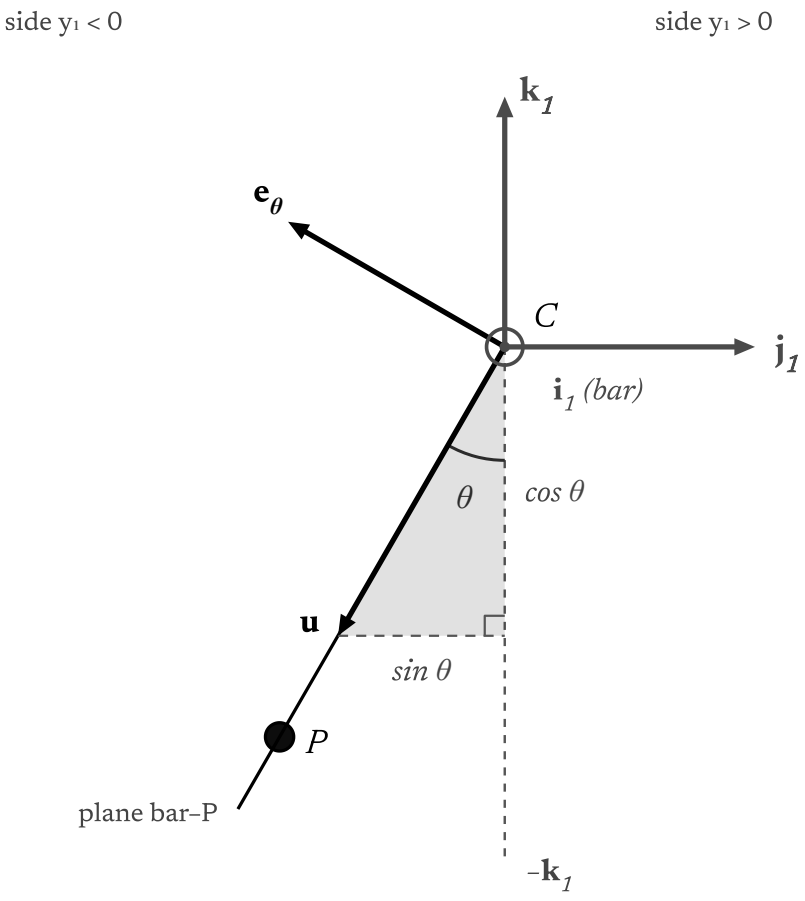


Figure 3. View along the bar, with $\mathbf{i}_1$ out of the paper ($\odot$) at C, so $\mathbf{j}_1 = \mathbf{k}_1 \times \mathbf{i}_1$ points right. The plane bar–P is seen edge-on, at the angle θ from the downward vertical $-\mathbf{k}_1$. Its unit vector $\mathbf{u}$ is the hypotenuse of the shaded triangle: leg $\cos\theta$ along $-\mathbf{k}_1$ and leg $\sin\theta$ along $-\mathbf{j}_1$; $\mathbf{e}_\theta = d\mathbf{u}/d\theta$ is perpendicular to it.

The second rotation is about the bar, so we now look along $\mathbf{i}_1$, with $\mathbf{i}_1$ pointing out of the paper ($\odot$) and the bar reduced to the point C. Take $\mathbf{k}_1$ up. Then $\mathbf{j}_1 = \mathbf{k}_1 \times \mathbf{i}_1$ points to the right: the fingers go from $\mathbf{k}_1$ (up) to $\mathbf{i}_1$ (towards the reader) and the thumb points right. The plane that contains the bar and P contains the line of sight, so in this view it is a straight line

through C. θ is its angle with the downward vertical $-\mathbf{k}_1$ (the trace of the plane Ox_1z_1), and $\theta > 0$ puts P on the side $y_1 < 0$, the side of A for $0 < \omega t < \pi$ (this is the sign convention of the official key); in the drawing that is the left.

Let $\mathbf{u}$ be the unit vector along that line, from C towards the side of P (an auxiliary vector, perpendicular to the bar). It is the hypotenuse (length 1) of the shaded right triangle: the leg adjacent to θ lies along $-\mathbf{k}_1$ and measures $\cos\theta$; the leg opposite to θ lies along $-\mathbf{j}_1$ and measures $\sin\theta$. Hence

$$\mathbf{u} = -\sin\theta\,\mathbf{j}_1 - \cos\theta\,\mathbf{k}_1$$

Both components are negative because, for $0 < \theta < \pi/2$, $\mathbf{u}$ points down and to the left. Differentiating with respect to θ ($\frac{d}{d\theta}(-\sin\theta) = -\cos\theta$, $\frac{d}{d\theta}(-\cos\theta) = \sin\theta$) gives a vector of length $\sqrt{\cos^2\theta + \sin^2\theta} = 1$, perpendicular to $\mathbf{u}$ in this view and pointing in the sense of increasing θ:

$$\mathbf{e}_\theta = \frac{d\mathbf{u}}{d\theta} = -\cos\theta\,\mathbf{j}_1 + \sin\theta\,\mathbf{k}_1$$

Relation with the standard azimuth. The usual azimuth ψ about the polar axis $\mathbf{i}_1$ is measured from $\mathbf{j}_1$ towards $\mathbf{k}_1$, with perpendicular unit vector $\cos\psi\,\mathbf{j}_1 + \sin\psi\,\mathbf{k}_1$. Matching $\mathbf{u} = -\sin\theta\,\mathbf{j}_1 - \cos\theta\,\mathbf{k}_1$ requires $\cos\psi = -\sin\theta$ and $\sin\psi = -\cos\theta$, which holds for $\psi = -\theta - \frac{\pi}{2}$:

$$\cos\left(-\theta - \tfrac{\pi}{2}\right) = \cos\left(\theta + \tfrac{\pi}{2}\right) = \cos\theta\cos\tfrac{\pi}{2} - \sin\theta\sin\tfrac{\pi}{2} = -\sin\theta, \qquad \sin\left(-\theta - \tfrac{\pi}{2}\right) = -\left(\sin\theta\cos\tfrac{\pi}{2} + \cos\theta\sin\tfrac{\pi}{2}\right) = -\cos\theta$$

(with $\cos\frac{\pi}{2}=0$, $\sin\frac{\pi}{2}=1$, and cosine even, sine odd). So θ is measured from the downward vertical instead of $\mathbf{j}_1$, and in the opposite sense; it is chosen so that $\theta=0$ is the hanging position, in the plane Ox_1z_1.

The segment CP inside the plane bar–P

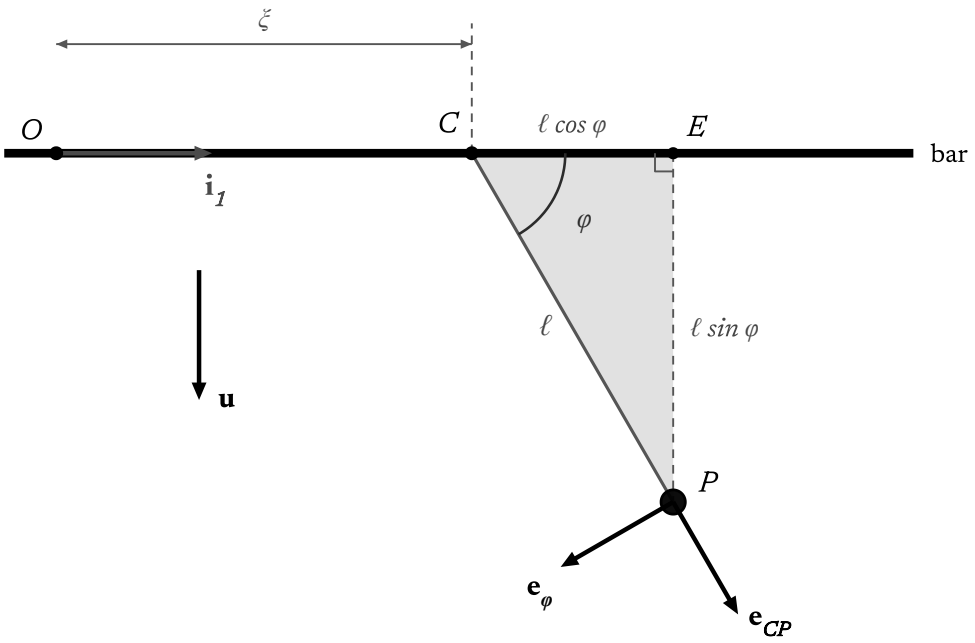


Figure 4. True view of the plane bar–P, which contains $\mathbf{i}_1$ (along the bar) and $\mathbf{u}$ (down the page). In the shaded right triangle CEP the hypotenuse is ℓ, the leg along the bar is $\ell\cos\varphi$ and the leg along $\mathbf{u}$ is $\ell\sin\varphi$; $OP=\xi\,\mathbf{i}_1+\ell\,\mathbf{e}_{CP}$. $\mathbf{e}_{CP}$ and $\mathbf{e}_\varphi=\partial\mathbf{e}_{CP}/\partial\varphi$ in orange.

This drawing is the true view of the plane that contains the bar ($\mathbf{i}_1$, to the right) and $\mathbf{u}$ (down the page). The segment CP makes the angle φ with the bar, with $\varphi\in(0,\pi)$. Let E be the foot of the perpendicular from P to the line of the bar. In the shaded right triangle CEP (right angle at E) the hypotenuse is $CP=\ell$, the leg adjacent to φ is $CE=\ell\cos\varphi$, along $\mathbf{i}_1$, and the opposite leg is $EP=\ell\sin\varphi$, along $\mathbf{u}$. Therefore $\overrightarrow{CP}=\ell\cos\varphi\,\mathbf{i}_1+\ell\sin\varphi\,\mathbf{u}$, and dividing by ℓ and substituting $\mathbf{u}$:

$$\mathbf{e}_{CP}=\cos\varphi\,\mathbf{i}_1+\sin\varphi\,\mathbf{u}=\cos\varphi\,\mathbf{i}_1-\sin\varphi\sin\theta\,\mathbf{j}_1-\sin\varphi\cos\theta\,\mathbf{k}_1$$

The other two unit vectors of $\mathcal{B}_2$ come from differentiating $\mathbf{e}_{CP}$ with respect to each angle. With respect to φ ($\frac{\partial}{\partial\varphi}\cos\varphi=-\sin\varphi$, $\frac{\partial}{\partial\varphi}\sin\varphi=\cos\varphi$) we get the unit vector perpendicular to CP in this plane, drawn at P:

$$\mathbf{e}_\varphi = \frac{\partial \mathbf{e}_{CP}}{\partial \varphi} = -\sin\varphi\,\mathbf{i}_1 + \cos\varphi\,\mathbf{u} = -\sin\varphi\,\mathbf{i}_1 - \cos\varphi\sin\theta\,\mathbf{j}_1 - \cos\varphi\cos\theta\,\mathbf{k}_1$$

With respect to θ, only $\mathbf{u}$ depends on it, so $\frac{\partial \mathbf{e}_{CP}}{\partial \theta} = \sin\varphi\,\frac{d\mathbf{u}}{d\theta} = \sin\varphi\,\mathbf{e}_\theta$, which has length $\sin\varphi$:

$$\mathbf{e}_\theta = \frac{1}{\sin\varphi}\frac{\partial \mathbf{e}_{CP}}{\partial \theta} = -\cos\theta\,\mathbf{j}_1 + \sin\theta\,\mathbf{k}_1$$

The matrix with columns $\mathbf{e}_{CP}, \mathbf{e}_\theta, \mathbf{e}_\varphi$ in $\mathcal{B}_1$ is

$$[{}_1R_2] = \begin{pmatrix} \cos\varphi & 0 & -\sin\varphi \\ -\sin\varphi\sin\theta & -\cos\theta & -\cos\varphi\sin\theta \\ -\sin\varphi\cos\theta & \sin\theta & -\cos\varphi\cos\theta \end{pmatrix}$$

Expanding along the first row: $\det[{}_1R_2] = \cos\varphi\left(\cos\varphi\cos^2\theta + \cos\varphi\sin^2\theta\right) - \sin\varphi\left(-\sin\varphi\sin^2\theta - \sin\varphi\cos^2\theta\right) = \cos^2\varphi + \sin^2\varphi = 1$. Each column has unit norm, and $\mathbf{e}_{CP}\cdot\mathbf{e}_\theta = \sin\varphi\sin\theta\cos\theta - \sin\varphi\cos\theta\sin\theta = 0$, $\mathbf{e}_{CP}\cdot\mathbf{e}_\varphi = -\cos\varphi\sin\varphi + \sin\varphi\cos\varphi\left(\sin^2\theta + \cos^2\theta\right) = 0$, $\mathbf{e}_\theta\cdot\mathbf{e}_\varphi = \cos\theta\cos\varphi\sin\theta - \sin\theta\cos\varphi\cos\theta = 0$, so $[{}_1R_2][{}_1R_2]^T = I$ and $\mathbf{e}_{CP}\times\mathbf{e}_\theta = \mathbf{e}_\varphi$.

Derivatives of $\mathcal{B}_2$ in S_1 (chain rule, $\frac{d}{dt}\mathbf{e}(\varphi(t),\theta(t)) = \frac{\partial \mathbf{e}}{\partial\varphi}\dot\varphi + \frac{\partial \mathbf{e}}{\partial\theta}\dot\theta$, with $\mathbf{i}_1,\mathbf{j}_1,\mathbf{k}_1$ constant in S_1). Using $\partial\mathbf{e}_\varphi/\partial\varphi = \cos\varphi\,(-\mathbf{i}_1) + \sin\varphi\sin\theta\,\mathbf{j}_1 + \sin\varphi\cos\theta\,\mathbf{k}_1 = -\mathbf{e}_{CP}$, $\partial\mathbf{e}_\varphi/\partial\theta = -\cos\varphi\cos\theta\,\mathbf{j}_1 + \cos\varphi\sin\theta\,\mathbf{k}_1 = \cos\varphi\,\mathbf{e}_\theta$ and $\partial\mathbf{e}_\theta/\partial\theta = \sin\theta\,\mathbf{j}_1 + \cos\theta\,\mathbf{k}_1 = -\left(\sin\varphi\,\mathbf{e}_{CP} + \cos\varphi\,\mathbf{e}_\varphi\right)$:

$$\left.\frac{d\mathbf{e}_{CP}}{dt}\right|_1 = \dot\varphi\,\mathbf{e}_\varphi + \dot\theta\sin\varphi\,\mathbf{e}_\theta, \qquad \left.\frac{d\mathbf{e}_\varphi}{dt}\right|_1 = -\dot\varphi\,\mathbf{e}_{CP} + \dot\theta\cos\varphi\,\mathbf{e}_\theta, \qquad \left.\frac{d\mathbf{e}_\theta}{dt}\right|_1 = -\dot\theta\left(\sin\varphi\,\mathbf{e}_{CP} + \cos\varphi\,\mathbf{e}_\varphi\right)$$

Position of P in S_1. To go from O to P we pass through the contact point, which is on the bar: $\overrightarrow{OP} = \overrightarrow{OC} + \overrightarrow{CP}$, with $\overrightarrow{OC} = \xi\,\mathbf{i}_1$ (the dimension ξ of the drawing) and $\overrightarrow{CP} = \ell\,\mathbf{e}_{CP}$:

$$\mathbf{r}_1^P = \xi\,\mathbf{i}_1 + \ell\,\mathbf{e}_{CP} = (\xi + \ell\cos\varphi)\,\mathbf{i}_1 - \ell\sin\varphi\sin\theta\,\mathbf{j}_1 - \ell\sin\varphi\cos\theta\,\mathbf{k}_1$$

Kinematics

(a) Position and velocity of A in S_1

> **Pedagogical Justification:** A is fixed in S_0; its coordinates in S_1 were obtained above from the top view and from $\mathbf{r}_1^A = [{}_0R_1]^T \mathbf{r}_0^A$; its relative velocity is the derivative of those coordinates in S_1.

$$\boxed{\mathbf{r}_1^A = R\cos\omega t\,\mathbf{i}_1 - R\sin\omega t\,\mathbf{j}_1 + h\,\mathbf{k}_1}$$

With the chain rule $\frac{d}{dt}\cos\omega t = -\omega\sin\omega t$ and $\frac{d}{dt}\sin\omega t = \omega\cos\omega t$:

$$\boxed{\mathbf{v}_1^A = \left.\frac{d\mathbf{r}_1^A}{dt}\right|_1 = -R\omega\sin\omega t\,\mathbf{i}_1 - R\omega\cos\omega t\,\mathbf{j}_1}$$

Check with the velocity composition (Notes Eq. 3.26): $\mathbf{0} = \mathbf{v}_0^A = \mathbf{v}_1^A + \boldsymbol{\omega}_{10}\times\mathbf{r}_1^A$, and $\omega\mathbf{k}_1\times\mathbf{r}_1^A = R\omega\cos\omega t\,\mathbf{j}_1 + R\omega\sin\omega t\,\mathbf{i}_1 = -\mathbf{v}_1^A$. In S_1, A describes a circle of radius R at height h with angular velocity $-\omega$.

(b) Position, velocity and acceleration of P in S_1

> **Pedagogical Justification:** $\overrightarrow{OP} = \overrightarrow{OC} + \overrightarrow{CP}$ with $\overrightarrow{OC} = \xi\,\mathbf{i}_1$ along the bar and $\overrightarrow{CP} = \ell\,\mathbf{e}_{CP}$. Derivatives are taken in S_1 with the product rule and the basis derivatives of the change-of-basis section.

Position (built step by step in the previous section):

$$\boxed{\mathbf{r}_1^P = \xi\,\mathbf{i}_1 + \ell\,\mathbf{e}_{CP} = (\xi + \ell\cos\varphi)\,\mathbf{i}_1 - \ell\sin\varphi\sin\theta\,\mathbf{j}_1 - \ell\sin\varphi\cos\theta\,\mathbf{k}_1}$$

Velocity. Product rule, $\frac{d}{dt}(\ell\,\mathbf{e}_{CP}) = \dot\ell\,\mathbf{e}_{CP} + \ell\left(\dot\varphi\,\mathbf{e}_\varphi + \dot\theta\sin\varphi\,\mathbf{e}_\theta\right)$:

$$\boxed{\mathbf{v}_1^P = \dot\xi\,\mathbf{i}_1 + \dot\ell\,\mathbf{e}_{CP} + \ell\dot\varphi\,\mathbf{e}_\varphi + \ell\dot\theta\sin\varphi\,\mathbf{e}_\theta}$$

In $\mathcal{B}_1$ (differentiating each component; e.g. $\frac{d}{dt}(-\ell\sin\varphi\sin\theta) = -\dot\ell\sin\varphi\sin\theta - \ell\dot\varphi\cos\varphi\sin\theta - \ell\dot\theta\sin\varphi\cos\theta$):

$$\mathbf{v}_1^P = \left(\dot\xi + \dot\ell\cos\varphi - \ell\dot\varphi\sin\varphi\right)\mathbf{i}_1 - \left(\dot\ell\sin\varphi\sin\theta + \ell\dot\varphi\cos\varphi\sin\theta + \ell\dot\theta\sin\varphi\cos\theta\right)\mathbf{j}_1 - \left(\dot\ell\sin\varphi\cos\theta + \ell\dot\varphi\cos\varphi\cos\theta - \ell\dot\theta\sin\varphi\sin\theta\right)\mathbf{k}_1$$

Acceleration. Differentiate term by term:

- $\frac{d}{dt}(\dot\xi\,\mathbf{i}_1) = \ddot\xi\,\mathbf{i}_1$;
- $\frac{d}{dt}(\dot\ell\,\mathbf{e}_{CP}) = \ddot\ell\,\mathbf{e}_{CP} + \dot\ell\dot\varphi\,\mathbf{e}_\varphi + \dot\ell\dot\theta\sin\varphi\,\mathbf{e}_\theta$;
- $\frac{d}{dt}(\ell\dot\varphi\,\mathbf{e}_\varphi) = (\dot\ell\dot\varphi + \ell\ddot\varphi)\,\mathbf{e}_\varphi + \ell\dot\varphi\left(-\dot\varphi\,\mathbf{e}_{CP} + \dot\theta\cos\varphi\,\mathbf{e}_\theta\right)$;
- $\frac{d}{dt}(\ell\dot\theta\sin\varphi\,\mathbf{e}_\theta) = \left(\dot\ell\dot\theta\sin\varphi + \ell\ddot\theta\sin\varphi + \ell\dot\theta\dot\varphi\cos\varphi\right)\mathbf{e}_\theta - \ell\dot\theta^2\sin\varphi\,(\sin\varphi\,\mathbf{e}_{CP} + \cos\varphi\,\mathbf{e}_\varphi)$.

Collecting along each unit vector:

$$\boxed{\mathbf{a}_1^P = \ddot\xi\,\mathbf{i}_1 + \left(\ddot\ell - \ell\dot\varphi^2 - \ell\dot\theta^2\sin^2\varphi\right)\mathbf{e}_{CP} + \left(\ell\ddot\varphi + 2\dot\ell\dot\varphi - \ell\dot\theta^2\sin\varphi\cos\varphi\right)\mathbf{e}_\varphi + \left(\ell\ddot\theta\sin\varphi + 2\dot\ell\dot\theta\sin\varphi + 2\ell\dot\theta\dot\varphi\cos\varphi\right)\mathbf{e}_\theta}$$

with $\mathbf{e}_{CP}, \mathbf{e}_\varphi, \mathbf{e}_\theta$ given in $\mathcal{B}_1$ in the change-of-basis section. The bracketed terms are the spherical-coordinate acceleration of P about the moving centre C, plus the acceleration $\ddot\xi\,\mathbf{i}_1$ of the centre along the bar. Time enters through $\xi, \ell, \varphi, \theta$ only; the explicit time dependence appears when ℓ and φ are replaced by the constraints of part (c).

(c) AC, ℓ and φ as functions of ξ and t

> **Pedagogical Justification:** The string has constant length, which gives ℓ. The angle φ follows from the frictionless contact: the massless piece of string at C must be in equilibrium under the two tensions and the bar's reaction, which has no component along the bar.

Vector $\overrightarrow{AC}$: with $\mathbf{r}_1^C = \xi\,\mathbf{i}_1$ and part (a),

$$\boxed{\overrightarrow{AC} = \mathbf{r}_1^C - \mathbf{r}_1^A = (\xi - R\cos\omega t)\,\mathbf{i}_1 + R\sin\omega t\,\mathbf{j}_1 - h\,\mathbf{k}_1}$$

$$AC^2 = (\xi - R\cos\omega t)^2 + R^2\sin^2\omega t + h^2 = \xi^2 - 2R\xi\cos\omega t + R^2\left(\cos^2\omega t + \sin^2\omega t\right) + h^2 \quad\Longrightarrow\quad \boxed{AC = \sqrt{\xi^2 - 2R\xi\cos\omega t + R^2 + h^2}}$$

Length ℓ. $AC + CP = L$:

$$\boxed{\ell = L - \sqrt{\xi^2 - 2R\xi\cos\omega t + R^2 + h^2}}$$

Angle φ. With no friction the tension is the same, T, on both sides of C. Equilibrium of the massless string element at C: $T\,\mathbf{e}_{CA} + T\,\mathbf{e}_{CP} + \mathbf{N}_b = \mathbf{0}$ with $\mathbf{N}_b \cdot \mathbf{i}_1 = 0$. Projecting on $\mathbf{i}_1$: $\mathbf{e}_{CP}\cdot\mathbf{i}_1 = -\mathbf{e}_{CA}\cdot\mathbf{i}_1$, where $\mathbf{e}_{CA} = -\overrightarrow{AC}/AC$. Hence

$$\cos\varphi = \frac{\xi - R\cos\omega t}{AC} \quad\Longrightarrow\quad \boxed{\varphi = \arccos\frac{\xi - R\cos\omega t}{\sqrt{\xi^2 - 2R\xi\cos\omega t + R^2 + h^2}}}, \qquad \sin\varphi = \frac{\sqrt{R^2\sin^2\omega t + h^2}}{AC}$$

($\varphi \in (0,\pi)$, so $\sin\varphi > 0$). The two segments make equal angles with the bar, on opposite sides of the plane perpendicular to it. (Extension: this is also the condition for the path $A \to C \to P$ to be the shortest one for fixed P, $\frac{\partial}{\partial\xi}(AC + CP) = \cos\angle(\overrightarrow{AC}, \mathbf{i}_1) - \cos\varphi = 0$.)

Compact notation for part (e). Let $q = \xi - R\cos\omega t$, $\rho = \sqrt{R^2\sin^2\omega t + h^2}$ and $D = AC = \sqrt{q^2 + \rho^2}$. Then $q = D\cos\varphi$, $\rho = D\sin\varphi$: (D,φ) are the polar coordinates of the point (q,ρ), and $\ell = L - D$. Their derivatives:

- $\dot q = \dot\xi + R\omega\sin\omega t$ and $\ddot q = \ddot\xi + R\omega^2\cos\omega t$ (chain rule on $\cos\omega t$).
- From $\rho^2 = R^2\sin^2\omega t + h^2$: $2\rho\dot\rho = 2R^2\omega\sin\omega t\cos\omega t$, so $\dot\rho = \dfrac{R^2\omega\sin 2\omega t}{2\rho}$.
  Differentiating $\rho\dot\rho = \frac12 R^2\omega\sin 2\omega t$ (product rule): $\dot\rho^2 + \rho\ddot\rho = R^2\omega^2\cos 2\omega t$, so $\ddot\rho = \dfrac{R^2\omega^2\cos 2\omega t - \dot\rho^2}{\rho}$.
- Differentiating $q = D\cos\varphi$, $\rho = D\sin\varphi$: $\dot q = \dot D\cos\varphi - D\dot\varphi\sin\varphi$, $\dot\rho = \dot D\sin\varphi + D\dot\varphi\cos\varphi$. Solving this rotation-type system:

$$\dot D = \dot q\cos\varphi + \dot\rho\sin\varphi, \qquad D\dot\varphi = \dot\rho\cos\varphi - \dot q\sin\varphi$$

- Differentiating once more, $\ddot q = (\ddot D - D\dot\varphi^2)\cos\varphi - (D\ddot\varphi + 2\dot D\dot\varphi)\sin\varphi$ and $\ddot\rho = (\ddot D - D\dot\varphi^2)\sin\varphi + (D\ddot\varphi + 2\dot D\dot\varphi)\cos\varphi$, so

$$\ddot D - D\dot\varphi^2 = \ddot q\cos\varphi + \ddot\rho\sin\varphi, \qquad D\ddot\varphi + 2\dot D\dot\varphi = \ddot\rho\cos\varphi - \ddot q\sin\varphi$$

- Finally $\dot\ell = -\dot D$, $\ddot\ell = -\ddot D$. Since $\dot\rho\sin\varphi = \dfrac{R^2\omega\sin\omega t\cos\omega t}{\rho}\,\dfrac{\rho}{D}$ and $\cos\varphi = q/D$,

$$\dot\ell = -(\dot\xi + R\omega\sin\omega t)\frac{\xi - R\cos\omega t}{D} - \frac{R^2\omega\sin\omega t\cos\omega t}{D} \quad\Longrightarrow\quad \boxed{\dot\ell + \dot\xi\cos\varphi = -\frac{R\omega\,\xi\sin\omega t}{D}}$$

Forces and Equations of Motion

(d) Forces acting on P

> **Pedagogical Justification:** In the non-inertial frame S_1, $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eqs. 4.62–4.63). The actual forces on P are only the weight and the pull of the string (the bar touches the string, not P). The inertia forces are those of Slide 28 with $\mathbf{a}_0^O = \mathbf{0}$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$, $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$.

Actual forces:

- Weight: $\mathbf{W} = -mg\,\mathbf{k}_1$.
- Tension, along the string towards C: $\mathbf{T} = -T\,\mathbf{e}_{CP} = -T\cos\varphi\,\mathbf{i}_1 + T\sin\varphi\sin\theta\,\mathbf{j}_1 + T\sin\varphi\cos\theta\,\mathbf{k}_1$, with $T > 0$.

Inertia forces: write $\mathbf{r}_1^P = x\,\mathbf{i}_1 + y\,\mathbf{j}_1 + z\,\mathbf{k}_1$ with $x = \xi + \ell\cos\varphi$, $y = -\ell\sin\varphi\sin\theta$, and $\mathbf{v}_1^P = v_x\mathbf{i}_1 + v_y\mathbf{j}_1 + v_z\mathbf{k}_1$ from part (b). With $\mathbf{k}_1\times\mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{k}_1\times\mathbf{j}_1 = -\mathbf{i}_1$, $\mathbf{k}_1\times\mathbf{k}_1 = \mathbf{0}$:

- Origin and Euler: $\mathbf{F}_A = -m\mathbf{a}_0^O = \mathbf{0}$, $\mathbf{F}_E = -m\boldsymbol{\alpha}_{10}\times\mathbf{r}_1^P = \mathbf{0}$.
- Centrifugal: $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = \omega\,(x\,\mathbf{j}_1 - y\,\mathbf{i}_1)$, $\boldsymbol{\omega}_{10}\times\left(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P\right) = \omega^2\,(-x\,\mathbf{i}_1 - y\,\mathbf{j}_1)$, so (Slide 30, $\mathbf{r}_\perp = x\,\mathbf{i}_1 + y\,\mathbf{j}_1$)

$$\boxed{\mathbf{F}_C = m\omega^2(\xi + \ell\cos\varphi)\,\mathbf{i}_1 - m\omega^2\ell\sin\varphi\sin\theta\,\mathbf{j}_1}$$

- Coriolis: $\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P = \omega\,(v_x\,\mathbf{j}_1 - v_y\,\mathbf{i}_1)$, so $\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P = 2m\omega v_y\,\mathbf{i}_1 - 2m\omega v_x\,\mathbf{j}_1$:

$$\boxed{\mathbf{F}_{COR} = -2m\omega\left(\dot\ell\sin\varphi\sin\theta + \ell\dot\varphi\cos\varphi\sin\theta + \ell\dot\theta\sin\varphi\cos\theta\right)\mathbf{i}_1 - 2m\omega\left(\dot\xi + \dot\ell\cos\varphi - \ell\dot\varphi\sin\varphi\right)\mathbf{j}_1}$$

Both inertia forces agree with the official key.

(e) Equations of motion

> **Pedagogical Justification:** Project $m\mathbf{a}_1^P = \mathbf{W} + \mathbf{T} + \mathbf{F}_C + \mathbf{F}_{COR}$ on $\mathcal{B}_2$. The tension is along $\mathbf{e}_{CP}$, so the projections on $\mathbf{e}_\theta$ and $\mathbf{e}_\varphi$ are free of the unknown T: they are the two equations of motion for the two degrees of freedom ξ, θ. The projection on $\mathbf{e}_{CP}$ then gives T .

Projections of each term (using $\mathbf{i}_1 \cdot \mathbf{e}_{CP} = \cos\varphi$, $\mathbf{i}_1 \cdot \mathbf{e}_\varphi = -\sin\varphi$, $\mathbf{i}_1 \cdot \mathbf{e}_\theta = 0$, and the components of the change-of-basis section):

Term	$\cdot\,\mathbf{e}_\theta$	$\cdot\,\mathbf{e}_\varphi$	$\cdot\,\mathbf{e}_{CP}$
$\mathbf{a}_1^P$ (multiply by m)	$\ell\ddot{\theta}\sin\varphi + 2\dot{\ell}\dot{\theta}\sin\varphi + 2\ell\dot{\theta}\dot{\varphi}\cos\varphi$	$\ell\ddot{\varphi} + 2\dot{\ell}\dot{\varphi} - \ddot{\xi}\sin\varphi - \ell\dot{\theta}^2\sin\varphi\cos\varphi$	$\ddot{\xi}\cos\varphi + \ddot{\ell} - \ell\dot{\varphi}^2 - \ell\dot{\theta}^2\sin^2\varphi$
$\mathbf{W}$	$-mg\sin\theta$	$mg\cos\varphi\cos\theta$	$mg\sin\varphi\cos\theta$
$\mathbf{T}$	0	0	$-T$
$\mathbf{F}_C$	$m\omega^2\ell\sin\varphi\sin\theta\cos\theta$	$-m\omega^2\sin\varphi\left(\xi + \ell\cos\varphi\cos^2\theta\right)$	$m\omega^2\left[(\xi + \ell\cos\varphi)\cos\varphi + \ell\sin^2\varphi\sin^2\theta\right]$
$\mathbf{F}_{COR}$	$2m\omega\cos\theta\, v_x$	$2m\omega\left[\sin\theta(\dot{\ell} + \dot{\xi}\cos\varphi) + \ell\dot{\theta}\sin^2\varphi\cos\theta\right]$	$2m\omega\left[\sin\theta(\dot{\xi}\sin\varphi - \ell\dot{\varphi}) - \ell\dot{\theta}\sin\varphi\cos\varphi\cos\theta\right]$

Details of the less obvious entries:

- $\mathbf{F}_C \cdot \mathbf{e}_\varphi = m\omega^2\left[-x\sin\varphi - y\cos\varphi\sin\theta\right] = m\omega^2\left[-(\xi + \ell\cos\varphi)\sin\varphi + \ell\sin\varphi\cos\varphi\sin^2\theta\right] = -m\omega^2\sin\varphi\left[\xi + \ell\cos\varphi(1 - \sin^2\theta)\right]$.
- $\mathbf{F}_{COR} \cdot \mathbf{e}_\varphi = 2m\omega\left[-v_y\sin\varphi + v_x\cos\varphi\sin\theta\right]$, where $-v_y\sin\varphi = \dot{\ell}\sin^2\varphi\sin\theta + \ell\dot{\varphi}\sin\varphi\cos\varphi\sin\theta + \ell\dot{\theta}\sin^2\varphi\cos\theta$ and $v_x\cos\varphi\sin\theta = \dot{\xi}\cos\varphi\sin\theta + \dot{\ell}\cos^2\varphi\sin\theta - \ell\dot{\varphi}\sin\varphi\cos\varphi\sin\theta$; the $\dot{\varphi}$ terms cancel and $\sin^2\varphi + \cos^2\varphi = 1$.
- $\mathbf{F}_{COR} \cdot \mathbf{e}_{CP} = 2m\omega\left[v_y\cos\varphi + v_x\sin\varphi\sin\theta\right]$: the $\dot{\ell}$ terms cancel and the $\dot{\varphi}$ terms add up to $-\ell\dot{\varphi}\sin\theta\left(\cos^2\varphi + \sin^2\varphi\right)$.
- $\mathbf{F}_{COR} \cdot \mathbf{e}_\theta = 2m\omega\left[-v_x\left(-\cos\theta\right)\right] = 2m\omega\cos\theta\, v_x$, with $v_x = \dot{\xi} + \dot{\ell}\cos\varphi - \ell\dot{\varphi}\sin\varphi$.

Equation along $\mathbf{e}_\theta$ (divided by m):

$$\boxed{\ell\sin\varphi\,\ddot{\theta} + 2\dot{\ell}\dot{\theta}\sin\varphi + 2\ell\dot{\theta}\dot{\varphi}\cos\varphi = -g\sin\theta + \omega^2\ell\sin\varphi\sin\theta\cos\theta + 2\omega\cos\theta\left(\dot{\xi} + \dot{\ell}\cos\varphi - \ell\dot{\varphi}\sin\varphi\right)}$$

Equation along $\mathbf{e}_\varphi$ (divided by m):

$$\boxed{\ell\ddot{\varphi} + 2\dot{\ell}\dot{\varphi} - \ddot{\xi}\sin\varphi - \ell\dot{\theta}^2\sin\varphi\cos\varphi = g\cos\varphi\cos\theta - \omega^2\sin\varphi\left(\xi + \ell\cos\varphi\cos^2\theta\right) + 2\omega\left[\sin\theta\left(\dot{\ell} + \dot{\xi}\cos\varphi\right) + \ell\dot{\theta}\sin^2\varphi\cos\theta\right]}$$

Closing the system. ℓ, φ and their derivatives are the functions of $(\xi, \dot\xi, \ddot\xi, t)$ of part (c). In particular $\ell\ddot\varphi = \frac{\ell}{D}\left(\ddot\rho\cos\varphi - \ddot q\sin\varphi - 2\dot D\dot\varphi\right)$ with $\ddot q = \ddot\xi + R\omega^2\cos\omega t$, and $2\dot\ell\dot\varphi = -2\dot D\dot\varphi$, so with $\ell + D = L$ the left-hand side of the $\mathbf{e}_\varphi$ equation becomes

$$\ell\ddot\varphi + 2\dot\ell\dot\varphi - \ddot\xi\sin\varphi = -\frac{L}{D}\sin\varphi\,\ddot\xi + \frac{\ell}{D}\left(\ddot\rho\cos\varphi - R\omega^2\cos\omega t\sin\varphi\right) - \frac{2L}{D}\dot D\dot\varphi$$

The coefficient of $\ddot\xi$, $-(L/D)\sin\varphi$, is non-zero while the string is bent ($\sin\varphi > 0$), so the $\mathbf{e}_\varphi$ equation gives $\ddot\xi$ and the $\mathbf{e}_\theta$ equation gives $\ddot\theta$: **two second-order ODEs for $\xi(t)$ and $\theta(t)$**, non-autonomous through ωt (the motion of A in S_1). In the $\mathbf{e}_\varphi$ equation the Coriolis term can also be written with $\dot\ell + \dot\xi\cos\varphi = -R\omega\xi\sin\omega t/D$.

Tension (projection on $\mathbf{e}_{CP}$, solved for T):

$$\boxed{T = -m\left(\ddot\xi\cos\varphi + \ddot\ell - \ell\dot\varphi^2 - \ell\dot\theta^2\sin^2\varphi\right) + mg\sin\varphi\cos\theta + m\omega^2\left[(\xi + \ell\cos\varphi)\cos\varphi + \ell\sin^2\varphi\sin^2\theta\right] + 2m\omega\left[\sin\theta\left(\dot\xi\sin\varphi - \ell\dot\varphi\right) - \ell\dot\theta\sin\varphi\cos\varphi\cos\theta\right]}$$

The solution is valid while $T > 0$ (string tense).

Answers and Checks

(a) Position and velocity of A

$\mathbf{r}_1^A = R\cos\omega t\,\mathbf{i}_1 - R\sin\omega t\,\mathbf{j}_1 + h\,\mathbf{k}_1$ and $\mathbf{v}_1^A = -R\omega\sin\omega t\,\mathbf{i}_1 - R\omega\cos\omega t\,\mathbf{j}_1$: in S_1, A goes round a circle of radius R at height h with angular velocity $-\omega$.

(b) Position, velocity and acceleration of P

$\mathbf{r}_1^P = \xi\,\mathbf{i}_1 + \ell\,\mathbf{e}_{CP}$, $\mathbf{v}_1^P = \dot\xi\,\mathbf{i}_1 + \dot\ell\,\mathbf{e}_{CP} + \ell\dot\varphi\,\mathbf{e}_\varphi + \ell\dot\theta\sin\varphi\,\mathbf{e}_\theta$ and $\mathbf{a}_1^P$ as boxed in the kinematics section, with $\mathbf{e}_{CP}$, $\mathbf{e}_\varphi$, $\mathbf{e}_\theta$ given in $\mathcal{B}_1$.

(c) AC, ℓ and φ

$AC = \sqrt{\xi^2 - 2R\xi\cos\omega t + R^2 + h^2}$, $\ell = L - AC$ and $\cos\varphi = (\xi - R\cos\omega t)/AC$.

(d) Forces

Weight $-mg\,\mathbf{k}_1$, tension $-T\,\mathbf{e}_{CP}$, and the centrifugal and Coriolis forces boxed above; the Euler force and the force of the origin's acceleration are zero.

(e) Equations of motion

The $\mathbf{e}_\theta$ and $\mathbf{e}_\varphi$ equations boxed above, closed with part (c), are two second-order ODEs for $\xi(t)$ and $\theta(t)$; the $\mathbf{e}_{CP}$ projection gives T.

(f) Initial conditions at $t = 0$

> **Pedagogical Justification:** At $t = 0^-$ the particle hangs at rest below A, and the string is straight and vertical. The bar touches it at the point of the string at the height of the bar. No impulsive force acts on P: the string is perpendicular to the bar, so displacing C sideways does not change the path length to first order ($\dot{D}(0) = 0$ below). Position and velocity of P are therefore continuous at $t = 0$.

Positions. At $t = 0$ the bar lies along $\mathbf{i}_0$, and the string meets it directly below A: $\xi(0) = R$. Then $q(0) = R - R = 0$, $\rho(0) = h$, $D(0) = h$, so

$$\xi(0) = R, \qquad \ell(0) = L - h, \qquad \cos\varphi(0) = 0 \;\Rightarrow\; \varphi(0) = \frac{\pi}{2}, \qquad \theta(0) = 0$$

(P is directly below the bar in the plane Ox_1z_1). Check: $\mathbf{r}_1^P(0) = R\,\mathbf{i}_1 - (L - h)\,\mathbf{k}_1$, i.e. P at depth $L - h$ below the bar, L below A.

Velocities. $\mathbf{v}_0^P(0) = \mathbf{0}$, and by Notes Eq. 3.26

$$\mathbf{v}_1^P(0) = \mathbf{v}_0^P - \boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = -\omega\,\mathbf{k}_1 \times [R\,\mathbf{i}_1 - (L - h)\,\mathbf{k}_1] = -\omega R\,\mathbf{j}_1$$

At $\varphi = \pi/2, \theta = 0$: $\mathbf{e}_{CP} = -\mathbf{k}_1$, $\mathbf{e}_\varphi = -\mathbf{i}_1$, $\mathbf{e}_\theta = -\mathbf{j}_1$, so the velocity of part (b) is $\mathbf{v}_1^P(0) = \left(\dot\xi - \ell\dot\varphi\right)\mathbf{i}_1 - \ell\dot\theta\,\mathbf{j}_1 - \dot\ell\,\mathbf{k}_1$. Matching components:

- $\mathbf{k}_1$: $\dot\ell(0) = 0$, consistent with the constraint: $\dot D(0) = \dot q\cos\frac{\pi}{2} + \dot\rho(0)\sin\frac{\pi}{2} = \dot\rho(0) = \dfrac{R^2\omega\sin 0}{2h} = 0$.
- $\mathbf{j}_1$: $(L - h)\,\dot\theta(0) = \omega R$, so $\dot\theta(0) = \dfrac{\omega R}{L - h}$.
- $\mathbf{i}_1$: $\dot\xi(0) = (L - h)\,\dot\varphi(0)$. From part (c), $D\dot\varphi = \dot\rho\cos\varphi - \dot q\sin\varphi$ gives $h\,\dot\varphi(0) = -\dot q(0) = -\left(\dot\xi(0) + R\omega\sin 0\right) = -\dot\xi(0)$. Substituting, $\dot\xi(0)\left(1 + \dfrac{L - h}{h}\right) = \dot\xi(0)\dfrac{L}{h} = 0$, so $\dot\xi(0) = 0$ and $\dot\varphi(0) = 0$.

$$\boxed{\xi(0) = R, \quad \dot\xi(0) = 0, \quad \theta(0) = 0, \quad \dot\theta(0) = \frac{\omega R}{L - h}} \qquad \left(\ell(0) = L - h,\ \dot\ell(0) = 0,\ \varphi(0) = \frac{\pi}{2},\ \dot\varphi(0) = 0\right)$$

In S_1 the particle starts moving backwards ($-\mathbf{j}_1$) with the speed ωR of the point of the bar that hits the string: P lags behind the bar, θ grows from zero.

(g) Is the mechanical energy conserved?

> **Pedagogical Justification:** By Slide 29, E_1^P is conserved only if every force that does work in S_1 derives from a potential that does not depend explicitly on time. Here $\mathbf{W}$ and $\mathbf{F}_C$ have potentials (mgz and $-\frac{1}{2}m\omega^2 r_\perp^2$, ω constant, Slide 30) and $\mathbf{F}_{COR}$ does no work (Slide 31). The question is the power of the tension, which is a constraint force of a constraint that moves.

In S_1. Define $E_1^P = \frac{1}{2}m\left|\mathbf{v}_1^P\right|^2 + mgz - \frac{1}{2}m\omega^2\left(x^2+y^2\right)$. By the work–energy theorem in S_1, $\dfrac{dE_1^P}{dt} = \mathbf{T}\cdot\mathbf{v}_1^P = -T\,\mathbf{e}_{CP}\cdot\mathbf{v}_1^P$. From part (b), $\mathbf{e}_{CP}\cdot\mathbf{v}_1^P = \dot\xi\cos\varphi + \dot\ell$, and part (c) gives $\dot\ell + \dot\xi\cos\varphi = -R\omega\xi\sin\omega t/D$. Therefore

$$\boxed{\frac{dE_1^P}{dt} = \frac{T\,R\,\omega\,\xi\sin\omega t}{\sqrt{\xi^2 - 2R\xi\cos\omega t + R^2 + h^2}}}$$

which is positive for $0 < \omega t < \pi$ (with $T>0, \xi>0$): E_1^P **is not conserved.** The reason is that the end A of the string moves in S_1 (part (a)): the constraint "string tense" depends explicitly on time in S_1, and the tension does work on P.

In S_0. $E_0^P = \frac{1}{2}m\left|\mathbf{v}_0^P\right|^2 + mgz$ changes at the rate $\mathbf{T}\cdot\mathbf{v}_0^P$ with $\mathbf{v}_0^P = \mathbf{v}_1^P + \boldsymbol{\omega}_{10}\times\mathbf{r}_1^P$ (Notes Eq. 3.26). The extra term is $-T\,\mathbf{e}_{CP}\cdot\omega\,(x\,\mathbf{j}_1 - y\,\mathbf{i}_1) = -T\omega\,(-x\sin\varphi\sin\theta - y\cos\varphi) = T\omega\sin\varphi\sin\theta\,(\xi + \ell\cos\varphi - \ell\cos\varphi) = T\omega\xi\sin\varphi\sin\theta$, so

$$\frac{dE_0^P}{dt} = T\,\omega\,\xi\left(\frac{R\sin\omega t}{D} + \sin\varphi\sin\theta\right) \neq 0$$

No: the mechanical energy of the particle is not conserved in either frame. The motor that keeps the bar at constant ω feeds energy into the particle through the string.

(h) Separation condition

> **Pedagogical Justification:** The string can only be pushed by the bar. The bar's force on the string, $\mathbf{N}_b = -T\,(\mathbf{e}_{CA} + \mathbf{e}_{CP})$, vanishes when the string is straight; beyond that instant the bend would have to reverse, which a bar pushing from one side cannot impose. Separation is therefore the instant at which A, C and P become collinear: $\mathbf{e}_{CP} = -\mathbf{e}_{CA}$.

From part (c), $-\mathbf{e}_{CA} = \overrightarrow{AC}/D = \dfrac{1}{D}\left[(\xi - R\cos\omega t)\,\mathbf{i}_1 + R\sin\omega t\,\mathbf{j}_1 - h\,\mathbf{k}_1\right]$. The $\mathbf{i}_1$ components of $\mathbf{e}_{CP}$ and $-\mathbf{e}_{CA}$ are already equal (frictionless condition, $\cos\varphi = q/D$). Equating the other two, with $\sin\varphi = \rho/D$:

$$-\frac{\rho}{D}\sin\theta = \frac{R\sin\omega t}{D}, \qquad -\frac{\rho}{D}\cos\theta = -\frac{h}{D} \quad\Longrightarrow\quad \sin\theta = -\frac{R\sin\omega t}{\rho}, \quad \cos\theta = \frac{h}{\rho}$$

Dividing the two relations ($\cos\theta = h/\rho > 0$):

$$\boxed{\tan\theta = -\frac{R\sin\omega t}{h}} \qquad \text{i.e.} \quad \theta_{sep}(t) = -\arctan\left(\frac{R\sin\omega t}{h}\right)$$

The string leaves the bar at the first instant at which $\theta(t)$, obtained from the equations of part (e), reaches this value. For $0 < \omega t < \pi$ it is negative: P must have swung forward, past the vertical plane of the bar, to the side opposite to A, so that the straight line from A through C passes through P.

Checks

- **Mechanism.** In S_1 the anchor A circles backwards at radius R and speed ωR, while the string is held on the rotating bar. The string's end at A and the bar together act as a moving constraint that pumps energy into the pendulum ($dE_1^P/dt > 0$ for $0 < \omega t < \pi$). The centrifugal force pushes P outward along the bar ($+\mathbf{i}_1$); the Coriolis force deflects it sideways and couples ξ and θ.
- **Initial kick.** $\dot\theta(0) = \omega R/(L-h)$: in S_1 the particle starts swinging backwards about the contact point with the speed of the point of the bar that hits the string, a pendulum of length $L - h$.
- **Limits.** $R \to 0$: A lies on the rotation axis and is fixed in S_1, the constraint becomes time-independent, and $dE_1^P/dt = TR\omega\xi\sin\omega t/D \to 0$: E_1^P becomes a first integral (Slide 29). $\omega \to 0$: the inertia forces vanish and the bar would never reach the string; the problem reduces to a pendulum at rest. Separation: $h \to \infty$ gives $\theta_{sep} \to 0$; $\omega t = \pi$ (anchor on the opposite side of the bar) gives $\theta_{sep} = 0$. Basis: $\theta = 0$, $\varphi = \pi/2$ give $\mathbf{u} = -\mathbf{k}_1$ and $\mathbf{e}_{CP} = -\mathbf{k}_1$, the hanging string.
- **Inertial check.** Differentiating $[{}_0R_1]\mathbf{r}_1^P$ twice reproduces Notes Eq. 3.27; at random states, with $\ddot\xi, \ddot\theta$ taken from the two equations of part (e), $m\mathbf{a}_0^P - \mathbf{W}$ is parallel to $\mathbf{e}_{CP}$ (only the tension is left), and $dE_1^P/dt = -T\,\mathbf{e}_{CP}\cdot\mathbf{v}_1^P$; the constraint derivatives of part (c), the initial conditions and the separation condition were also checked symbolically with sympy.
- **Dimensions.** $[AC] = \sqrt{\mathrm{m}^2} = \mathrm{m}$; $[\omega^2\ell] = [\omega\dot\xi] = [g] = \mathrm{m}\cdot\mathrm{s}^{-2}$, the same as $[\ell\ddot\theta] = [\ddot\xi]$ in the equations of part (e); $[m\omega^2\xi] = [m\omega\dot\ell] = \mathrm{kg}\cdot\mathrm{m}\cdot\mathrm{s}^{-2} = \mathrm{N}$; $[\dot\theta(0)] = \dfrac{\mathrm{s}^{-1}\cdot\mathrm{m}}{\mathrm{m}} = \mathrm{s}^{-1}$; $[TR\omega\xi/D] = \dfrac{\mathrm{N}\cdot\mathrm{m}\cdot\mathrm{s}^{-1}\cdot\mathrm{m}}{\mathrm{m}} = \mathrm{W}$, a power; $[R\sin\omega t/h] = 1$, as $\tan\theta$ must be.

- **Check against the official key:** the centrifugal and Coriolis forces of part (d) coincide term by term with the official key (the key gives only these two forces).

Source: Problems.pdf (PDF page 24); key on PDF page 75

PROBLEM 37

Particle on a Spinning Elliptic Cone

Original statement · Problems.pdf, p. 25

Rotating Constraint Surface

A heavy point particle P with mass m moves without friction on a cone of elliptic cross-section as shown in the figure. We define an auxiliary reference frame $S_1 : \{O; \mathcal{B}_1\}$ attached to the cone. The cone is defined in term [sic] of two parameters ζ, β as

$$\text{Elliptic cone:} \quad \begin{cases} x_1 = \zeta \cos\beta \\ y_1 = 2\zeta \sin\beta \\ z_1 = \zeta \end{cases} \qquad (5)$$

The cone spins about its Oz_1 axis at a constant angular rate Ω with respect to the inertial reference frame $S_0 : \{O; \mathcal{B}_0\}$. At time $t = 0$, the reference frame S_1 coincides with S_0, the particle P is located at $\mathbf{r}^P_{1,0} = 2\mathbf{j}_1 + \mathbf{k}_1$, and has a velocity $\mathbf{v}^P_{1,0} = \mathbf{i}_1$.

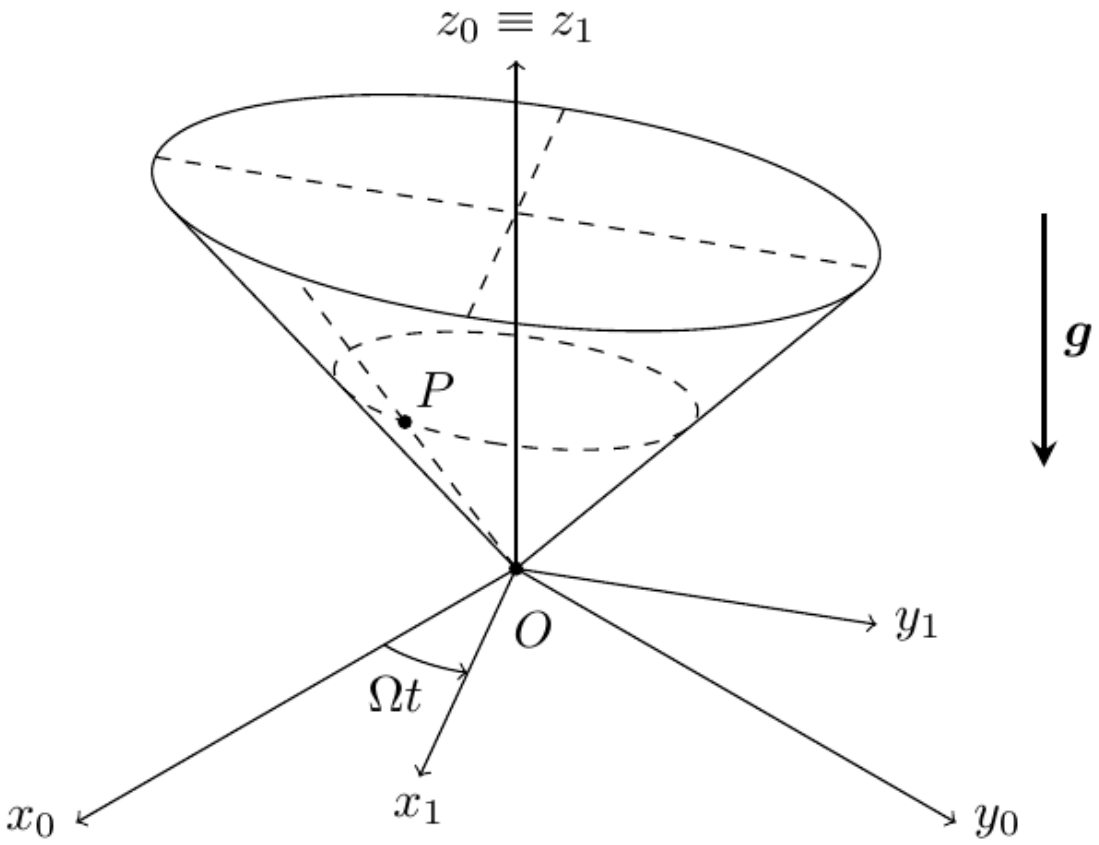


Figure: Elliptic cone with vertex O and vertical axis z0 = z1, frame S1 rotated by the angle Omega t about z0, particle P on the cone, gravity g downward (Problems.pdf, p. 25).

(a) How many degrees of freedom does the particle motion have?

(b) Obtain the general expression for the position, velocity, and acceleration of P **in reference frame** S_1 as a function of ζ, β and their derivatives. Remember to define any unit vectors that you use in terms of the vectors of $\mathcal{B}_0$.

(c) Identify and express mathematically all the forces acting on P as seen from S_1.

(d) Obtain two second-order differential equations for the evolution of ζ, β in time, which do not depend on any other unknowns in the problem.

(e) Find, justify, and write down one conservation law in this problem, including the value of any constants there might be.

(f) Find the equilibrium positions for P in S_1 for $\zeta > 0$ as a function of the constants of the problem.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a heavy point particle P of mass m slides on the inner surface of an elliptic cone with vertex O and vertical axis $Oz_1 \equiv Oz_0$. The cone spins about that axis with constant angular rate Ω relative to the inertial frame S_0.
- **Hypotheses:**
  1. $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ is inertial; gravity is $\mathbf{g} = -g\,\mathbf{k}_0$.
  2. The contact is smooth (no friction) and bilateral: the reaction of the cone is normal to its surface.
  3. Ω is constant and the vertex O is fixed, so O is a common origin of S_0 and S_1.
  4. The data $\mathbf{r}^P_{1,0} = 2\mathbf{j}_1 + \mathbf{k}_1$ and $\mathbf{v}^P_{1,0} = \mathbf{i}_1$ are read in SI units (m and m/s).
- **Why a moving frame:** in S_0 the surface of the cone moves, so the constraint depends explicitly on time. In the frame S_1 attached to the cone the surface is fixed and is described by the two parameters (ζ, β) of Eq. (5). The price is that S_1 is not inertial, so Newton's law in S_1 needs the inertia forces of Notes Eqs. 4.62–4.63 (Slides 27–28); they are known functions of the relative position and velocity.
- **Fixed (inertial) frame:** $S_0 = \{O; \mathcal{B}_0\}, \mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, with $\mathbf{k}_0$ vertical and upward.
- **Cone frame:** $S_1 = \{O; \mathcal{B}_1\}, \mathcal{B}_1 = \{\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, attached to the cone, with $\mathbf{k}_1 = \mathbf{k}_0$ along its axis and $\mathbf{i}_1, \mathbf{j}_1$ along the two axes of its elliptic sections. It turns the angle Ωt about $\mathbf{k}_0$ (the figure of the statement measures Ωt from x_0 to x_1) and coincides with S_0 at $t = 0$.
- **Surface basis:** the two tangent vectors $\mathbf{e}_\zeta, \mathbf{e}_\beta$ of the cone at P and its unit normal $\mathbf{n}$, deduced in the next section.
- **Coordinates:** (ζ, β), the parameters of Eq. (5): ζ is the height of P above the vertex and β the angular parameter of the elliptic section through P.

Symbol	Meaning	SI unit
m	mass of P	kg
g	gravitational acceleration	m/s^2
Ω	constant spin rate of the cone	rad/s
ζ	height of P above the vertex ($z_1 = \zeta$)	m
β	angular parameter of the elliptic section	rad
N	magnitude of the normal reaction	N

(a) Degrees of freedom

Pedagogical Justification: a free particle has 3 coordinates; each independent holonomic constraint removes one.

Eliminating the parameters of Eq. (5): $\cos\beta = x_1/\zeta$, $\sin\beta = y_1/(2\zeta)$ and $\zeta = z_1$, so $\cos^2\beta + \sin^2\beta = 1$ gives the single constraint equation of the surface, fixed in S_1:

$$\frac{x_1^2}{z_1^2} + \frac{y_1^2}{4z_1^2} = 1 \iff x_1^2 + \frac{y_1^2}{4} - z_1^2 = 0$$

Hence $\mathrm{DOF} = 3 - 1 = 2$, and (ζ, β) are suitable generalized coordinates. Each horizontal section $z_1 = \zeta$ is an ellipse with semi-axes ζ (along $\mathbf{i}_1$) and 2ζ (along $\mathbf{j}_1$).

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from above

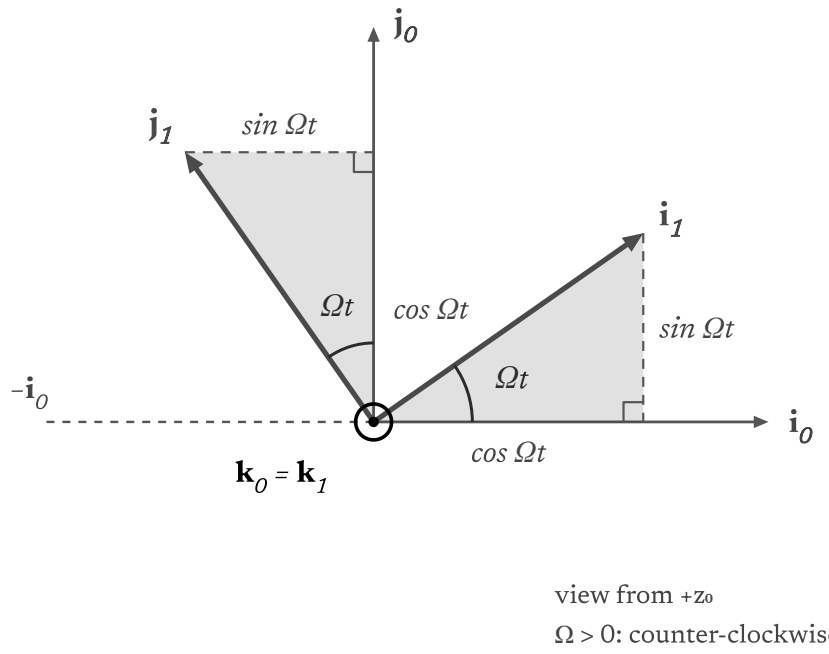


Figure 1. View from $+z_0$, with $\mathbf{k}_0 = \mathbf{k}_1$ out of the paper ($\odot$), so $\mathbf{j}_0 = \mathbf{k}_0 \times \mathbf{i}_0$ points up. $\mathbf{i}_1, \mathbf{j}_1$ are $\mathbf{i}_0, \mathbf{j}_0$ turned counter-clockwise by Ωt: $\mathbf{i}_1$ has legs $\cos\Omega t$ along $\mathbf{i}_0$ and $\sin\Omega t$ along $\mathbf{j}_0$; $\mathbf{j}_1$ has legs $\cos\Omega t$ along $\mathbf{j}_0$ and $\sin\Omega t$ along $-\mathbf{i}_0$, hence its minus sign.

The cone spins about $\mathbf{k}_0$, so we look down the axis from $+z_0$: $\mathbf{k}_0 = \mathbf{k}_1$ points out of the paper, towards the reader ($\odot$). Take $\mathbf{i}_0$ to the right; then $\mathbf{j}_0 = \mathbf{k}_0 \times \mathbf{i}_0$ points up the page (right-hand rule: fingers from $\mathbf{k}_0$, towards the reader, to $\mathbf{i}_0$, right; the thumb points up). With $\Omega > 0$ the cone turns counter-clockwise in this view (thumb along $\mathbf{k}_0$), and at time t it has turned Ωt from $\mathbf{i}_0$:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of the shaded right triangle on the right: the leg adjacent to Ωt lies along $\mathbf{i}_0$ and measures $\cos \Omega t$; the opposite leg lies along $+\mathbf{j}_0$ and measures $\sin \Omega t$.
- $\mathbf{j}_1$ starts on $\mathbf{j}_0$ (up) and turns the same angle counter-clockwise, so its tip leans to the left, towards $-\mathbf{i}_0$: its leg along $\mathbf{j}_0$ (adjacent) is $\cos \Omega t$ and its leg along $-\mathbf{i}_0$ (opposite) is $\sin \Omega t$. **That is where the minus sign of $\mathbf{j}_1$ comes from.**
- $\mathbf{k}_1 = \mathbf{k}_0$, the axis of the rotation.

$$\mathbf{i}_1 = \cos \Omega t\, \mathbf{i}_0 + \sin \Omega t\, \mathbf{j}_0, \qquad \mathbf{j}_1 = -\sin \Omega t\, \mathbf{i}_0 + \cos \Omega t\, \mathbf{j}_0, \qquad \mathbf{k}_1 = \mathbf{k}_0$$

The columns of the change-of-basis matrix are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos \Omega t & -\sin \Omega t & 0 \\ \sin \Omega t & \cos \Omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det [{}_0R_1] = \cos \Omega t \cos \Omega t - (-\sin \Omega t) \sin \Omega t = \cos^2 \Omega t + \sin^2 \Omega t = 1$$

In the product $[{}_0R_1][{}_0R_1]^T$ the diagonal entries are $\cos^2 \Omega t + \sin^2 \Omega t = 1$ (first and second rows) and 1 (third row), and the off-diagonal ones are $\cos \Omega t \sin \Omega t - \sin \Omega t \cos \Omega t = 0$ (first row times second) and 0 (any row times the third), so $[{}_0R_1][{}_0R_1]^T = I$.

Motion of S_1: the origin is fixed, $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$. For a simple rotation about the fixed vector $\mathbf{k}_0$ (Slide 16), $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{k}_1$. Check with the chain rule: $\left.\dfrac{d\mathbf{i}_1}{dt}\right|_0 = \dfrac{d\cos\Omega t}{d(\Omega t)}\,\Omega\,\mathbf{i}_0 + \dfrac{d\sin\Omega t}{d(\Omega t)}\,\Omega\,\mathbf{j}_0 = \Omega\,(-\sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{j}_0) = \Omega\,\mathbf{j}_1 = \Omega\,\mathbf{k}_1 \times \mathbf{i}_1$. Since Ω and $\mathbf{k}_1 = \mathbf{k}_0$ are constant, $\boldsymbol{\alpha}_{10} = \left.\dfrac{d\boldsymbol{\omega}_{10}}{dt}\right|_0 = \mathbf{0}$.

The elliptic section through P

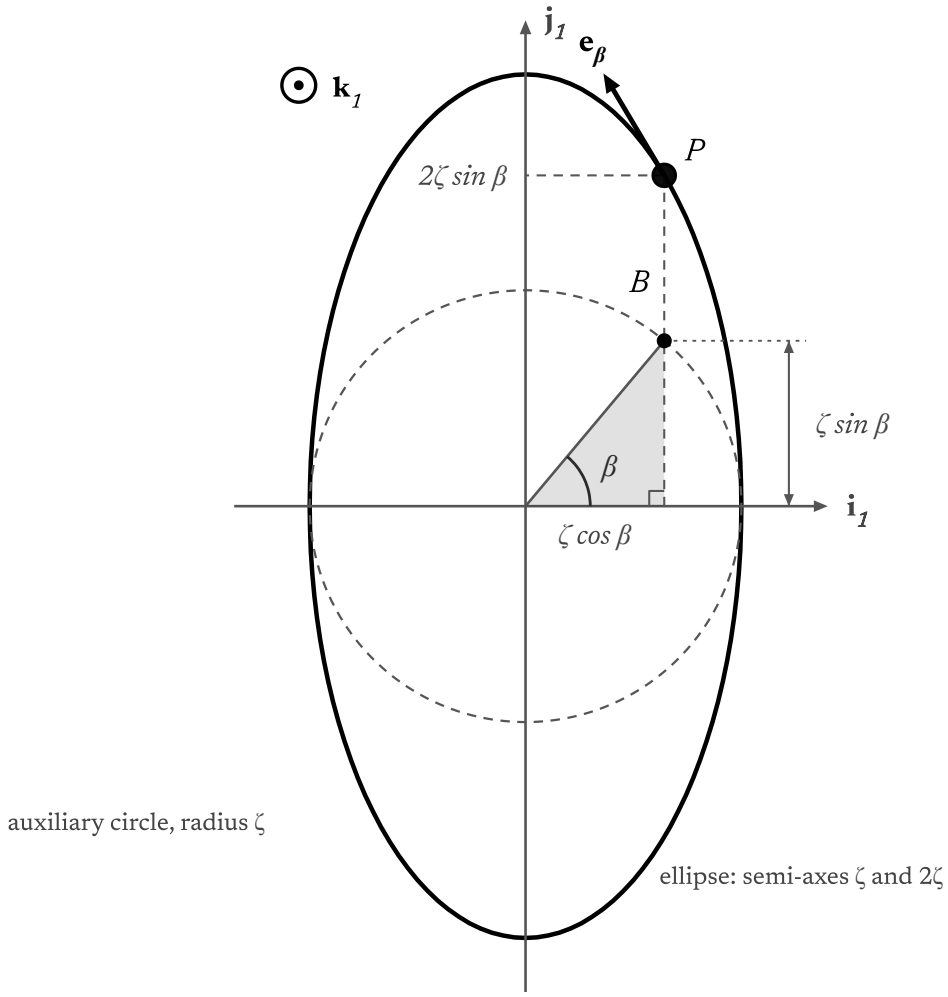


Figure 2. Section $z_1 = \zeta$ seen from above ($\mathbf{k}_1$ out of the paper). β is the polar angle of the point B of the dashed auxiliary circle of radius ζ; in the shaded triangle the legs are $\zeta\cos\beta$ and $\zeta\sin\beta$. The projection of P has the same $x_1 = \zeta\cos\beta$ and twice the height, $y_1 = 2\zeta\sin\beta$; $\mathbf{e}_\beta$ (orange) is tangent to the ellipse.

O is the origin of S_1, so $\mathbf{r}_1^P = \overrightarrow{OP}$ (Notes Eq. 3.16 with $\mathbf{r}_0^O = \mathbf{0}$). We build it in two steps: up the axis to the height of P, then across the horizontal section. The section $z_1 = \zeta$, seen from above, is the ellipse of semi-axes ζ along $\mathbf{i}_1$ and 2ζ along $\mathbf{j}_1$.

What β is. β is not the polar angle of P. Draw the auxiliary circle of radius ζ (dashed) and the point B of that circle at the polar angle β. In the shaded right triangle with vertex O and hypotenuse $OB = \zeta$, the leg adjacent to β lies along $\mathbf{i}_1$ and measures $\zeta\cos\beta$, and the

opposite leg measures $\zeta \sin\beta$. The point of the ellipse has the same $x_1 = \zeta\cos\beta$ and twice the height, $y_1 = 2\zeta\sin\beta$: the ellipse is the circle stretched by a factor 2 along $\mathbf{j}_1$. So β is the polar angle of B, not of P; the polar angle α of the horizontal projection of P satisfies

$$\tan\alpha = \frac{y_1}{x_1} = \frac{2\zeta\sin\beta}{\zeta\cos\beta} = 2\tan\beta$$

and $\alpha = \beta$ only at $\beta = 0, \pi/2, \pi, 3\pi/2$, on the axes of the ellipse. Adding the vertical part $\zeta\,\mathbf{k}_1$:

$$\mathbf{r}_1^P = \zeta\cos\beta\,\mathbf{i}_1 + 2\zeta\sin\beta\,\mathbf{j}_1 + \zeta\,\mathbf{k}_1$$

Check with the initial datum: $\zeta = 1, \beta = \pi/2$ give $2\mathbf{j}_1 + \mathbf{k}_1 = \mathbf{r}_{1,0}^P$.

Tangent vectors and normal of the cone. Two tangent vectors at P are the partial derivatives of $\mathbf{r}_1^P$ with respect to each parameter (they are not unit vectors). With $\frac{\partial}{\partial\beta}\cos\beta = -\sin\beta$ and $\frac{\partial}{\partial\beta}\sin\beta = \cos\beta$:

$$\mathbf{e}_\zeta \equiv \frac{\partial\mathbf{r}_1^P}{\partial\zeta} = \cos\beta\,\mathbf{i}_1 + 2\sin\beta\,\mathbf{j}_1 + \mathbf{k}_1, \qquad \frac{\partial\mathbf{r}_1^P}{\partial\beta} = \zeta\,\mathbf{e}_\beta, \quad \mathbf{e}_\beta \equiv -\sin\beta\,\mathbf{i}_1 + 2\cos\beta\,\mathbf{j}_1$$

$\mathbf{e}_\zeta$ runs along the generator through P (the straight line from the vertex) and $\mathbf{e}_\beta$ is tangent to the ellipse, drawn at P in the figure. A normal vector is their cross product, expanded component by component:

$$\mathbf{e}_\zeta \times \mathbf{e}_\beta = \begin{vmatrix} \mathbf{i}_1 & \mathbf{j}_1 & \mathbf{k}_1 \\ \cos\beta & 2\sin\beta & 1 \\ -\sin\beta & 2\cos\beta & 0 \end{vmatrix} = (0 - 2\cos\beta)\,\mathbf{i}_1 - (0 + \sin\beta)\,\mathbf{j}_1 + \left(2\cos^2\beta + 2\sin^2\beta\right)\mathbf{k}_1 = -2\cos\beta\,\mathbf{i}_1 - \sin\beta\,\mathbf{j}_1 + 2\,\mathbf{k}_1$$

Its squared norm is $4\cos^2\beta + \sin^2\beta + 4 = 5 + 3\cos^2\beta$, so the unit normal (pointing into the cone, upward) is

$$\mathbf{n} = \frac{-2\cos\beta\,\mathbf{i}_1 - \sin\beta\,\mathbf{j}_1 + 2\,\mathbf{k}_1}{\sqrt{5 + 3\cos^2\beta}}$$

(Check: the gradient of $x_1^2 + y_1^2/4 - z_1^2$ is $(2x_1, y_1/2, -2z_1) = \zeta(2\cos\beta, \sin\beta, -2)$, parallel to $\mathbf{n}$.)

The two principal vertical sections

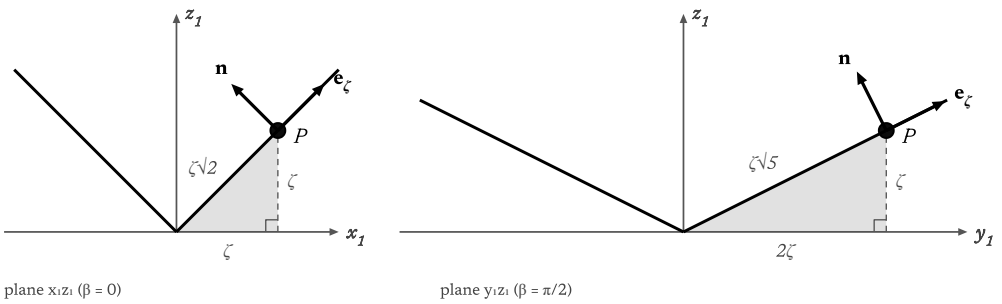


Figure 3. Principal vertical sections. Left, plane x_1z_1 ($\beta = 0$): legs ζ and ζ, hypotenuse $\zeta\sqrt{2}$, $\mathbf{e}_\zeta = \mathbf{i}_1 + \mathbf{k}_1$. Right, plane y_1z_1 ($\beta = \pi/2$): legs 2ζ and ζ, hypotenuse $\zeta\sqrt{5}$, $\mathbf{e}_\zeta = 2\mathbf{j}_1 + \mathbf{k}_1$. The normal $\mathbf{n}$ is perpendicular to the generator in each plane.

In the vertical plane x_1z_1 ($\beta = 0$ or π) the generators are the lines $z_1 = |x_1|$: in the shaded right triangle the horizontal leg is ζ and the vertical leg is ζ, so the hypotenuse is $\zeta\sqrt{2}$ and the generator makes 45° with the horizontal. In the plane y_1z_1 ($\beta = \pm\pi/2$) the generators are $z_1 = |y_1|/2$: horizontal leg 2ζ, vertical leg ζ, hypotenuse $\sqrt{4\zeta^2 + \zeta^2} = \zeta\sqrt{5}$. At $\beta = 0$, $\mathbf{e}_\zeta = \mathbf{i}_1 + \mathbf{k}_1$ and $\mathbf{n} = (-\mathbf{i}_1 + \mathbf{k}_1)/\sqrt{2}$; at $\beta = \pi/2$, $\mathbf{e}_\zeta = 2\mathbf{j}_1 + \mathbf{k}_1$ and $\mathbf{n} = (-\mathbf{j}_1 + 2\mathbf{k}_1)/\sqrt{5}$, perpendicular to the generator in each drawing. These sections are where the equilibria of part (f) lie.

Kinematics

(b) Position, velocity and acceleration in S_1

Pedagogical Justification: the relative derivatives $\frac{d}{dt}\big|_1$ treat $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ as constants, so only the components of $\mathbf{r}_1^P$ are differentiated (product and chain rules).

$$\mathbf{r}_1^P = \zeta\cos\beta\,\mathbf{i}_1 + 2\zeta\sin\beta\,\mathbf{j}_1 + \zeta\,\mathbf{k}_1$$

Component by component, with $\frac{d}{dt}\cos\beta = \frac{d\cos\beta}{d\beta}\dot\beta = -\dot\beta\sin\beta$ and $\frac{d}{dt}\sin\beta = \dot\beta\cos\beta$:

$$\frac{d}{dt}(\zeta\cos\beta) = \dot\zeta\cos\beta - \zeta\dot\beta\sin\beta, \qquad \frac{d}{dt}(2\zeta\sin\beta) = 2\dot\zeta\sin\beta + 2\zeta\dot\beta\cos\beta, \qquad \frac{d}{dt}\zeta = \dot\zeta$$

$$\mathbf{v}_1^P = \left(\dot\zeta\cos\beta - \zeta\dot\beta\sin\beta\right)\mathbf{i}_1 + 2\left(\dot\zeta\sin\beta + \zeta\dot\beta\cos\beta\right)\mathbf{j}_1 + \dot\zeta\,\mathbf{k}_1$$

Equivalently $\mathbf{v}_1^P = \dot\zeta\,\mathbf{e}_\zeta + \zeta\dot\beta\,\mathbf{e}_\beta$: the velocity is tangent to the cone.

Differentiating again. For the $\mathbf{i}_1$ component, $\frac{d}{dt}(\dot\zeta\cos\beta) = \ddot\zeta\cos\beta - \dot\zeta\dot\beta\sin\beta$ and $\frac{d}{dt}(\zeta\dot\beta\sin\beta) = \dot\zeta\dot\beta\sin\beta + \zeta\ddot\beta\sin\beta + \zeta\dot\beta^2\cos\beta$; for the $\mathbf{j}_1$ component, $\frac{d}{dt}(\dot\zeta\sin\beta) = \ddot\zeta\sin\beta + \dot\zeta\dot\beta\cos\beta$ and $\frac{d}{dt}(\zeta\dot\beta\cos\beta) = \dot\zeta\dot\beta\cos\beta + \zeta\ddot\beta\cos\beta - \zeta\dot\beta^2\sin\beta$. Collecting terms:

$$\mathbf{a}_1^P = \left[\left(\ddot\zeta - \zeta\dot\beta^2\right)\cos\beta - \left(2\dot\zeta\dot\beta + \zeta\ddot\beta\right)\sin\beta\right]\mathbf{i}_1 + 2\left[\left(\ddot\zeta - \zeta\dot\beta^2\right)\sin\beta + \left(2\dot\zeta\dot\beta + \zeta\ddot\beta\right)\cos\beta\right]\mathbf{j}_1 + \ddot\zeta\,\mathbf{k}_1$$

The unit vectors $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ are defined in terms of $\mathcal{B}_0$ in the change-of-basis section. For later use we write $\mathbf{r}_1^P = x_1\mathbf{i}_1 + y_1\mathbf{j}_1 + z_1\mathbf{k}_1$ and $\mathbf{v}_1^P = \dot x_1\mathbf{i}_1 + \dot y_1\mathbf{j}_1 + \dot z_1\mathbf{k}_1$ with the components above.

Forces and Equations of Motion

(c) Forces acting on P as seen from S_1

> **Pedagogical Justification:** in the non-inertial frame S_1 Newton's law reads $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63), where $\mathbf{F}$ are the actual forces and $\mathbf{F}_i$ the inertia forces of Notes Eq. 4.62 (Slide 28).

1. Weight: $\mathbf{W} = -mg\,\mathbf{k}_0 = -mg\,\mathbf{k}_1$.

2. Normal reaction. The contact is smooth, so the reaction is along the unit normal of the change-of-basis section:

$$\mathbf{N} = N\,\mathbf{n}, \qquad \mathbf{n} = \frac{-2\cos\beta\,\mathbf{i}_1 - \sin\beta\,\mathbf{j}_1 + 2\,\mathbf{k}_1}{\sqrt{5 + 3\cos^2\beta}}$$

with N an unknown scalar.

3. Inertia forces (Notes Eq. 4.62). Since $\mathbf{a}_0^O = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$, the force due to the acceleration of the origin and the Euler force vanish: $\mathbf{F}_A = -m\mathbf{a}_0^O = \mathbf{0}$, $\mathbf{F}_E = -m\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P = \mathbf{0}$.

Centrifugal force. With $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$, $\mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$:

$$\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = \Omega\,\mathbf{k}_1 \times (x_1\mathbf{i}_1 + y_1\mathbf{j}_1 + z_1\mathbf{k}_1) = \Omega\,(x_1\mathbf{j}_1 - y_1\mathbf{i}_1)$$

$$\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) = \Omega\,\mathbf{k}_1 \times \Omega\,(x_1\mathbf{j}_1 - y_1\mathbf{i}_1) = -\Omega^2\,(x_1\mathbf{i}_1 + y_1\mathbf{j}_1)$$

$$\mathbf{F}_C = -m\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) = m\Omega^2\,(x_1\mathbf{i}_1 + y_1\mathbf{j}_1) = m\Omega^2\zeta\,(\cos\beta\,\mathbf{i}_1 + 2\sin\beta\,\mathbf{j}_1)$$

which is $m\Omega^2\mathbf{r}_\perp$, radially outward from the axis (Slide 30).

Coriolis force.

$$\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = -2m\Omega\,\mathbf{k}_1 \times (\dot{x}_1\mathbf{i}_1 + \dot{y}_1\mathbf{j}_1 + \dot{z}_1\mathbf{k}_1) = 2m\Omega\,(\dot{y}_1\,\mathbf{i}_1 - \dot{x}_1\,\mathbf{j}_1)$$

$$\mathbf{F}_{COR} = 4m\Omega\left(\dot{\zeta}\sin\beta + \zeta\dot{\beta}\cos\beta\right)\mathbf{i}_1 - 2m\Omega\left(\dot{\zeta}\cos\beta - \zeta\dot{\beta}\sin\beta\right)\mathbf{j}_1$$

Equation of motion in S_1 (Notes Eq. 4.63):

$$m\mathbf{a}_1^P = -mg\,\mathbf{k}_1 + N\,\mathbf{n} + \mathbf{F}_C + \mathbf{F}_{COR}$$

(d) Two equations of motion free of the reaction

> **Pedagogical Justification:** the only unknown besides ζ and β is N, and $\mathbf{N}$ is perpendicular to the surface. Projecting the vector equation on the two tangent vectors $\mathbf{e}_\zeta$ and $\mathbf{e}_\beta$ eliminates it: $(-2\cos\beta)\cos\beta + (-\sin\beta)(2\sin\beta) + 2 = 0$ and $(-2\cos\beta)(-\sin\beta) + (-\sin\beta)(2\cos\beta) = 0$.

Projection on $\mathbf{e}_\zeta = (\cos\beta, 2\sin\beta, 1)$. Left-hand side:

$$\mathbf{a}_1^P \cdot \mathbf{e}_\zeta = \left(\ddot{\zeta} - \zeta\dot{\beta}^2\right)\left(\cos^2\beta + 4\sin^2\beta\right) + \left(2\dot{\zeta}\dot{\beta} + \zeta\ddot{\beta}\right)\left(-\sin\beta\cos\beta + 4\sin\beta\cos\beta\right) + \ddot{\zeta}$$

$$= \ddot{\zeta} + \left(\ddot{\zeta} - \zeta\dot{\beta}^2\right)\left(1 + 3\sin^2\beta\right) + 3\left(2\dot{\zeta}\dot{\beta} + \zeta\ddot{\beta}\right)\sin\beta\cos\beta$$

Right-hand side, term by term: $-mg\,\mathbf{k}_1 \cdot \mathbf{e}_\zeta = -mg$; $\mathbf{F}_C \cdot \mathbf{e}_\zeta = m\Omega^2\zeta\left(\cos^2\beta + 4\sin^2\beta\right) = m\Omega^2\zeta\left(1 + 3\sin^2\beta\right)$;

$$\mathbf{F}_{COR} \cdot \mathbf{e}_\zeta = 4m\Omega\left(\dot{\zeta}\sin\beta + \zeta\dot{\beta}\cos\beta\right)\cos\beta - 4m\Omega\left(\dot{\zeta}\cos\beta - \zeta\dot{\beta}\sin\beta\right)\sin\beta = 4m\Omega\zeta\dot{\beta}\left(\cos^2\beta + \sin^2\beta\right) = 4m\Omega\zeta\dot{\beta}$$

Dividing by m:

$$\boxed{\ddot{\zeta} + \left(\ddot{\zeta} - \zeta\dot{\beta}^2\right)\left(1 + 3\sin^2\beta\right) + 3\left(2\dot{\zeta}\dot{\beta} + \zeta\ddot{\beta}\right)\cos\beta\sin\beta = -g + \Omega^2\zeta\left(1 + 3\sin^2\beta\right) + 4\Omega\zeta\dot{\beta}}$$

Projection on $\mathbf{e}_\beta = (-\sin\beta, 2\cos\beta, 0)$. Left-hand side:

$$\mathbf{a}_1^P \cdot \mathbf{e}_\beta = \left(\ddot{\zeta} - \zeta\dot{\beta}^2\right)\left(-\sin\beta\cos\beta + 4\sin\beta\cos\beta\right) + \left(2\dot{\zeta}\dot{\beta} + \zeta\ddot{\beta}\right)\left(\sin^2\beta + 4\cos^2\beta\right) = 3\left(\ddot{\zeta} - \zeta\dot{\beta}^2\right)\sin\beta\cos\beta + \left(2\dot{\zeta}\dot{\beta} + \zeta\ddot{\beta}\right)\left(1 + 3\cos^2\beta\right)$$

Right-hand side: $-mg\,\mathbf{k}_1 \cdot \mathbf{e}_\beta = 0$; $\mathbf{F}_C \cdot \mathbf{e}_\beta = m\Omega^2\zeta\left(-\cos\beta\sin\beta + 4\sin\beta\cos\beta\right) = 3m\Omega^2\zeta\sin\beta\cos\beta$;

$$\mathbf{F}_{COR} \cdot \mathbf{e}_\beta = -4m\Omega\left(\dot{\zeta}\sin\beta + \zeta\dot{\beta}\cos\beta\right)\sin\beta - 4m\Omega\left(\dot{\zeta}\cos\beta - \zeta\dot{\beta}\sin\beta\right)\cos\beta = -4m\Omega\dot{\zeta}$$

Dividing by m:

$$\boxed{3\left(\ddot{\zeta} - \zeta\dot{\beta}^2\right)\cos\beta\sin\beta + \left(2\dot{\zeta}\dot{\beta} + \zeta\ddot{\beta}\right)\left(1 + 3\cos^2\beta\right) = 3\Omega^2\zeta\cos\beta\sin\beta - 4\Omega\dot{\zeta}}$$

These two equations contain only ζ, β and their derivatives. The initial conditions follow from the data: $\mathbf{r}_{1,0}^P = 2\mathbf{j}_1 + \mathbf{k}_1$ gives $\zeta_0 = 1$, $\zeta_0 \cos\beta_0 = 0$ and $2\zeta_0 \sin\beta_0 = 2$, so $\beta_0 = \pi/2$; $\mathbf{v}_{1,0}^P = \mathbf{i}_1$ gives $\dot\zeta_0 = 0$ ($\mathbf{k}_1$ component) and $\dot\zeta_0 \cos\beta_0 - \zeta_0\dot\beta_0 \sin\beta_0 = -\dot\beta_0 = 1$, so $\dot\beta_0 = -1$ rad/s; the $\mathbf{j}_1$ component $2(\dot\zeta_0 \sin\beta_0 + \zeta_0\dot\beta_0 \cos\beta_0) = 0$ is consistent (the initial velocity is tangent to the cone).

Answers and Checks

(a) Degrees of freedom

$\mathrm{DOF} = 2$, with generalized coordinates (ζ, β).

(b) Kinematics in S_1

$\mathbf{r}_1^P$, $\mathbf{v}_1^P$ and $\mathbf{a}_1^P$ as in the kinematics section, with $\mathbf{i}_1 = \cos\Omega t\,\mathbf{i}_0 + \sin\Omega t\,\mathbf{j}_0, \mathbf{j}_1 = -\sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{j}_0, \mathbf{k}_1 = \mathbf{k}_0$.

(c) Forces

Weight $-mg\,\mathbf{k}_1$, normal reaction $N\,\mathbf{n}$, centrifugal force $m\Omega^2\zeta(\cos\beta\,\mathbf{i}_1 + 2\sin\beta\,\mathbf{j}_1)$ and Coriolis force $2m\Omega(\dot y_1\,\mathbf{i}_1 - \dot x_1\,\mathbf{j}_1)$; the Euler force and the force of the origin's acceleration are zero.

(d) Equations of motion

The two boxed equations of the previous section, with $\zeta_0 = 1$ m, $\beta_0 = \pi/2$, $\dot\zeta_0 = 0$, $\dot\beta_0 = -1$ rad/s.

(e) Conservation law: mechanical energy in S_1

> **Pedagogical Justification:** E_1^P is conserved when every force that does work in S_1, inertia forces included, derives from a potential (Slide 29). Here: $\mathbf{N}\cdot\mathbf{v}_1^P = 0$ because the cone is fixed in S_1 and $\mathbf{v}_1^P$ is tangent to it; $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P = 0$ always (Slide 31); $\mathbf{F}_A = \mathbf{F}_E = \mathbf{0}$; the weight is a constant force in S_1 ($\mathbf{k}_1 = \mathbf{k}_0$) with potential mgz_1; and, since $\boldsymbol{\omega}_{10}$ is constant, the centrifugal force has potential $-\frac{1}{2}m\Omega^2 r_\perp^2$ (Slide 30).

Kinetic energy in S_1, expanding the squares of the components of $\mathbf{v}_1^P$:

$$|\mathbf{v}_1^P|^2 = \left(\dot\zeta^2\cos^2\beta - 2\zeta\dot\zeta\dot\beta\sin\beta\cos\beta + \zeta^2\dot\beta^2\sin^2\beta\right) + 4\left(\dot\zeta^2\sin^2\beta + 2\zeta\dot\zeta\dot\beta\sin\beta\cos\beta + \zeta^2\dot\beta^2\cos^2\beta\right) + \dot\zeta^2$$

$$T_1^P = \frac{m}{2}\left[\left(2 + 3\sin^2\beta\right)\dot\zeta^2 + \left(1 + 3\cos^2\beta\right)\zeta^2\dot\beta^2 + 6\zeta\dot\zeta\dot\beta\sin\beta\cos\beta\right]$$

Potential in S_1, with $z_1 = \zeta$ and $r_\perp^2 = x_1^2 + y_1^2 = \zeta^2\left(\cos^2\beta + 4\sin^2\beta\right) = \zeta^2\left(1 + 3\sin^2\beta\right)$:

$$V_1 = mg\zeta - \frac{1}{2}m\Omega^2\zeta^2\left(1+3\sin^2\beta\right)$$

Direct check that $E_1^P = T_1^P + V_1$ is constant. Since $\mathbf{v}_1^P = \dot\zeta\,\mathbf{e}_\zeta + \zeta\dot\beta\,\mathbf{e}_\beta$ and $\dfrac{dV_1}{dt} = -\left(\mathbf{W}+\mathbf{F}_C\right)\cdot\mathbf{v}_1^P$ (chain rule on $V_1(\mathbf{r}_1^P(t))$):

$$\frac{dE_1^P}{dt} = \left(m\mathbf{a}_1^P - \mathbf{W} - \mathbf{F}_C\right)\cdot\mathbf{v}_1^P = \dot\zeta\left(m\mathbf{a}_1^P - \mathbf{W} - \mathbf{F}_C\right)\cdot\mathbf{e}_\zeta + \zeta\dot\beta\left(m\mathbf{a}_1^P - \mathbf{W} - \mathbf{F}_C\right)\cdot\mathbf{e}_\beta$$

and by the two equations of part (d) the brackets equal $4m\Omega\zeta\dot\beta$ and $-4m\Omega\dot\zeta$, so

$$\frac{dE_1^P}{dt} = \dot\zeta\left(4m\Omega\zeta\dot\beta\right) + \zeta\dot\beta\left(-4m\Omega\dot\zeta\right) = 0$$

Conservation law:

$$\boxed{\tfrac{m}{2}\left[\left(2+3\sin^2\beta\right)\dot\zeta^2 + \left(1+3\cos^2\beta\right)\zeta^2\dot\beta^2 + 6\zeta\dot\zeta\dot\beta\sin\beta\cos\beta\right] + mg\zeta - \tfrac{m}{2}\Omega^2\zeta^2\left(1+3\sin^2\beta\right) = E_1^P = m\left(\tfrac{1}{2}v_0^2 + g\zeta_0 - 2\Omega^2\zeta_0^2\right)}$$

with $v_0 = |\mathbf{v}_{1,0}^P| = 1$ m/s and $\zeta_0 = 1$ m, so numerically $E_1^P = m\left(\tfrac{1}{2} + g - 2\Omega^2\right)$ in SI units.

The constant comes from $t=0$: $\zeta_0 = 1$ m, $\beta_0 = \pi/2$, $\dot\zeta_0 = 0$, $\dot\beta_0 = -1$ rad/s give $T_1^P(0) = \tfrac{m}{2}\,(1+0)\,(1)^2(-1)^2 = \tfrac{m}{2}$ (that is, $\tfrac{1}{2}m|\mathbf{v}_{1,0}^P|^2$ with $|\mathbf{v}_{1,0}^P| = 1$ m/s) and $V_1(0) = mg(1) - \tfrac{1}{2}m\Omega^2(1)^2(1+3) = m\left(g - 2\Omega^2\right)$, in joules when m, g and Ω are in SI units.

The angular momentum about Oz_0 is **not** conserved: the moment of $\mathbf{N}$ about the axis is $\left(\mathbf{r}_1^P\times\mathbf{n}\right)\cdot\mathbf{k}_1\,N = (x_1 n_y - y_1 n_x)\,N = \dfrac{\zeta\left(-\cos\beta\sin\beta + 4\sin\beta\cos\beta\right)}{\sqrt{5+3\cos^2\beta}}N = \dfrac{3\zeta\sin\beta\cos\beta}{\sqrt{5+3\cos^2\beta}}N$, which does not vanish in general because the section is not circular.

(f) Equilibrium positions in S_1

> **Pedagogical Justification:** an equilibrium in S_1 is a solution with ζ and β constant, so $\dot\zeta = \dot\beta = \ddot\zeta = \ddot\beta = 0$ in the two equations of part (d); the Coriolis terms vanish automatically.

The $\mathbf{e}_\beta$ equation reduces to $0 = 3\Omega^2\zeta\cos\beta\sin\beta = \tfrac{3}{2}\Omega^2\zeta\sin 2\beta$. With $\zeta > 0$ and $\Omega \neq 0$, $\sin 2\beta = 0$:

$$\beta \in \left\{0,\ \frac{\pi}{2},\ \pi,\ \frac{3\pi}{2}\right\}$$

The $\mathbf{e}_\zeta$ equation reduces to $0 = -g + \Omega^2 \zeta\left(1 + 3\sin^2\beta\right)$, so

$$\zeta_{eq} = \frac{g}{\Omega^2\left(1 + 3\sin^2\beta\right)}$$

β	ζ_{eq}	**Position** $\mathbf{r}_1^P$
0	g/Ω^2	$\frac{g}{\Omega^2}(\mathbf{i}_1 + \mathbf{k}_1)$
π	g/Ω^2	$\frac{g}{\Omega^2}(-\mathbf{i}_1 + \mathbf{k}_1)$
$\pi/2$	$g/(4\Omega^2)$	$\frac{g}{2\Omega^2}\mathbf{j}_1 + \frac{g}{4\Omega^2}\mathbf{k}_1$
$3\pi/2$	$g/(4\Omega^2)$	$-\frac{g}{2\Omega^2}\mathbf{j}_1 + \frac{g}{4\Omega^2}\mathbf{k}_1$

Force-balance check at $\beta = 0$: $\mathbf{n} = (-2, 0, 2)/\sqrt{8}$, so $\mathbf{0} = -mg\,\mathbf{k}_1 + m\Omega^2\zeta\,\mathbf{i}_1 + \frac{N}{\sqrt{2}}(-\mathbf{i}_1 + \mathbf{k}_1)$ gives $N = \sqrt{2}\,mg$ and $\Omega^2\zeta = g$. **At $\beta = \pi/2$:** $\mathbf{n} = (0, -1, 2)/\sqrt{5}$, so $\mathbf{0} = -mg\,\mathbf{k}_1 + 2m\Omega^2\zeta\,\mathbf{j}_1 + \frac{N}{\sqrt{5}}(-\mathbf{j}_1 + 2\mathbf{k}_1)$ gives $N = \frac{\sqrt{5}}{2}mg$ and $2\Omega^2\zeta = \frac{g}{2}$, i.e. $\zeta = g/(4\Omega^2)$. Both agree with the table.

Extension (stability, not asked): the Hessian of $V_1(\zeta, \beta)$ is diag $\left(-m\Omega^2,\ -3mg^2/\Omega^2\right)$ at $\beta = 0, \pi$ and diag $\left(-4m\Omega^2,\ 3mg^2/(16\Omega^2)\right)$ at $\beta = \pi/2, 3\pi/2$. V_1 has no minimum (along each generator it is a downward parabola in ζ), so the energy criterion cannot certify the stability of any of these equilibria.

Checks

- **Meaning of the equilibria:** they lie on the four generators contained in the principal planes x_1z_1 and y_1z_1 of the cone (the sections drawn above). Only there does the normal $\mathbf{n}$ lie in the vertical plane through the axis, so that it can balance the weight and the (radial) centrifugal force without a tangential remainder. On the steeper generators ($\beta = 0, \pi$, slope 45°) the particle must be higher, $\zeta = g/\Omega^2$, than on the flatter ones ($\beta = \pm\pi/2$, $\zeta = g/(4\Omega^2)$).
- **Gyroscopic coupling:** the Coriolis terms $4\Omega\zeta\dot\beta$ and $-4\Omega\dot\zeta$ couple the two equations antisymmetrically, which is why they cancel in dE_1^P/dt (Coriolis never does work).
- **Limit $\Omega \to 0$:** the centrifugal and Coriolis terms vanish, E_1^P becomes the ordinary energy $\frac{1}{2}mv^2 + mgz$, and $\zeta_{eq} \to \infty$: a fixed, vertex-down smooth cone has no equilibrium position with $\zeta > 0$. Basis: $t = 0$ gives $[{}_0R_1] = I$, the two frames coincide as stated.

- **Orthogonality:** $\mathbf{n}\cdot\mathbf{e}_\zeta = \mathbf{n}\cdot\mathbf{e}_\beta = 0$ (computed in part (d)); $\mathbf{e}_\zeta$ and $\mathbf{e}_\beta$ are not orthogonal in general, $\mathbf{e}_\zeta\cdot\mathbf{e}_\beta = 3\sin\beta\cos\beta$, which is why the projections of part (d) mix the two accelerations.
- **Dimensions:** in the $\mathbf{e}_\zeta$ equation, $[\ddot{\zeta}] = [\zeta\dot{\beta}^2] = [\dot{\zeta}\dot{\beta}] = [\zeta\ddot{\beta}] = \mathrm{m/s^2}$, $[g] = \mathrm{m/s^2}$, $[\Omega^2\zeta] = \mathrm{s^{-2}\,m}$, $[\Omega\zeta\dot{\beta}] = \mathrm{s^{-1}\,m\,s^{-1}}$; the $\mathbf{e}_\beta$ equation has the same terms. $[E_1^P] = \mathrm{kg\,m^2/s^2 = J}$; $[g/\Omega^2] = \mathrm{(m/s^2)/s^{-2} = m}$.
- **Symbolic check (sympy):** $\mathbf{a}_0^P$ obtained by differentiating $[{}_0R_1]\,\mathbf{r}_1^P$ twice equals $\mathbf{a}_1^P + \boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P) + 2\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P$ (Notes Eq. 3.27); both projections match the key; $dE_1^P/dt = 0$ along the equations of motion; the four equilibria satisfy both equations.
- **Check against the official key:** both equations of part (d) coincide with the official key (Problems.pdf, p. 75) term by term.

Source: Problems.pdf (PDF page 25); key on PDF page 75

PROBLEM 40

Spring-Coupled Particles on a Vertical Axis and a Revolving Ring

Original statement · Problems.pdf, p. 28

Two Particles in a Rotating Frame

Two heavy particles P and Q, each of mass m, are connected with a spring of stiffness k as shown in the figure. An inertial reference frame S_0 is defined with origin O and vector basis $\mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$. Particle P slides without friction along the vertical axis, while particle Q slides without friction on a ring of radius R contained in the horizontal plane defined by O, $\mathbf{i}_0$ and $\mathbf{j}_0$. The center D of this ring is located a distance ℓ from O and rotates around the vertical axis of S_0 with a known constant angular rate Ω. A rotating reference frame S_1 is defined so that the $\mathbf{j}_1$ vector follows the position of D. We define the generalized coordinates ζ (vertical position of P) and ϕ (angle of Q on the circle from $\mathbf{j}_1$) as shown.

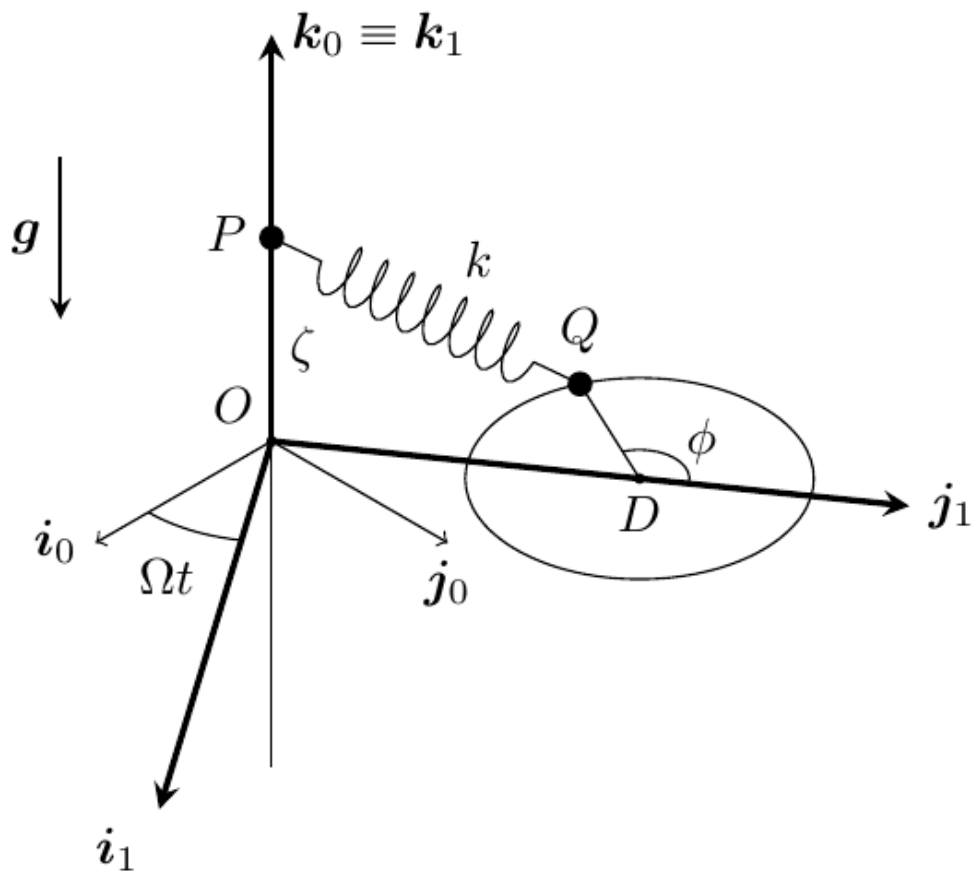


Figure: Particle P on the vertical axis k0 = k1, particle Q on a horizontal ring of centre D on the j1 axis, spring k between P and Q, frame S1 rotated by Omega t about k0 (Problems.pdf, p. 28).

(a) How many degrees of freedom does the system have? Define the vectors $\mathbf{i}_1, \mathbf{j}_1$ and $\mathbf{k}_1$ in terms of the vectors of $\mathcal{B}_0$.

(b) Write down the general expression of the position, velocity, and acceleration of P and Q with respect to S_1 in terms of ζ, ϕ and their derivatives.

(c) Identify and write down the general expression of all the forces acting on P and Q.

(d) Using S_1 for your analysis, write down the scalar equations of motion of P and Q.

(e) Starting from these equations, write down a system of two differential equations for ζ and ϕ, that do not depend on any other unknowns of the problem.

(f) Find one conservation law of the problem (i.e., a first integral of motion) in reference frame S_1.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** two heavy particles of mass m. P slides on the fixed vertical axis Oz_0; Q slides on a horizontal ring of radius R in the plane $z_0 = 0$ whose centre D, at distance ℓ from O, revolves about Oz_0 at constant rate Ω. A spring of stiffness k joins P and Q.
- **Hypotheses:**
  1. $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ is inertial, $\mathbf{k}_0$ vertical upward, gravity $-g\,\mathbf{k}_0$.
  2. Both contacts are smooth and bilateral: the reactions have no component along the direction in which each particle can move.
  3. The spring is linear and massless, with zero natural length (no natural length is given), so the force it exerts on P is $k\left(\mathbf{r}^Q - \mathbf{r}^P\right)$ and on Q the opposite.
  4. The ring moves rigidly with S_1 (only its centre's motion is prescribed; it does not spin relative to S_1, which is irrelevant anyway for a smooth circular wire).
  5. ϕ is measured from $\mathbf{j}_1$ in the positive sense about $\mathbf{k}_1$ (from $\mathbf{j}_1$ towards $-\mathbf{i}_1$), as in the figure; this is also the convention of the official key, whose ring reaction contains the radial vector $-\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{j}_1$.

- **Why S_1:** in S_1 the ring is fixed (centre at $\ell\,\mathbf{j}_1$), so Q moves on a fixed circle described by ϕ alone and P stays on the axis. The inertia forces (Notes Eq. 4.62) are the price; for P, on the rotation axis, they vanish.
- **Fixed (inertial) frame:** $S_0 = \{O; \mathcal{B}_0\}, \mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$.
- **Moving frame:** $S_1 = \{O; \mathcal{B}_1\}$ with the same origin, $\mathbf{k}_1 = \mathbf{k}_0$ and $\mathbf{j}_1$ pointing from O to D. S_1 is turned Ωt about $\mathbf{k}_0$ (the figure of the statement measures Ωt from $\mathbf{i}_0$ to $\mathbf{i}_1$), and we take $S_1 \equiv S_0$ at $t = 0$.
- **Ring basis:** $\{\mathbf{e}_r, \mathbf{e}_\phi, \mathbf{k}_1\}$ attached to Q: $\mathbf{e}_r$ from D to Q, $\mathbf{e}_\phi$ tangent to the ring; deduced in the next section.
- **Coordinates:** ζ, the height of P on the axis, and ϕ, the angle of Q on the ring.

Symbol	Meaning	SI unit
m	mass of each particle	kg
k	spring stiffness	N/m
R	radius of the ring	m
ℓ	distance OD	m
Ω	constant angular rate of D	rad/s
g	gravitational acceleration	m/s^2
ζ, ϕ	height of P, angle of Q on the ring	m, rad
$N_1, \dots, N_4$	reaction components	N

(a) Degrees of freedom and the basis $\mathcal{B}_1$

> **Pedagogical Justification:** count coordinates minus independent constraints for each particle.

P is free in space (3 coordinates) but constrained to the vertical axis ($x = y = 0$): 1 DOF, the height ζ. Q (3 coordinates) is constrained to the plane $z_1 = 0$ and to the circle $\left|\mathbf{r}^Q - \overrightarrow{OD}\right| = R$: 1 DOF, the angle ϕ. The motion of D is prescribed, so it adds no unknown. Hence

$$\text{DOF} = 6 - 4 = 2, \qquad \text{generalized coordinates } (\zeta, \phi)$$

The vectors of $\mathcal{B}_1$ in terms of $\mathcal{B}_0$ are deduced in the next section.

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from above

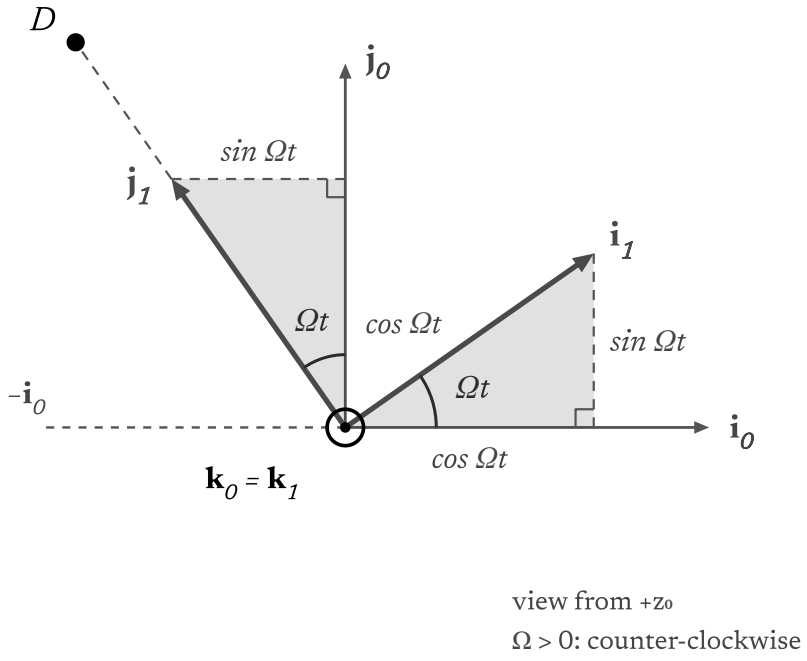


Figure 1. View from $+z_0$, with $\mathbf{k}_0 = \mathbf{k}_1$ out of the paper ($\odot$), so $\mathbf{j}_0 = \mathbf{k}_0 \times \mathbf{i}_0$ points up. $\mathbf{i}_1, \mathbf{j}_1$ are $\mathbf{i}_0, \mathbf{j}_0$ turned counter-clockwise by Ωt, and $\mathbf{j}_1$ points from O to the centre D of the ring. $\mathbf{j}_1$ has legs $\cos \Omega t$ along $\mathbf{j}_0$ and $\sin \Omega t$ along $-\mathbf{i}_0$, hence its minus sign.

S_1 turns about the vertical, so we look down the axis from $+z_0$: $\mathbf{k}_0 = \mathbf{k}_1$ points out of the paper ($\odot$). With $\mathbf{i}_0$ to the right, $\mathbf{j}_0 = \mathbf{k}_0 \times \mathbf{i}_0$ points up the page (right-hand rule: fingers from $\mathbf{k}_0$, towards the reader, to $\mathbf{i}_0$; thumb up). With $\Omega > 0$ the frame turns counter-clockwise in this view, and at time t it has turned Ωt:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of the right triangle whose leg adjacent to Ωt lies along $\mathbf{i}_0$ ($\cos \Omega t$) and whose opposite leg lies along $+\mathbf{j}_0$ ($\sin \Omega t$).
- $\mathbf{j}_1$, the direction of D, starts on $\mathbf{j}_0$ and turns the same angle counter-clockwise, so it leans to the left: adjacent leg $\cos \Omega t$ along $\mathbf{j}_0$, opposite leg $\sin \Omega t$ along $-\mathbf{i}_0$. **The minus sign of $\mathbf{j}_1$ is that leftward lean.**
- $\mathbf{k}_1 = \mathbf{k}_0$, the axis of the rotation.

$$\mathbf{i}_1 = \cos\Omega t\,\mathbf{i}_0 + \sin\Omega t\,\mathbf{j}_0, \qquad \mathbf{j}_1 = -\sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{j}_0, \qquad \mathbf{k}_1 = \mathbf{k}_0$$

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & -\sin\Omega t & 0 \\ \sin\Omega t & \cos\Omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos\Omega t\cos\Omega t - (-\sin\Omega t)\sin\Omega t = \cos^2\Omega t + \sin^2\Omega t = 1$$

Orthogonality: in $[{}_0R_1][{}_0R_1]^T$ each diagonal entry is a squared row norm, $\cos^2\Omega t + \sin^2\Omega t = 1$ (or 1 for the third row), and the first two rows have dot product $\cos\Omega t\sin\Omega t - \sin\Omega t\cos\Omega t = 0$ (the third row is orthogonal to both), so $[{}_0R_1][{}_0R_1]^T = I$.

Motion of S_1: $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$; simple rotation about $\mathbf{k}_0$ (Slide 16): $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{k}_1$, and $\boldsymbol{\alpha}_{10} = \mathbf{0}$ because Ω and $\mathbf{k}_0$ are constant. Check with the chain rule: $\left.\frac{d\mathbf{j}_1}{dt}\right|_0 = -\Omega\cos\Omega t\,\mathbf{i}_0 - \Omega\sin\Omega t\,\mathbf{j}_0 = -\Omega\,\mathbf{i}_1 = \Omega\,\mathbf{k}_1 \times \mathbf{j}_1$.

The ring and the position of Q

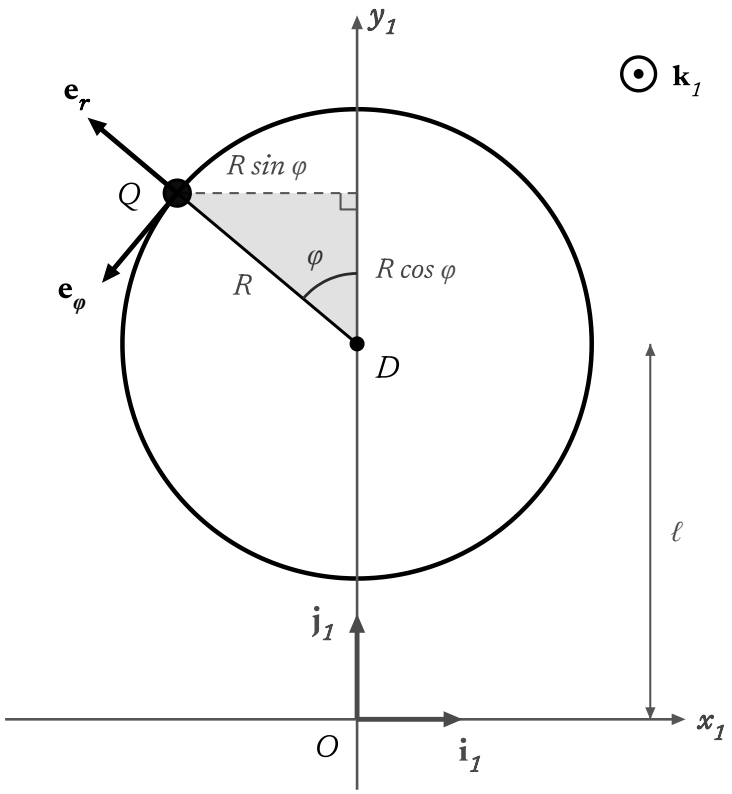


Figure 2. Plane of the ring, fixed in S_1, seen from above ($\mathbf{k}_1$ out of the paper). $OQ = \ell\,\mathbf{j}_1 + R\,\mathbf{e}_r$. In the shaded right triangle DFQ the hypotenuse is R, the leg along $\mathbf{j}_1$ is $R\cos\phi$ and the leg along $-\mathbf{i}_1$ is $R\sin\phi$. $\mathbf{i}_1, \mathbf{j}_1$ in blue, $\mathbf{e}_r, \mathbf{e}_\phi$ in orange.

The plane of the ring is fixed in S_1; we draw it from above ($\mathbf{k}_1$ out of the paper), with $\mathbf{i}_1$ to the right and $\mathbf{j}_1$ up, towards D. To go from O to Q we pass through the centre of the ring, because the distance from D is the one that stays constant:

$$\overrightarrow{OQ} = \overrightarrow{OD} + \overrightarrow{DQ}, \qquad \overrightarrow{OD} = \ell\,\mathbf{j}_1$$

$\overrightarrow{DQ}$ has length R and makes the angle ϕ with $\mathbf{j}_1$, measured counter-clockwise (towards $-\mathbf{i}_1$). Let F be the point of the line $D + s\,\mathbf{j}_1$ level with Q. In the shaded right triangle DFQ (right angle at F) the hypotenuse is $DQ = R$, the leg adjacent to ϕ is $DF = R\cos\phi$, along $+\mathbf{j}_1$, and the opposite leg is $FQ = R\sin\phi$, along $-\mathbf{i}_1$ because a counter-clockwise turn from $\mathbf{j}_1$ carries the point to the left. Hence the unit radial vector of the ring is

$$\mathbf{e}_r = \frac{\overrightarrow{DQ}}{R} = -\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{j}_1, \qquad \mathbf{r}_1^Q = \ell\,\mathbf{j}_1 + R\,\mathbf{e}_r = -R\sin\phi\,\mathbf{i}_1 + (\ell + R\cos\phi)\,\mathbf{j}_1$$

Relation with the standard polar angle. The usual polar angle ψ about D is measured from $\mathbf{i}_1$, with $\mathbf{e}_r = \cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1$. Matching requires $\cos\psi = -\sin\phi$ and $\sin\psi = \cos\phi$, which holds for $\psi = \phi + \frac{\pi}{2}$:

$$\cos\left(\phi + \tfrac{\pi}{2}\right) = \cos\phi\cos\tfrac{\pi}{2} - \sin\phi\sin\tfrac{\pi}{2} = -\sin\phi, \qquad \sin\left(\phi + \tfrac{\pi}{2}\right) = \sin\phi\cos\tfrac{\pi}{2} + \cos\phi\sin\tfrac{\pi}{2} = \cos\phi$$

(with $\cos\frac{\pi}{2} = 0$, $\sin\frac{\pi}{2} = 1$). So ϕ is the polar angle measured from $\mathbf{j}_1$, the direction $O \to D$; $\phi = 0$ is the point of the ring farthest from the axis and $\phi = \pi$ the nearest.

Ring basis. Differentiating $\mathbf{e}_r$ with respect to ϕ ($\frac{d}{d\phi}(-\sin\phi) = -\cos\phi$, $\frac{d}{d\phi}\cos\phi = -\sin\phi$) gives a vector of length $\sqrt{\cos^2\phi + \sin^2\phi} = 1$ tangent to the ring, pointing in the sense of increasing ϕ:

$$\mathbf{e}_\phi = \frac{d\mathbf{e}_r}{d\phi} = -\cos\phi\,\mathbf{i}_1 - \sin\phi\,\mathbf{j}_1, \qquad \mathbf{k}_1$$

Check: $\mathbf{e}_r \times \mathbf{e}_\phi = [(-\sin\phi)(-\sin\phi) - (\cos\phi)(-\cos\phi)]\,\mathbf{k}_1 = \mathbf{k}_1$, so $\{\mathbf{e}_r, \mathbf{e}_\phi, \mathbf{k}_1\}$ is right-handed and orthonormal ($\mathbf{e}_r \cdot \mathbf{e}_\phi = \sin\phi\cos\phi - \cos\phi\sin\phi = 0$). The inverse relations used below are $\mathbf{j}_1 = \cos\phi\,\mathbf{e}_r - \sin\phi\,\mathbf{e}_\phi$ and $\mathbf{i}_1 = -\sin\phi\,\mathbf{e}_r - \cos\phi\,\mathbf{e}_\phi$.

Position of P. P is on the axis at height ζ: $\mathbf{r}_1^P = \zeta\,\mathbf{k}_1$. The spring vector is then

$$\mathbf{r}^Q - \mathbf{r}^P = \ell\,\mathbf{j}_1 + R\,\mathbf{e}_r - \zeta\,\mathbf{k}_1 = -R\sin\phi\,\mathbf{i}_1 + (\ell + R\cos\phi)\,\mathbf{j}_1 - \zeta\,\mathbf{k}_1$$

Kinematics

(b) Relative kinematics of P and Q in S_1

Pedagogical Justification: O is the origin of S_1, so $\mathbf{r}_1 = \overrightarrow{OP}$ or $\overrightarrow{OQ}$ (Notes Eq. 3.16). The vectors $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ are constant in S_1; $\mathbf{e}_r$ and $\mathbf{e}_\phi$ change only through $\phi(t)$, so the chain rule gives $\left.\dfrac{d\mathbf{e}_r}{dt}\right|_1 = \dfrac{d\mathbf{e}_r}{d\phi}\dot\phi = \dot\phi\,\mathbf{e}_\phi$ and $\left.\dfrac{d\mathbf{e}_\phi}{dt}\right|_1 = \dfrac{d\mathbf{e}_\phi}{d\phi}\dot\phi = \dot\phi\,(\sin\phi\,\mathbf{i}_1 - \cos\phi\,\mathbf{j}_1) = -\dot\phi\,\mathbf{e}_r$.

Particle P:

$$\mathbf{r}_1^P = \zeta\,\mathbf{k}_1, \qquad \mathbf{v}_1^P = \dot\zeta\,\mathbf{k}_1, \qquad \mathbf{a}_1^P = \ddot\zeta\,\mathbf{k}_1$$

Particle Q: with $\overrightarrow{OD} = \ell\,\mathbf{j}_1$ constant in S_1,

$$\mathbf{r}_1^Q = \ell\,\mathbf{j}_1 + R\,\mathbf{e}_r = -R\sin\phi\,\mathbf{i}_1 + (\ell + R\cos\phi)\,\mathbf{j}_1$$

$$\mathbf{v}_1^Q = R\left.\frac{d\mathbf{e}_r}{dt}\right|_1 = R\dot\phi\,\mathbf{e}_\phi = -R\dot\phi\cos\phi\,\mathbf{i}_1 - R\dot\phi\sin\phi\,\mathbf{j}_1$$

By the product rule, $\mathbf{a}_1^Q = R\ddot\phi\,\mathbf{e}_\phi + R\dot\phi\left.\dfrac{d\mathbf{e}_\phi}{dt}\right|_1$:

$$\mathbf{a}_1^Q = R\ddot\phi\,\mathbf{e}_\phi - R\dot\phi^2\,\mathbf{e}_r$$

Forces and Equations of Motion

(c) Forces on P and Q (actual and inertia forces in S_1)

Pedagogical Justification: in S_1 we apply $m\mathbf{a}_1 = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63) with $\mathbf{F}_i$ from Notes Eq. 4.62. Here $\mathbf{a}_0^O = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$, so only the centrifugal and Coriolis forces survive (Slide 28).

Spring. With the spring vector of the change-of-basis section, the spring force on P is $\mathbf{F}_s^P = k\left(\mathbf{r}^Q - \mathbf{r}^P\right)$ and on Q it is $\mathbf{F}_s^Q = -k\left(\mathbf{r}^Q - \mathbf{r}^P\right)$. In the ring basis, using $\mathbf{j}_1 = \cos\phi\,\mathbf{e}_r - \sin\phi\,\mathbf{e}_\phi$:

$$\mathbf{F}_s^Q = -k\left(\ell\cos\phi + R\right)\mathbf{e}_r + k\ell\sin\phi\,\mathbf{e}_\phi + k\zeta\,\mathbf{k}_1$$

Forces on P:

- Weight $-mg\,\mathbf{k}_1$.
- Reaction of the smooth axis, perpendicular to $\mathbf{k}_1$: $\mathbf{N}_P = N_1\,\mathbf{i}_1 + N_2\,\mathbf{j}_1$.
- Spring $\mathbf{F}_s^P = k\left[-R\sin\phi\,\mathbf{i}_1 + (\ell + R\cos\phi)\,\mathbf{j}_1 - \zeta\,\mathbf{k}_1\right]$.
- Inertia forces: P is on the rotation axis. $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = \Omega\zeta\,\mathbf{k}_1\times\mathbf{k}_1 = \mathbf{0}$, so $\mathbf{F}_C^P = \mathbf{0}$; $\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P = \Omega\dot{\zeta}\,\mathbf{k}_1\times\mathbf{k}_1 = \mathbf{0}$, so $\mathbf{F}_{COR}^P = \mathbf{0}$.

Forces on Q:

- Weight $-mg\,\mathbf{k}_1$.
- Reaction of the smooth ring, perpendicular to the tangent $\mathbf{e}_\phi$: $\mathbf{N}_Q = N_3\,\mathbf{e}_r + N_4\,\mathbf{k}_1 = N_3\left(-\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{j}_1\right) + N_4\,\mathbf{k}_1$.
- Spring $\mathbf{F}_s^Q$ above.
- Centrifugal force. Since $\mathbf{r}_1^Q$ has no $\mathbf{k}_1$ component, $\boldsymbol{\omega}_{10}\times\left(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^Q\right) = -\Omega^2\mathbf{r}_1^Q$ (from $\mathbf{k}_1\times(\mathbf{k}_1\times\mathbf{i}_1) = -\mathbf{i}_1$ and $\mathbf{k}_1\times(\mathbf{k}_1\times\mathbf{j}_1) = -\mathbf{j}_1$), hence

$$\mathbf{F}_C^Q = m\Omega^2\mathbf{r}_1^Q = m\Omega^2\left(\ell\,\mathbf{j}_1 + R\,\mathbf{e}_r\right) = m\Omega^2\left(\ell\cos\phi + R\right)\mathbf{e}_r - m\Omega^2\ell\sin\phi\,\mathbf{e}_\phi$$

- Coriolis force. From $\mathbf{e}_r\times\mathbf{e}_\phi = \mathbf{k}_1$ (cyclic: $\mathbf{e}_\phi\times\mathbf{k}_1 = \mathbf{e}_r$), $\mathbf{k}_1\times\mathbf{e}_\phi = -\mathbf{e}_r$:

$$\mathbf{F}_{COR}^Q = -2m\Omega\,\mathbf{k}_1\times R\dot{\phi}\,\mathbf{e}_\phi = 2m\Omega R\dot{\phi}\,\mathbf{e}_r$$

- Euler force and force due to the origin's acceleration: zero for both particles.

(d) Scalar equations of motion in S_1

> **Pedagogical Justification:** project $m\mathbf{a}_1 = \mathbf{F} + \mathbf{F}_i$ for each particle on an orthonormal basis adapted to its constraint: $\{\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ for P and $\{\mathbf{e}_r, \mathbf{e}_\phi, \mathbf{k}_1\}$ for Q. This gives $3 + 3 = 6$ scalar equations for the 6 unknowns $\zeta, \phi, N_1, N_2, N_3, N_4$.

Particle P ($m\ddot{\zeta}\,\mathbf{k}_1 = -mg\,\mathbf{k}_1 + N_1\mathbf{i}_1 + N_2\mathbf{j}_1 + \mathbf{F}_s^P$):

$$\mathbf{i}_1: \quad 0 = N_1 - kR\sin\phi$$

$$\mathbf{j}_1: \quad 0 = N_2 + k\left(\ell + R\cos\phi\right)$$

$$\mathbf{k}_1: \quad m\ddot{\zeta} = -mg - k\zeta$$

Particle Q ($m\left(R\ddot{\phi}\,\mathbf{e}_\phi - R\dot{\phi}^2\mathbf{e}_r\right) = -mg\,\mathbf{k}_1 + N_3\mathbf{e}_r + N_4\mathbf{k}_1 + \mathbf{F}_s^Q + \mathbf{F}_C^Q + \mathbf{F}_{COR}^Q$):

$$\mathbf{e}_r: \quad -mR\dot{\phi}^2 = N_3 - k\left(\ell\cos\phi + R\right) + m\Omega^2\left(\ell\cos\phi + R\right) + 2m\Omega R\dot{\phi}$$

$$\mathbf{e}_\phi: \quad mR\ddot{\phi} = k\ell\sin\phi - m\Omega^2\ell\sin\phi$$

$$\mathbf{k}_1: \quad 0 = -mg + N_4 + k\zeta$$

Answers and Checks

(a) Degrees of freedom and $\mathcal{B}_1$

$\mathrm{DOF} = 2$, coordinates (ζ, ϕ); $\mathbf{i}_1 = \cos\Omega t\,\mathbf{i}_0 + \sin\Omega t\,\mathbf{j}_0, \mathbf{j}_1 = -\sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{j}_0, \mathbf{k}_1 = \mathbf{k}_0$.

(b) Kinematics in S_1

$\mathbf{r}_1^P = \zeta\,\mathbf{k}_1, \mathbf{v}_1^P = \dot{\zeta}\,\mathbf{k}_1, \mathbf{a}_1^P = \ddot{\zeta}\,\mathbf{k}_1$; $\mathbf{r}_1^Q = \ell\,\mathbf{j}_1 + R\,\mathbf{e}_r, \mathbf{v}_1^Q = R\dot{\phi}\,\mathbf{e}_\phi, \mathbf{a}_1^Q = R\ddot{\phi}\,\mathbf{e}_\phi - R\dot{\phi}^2\,\mathbf{e}_r$.

(c) Forces

As listed in the previous section: weights, reactions $\mathbf{N}_P = N_1\mathbf{i}_1 + N_2\mathbf{j}_1$ and $\mathbf{N}_Q = N_3\mathbf{e}_r + N_4\mathbf{k}_1$, spring forces $\pm k(\mathbf{r}^Q - \mathbf{r}^P)$, and on Q only the centrifugal force $m\Omega^2\mathbf{r}_1^Q$ and the Coriolis force $2m\Omega R\dot{\phi}\,\mathbf{e}_r$.

(d) Scalar equations of motion

The six projections of the previous section.

(e) Two equations free of the reactions

> **Pedagogical Justification:** the reactions appear only in the projections perpendicular to the allowed motions; the projections along the allowed motions ($\mathbf{k}_1$ for P, $\mathbf{e}_\phi$ for Q) are free of them.

Dividing the $\mathbf{k}_1$ equation of P and the $\mathbf{e}_\phi$ equation of Q by m and mR:

$$\boxed{\ddot{\zeta} + \frac{k}{m}\zeta = -g, \qquad \ddot{\phi} = \frac{\ell}{R}\left(\frac{k}{m} - \Omega^2\right)\sin\phi}$$

The other four equations then give the reactions once $\zeta(t), \phi(t)$ are known: $N_1 = kR\sin\phi$, $N_2 = -k\left(\ell + R\cos\phi\right)$, $N_3 = -mR\dot{\phi}^2 + \left(k - m\Omega^2\right)\left(R + \ell\cos\phi\right) - 2m\Omega R\dot{\phi}$, $N_4 = mg - k\zeta$. In particular the ring reaction has the form $\mathbf{N}_Q = N_3\left(-\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{j}_1\right) + N_4\,\mathbf{k}_1$ of the official key.

The two equations are **decoupled**: the vertical spring component acts only on P's coordinate, and the horizontal one only on Q's, because the spring force is linear in $\mathbf{r}^Q - \mathbf{r}^P$ and P moves along $\mathbf{k}_1$ while Q moves in the plane $\perp \mathbf{k}_1$.

(f) A conservation law in S_1

> **Pedagogical Justification:** in S_1 the constraints (axis and ring) are fixed and smooth, so the reactions do no work; the Coriolis force on Q is perpendicular to $\mathbf{v}_1^Q$ (Slide 31); the weight and spring are conservative; and with $\boldsymbol{\omega}_{10}$ constant the centrifugal force has potential $-\tfrac{1}{2}m\Omega^2 r_\perp^2$ (Slide 30). Hence the mechanical energy of the system in S_1 is conserved (Slide 29).

Potential energies in S_1: weight of P, $mg\zeta$ (weight of Q: $mg\cdot 0$); spring $\tfrac{1}{2}k\left|\mathbf{r}^Q - \mathbf{r}^P\right|^2$; centrifugal on Q, $-\tfrac{1}{2}m\Omega^2\left|\mathbf{r}_1^Q\right|^2$ (here $r_\perp^Q = |\mathbf{r}_1^Q|$). With

$$\left|\mathbf{r}^Q - \mathbf{r}^P\right|^2 = R^2\sin^2\phi + (\ell + R\cos\phi)^2 + \zeta^2 = \ell^2 + R^2 + 2\ell R\cos\phi + \zeta^2, \qquad \left|\mathbf{r}_1^Q\right|^2 = \ell^2 + R^2 + 2\ell R\cos\phi$$

and dropping the constant $\tfrac{1}{2}\left(k - m\Omega^2\right)\left(\ell^2 + R^2\right)$:

$$\boxed{E_1 = \frac{1}{2}m\left(\dot\zeta^2 + R^2\dot\phi^2\right) + mg\zeta + \frac{1}{2}k\zeta^2 + \left(k - m\Omega^2\right)\ell R\cos\phi = \text{const}}$$

Check with the equations of part (e) (chain rule $\frac{d}{dt}\cos\phi = -\dot\phi\sin\phi$, $\frac{d}{dt}\zeta^2 = 2\zeta\dot\zeta$):

$$\frac{dE_1}{dt} = m\dot\zeta\ddot\zeta + mR^2\dot\phi\ddot\phi + mg\dot\zeta + k\zeta\dot\zeta - \left(k - m\Omega^2\right)\ell R\dot\phi\sin\phi = m\dot\zeta\left(\ddot\zeta + g + \frac{k}{m}\zeta\right) + mR\dot\phi\left(R\ddot\phi - \left(\frac{k}{m} - \Omega^2\right)\ell\sin\phi\right) = 0$$

Because the equations are decoupled, each bracket vanishes separately, so in fact two first integrals exist: $\frac{1}{2}m\dot\zeta^2 + mg\zeta + \frac{1}{2}k\zeta^2 = \text{const}$ and $\frac{1}{2}mR^2\dot\phi^2 + \left(k - m\Omega^2\right)\ell R\cos\phi = \text{const}$. The constants are fixed by initial conditions, which the statement does not give.

Checks

- **Particle P:** $\ddot\zeta + (k/m)\zeta = -g$ is a harmonic oscillator about $\zeta_{eq} = -mg/k$ with angular frequency $\sqrt{k/m}$, unaffected by the rotation (it lies on the axis, where centrifugal and Coriolis forces vanish).
- **Particle Q:** $\ddot\phi = \frac{\ell}{R}\left(\frac{k}{m} - \Omega^2\right)\sin\phi$ is a pendulum equation. The spring pulls Q towards the axis (towards $\phi = \pi$, the point of the ring nearest O) and the centrifugal force pushes it outward (towards $\phi = 0$). If $\Omega^2 < k/m$ the stable position is $\phi = \pi$; if $\Omega^2 > k/m$ it is $\phi = 0$; at $\Omega^2 = k/m$ the two effects cancel exactly and $\dot\phi$ is constant. The Coriolis force only loads the ring radially (N_3).
- **Limits:** $\Omega \to 0$ gives a fixed ring and $\ddot\phi = \frac{k\ell}{mR}\sin\phi$; $\ell \to 0$ (ring centred on the axis) gives $\ddot\phi = 0$, uniform motion, since every point of the ring is then equidistant from the axis. Geometry: $\phi = 0$ gives $\mathbf{r}_1^Q = (\ell + R)\,\mathbf{j}_1$, the far point of the ring, and $\phi = \pi$ gives $(\ell - R)\,\mathbf{j}_1$.
- **Dimensions:** $[k/m] = \mathrm{N/(m\,kg)} = \mathrm{s^{-2}}$, so $[(k/m)\zeta] = \mathrm{m/s^2} = [g] = [\ddot\zeta]$; $[(\ell/R)(k/m - \Omega^2)] = \mathrm{s^{-2}} = [\ddot\phi]$; $[N_3]$: $[mR\dot\phi^2] = [kR] = [m\Omega R\dot\phi] = \mathrm{N}$; $[E_1] = \mathrm{kg\,m^2/s^2} = [k\zeta^2] = [k\ell R] = \mathrm{J}$.
- **Symbolic check (sympy):** $\mathbf{a}_0$ of both particles obtained by differentiating $[{}_0R_1]\,\mathbf{r}_1$ twice matches Notes Eq. 3.27; the six scalar equations, the four reactions, and $dE_1/dt = 0$ along the motion are confirmed.
- **Check against the official key** (Problems.pdf, p. 76): "One of the forces acting on Q is the reaction of the ring, $\mathbf{N}_Q = N_3(-\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{j}_1) + N_4\,\mathbf{k}_1$", which matches part (c).

Source: Problems.pdf (PDF page 28); key on PDF page 76

PROBLEM 41

Satellite Solar Panel: Composition of Rotations

Original statement · Problems.pdf, p. 29

Relative Kinematics

The angle θ describing the rotation of the solar panels of a reconnaissance satellite about the body-fixed axis x is an arbitrary function of time. The satellite spins about the z axis at the constant angular rate ω. Derive expressions for the absolute velocity and acceleration of point B in the figure relative to an inertial reference frame centered in A.

(Taken from Ginsberg, "Engineering Dynamics", Cambridge University Press, 2008.)

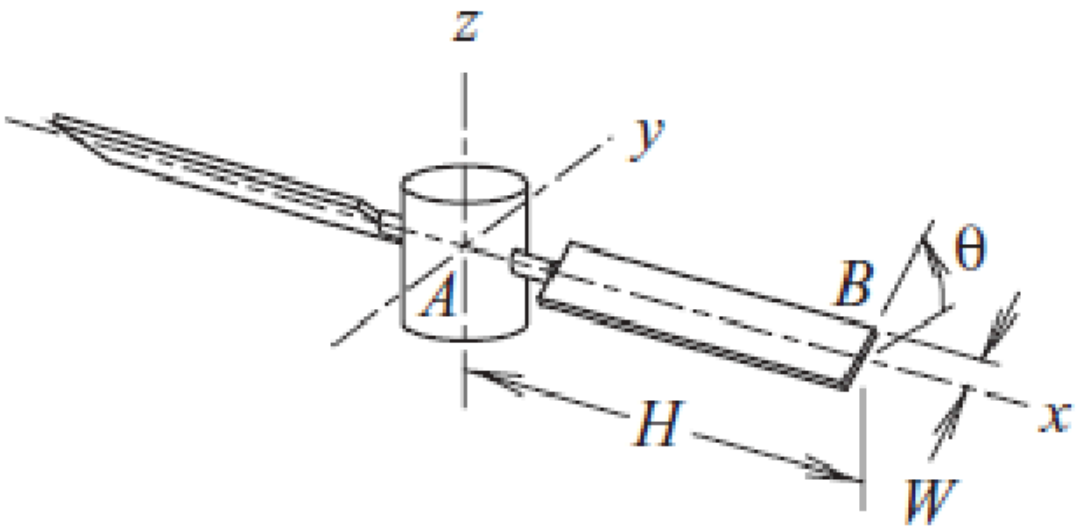


Figure: Satellite body spinning about z with a solar panel rotated by the angle θ about the body-fixed x axis, points A and B, dimensions H and W (Problems.pdf, p. 29).

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a satellite body spins about its z axis at constant rate ω; a solar panel turns about the body-fixed x axis through the prescribed angle $\theta(t)$. Point B is a corner of the panel.

- **Why relative motion.** The motion of B is the superposition of two simple motions: in the body frame S_1, B just describes a circle of radius W about the x axis; and S_1 itself just spins about z. Notes Eqs. 3.26–3.27 combine the two without writing the inertial position explicitly. A second route uses the panel frame S_2, in which B is fixed, and the composition of angular velocities (Notes Eqs. 3.13–3.15).
- **Hypotheses:**
  1. The frame centred at A with non-rotating axes is inertial, so the centre A of the satellite is fixed: $\mathbf{v}_0^A = \mathbf{a}_0^A = \mathbf{0}$.
  2. Body and panel are rigid; ω is constant; $\theta(t)$ is an arbitrary (known) function.
  3. **Geometry (read from the figure):** B is at distance H from A measured along x, and at distance W from the x axis; θ is measured about $+x$ from the body axis y, so for $\theta = 0$ the segment from the hinge axis to B points along $+y$.
- **Inertial frame:** $S_0 = \{A; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, basis $\mathcal{B}_0$.
- **Body frame:** $S_1 = \{A; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ (the axes x, y, z of the figure), basis $\mathcal{B}_1$, turning with the satellite about $\mathbf{k}_0 = \mathbf{k}_1$. It coincides with S_0 at $t = 0$.
- **Panel frame:** $S_2 = \{C; \mathbf{i}_2, \mathbf{j}_2, \mathbf{k}_2\}$, basis $\mathcal{B}_2$, attached to the panel, where $C = A + H\,\mathbf{i}_1$ is the point of the hinge axis at the abscissa of B; $\mathbf{i}_2 = \mathbf{i}_1$ is the hinge axis and $\mathbf{j}_2$ points from C to B.
- **Degrees of freedom:** none to determine; the motion is prescribed by ωt and $\theta(t)$ (kinematics only).

Symbol	Meaning	SI unit
H	axial distance from A to B (along x)	m
W	distance from B to the hinge axis x	m
ω	constant spin rate of the satellite about z	rad/s
$\theta(t)$	panel angle about x	rad

Change of Basis and Position Vector

First rotation: the body frame S_1 about $\mathbf{k}_0$

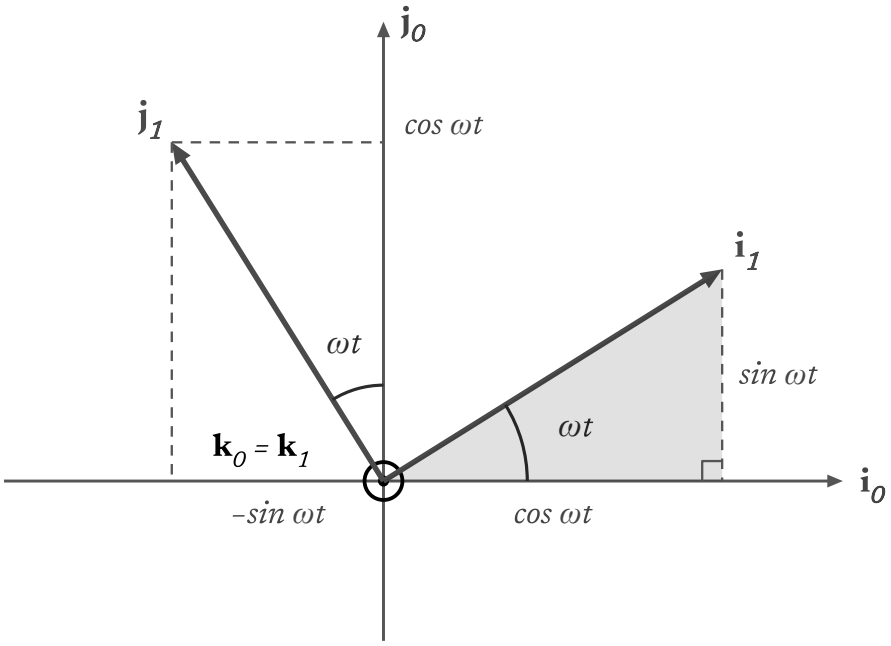


Figure 1. First rotation, seen from $+z$ with $\mathbf{k}_0 = \mathbf{k}_1$ out of the paper ($\odot$): $\mathbf{i}_1, \mathbf{j}_1$ are $\mathbf{i}_0, \mathbf{j}_0$ turned counter-clockwise by ωt. $\mathbf{i}_1$ has components $\cos\omega t$ along $\mathbf{i}_0$ and $\sin\omega t$ along $\mathbf{j}_0$; the tip of $\mathbf{j}_1$ moves to the side of $-\mathbf{i}_0$, hence its component $-\sin\omega t$.

The spin is about $\mathbf{k}_0$, so we look down that axis, from $+z$: $\mathbf{k}_0 = \mathbf{k}_1$ points out of the paper ($\odot$). With $\mathbf{i}_0$ to the right, the right-hand rule fixes $\mathbf{j}_0$: $\mathbf{i}_0 \times \mathbf{j}_0 = \mathbf{k}_0$ means that the fingers turning from $\mathbf{i}_0$ to $\mathbf{j}_0$ must curl counter-clockwise for the thumb to point at the reader, so $\mathbf{j}_0$ points up the page. A positive spin $\omega\,\mathbf{k}_0$ turns the body counter-clockwise in this view, and after a time t the body axes have turned the angle ωt:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of the shaded right triangle. The leg adjacent to ωt lies along $\mathbf{i}_0$ and measures $\cos\omega t$; the leg opposite to ωt lies along $+\mathbf{j}_0$ and measures $\sin\omega t$. So $\mathbf{i}_1 = \cos\omega t\,\mathbf{i}_0 + \sin\omega t\,\mathbf{j}_0$.
- $\mathbf{j}_1$ starts on $\mathbf{j}_0$ (up) and turns the same angle counter-clockwise, which moves its tip to the left, to the side of $-\mathbf{i}_0$. Its leg along $\mathbf{j}_0$ (adjacent) is $\cos\omega t$ and its leg along $\mathbf{i}_0$ (opposite) is $\sin\omega t$ pointing to $-\mathbf{i}_0$: $\mathbf{j}_1 = -\sin\omega t\,\mathbf{i}_0 + \cos\omega t\,\mathbf{j}_0$. **The minus sign comes from the tip of $\mathbf{j}_1$ moving towards $-\mathbf{i}_0$.**
- $\mathbf{k}_1 = \mathbf{k}_0$, the axis of the rotation.

The columns of the matrix are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1\cdot(\cos\omega t\cos\omega t - (-\sin\omega t)\sin\omega t) = \cos^2\omega t + \sin^2\omega t = 1$$

(expansion along the third row). The columns have unit norm, $\cos^2\omega t + \sin^2\omega t = 1$, and the first two are orthogonal, $\cos\omega t\,(-\sin\omega t) + \sin\omega t\cos\omega t = 0$, so $[{}_0R_1][{}_0R_1]^T = I$.

Motion of S_1. $\boldsymbol{\omega}_{10} = \omega\,\mathbf{k}_1$ and, ω and $\mathbf{k}_1 = \mathbf{k}_0$ being constant, $\boldsymbol{\alpha}_{10} = \mathbf{0}$. Check with Poisson's formula: by the chain rule, $\left.\frac{d\mathbf{i}_1}{dt}\right|_0 = -\omega\sin\omega t\,\mathbf{i}_0 + \omega\cos\omega t\,\mathbf{j}_0 = \omega\,\mathbf{j}_1$, and indeed $\boldsymbol{\omega}_{10}\times\mathbf{i}_1 = \omega\,\mathbf{k}_1\times\mathbf{i}_1 = \omega\,\mathbf{j}_1$.

Second rotation: the panel frame S_2 about the hinge $\mathbf{i}_1$

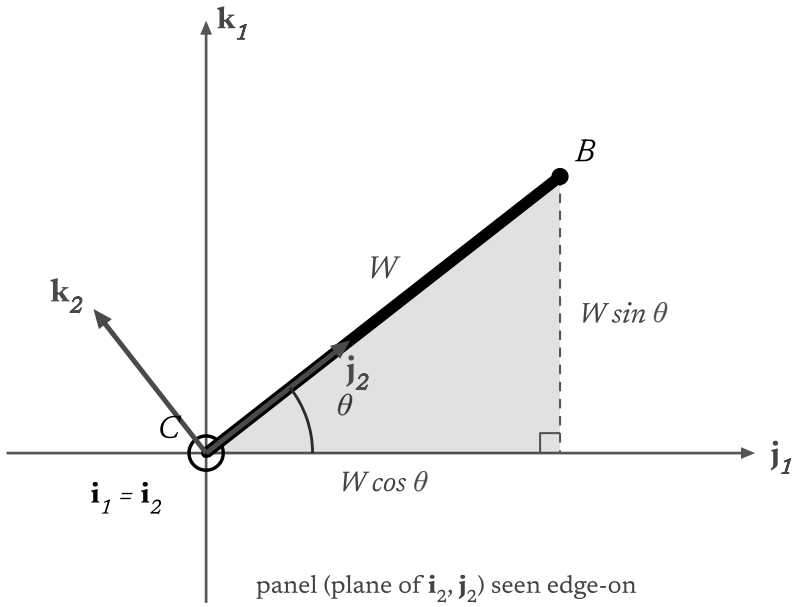


Figure 2. Second rotation, seen along the hinge with $\mathbf{i}_1 = \mathbf{i}_2$ out of the paper ($\odot$), so $\mathbf{j}_1$ points right and $\mathbf{k}_1 = \mathbf{i}_1\times\mathbf{j}_1$ up. The panel turns θ counter-clockwise: $CB = W\,\mathbf{j}_2$ is the hypotenuse of the shaded triangle, with legs $W\cos\theta$ along $\mathbf{j}_1$ and $W\sin\theta$ along $\mathbf{k}_1$.

The panel turns about the hinge axis $\mathbf{i}_1 = \mathbf{i}_2$, so we look along that axis with $\mathbf{i}_1$ out of the paper ($\odot$). With $\mathbf{j}_1$ to the right, $\mathbf{k}_1 = \mathbf{i}_1\times\mathbf{j}_1$: the fingers go from $\mathbf{i}_1$ (towards the reader) to $\mathbf{j}_1$ (right) and the thumb points up the page, so $\mathbf{k}_1$ points up. The angle θ is measured about $+\mathbf{i}_1$, which is counter-clockwise in this view, from $\mathbf{j}_1$:

- $\mathbf{j}_2$ has the leg $\cos\theta$ along $\mathbf{j}_1$ (adjacent) and the leg $\sin\theta$ along $+\mathbf{k}_1$ (opposite): $\mathbf{j}_2 = \cos\theta\,\mathbf{j}_1 + \sin\theta\,\mathbf{k}_1$.
- $\mathbf{k}_2$ starts on $\mathbf{k}_1$ and its tip moves to the side of $-\mathbf{j}_1$: $\mathbf{k}_2 = -\sin\theta\,\mathbf{j}_1 + \cos\theta\,\mathbf{k}_1$.
- $\mathbf{i}_2 = \mathbf{i}_1$.

$$[{}_1R_2] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}, \qquad \det[{}_1R_2] = 1 \cdot \left(\cos^2\theta + \sin^2\theta\right) = 1, \qquad \boldsymbol{\omega}_{21} = \dot{\theta}\,\mathbf{i}_1$$

The columns are orthonormal (norms $\cos^2\theta + \sin^2\theta = 1$, scalar product of the last two $-\cos\theta\sin\theta + \sin\theta\cos\theta = 0$), so $[{}_1R_2][{}_1R_2]^T = I$; hence $[{}_0R_2] = [{}_0R_1][{}_1R_2]$ is also a proper rotation, with $\det[{}_0R_2] = 1 \cdot 1 = 1$ (Notes Eq. 1.16).

Position vector of B

To go from A to B we pass through the hinge point C, because the distance from the hinge axis is the one that stays constant:

$$\mathbf{r}_1^B = \overrightarrow{AC} + \overrightarrow{CB}$$

- $\overrightarrow{AC} = H\,\mathbf{i}_1$: C is on the hinge axis, at the abscissa H of B.
- $\overrightarrow{CB} = W\,\mathbf{j}_2$ is fixed in S_2. In the shaded triangle of the hinge view the hypotenuse is $|CB| = W$, the leg adjacent to θ is $W\cos\theta$ along $\mathbf{j}_1$ and the leg opposite to θ is $W\sin\theta$ along $\mathbf{k}_1$; this is the second column of $[{}_1R_2]$ times W.

$$\mathbf{r}_1^B = H\,\mathbf{i}_1 + W\cos\theta\,\mathbf{j}_1 + W\sin\theta\,\mathbf{k}_1$$

Checks: $\theta = 0$ gives $H\,\mathbf{i}_1 + W\,\mathbf{j}_1$ (panel along $+y$, as in the reading of the figure); $\theta = \pi/2$ gives $H\,\mathbf{i}_1 + W\,\mathbf{k}_1$ (panel turned up, along $+z$).

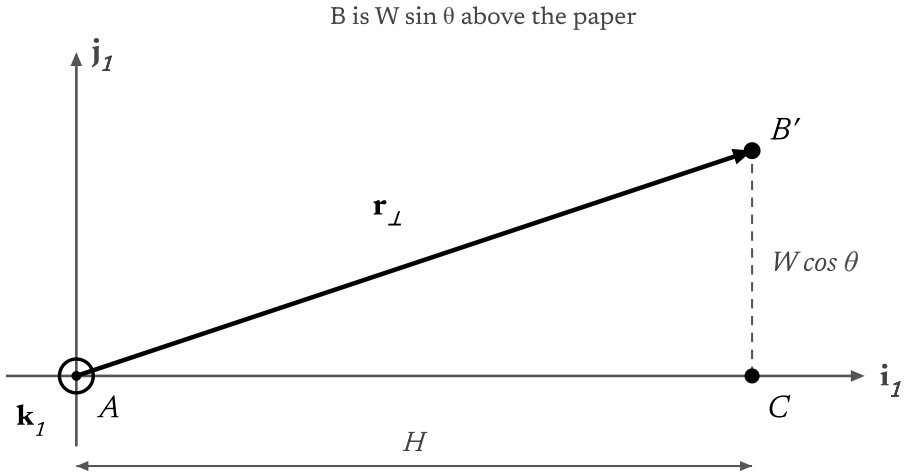


Figure 3. Body plane seen from $+z$ ($\mathbf{k}_1$ out of the paper): C is at H along $\mathbf{i}_1$ and the corner B projects onto the point B' at $W\cos\theta$ along $\mathbf{j}_1$ (B itself is $W\sin\theta$ above the paper). $\mathbf{r}_\perp = H\,\mathbf{i}_1 + W\cos\theta\,\mathbf{j}_1$ is the distance vector from the spin axis, the one that enters the centripetal term.

Distance to the spin axis. The spin is about $\mathbf{k}_1$, so the part of $\mathbf{r}_1^B$ that matters for the rotation is the one perpendicular to $\mathbf{k}_1$: dropping the component $W\sin\theta\,\mathbf{k}_1$ (the height of B above the body plane), $\mathbf{r}_\perp = H\,\mathbf{i}_1 + W\cos\theta\,\mathbf{j}_1$, the vector from the axis to the projection B' in the figure above. It reappears in the centripetal term.

Kinematics

1. Relative motion of B in the body frame S_1

Pedagogical Justification: the body frame rotates at constant rate with fixed origin, so Notes Eq. 3.27 keeps only three terms; the relative quantities are derivatives in S_1, where $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ are constant.

Only the components of $\mathbf{r}_1^B$ change. By the chain rule, $\frac{d}{dt}(W\cos\theta) = -W\sin\theta\,\dot\theta$ and $\frac{d}{dt}(W\sin\theta) = W\cos\theta\,\dot\theta$ (H is constant):

$$\mathbf{v}_1^B = \left.\frac{d\mathbf{r}_1^B}{dt}\right|_1 = -W\dot\theta\sin\theta\,\mathbf{j}_1 + W\dot\theta\cos\theta\,\mathbf{k}_1$$

Differentiating again with the product rule, $\frac{d}{dt}(\dot\theta\sin\theta) = \ddot\theta\sin\theta + \dot\theta\,\dot\theta\cos\theta$ and $\frac{d}{dt}(\dot\theta\cos\theta) = \ddot\theta\cos\theta - \dot\theta\,\dot\theta\sin\theta$:

$$\mathbf{a}_1^B = \left.\frac{d\mathbf{v}_1^B}{dt}\right|_1 = -W\left(\ddot{\theta}\sin\theta + \dot{\theta}^2\cos\theta\right)\mathbf{j}_1 + W\left(\ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta\right)\mathbf{k}_1$$

(circular motion of radius W about the x axis: tangential part $W\ddot{\theta}\,\mathbf{k}_2$ and normal part $-W\dot{\theta}^2\,\mathbf{j}_2$).

2. Absolute velocity of B

> **Pedagogical Justification:** Notes Eq. 3.26 (Slide 24), $\mathbf{v}_0^B = \mathbf{v}_0^A + \mathbf{v}_1^B + \boldsymbol{\omega}_{10}\times\mathbf{r}_1^B$, with $\mathbf{v}_0^A = \mathbf{0}$.

Cross products: $\mathbf{k}_1\times\mathbf{i}_1 = \mathbf{j}_1, \mathbf{k}_1\times\mathbf{j}_1 = -\mathbf{i}_1, \mathbf{k}_1\times\mathbf{k}_1 = \mathbf{0}$:

$$\boldsymbol{\omega}_{10}\times\mathbf{r}_1^B = \omega\mathbf{k}_1\times(H\,\mathbf{i}_1 + W\cos\theta\,\mathbf{j}_1 + W\sin\theta\,\mathbf{k}_1) = H\omega\,\mathbf{j}_1 - W\omega\cos\theta\,\mathbf{i}_1$$

(only $\mathbf{r}_\perp$ contributes). Adding $\mathbf{v}_1^B$ component by component:

$$\boxed{\mathbf{v}_0^B = -W\omega\cos\theta\,\mathbf{i}_1 + \left(H\omega - W\dot{\theta}\sin\theta\right)\mathbf{j}_1 + W\dot{\theta}\cos\theta\,\mathbf{k}_1}$$

The components are given in the body basis $\mathcal{B}_1$; in $\mathcal{B}_0$ they follow from $\{\mathbf{v}_0^B\}_0 = [{}_0R_1]\{\mathbf{v}_0^B\}_1$.

3. Absolute acceleration of B

> **Pedagogical Justification:** Notes Eq. 3.27 (Slide 25) with $\mathbf{a}_0^A = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$: $\mathbf{a}_0^B = \mathbf{a}_1^B + \boldsymbol{\omega}_{10}\times\left(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^B\right) + 2\boldsymbol{\omega}_{10}\times\mathbf{v}_1^B$.

Centripetal term (reusing $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^B$ from step 2):

$$\boldsymbol{\omega}_{10}\times\left(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^B\right) = \omega\mathbf{k}_1\times(-W\omega\cos\theta\,\mathbf{i}_1 + H\omega\,\mathbf{j}_1) = -W\omega^2\cos\theta\,\mathbf{j}_1 - H\omega^2\,\mathbf{i}_1$$

i.e. $-\omega^2\mathbf{r}_\perp$ with $\mathbf{r}_\perp = H\,\mathbf{i}_1 + W\cos\theta\,\mathbf{j}_1$, the vector of the body-plane figure.

Coriolis term:

$$2\boldsymbol{\omega}_{10}\times\mathbf{v}_1^B = 2\omega\mathbf{k}_1\times\left(-W\dot{\theta}\sin\theta\,\mathbf{j}_1 + W\dot{\theta}\cos\theta\,\mathbf{k}_1\right) = -2W\omega\dot{\theta}\sin\theta\,(\mathbf{k}_1\times\mathbf{j}_1) = 2W\omega\dot{\theta}\sin\theta\,\mathbf{i}_1$$

Sum, collecting the $\mathbf{i}_1$, $\mathbf{j}_1$ and $\mathbf{k}_1$ components of $\mathbf{a}_1^B$, the centripetal term and the Coriolis term:

$$\boxed{\mathbf{a}_0^B = \left(2W\omega\dot\theta\sin\theta - H\omega^2\right)\mathbf{i}_1 - \left(W\ddot\theta\sin\theta + W\dot\theta^2\cos\theta + W\omega^2\cos\theta\right)\mathbf{j}_1 + \left(W\ddot\theta\cos\theta - W\dot\theta^2\sin\theta\right)\mathbf{k}_1}$$

4. Second route: panel frame and composition of angular velocities

> **Pedagogical Justification:** B is fixed in S_2, so $\mathbf{v}_2^B = \mathbf{a}_2^B = \mathbf{0}$ and Notes Eqs. 3.26–3.27 applied to S_2 (origin C) need only $\boldsymbol{\omega}_{20}$, $\boldsymbol{\alpha}_{20}$ and the motion of C. Angular velocities add (Notes Eqs. 3.13–3.15, Slide 19); angular accelerations do not simply add.

$$\boldsymbol{\omega}_{20} = \boldsymbol{\omega}_{21} + \boldsymbol{\omega}_{10} = \dot\theta\,\mathbf{i}_1 + \omega\,\mathbf{k}_1$$

Its derivative in S_0 by the Coriolis theorem (Notes Eq. 3.8), with $\frac{d}{dt}\big|_1\left(\dot\theta\,\mathbf{i}_1 + \omega\,\mathbf{k}_1\right) = \ddot\theta\,\mathbf{i}_1$ (ω constant):

$$\boldsymbol{\alpha}_{20} = \left.\frac{d\boldsymbol{\omega}_{20}}{dt}\right|_1 + \boldsymbol{\omega}_{10}\times\boldsymbol{\omega}_{20} = \ddot\theta\,\mathbf{i}_1 + \omega\mathbf{k}_1\times\dot\theta\,\mathbf{i}_1 = \ddot\theta\,\mathbf{i}_1 + \omega\dot\theta\,\mathbf{j}_1$$

Motion of C (fixed in S_1 at $H\mathbf{i}_1$): $\mathbf{v}_0^C = \omega\mathbf{k}_1\times H\mathbf{i}_1 = H\omega\,\mathbf{j}_1$, $\mathbf{a}_0^C = \omega\mathbf{k}_1\times H\omega\,\mathbf{j}_1 = -H\omega^2\,\mathbf{i}_1$. With $\mathbf{r}_2^B = \overrightarrow{CB} = W\cos\theta\,\mathbf{j}_1 + W\sin\theta\,\mathbf{k}_1$:

$$\boldsymbol{\omega}_{20}\times\mathbf{r}_2^B = \dot\theta W\cos\theta\,\mathbf{k}_1 - \dot\theta W\sin\theta\,\mathbf{j}_1 - \omega W\cos\theta\,\mathbf{i}_1$$

so $\mathbf{v}_0^B = \mathbf{v}_0^C + \boldsymbol{\omega}_{20}\times\mathbf{r}_2^B$ reproduces the boxed velocity. For the acceleration, $\mathbf{a}_0^B = \mathbf{a}_0^C + \boldsymbol{\alpha}_{20}\times\mathbf{r}_2^B + \boldsymbol{\omega}_{20}\times(\boldsymbol{\omega}_{20}\times\mathbf{r}_2^B)$:

$$\boldsymbol{\alpha}_{20}\times\mathbf{r}_2^B = \left(\ddot\theta\,\mathbf{i}_1 + \omega\dot\theta\,\mathbf{j}_1\right)\times(W\cos\theta\,\mathbf{j}_1 + W\sin\theta\,\mathbf{k}_1) = W\ddot\theta\cos\theta\,\mathbf{k}_1 - W\ddot\theta\sin\theta\,\mathbf{j}_1 + W\omega\dot\theta\sin\theta\,\mathbf{i}_1$$

$$\boldsymbol{\omega}_{20}\times\left(\boldsymbol{\omega}_{20}\times\mathbf{r}_2^B\right) = \left(\dot\theta\,\mathbf{i}_1 + \omega\,\mathbf{k}_1\right)\times\left(-\omega W\cos\theta\,\mathbf{i}_1 - \dot\theta W\sin\theta\,\mathbf{j}_1 + \dot\theta W\cos\theta\,\mathbf{k}_1\right)$$

$$= -\dot\theta^2 W \sin\theta\,\mathbf{k}_1 - \dot\theta^2 W\cos\theta\,\mathbf{j}_1 - \omega^2 W\cos\theta\,\mathbf{j}_1 + \omega\dot\theta W\sin\theta\,\mathbf{i}_1$$

(using $\mathbf{i}_1\times\mathbf{j}_1=\mathbf{k}_1, \mathbf{i}_1\times\mathbf{k}_1=-\mathbf{j}_1, \mathbf{k}_1\times\mathbf{i}_1=\mathbf{j}_1, \mathbf{k}_1\times\mathbf{j}_1=-\mathbf{i}_1$). Adding $\mathbf{a}_0^C = -H\omega^2\,\mathbf{i}_1$: the $\mathbf{i}_1$ component is $2W\omega\dot\theta\sin\theta - H\omega^2$, the $\mathbf{j}_1$ component $-W(\ddot\theta\sin\theta + \dot\theta^2\cos\theta + \omega^2\cos\theta)$, the $\mathbf{k}_1$ component $W(\ddot\theta\cos\theta - \dot\theta^2\sin\theta)$: the same boxed result. Note that the Coriolis term of route 1, $2W\omega\dot\theta\sin\theta\,\mathbf{i}_1$, appears in route 2 split in two equal halves, one from $\boldsymbol{\alpha}_{20}$ (the term $\omega\dot\theta\,\mathbf{j}_1$ that makes angular accelerations not add) and one from the double cross product.

Answers and Checks

Absolute velocity and acceleration of B

Both routes give, in the body basis $\mathcal{B}_1$ (to be multiplied by $[{}_0R_1]$ for the components in $\mathcal{B}_0$):

$$\mathbf{v}_0^B = -W\omega\cos\theta\,\mathbf{i}_1 + \Big(H\omega - W\dot\theta\sin\theta\Big)\,\mathbf{j}_1 + W\dot\theta\cos\theta\,\mathbf{k}_1$$

$$\mathbf{a}_0^B = \Big(2W\omega\dot\theta\sin\theta - H\omega^2\Big)\,\mathbf{i}_1 - \Big(W\ddot\theta\sin\theta + W\dot\theta^2\cos\theta + W\omega^2\cos\theta\Big)\,\mathbf{j}_1 + \Big(W\ddot\theta\cos\theta - W\dot\theta^2\sin\theta\Big)\,\mathbf{k}_1$$

No forces or masses are given, so the problem has no dynamics to write.

Checks

- **Reading of the terms.** $-H\omega^2\mathbf{i}_1 - W\omega^2\cos\theta\,\mathbf{j}_1 = -\omega^2\mathbf{r}_\perp$ is the centripetal acceleration of the spin; the $W\ddot\theta$ and $W\dot\theta^2$ terms are the tangential and normal accelerations of the panel rotation; $2W\omega\dot\theta\sin\theta\,\mathbf{i}_1$ is the Coriolis coupling of the two rotations, present only when the panel moves ($\dot\theta\neq 0$) with a velocity component normal to the spin axis.
- **Limits:** $\dot\theta = 0$ (panel locked) gives the rigid-body motion $\mathbf{v}_0^B = \boldsymbol{\omega}_{10}\times\mathbf{r}_1^B$, $\mathbf{a}_0^B = -\omega^2\mathbf{r}_\perp$; $\omega = 0$ (no spin) leaves the circular motion of radius W about the hinge.
- **Orthogonality:** $\mathbf{j}_2\cdot\mathbf{k}_2 = -\cos\theta\sin\theta + \sin\theta\cos\theta = 0$ and $\mathbf{i}_1\cdot\mathbf{j}_1 = -\cos\omega t\sin\omega t + \sin\omega t\cos\omega t = 0$.
- **Dimensions:** every velocity component is of the form $[W\omega] = [H\omega] = [W\dot\theta] = \mathrm{m\,s^{-1}}$; every acceleration component $[W\omega\dot\theta] = [H\omega^2] = [W\ddot\theta] = [W\dot\theta^2] = \mathrm{m\,s^{-2}}$.
- **Verification (sympy):** $[{}_0R_1]$, $[{}_1R_2]$, $[{}_0R_2]$ are proper rotations; the inertial first and second derivatives of $[{}_0R_1]\{\mathbf{r}_1^B\}$ equal the Notes Eq. 3.26 and 3.27 expressions; both routes give exactly the official key.

- **Check against the official key:** the key of `Problems.pdf` (p. 76) for Problem 41 (absolute velocity and acceleration of B) is reproduced exactly by both routes, as the sympy verification above records; no discrepancy.

Source: Problems.pdf (PDF page 29); key on PDF page 76

PROBLEM 47

Two Repelling Particles on Concentric Wires in a Rotating Plane

Original statement · Problems.pdf, p. 32

Relative Equilibrium

We consider a system of two heavy particles of equal mass m. The first particle P is constrained to move without friction along a circular wire of radius a. The second particle Q is constrained to move without friction along a circular wire of radius $2a$. Both wires are contained in a plane Ox_1y_1 that rotates at a constant, known angular rate Ω about the axis $Oy_1 \equiv Oy_0$ with respect to a fixed reference frame S_0 with axes $Ox_0y_0z_0$ (not shown in the figure). We also define a moving reference frame S_1 with axes $Ox_1y_1z_1$. The two particles repel each other with a force that is proportional to their distance, with constant of proportionality, k.

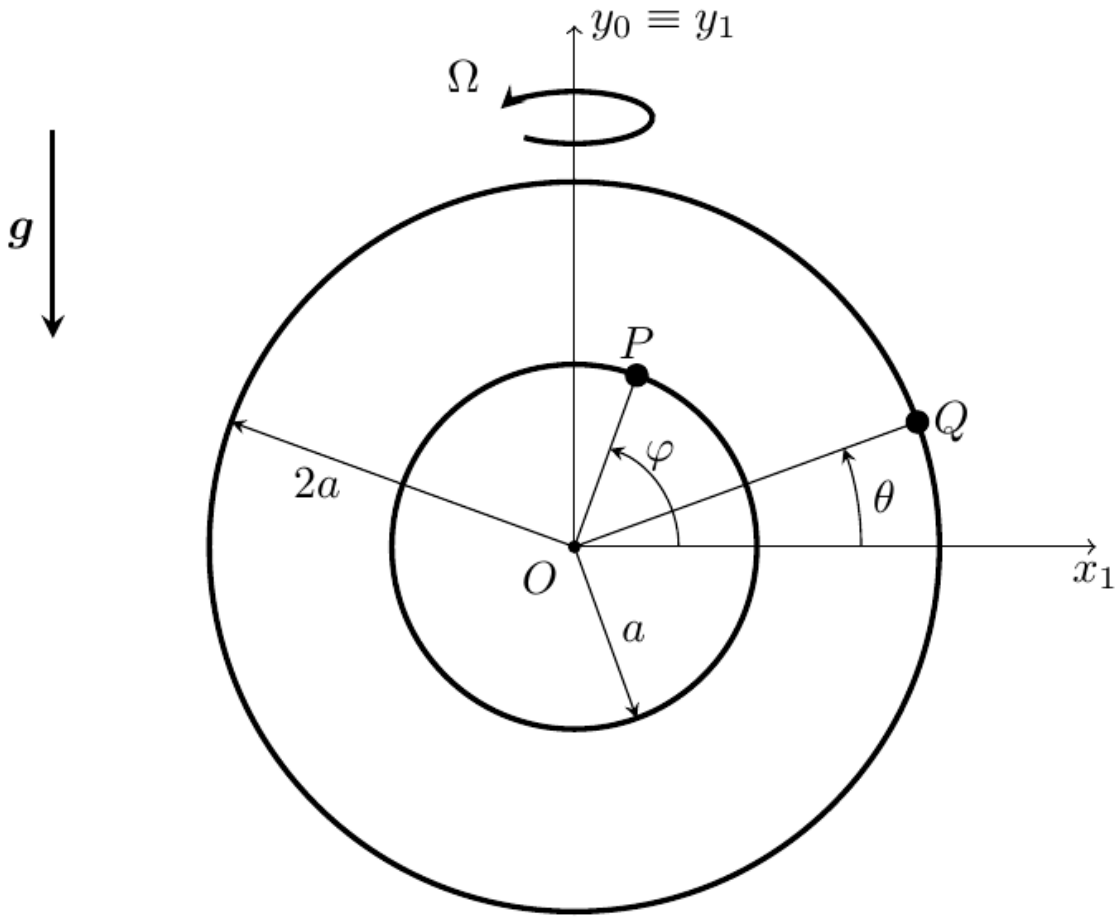


Figure: Concentric circular wires of radii a and 2a centred at O in the plane Ox1y1, particle P at angle varphi and particle Q at angle theta from x1, the plane spins with rate Omega about the vertical axis y0 = y1, gravity g downward (Problems.pdf, p. 32).

(a) How many configuration degrees of freedom has this system?

(b) How many independent components of the reaction forces on the particles need to be considered? Write down the reaction force on each particle.

(c) Write down the relative accelerations of P and Q with respect to S_1 as a function of the coordinates of the problem and their derivatives.

(d) Express the applied forces on the particles as a function of the coordinates of the problem.

(e) Express the fictitious forces of inertia on the particles as seen from S_1, as a function of the coordinates of the problem and their derivatives.

(f) Write down a system of ordinary differential equations to determine the evolution of the coordinates of the problem as a function of time, free of any other unknowns. You do not need to solve this system (since no initial conditions have been provided) but we assume that we know how to solve it, so that from this point we know the coordinates as a function of time.

(g) Determine the values of the reaction forces from the wires as a function of time.

(h) Is the mechanical energy of this system conserved in S_1? If so, write down the corresponding equation.

(i) Determine the values of $\alpha = \dfrac{ka}{mg}$ and $\beta = \dfrac{a\Omega^2}{g}$ so that $\varphi = 30^\circ, \theta = -60^\circ$ is an equilibrium position of the system in the relative motion.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** two heavy particles of mass m; P on a circular wire of radius a and Q on a concentric circular wire of radius $2a$, both centred at O and lying in the plane Ox_1y_1. The plane spins about the vertical axis $Oy_1 \equiv Oy_0$ with constant rate Ω. The particles

repel each other with a force k times their distance.

- **Hypotheses:**
  1. $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ is inertial; $\mathbf{j}_0$ is vertical upward and gravity is $-g\,\mathbf{j}_0$ (the figure draws $\mathbf{g}$ antiparallel to y_1).
  2. Smooth, bilateral wires: each reaction is perpendicular to the tangent of its wire.
  3. The repulsion acts along the line PQ: on P it is $k\left(\mathbf{r}^P - \mathbf{r}^Q\right)$, on Q it is $k\left(\mathbf{r}^Q - \mathbf{r}^P\right)$ (action and reaction).
  4. φ and θ are measured from Ox_1 in the positive sense about $\mathbf{k}_1$, as in the figure. The sense of Ω only affects the sign of the Coriolis terms; we take $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}_0$ with Ω a signed constant.

- **Why S_1:** both wires are fixed in S_1, so each particle has a single angular coordinate there. The inertia forces (Notes Eq. 4.62) are the price, and with $\boldsymbol{\omega}_{10}$ constant and O fixed only the centrifugal and Coriolis forces appear.
- **Frames:** $S_0 = \{O; \mathcal{B}_0\}$ (inertial) and $S_1 = \{O; \mathcal{B}_1\}$, $\mathcal{B}_1 = \{\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, attached to the plane of the wires, with common origin; $\mathbf{j}_1 = \mathbf{j}_0$ is the axis of rotation and $S_1 \equiv S_0$ at $t = 0$.
- **Polar bases:** $\{\mathbf{e}_r^P, \mathbf{e}_\varphi, \mathbf{k}_1\}$ at P and $\{\mathbf{e}_r^Q, \mathbf{e}_\theta, \mathbf{k}_1\}$ at Q, deduced in the next section.

Symbol	Meaning	SI unit
m	mass of each particle	kg
a	radius of the inner wire (outer: $2a$)	m
k	repulsion constant	N/m
Ω	constant rotation rate of the plane	rad/s
g	gravitational acceleration	m/s^2
φ, θ	angular positions of P and Q from Ox_1	rad
$\alpha = ka/(mg), \beta = a\Omega^2/g$	dimensionless parameters	1

(a) Configuration degrees of freedom

> **Pedagogical Justification:** each particle on a wire (a curve) loses 2 of its 3 coordinates.

P: 3 coordinates minus 2 constraints ($z_1 = 0, x_1^2 + y_1^2 = a^2$) = 1. Q: 3 minus 2 ($z_1 = 0, x_1^2 + y_1^2 = 4a^2$) = 1. The motion of the plane is prescribed. Hence

$$\text{DOF} = 2, \qquad \text{coordinates } (\varphi, \theta)$$

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from above

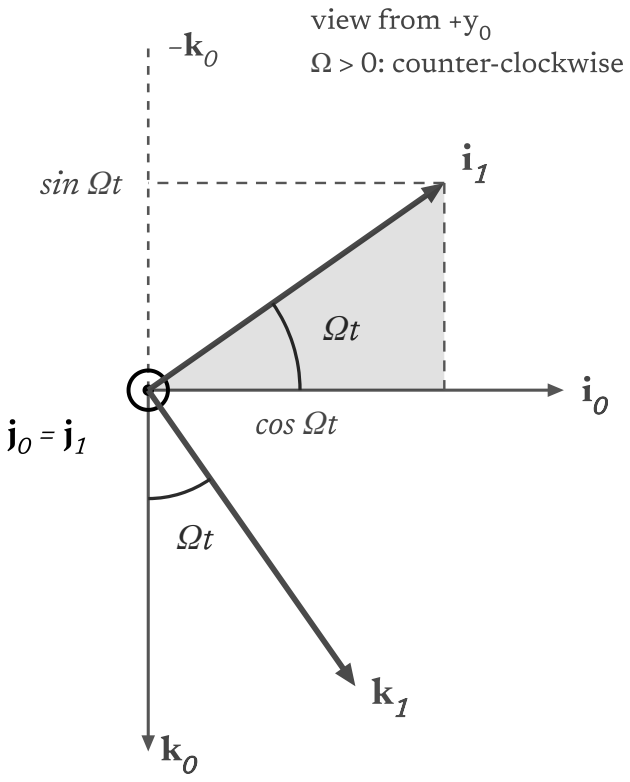


Figure 1. View from $+y_0$, with $\mathbf{j}_0 = \mathbf{j}_1$ out of the paper ($\odot$): $\mathbf{k}_0 = \mathbf{i}_0 \times \mathbf{j}_0$ points down the page. For $\Omega > 0$ the plane of the wires turns counter-clockwise by Ωt; $\mathbf{i}_1$ has the components $\cos \Omega t$ along $\mathbf{i}_0$ and $\sin \Omega t$ along $-\mathbf{k}_0$ (up the page), hence the minus sign.

The plane turns about $\mathbf{j}_0$, so we look down that axis, from $+y_0$: $\mathbf{j}_0 = \mathbf{j}_1$ points out of the paper ($\odot$). With $\mathbf{i}_0$ to the right, the right-hand rule gives $\mathbf{k}_0 = \mathbf{i}_0 \times \mathbf{j}_0$: the fingers turn from $\mathbf{i}_0$ (right) to $\mathbf{j}_0$ (towards the reader) and the thumb points down the page. So in this view $\mathbf{k}_0$ points down and $-\mathbf{k}_0$ up. For $\Omega > 0$ the plane turns counter-clockwise (thumb along $\mathbf{j}_0$, towards the reader), and after a time t the axis $\mathbf{i}_1$ has turned Ωt from $\mathbf{i}_0$:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of the shaded right triangle: the leg adjacent to Ωt is $\cos \Omega t$ along $\mathbf{i}_0$; the leg opposite to Ωt is $\sin \Omega t$ and points up the page, the direction $-\mathbf{k}_0$. Hence $\mathbf{i}_1 = \cos \Omega t\, \mathbf{i}_0 - \sin \Omega t\, \mathbf{k}_0$. **The minus sign comes from $\mathbf{k}_0$ pointing down the page in this view.**
- $\mathbf{k}_1$ starts on $\mathbf{k}_0$ (down) and turns Ωt counter-clockwise, which moves its tip towards $+\mathbf{i}_0$: leg $\cos \Omega t$ along $\mathbf{k}_0$ (adjacent) and $+ \sin \Omega t$ along $\mathbf{i}_0$ (opposite).
- $\mathbf{j}_1 = \mathbf{j}_0$, the axis of the rotation.

$$\mathbf{i}_1 = \cos\Omega t\,\mathbf{i}_0 - \sin\Omega t\,\mathbf{k}_0, \qquad \mathbf{j}_1 = \mathbf{j}_0, \qquad \mathbf{k}_1 = \sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{k}_0$$

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & 0 & \sin\Omega t \\ 0 & 1 & 0 \\ -\sin\Omega t & 0 & \cos\Omega t \end{pmatrix}$$

Checks: $\det[{}_0R_1] = \cos\Omega t\,(\cos\Omega t) + \sin\Omega t\,(\sin\Omega t) = 1$ (expansion along the second row). Rows: $\cos^2\Omega t + \sin^2\Omega t = 1, 1, \sin^2\Omega t + \cos^2\Omega t = 1$; first and third rows: $-\cos\Omega t\sin\Omega t + \sin\Omega t\cos\Omega t = 0$; so $[{}_0R_1][{}_0R_1]^T = I$. Also $\mathbf{i}_1 \times \mathbf{j}_1 = \sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{k}_0 = \mathbf{k}_1$ (right-handed).

Motion of S_1: $\mathbf{a}_0^O = \mathbf{0}$; $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}_1$. Check with Poisson's formula: by the chain rule, $\left.\dfrac{d\mathbf{i}_1}{dt}\right|_0 = -\Omega\sin\Omega t\,\mathbf{i}_0 - \Omega\cos\Omega t\,\mathbf{k}_0 = -\Omega\,\mathbf{k}_1$, and $\Omega\,\mathbf{j}_1 \times \mathbf{i}_1 = -\Omega\,\mathbf{k}_1$. Since Ω and $\mathbf{j}_0$ are constant, $\boldsymbol{\alpha}_{10} = \mathbf{0}$.

Position vectors in the plane of the wires

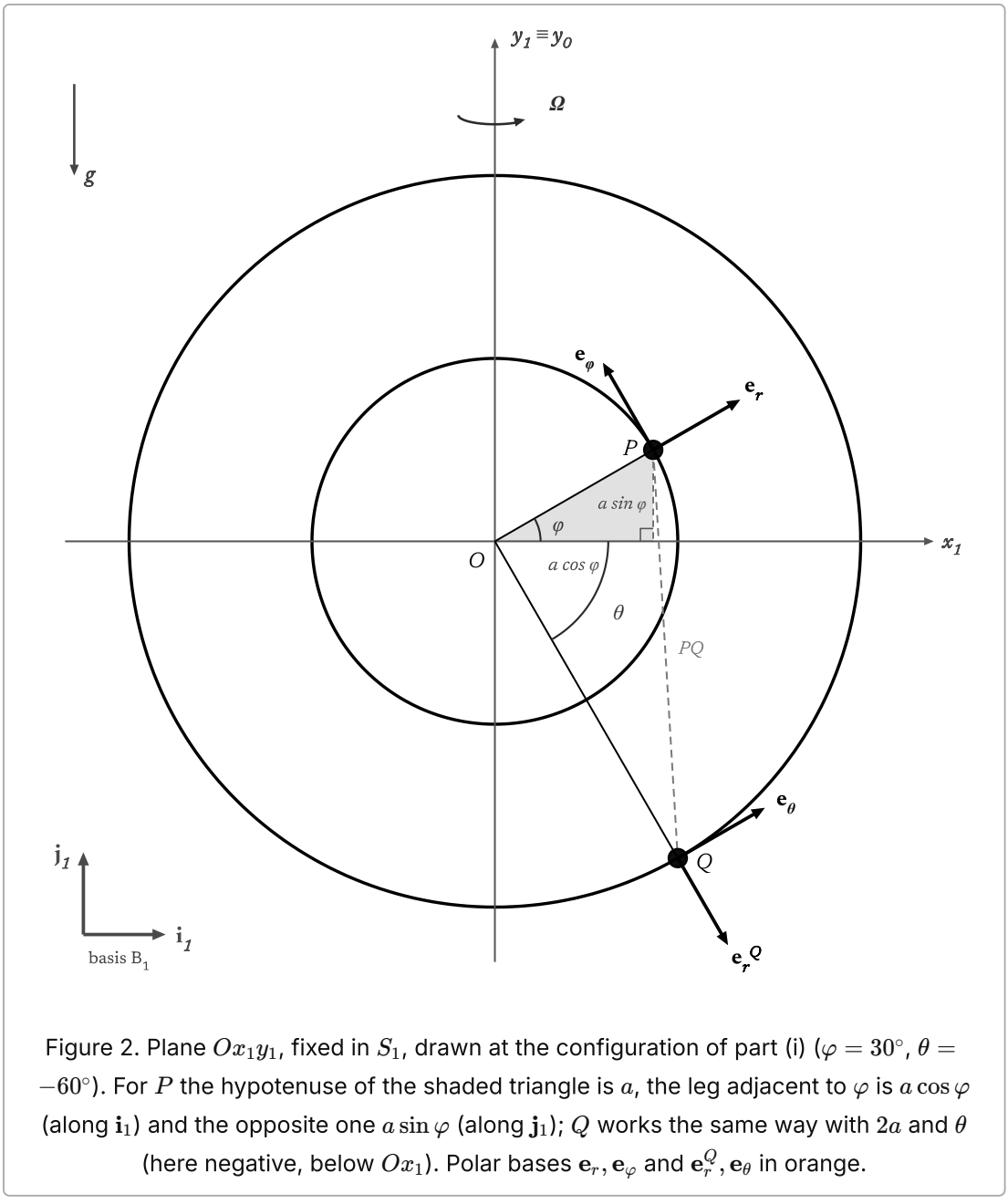


Figure 2. Plane Ox_1y_1, fixed in S_1, drawn at the configuration of part (i) ($\varphi = 30^\circ$, $\theta = -60^\circ$). For P the hypotenuse of the shaded triangle is a, the leg adjacent to φ is $a\cos\varphi$ (along $\mathbf{i}_1$) and the opposite one $a\sin\varphi$ (along $\mathbf{j}_1$); Q works the same way with $2a$ and θ (here negative, below Ox_1). Polar bases $\mathbf{e}_r, \mathbf{e}_\varphi$ and $\mathbf{e}_r^Q, \mathbf{e}_\theta$ in orange.

The plane Ox_1y_1 is fixed in S_1; it is drawn with $\mathbf{i}_1$ to the right, $\mathbf{j}_1$ up (and $\mathbf{k}_1 = \mathbf{i}_1 \times \mathbf{j}_1$ out of the paper), at the configuration of part (i).

- P is on the circle of radius a centred at O, at the angle φ from Ox_1. In the shaded right triangle the hypotenuse is $|OP| = a$, the leg adjacent to φ is $a\cos\varphi$ along $\mathbf{i}_1$ and the leg opposite to φ is $a\sin\varphi$ along $\mathbf{j}_1$: $\mathbf{r}_1^P = a\cos\varphi\,\mathbf{i}_1 + a\sin\varphi\,\mathbf{j}_1$.
- Q is on the circle of radius $2a$ at the angle θ; the same triangle with hypotenuse $2a$ gives $\mathbf{r}_1^Q = 2a\cos\theta\,\mathbf{i}_1 + 2a\sin\theta\,\mathbf{j}_1$. For $\theta < 0$ (as drawn) the leg $2a\sin\theta$ is negative: Q is below Ox_1.

These angles are the standard polar angles of the plane (measured from $\mathbf{i}_1$, counter-clockwise about $\mathbf{k}_1$), so no change of origin of the angle is needed.

Polar bases. The radial unit vector is the position divided by the radius, and the transverse one its derivative with respect to the angle ($\frac{d}{d\varphi}\cos\varphi = -\sin\varphi$, $\frac{d}{d\varphi}\sin\varphi = \cos\varphi$):

$$\mathbf{e}_r^P = \cos\varphi\,\mathbf{i}_1 + \sin\varphi\,\mathbf{j}_1, \quad \mathbf{e}_\varphi = \frac{d\mathbf{e}_r^P}{d\varphi} = -\sin\varphi\,\mathbf{i}_1 + \cos\varphi\,\mathbf{j}_1; \qquad \mathbf{e}_r^Q = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1, \quad \mathbf{e}_\theta = \frac{d\mathbf{e}_r^Q}{d\theta} = -\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1$$

so $\mathbf{r}_1^P = a\,\mathbf{e}_r^P$ and $\mathbf{r}_1^Q = 2a\,\mathbf{e}_r^Q$. Each triad $\{\mathbf{e}_r, \mathbf{e}_\varphi \text{ or } \mathbf{e}_\theta, \mathbf{k}_1\}$ is orthonormal and right-handed ($\mathbf{e}_r \times \mathbf{e}_\varphi = \cos^2\varphi\,(\mathbf{i}_1 \times \mathbf{j}_1) - \sin^2\varphi\,(\mathbf{j}_1 \times \mathbf{i}_1) = (\cos^2\varphi + \sin^2\varphi)\,\mathbf{k}_1 = \mathbf{k}_1$). Dot products used below, from the difference formulas $\cos\varphi\cos\theta + \sin\varphi\sin\theta = \cos(\varphi-\theta)$ and $-\sin\varphi\cos\theta + \cos\varphi\sin\theta = \sin(\theta-\varphi)$: $\mathbf{e}_r^Q \cdot \mathbf{e}_r^P = \cos(\varphi-\theta)$, $\mathbf{e}_r^Q \cdot \mathbf{e}_\varphi = \sin(\theta-\varphi)$, $\mathbf{e}_r^P \cdot \mathbf{e}_\theta = \sin(\varphi-\theta)$, $\mathbf{j}_1 \cdot \mathbf{e}_\varphi = \cos\varphi$, $\mathbf{j}_1 \cdot \mathbf{e}_r^P = \sin\varphi$, $\mathbf{i}_1 \cdot \mathbf{e}_\varphi = -\sin\varphi$, $\mathbf{i}_1 \cdot \mathbf{e}_r^P = \cos\varphi$ (and the same with θ for Q).

Kinematics

(c) Relative accelerations

Pedagogical Justification: $\mathbf{i}_1, \mathbf{j}_1$ are constant in S_1; by the chain rule $\left.\frac{d\mathbf{e}_r^P}{dt}\right|_1 = \frac{d\mathbf{e}_r^P}{d\varphi}\dot\varphi = \dot\varphi\,\mathbf{e}_\varphi$ and $\left.\frac{d\mathbf{e}_\varphi}{dt}\right|_1 = \dot\varphi\,(-\cos\varphi\,\mathbf{i}_1 - \sin\varphi\,\mathbf{j}_1) = -\dot\varphi\,\mathbf{e}_r^P$ (same for θ).

Product rule on $a\dot\varphi\,\mathbf{e}_\varphi$: $\frac{d}{dt}(a\dot\varphi) = a\ddot\varphi$ times $\mathbf{e}_\varphi$, plus $a\dot\varphi$ times $\left.\frac{d\mathbf{e}_\varphi}{dt}\right|_1$:

$$\mathbf{r}_1^P = a\,\mathbf{e}_r^P, \qquad \mathbf{v}_1^P = a\dot\varphi\,\mathbf{e}_\varphi, \qquad \mathbf{a}_1^P = a\ddot\varphi\,\mathbf{e}_\varphi + a\dot\varphi\,(-\dot\varphi\,\mathbf{e}_r^P) = a\ddot\varphi\,\mathbf{e}_\varphi - a\dot\varphi^2\,\mathbf{e}_r^P$$

$$\mathbf{r}_1^Q = 2a\,\mathbf{e}_r^Q, \qquad \mathbf{v}_1^Q = 2a\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{a}_1^Q = 2a\ddot\theta\,\mathbf{e}_\theta - 2a\dot\theta^2\,\mathbf{e}_r^Q$$

Forces and Equations of Motion

(b) Reaction forces

Pedagogical Justification: a smooth wire cannot push along its tangent, so its reaction lies in the normal plane of the wire, which is spanned by two independent directions; there is one unknown per removed coordinate.

Two components per particle, four in total:

$$\mathbf{N}^P = N_1\,\mathbf{e}_r^P + N_2\,\mathbf{k}_1, \qquad \mathbf{N}^Q = N_3\,\mathbf{e}_r^Q + N_4\,\mathbf{k}_1$$

(d) Applied forces

> **Pedagogical Justification:** the applied (actual, non-reaction) forces are the weights and the mutual repulsion; their components in each particle's polar basis are what enters the equations.

Weights: $-mg\,\mathbf{j}_1$ on each particle; with $\mathbf{j}_1\cdot\mathbf{e}_r^P = \sin\varphi$ and $\mathbf{j}_1\cdot\mathbf{e}_\varphi = \cos\varphi$, this is $-mg\sin\varphi\,\mathbf{e}_r^P - mg\cos\varphi\,\mathbf{e}_\varphi$ on P and $-mg\sin\theta\,\mathbf{e}_r^Q - mg\cos\theta\,\mathbf{e}_\theta$ on Q.

Repulsion, with $\mathbf{r}^P - \mathbf{r}^Q = a\,\mathbf{e}_r^P - 2a\,\mathbf{e}_r^Q = a\,(\cos\varphi - 2\cos\theta)\,\mathbf{i}_1 + a\,(\sin\varphi - 2\sin\theta)\,\mathbf{j}_1$ (the dashed segment PQ of the figure):

$$\mathbf{F}_k^P = k\left(\mathbf{r}^P - \mathbf{r}^Q\right) = ka\left[1 - 2\cos(\varphi-\theta)\right]\mathbf{e}_r^P + 2ka\sin(\varphi-\theta)\,\mathbf{e}_\varphi$$

$$\mathbf{F}_k^Q = k\left(\mathbf{r}^Q - \mathbf{r}^P\right) = ka\left[2 - \cos(\varphi-\theta)\right]\mathbf{e}_r^Q - ka\sin(\varphi-\theta)\,\mathbf{e}_\theta$$

(For P: $\mathbf{e}_r^Q\cdot\mathbf{e}_r^P = \cos(\varphi-\theta)$ and $-2a\,\mathbf{e}_r^Q\cdot\mathbf{e}_\varphi = -2a\sin(\theta-\varphi) = 2a\sin(\varphi-\theta)$. For Q: $-a\,\mathbf{e}_r^P\cdot\mathbf{e}_\theta = -a\sin(\varphi-\theta)$.)

(e) Inertia forces in S_1

> **Pedagogical Justification:** Notes Eq. 4.62 with $\mathbf{a}_0^O = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$: only the centrifugal force $-m\boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1)$ and the Coriolis force $-2m\boldsymbol{\omega}_{10}\times\mathbf{v}_1$ remain (Slide 28).

Cross products with $\mathbf{j}_1\times\mathbf{i}_1 = -\mathbf{k}_1$, $\mathbf{j}_1\times\mathbf{j}_1 = \mathbf{0}$, $\mathbf{j}_1\times\mathbf{k}_1 = \mathbf{i}_1$. For P, $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = \Omega a\,(\cos\varphi\,\mathbf{j}_1\times\mathbf{i}_1 + \sin\varphi\,\mathbf{j}_1\times\mathbf{j}_1) = -\Omega a\cos\varphi\,\mathbf{k}_1$ and $\boldsymbol{\omega}_{10}\times\left(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P\right) = -\Omega^2 a\cos\varphi\,\mathbf{j}_1\times\mathbf{k}_1 = -\Omega^2 a\cos\varphi\,\mathbf{i}_1$. Hence (it is $m\Omega^2\mathbf{r}_\perp$, Slide 30, with $\mathbf{r}_\perp = a\cos\varphi\,\mathbf{i}_1$ the horizontal leg of the triangle), using $\mathbf{i}_1 = \cos\varphi\,\mathbf{e}_r^P - \sin\varphi\,\mathbf{e}_\varphi$:

$$\mathbf{F}_C^P = m\Omega^2 a\cos\varphi\,\mathbf{i}_1 = m\Omega^2 a\cos\varphi\left(\cos\varphi\,\mathbf{e}_r^P - \sin\varphi\,\mathbf{e}_\varphi\right), \qquad \mathbf{F}_C^Q = 2m\Omega^2 a\cos\theta\left(\cos\theta\,\mathbf{e}_r^Q - \sin\theta\,\mathbf{e}_\theta\right)$$

For the Coriolis force, $\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P = \Omega a\dot\varphi\,(-\sin\varphi\,\mathbf{j}_1\times\mathbf{i}_1 + \cos\varphi\,\mathbf{j}_1\times\mathbf{j}_1) = \Omega a\dot\varphi\sin\varphi\,\mathbf{k}_1$, so

$$\mathbf{F}^P_{COR} = -2m\Omega a\dot\varphi\sin\varphi\,\mathbf{k}_1, \qquad \mathbf{F}^Q_{COR} = -4m\Omega a\dot\theta\sin\theta\,\mathbf{k}_1$$

The Coriolis forces are perpendicular to the plane of the wires: only the reactions N_2, N_4 can balance them.

(f) Equations of motion free of unknown reactions

> **Pedagogical Justification:** write $m\mathbf{a}_1 = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63) for each particle and project on its tangent ($\mathbf{e}_\varphi$, $\mathbf{e}_\theta$), which is perpendicular to both reaction components and to the Coriolis force.

P, along $\mathbf{e}_\varphi$: $ma\ddot\varphi = -mg\cos\varphi + 2ka\sin(\varphi-\theta) - m\Omega^2 a\sin\varphi\cos\varphi$.

Q, along $\mathbf{e}_\theta$: $2ma\ddot\theta = -mg\cos\theta - ka\sin(\varphi-\theta) - 2m\Omega^2 a\sin\theta\cos\theta$.

Dividing by ma and $2ma$:

$$\boxed{\ddot\varphi = -\frac{g}{a}\cos\varphi + \frac{2k}{m}\sin(\varphi-\theta) - \Omega^2\sin\varphi\cos\varphi, \qquad \ddot\theta = -\frac{g}{2a}\cos\theta - \frac{k}{2m}\sin(\varphi-\theta) - \Omega^2\sin\theta\cos\theta}$$

Remaining projections (they contain the reactions). P, along $\mathbf{e}^P_r$: $-ma\dot\varphi^2 = N_1 - mg\sin\varphi + ka\,[1-2\cos(\varphi-\theta)] + m\Omega^2 a\cos^2\varphi$. Along $\mathbf{k}_1$: $0 = N_2 - 2m\Omega a\dot\varphi\sin\varphi$.

Q, along $\mathbf{e}^Q_r$: $-2ma\dot\theta^2 = N_3 - mg\sin\theta + ka\,[2-\cos(\varphi-\theta)] + 2m\Omega^2 a\cos^2\theta$. Along $\mathbf{k}_1$: $0 = N_4 - 4m\Omega a\dot\theta\sin\theta$.

Answers and Checks

(a) Degrees of freedom

Two, with coordinates (φ, θ).

(b) Reactions

Four independent components: $\mathbf{N}^P = N_1\,\mathbf{e}^P_r + N_2\,\mathbf{k}_1$, $\mathbf{N}^Q = N_3\,\mathbf{e}^Q_r + N_4\,\mathbf{k}_1$.

(c) Relative accelerations

$\mathbf{a}^P_1 = a\ddot\varphi\,\mathbf{e}_\varphi - a\dot\varphi^2\,\mathbf{e}^P_r$, $\mathbf{a}^Q_1 = 2a\ddot\theta\,\mathbf{e}_\theta - 2a\dot\theta^2\,\mathbf{e}^Q_r$.

(d) Applied forces

Weights $-mg\,\mathbf{j}_1$ on each particle; repulsions $\mathbf{F}^P_k$ and $\mathbf{F}^Q_k$ as above.

(e) Inertia forces

$\mathbf{F}^P_C = m\Omega^2 a\cos\varphi\,\mathbf{i}_1$, $\mathbf{F}^Q_C = 2m\Omega^2 a\cos\theta\,\mathbf{i}_1$, $\mathbf{F}^P_{COR} = -2m\Omega a\dot\varphi\sin\varphi\,\mathbf{k}_1$, $\mathbf{F}^Q_{COR} = -4m\Omega a\dot\theta\sin\theta\,\mathbf{k}_1$.

(f) Equations of motion

The two boxed equations above, free of the reactions.

(g) Reactions as functions of time

> **Pedagogical Justification:** the remaining projections ($\mathbf{e}_r$ and $\mathbf{k}_1$) contain one reaction each; with $\varphi(t), \theta(t)$ known from (f), they give the reactions explicitly.

$$\boxed{\begin{aligned} N_1(t) &= -ma\dot{\varphi}^2 + mg\sin\varphi - ka\left[1 - 2\cos(\varphi-\theta)\right] - m\Omega^2 a\cos^2\varphi, \quad N_2(t) = 2m\Omega a\dot{\varphi}\sin\varphi \\ N_3(t) &= -2ma\dot{\theta}^2 + mg\sin\theta - ka\left[2 - \cos(\varphi-\theta)\right] - 2m\Omega^2 a\cos^2\theta, \quad N_4(t) = 4m\Omega a\dot{\theta}\sin\theta \end{aligned}}$$

(h) Mechanical energy in S_1

> **Pedagogical Justification:** yes, it is conserved. In S_1 the wires are fixed and smooth, so $\mathbf{N}\cdot\mathbf{v}_1 = 0$; the Coriolis forces do no work (Slide 31); the Euler force and $-m\mathbf{a}_0^Q$ vanish; the weights, the repulsion (a central force linear in the distance) and the centrifugal forces (constant $\boldsymbol{\omega}_{10}$, Slide 30) all derive from potentials.

Potentials in S_1: weights $mga\sin\varphi + 2mga\sin\theta$ (heights y_1); repulsion $-\frac{1}{2}k\left|\mathbf{r}^P - \mathbf{r}^Q\right|^2$ (minus sign: repulsive, so that $-\nabla_P$ gives $+k(\mathbf{r}^P - \mathbf{r}^Q)$), with

$$\left|\mathbf{r}^P - \mathbf{r}^Q\right|^2 = a^2 + 4a^2 - 4a^2\,\mathbf{e}_r^P\cdot\mathbf{e}_r^Q = 5a^2 - 4a^2\cos(\varphi-\theta)$$

and centrifugal $-\frac{1}{2}m\Omega^2\left(r_\perp^P\right)^2 - \frac{1}{2}m\Omega^2\left(r_\perp^Q\right)^2$ with $r_\perp = |x_1|$ (distance to the axis Oy_1, since $z_1 = 0$): $-\frac{1}{2}m\Omega^2 a^2\cos^2\varphi - 2m\Omega^2 a^2\cos^2\theta$. Kinetic energy: $\frac{1}{2}ma^2\dot{\varphi}^2 + \frac{1}{2}m\,4a^2\dot{\theta}^2$. Dropping the constant $-\frac{5}{2}ka^2$:

$$\boxed{E_1 = \frac{1}{2}ma^2\left(\dot{\varphi}^2 + 4\dot{\theta}^2\right) + mga\left(\sin\varphi + 2\sin\theta\right) + 2ka^2\cos(\varphi-\theta) - \frac{1}{2}m\Omega^2 a^2\left(\cos^2\varphi + 4\cos^2\theta\right) = \text{const}}$$

Check. By the chain rule ($\frac{d}{dt}\cos(\varphi-\theta) = -\sin(\varphi-\theta)(\dot{\varphi}-\dot{\theta})$, $\frac{d}{dt}\cos^2\varphi = -2\cos\varphi\sin\varphi\,\dot{\varphi}$):

$$\frac{dE_1}{dt} = \dot{\varphi}\left[ma^2\ddot{\varphi} + mga\cos\varphi - 2ka^2\sin(\varphi-\theta) + m\Omega^2 a^2\sin\varphi\cos\varphi\right] + \dot{\theta}\left[4ma^2\ddot{\theta} + 2mga\cos\theta + 2ka^2\sin(\varphi-\theta) + 4m\Omega^2 a^2\sin\theta\cos\theta\right]$$

and both brackets vanish by the two equations of (f) (multiplied by ma^2 and $4ma^2$), so $dE_1/dt = 0$.

(i) Parameters for the equilibrium $\varphi = 30^\circ, \theta = -60^\circ$

> **Pedagogical Justification:** a relative equilibrium has $\dot\varphi = \dot\theta = \ddot\varphi = \ddot\theta = 0$, so the right-hand sides of (f) must vanish. Dividing the first by g/a and the second by $g/(2a)$ introduces $\alpha = ka/(mg)$ and $\beta = a\Omega^2/g$.

$$0 = -\cos\varphi + 2\alpha\sin(\varphi-\theta) - \beta\sin\varphi\cos\varphi, \qquad 0 = -\cos\theta - \alpha\sin(\varphi-\theta) - 2\beta\sin\theta\cos\theta$$

With $\varphi - \theta = 90^\circ$ ($\sin = 1$), $\cos 30^\circ = \frac{\sqrt3}{2}$, $\sin 30^\circ = \frac12$, $\cos(-60^\circ) = \frac12$, $\sin(-60^\circ) = -\frac{\sqrt3}{2}$:

$$0 = -\frac{\sqrt3}{2} + 2\alpha - \frac{\sqrt3}{4}\beta, \qquad 0 = -\frac12 - \alpha + \frac{\sqrt3}{2}\beta$$

From the second, $\alpha = \frac{\sqrt3}{2}\beta - \frac12$. Substituting in the first: $-\frac{\sqrt3}{2} + \sqrt3\beta - 1 - \frac{\sqrt3}{4}\beta = 0$, i.e. $\frac{3\sqrt3}{4}\beta = 1 + \frac{\sqrt3}{2} = \frac{2+\sqrt3}{2}$, so

$$\beta = \frac{2\,(2+\sqrt3)}{3\sqrt3} = \frac{6+4\sqrt3}{9} \approx 1.4365, \qquad \alpha = \frac{\sqrt3}{2}\cdot\frac{6+4\sqrt3}{9} - \frac12 = \frac{12+6\sqrt3-9}{18} = \frac{1+2\sqrt3}{6} \approx 0.7440$$

$$\boxed{\alpha = \frac{ka}{mg} = \frac{1+2\sqrt3}{6} \approx 0.744, \qquad \beta = \frac{a\Omega^2}{g} = \frac{6+4\sqrt3}{9} \approx 1.4365}$$

Both values are positive, so the equilibrium is physically realisable.

Checks

- **Equilibrium:** at $\varphi - \theta = 90^\circ$ the radii OP and OQ are perpendicular (as in the wire-plane figure), so the tangential part of the repulsion is as large as it can be ($2ka$ on P, ka on Q), and it balances the tangential components of the weight and of the centrifugal force. P (at 30° above Ox_1) is pushed up its wire by the repulsion from Q, while gravity and the centrifugal force pull it back; Q (at 60° below Ox_1) is pushed down by the repulsion and by gravity, and the centrifugal force (which pulls it towards $\theta = 0$, away from the axis) holds it.

- **Coriolis forces:** they are normal to the rotating plane and only load the wires (N_2, N_4); they never enter the equations of motion or the energy.
- **Limits:** $\Omega \to 0$ removes the centrifugal terms and $N_2 = N_4 = 0$; $k \to 0$ decouples the two particles into two independent rotating-ring pendulums, $\ddot{\varphi} = -\frac{g}{a}\cos\varphi - \Omega^2 \sin\varphi\cos\varphi$.
- **Orthogonality:** $\mathbf{e}_r^P \cdot \mathbf{e}_\varphi = -\cos\varphi\sin\varphi + \sin\varphi\cos\varphi = 0$, and the same for Q.
- **Dimensions:** $[g/a] = [k/m] = [\Omega^2] = \mathrm{s}^{-2} = [\ddot{\varphi}]$; $[N_1]$: $[ma\dot{\varphi}^2] = [mg] = [ka] = [m\Omega^2 a] = \mathrm{N}$; $[N_2] = [m\Omega a\dot{\varphi}] = \mathrm{N}$; $[E_1] = \mathrm{kg\,m^2/s^2} = [mga] = [ka^2] = \mathrm{J}$; $\alpha = \mathrm{N\,m^{-1}\,m/N}$ and $\beta = \mathrm{m\,s^{-2}/(m\,s^{-2})}$ are dimensionless.
- **Symbolic check (sympy):** $\mathbf{a}_0$ of each particle obtained by differentiating $[{}_0R_1]\,\mathbf{r}_1$ twice equals the Notes Eq. 3.27 decomposition; the two equations of motion, the four reactions and $dE_1/dt = 0$ are confirmed, and solving the equilibrium conditions gives exactly the values above.
- **Check against the official key** (Problems.pdf, p. 76): $\alpha = 0.744$, $\beta = 1.4365$. Our exact values $\alpha = (1+2\sqrt{3})/6 = 0.74402$ and $\beta = (6+4\sqrt{3})/9 = 1.43647$ agree.

Source: Problems.pdf (PDF page 32); key on PDF page 76

PROBLEM 49

Particle on a Rotating Sphere Attracted to a Point

Original statement · Problems.pdf, p. 34

Relative Equilibria & Bifurcation

A point particle P of mass m slides without friction on a sphere of radius a centered at the origin of the inertial reference frame S_0 as shown in the figure. The particle is attracted by a point A in the equatorial plane, located a distance a from the surface of the sphere as shown. The force of attraction is linearly proportional [sic] the distance between this point and P, with known constant of proportionality $k > 0$. The sphere and the point A rotate rigidly about the vertical with a known angular velocity $\Omega \mathbf{k}_0$. We define the auxiliary reference frame S_1 that rotates with the sphere and point A, with $\mathbf{k}_1 = \mathbf{k}_0$ and $\mathbf{i}_1$ pointing in the direction of A. We introduce the angles ϕ and θ as shown to locate P on the sphere. Note that there is no gravity in this problem.

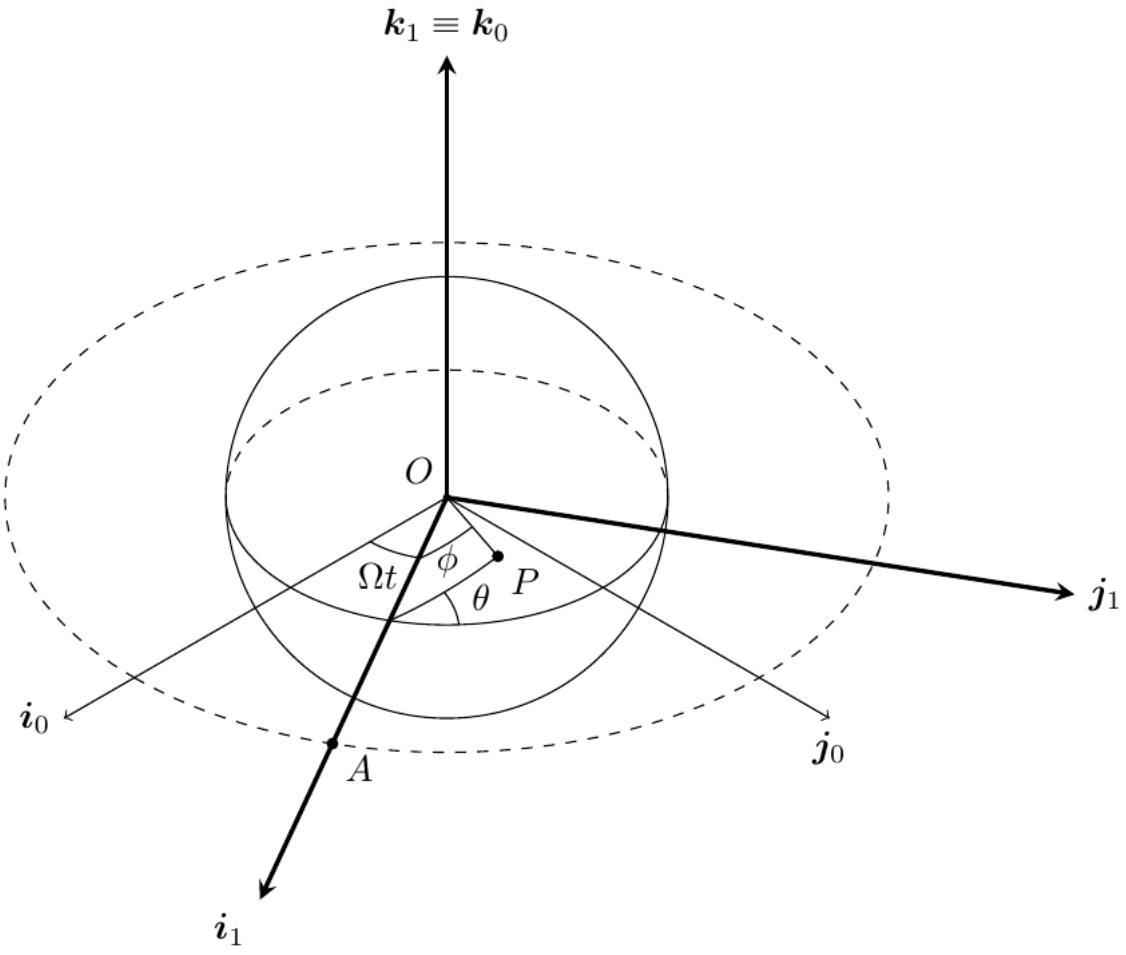


Figure: Sphere of radius a centred at O, point A on the i1 axis outside the sphere in the equatorial plane, particle P located by the angles phi and theta, frame S1 rotated by Omega t about k0 = k1 (Problems.pdf, p. 34).

(a) Express the position, velocity, and acceleration of P with respect to S_1 in terms of ϕ, θ and their derivatives. Remember to define any unit vectors that you want to use in terms of $\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0$ first.

(b) Write down the (vector) equation of motion of P in S_1, and derive from there a set of two scalar second-order differential equations for ϕ, θ that do not depend on any other unknowns. To do so, first identify and write down all the forces needed in your analysis.

(c) Is the mechanical energy of P conserved in S_1? Explain. If so, write down the corresponding conservation law.

(d) Identify all equilibrium configurations of the system, and discuss them as a function of the ratio $m\Omega^2/k$.

(e) Assuming the functions of time $\phi(t)$ and $\theta(t)$ are known, write down the expression of the reaction of the sphere on P.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a particle P of mass m slides on a smooth sphere of radius a centred at O. A point A of the equatorial plane, at distance a from the surface (so $OA = 2a$), attracts P with the force $k\,\overrightarrow{PA}$. Sphere and A rotate rigidly about the vertical with angular velocity $\Omega\,\mathbf{k}_0$. No gravity.
- **Hypotheses:**
  1. $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ is inertial; Ω is constant.
  2. The sphere is smooth and the contact bilateral: the reaction is radial, $\mathbf{N} = N\,\mathbf{e}_r$ with N of either sign.
  3. **Reading of the figure (angles):** ϕ is the angle at O between $\overrightarrow{OA}$ (i.e. $\mathbf{i}_1$) and $\overrightarrow{OP}$; θ is the angle, measured about the axis OA, between the plane containing $\mathbf{i}_1$ and $\overrightarrow{OP}$ and the equatorial plane (the figure draws it at the point where OA crosses the sphere, between the equator and the arc through P). So ϕ is a polar angle about the axis OA and θ the corresponding azimuth, with $\theta = 0$ in the equatorial plane on the side of $\mathbf{j}_1$ and $\theta = \pi/2$ in the upper hemisphere.

- **Frames:** $S_0 = \{O; \mathcal{B}_0\}$ inertial; $S_1 = \{O; \mathcal{B}_1\}$, $\mathcal{B}_1 = \{\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, rotating with the sphere, with $\mathbf{k}_1 = \mathbf{k}_0$ and $\mathbf{i}_1$ along $\overrightarrow{OA}$; the figure measures the angle Ωt from $\mathbf{i}_0$ to $\mathbf{i}_1$.
- **Spherical basis with polar axis OA:** $\{\mathbf{e}_r, \mathbf{e}_\phi, \mathbf{e}_\theta\}$ at P, deduced in the next section.
- **Constraints and degrees of freedom:** P has 3 coordinates and one constraint, $|\overrightarrow{OP}| = a$ (holonomic, scleronomic in S_1, smooth): CDOF $= 3 - 1 = 2$, with the angles (ϕ, θ); the unknown reaction is N along $\mathbf{e}_r$.
- **Why S_1:** in S_1 both the sphere and the attracting point A are fixed, so the attraction depends only on the position of P and the problem has no explicit time dependence; the inertia forces (Notes Eq. 4.62) take that role. With polar axis OA, the distance AP depends on ϕ only, which keeps the attraction simple.

Symbol	Meaning	SI unit
m	mass of P	kg
a	radius of the sphere ($OA = 2a$)	m
k	attraction constant	N/m
Ω	angular rate of sphere and A	rad/s
ϕ, θ	polar angle from OA, azimuth about OA	rad
N	normal reaction	N
$\Lambda = m\Omega^2/k$	ratio of centrifugal to attraction stiffness	1

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from above

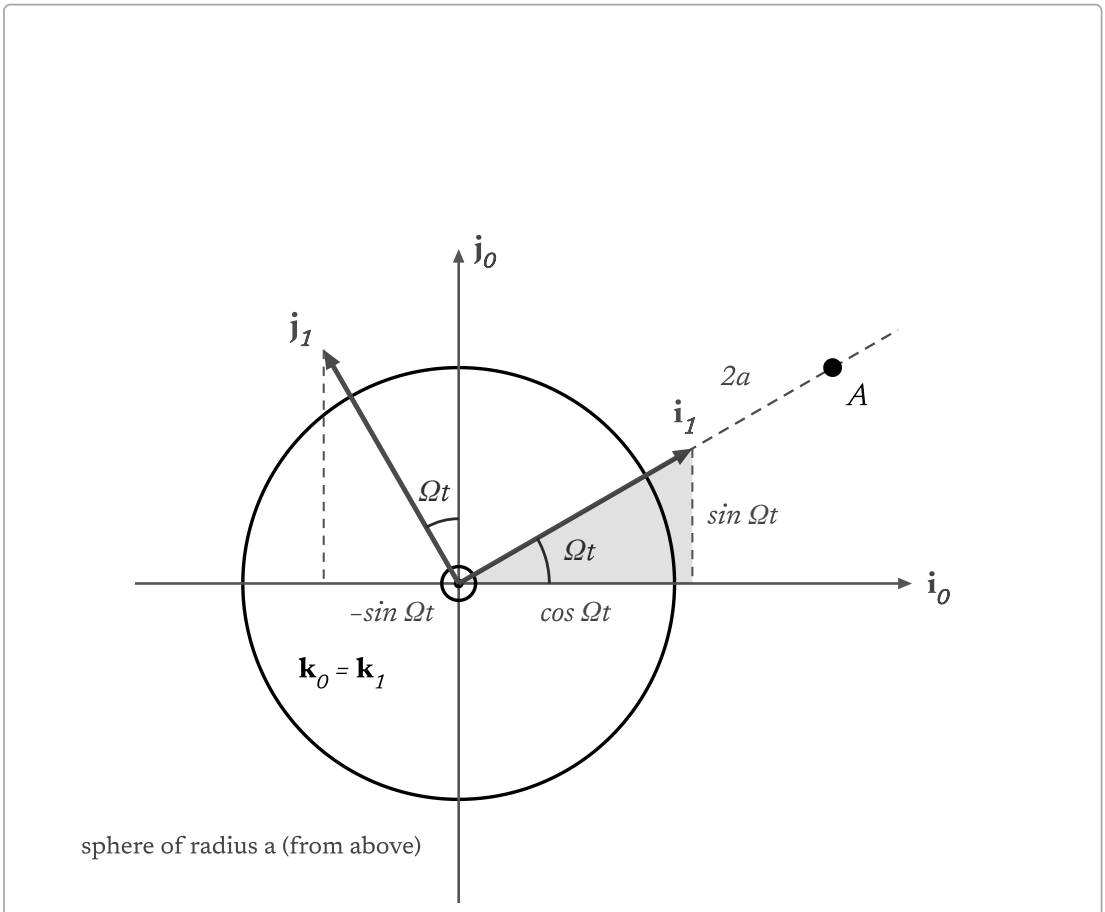


Figure 1. View from $+z$, with $\mathbf{k}_0 = \mathbf{k}_1$ out of the paper ($\odot$), so $\mathbf{j}_0$ is $\mathbf{i}_0$ turned 90° counter-clockwise. S_1 turns Ωt counter-clockwise: $\mathbf{i}_1$ has the legs $\cos\Omega t$ along $\mathbf{i}_0$ and $\sin\Omega t$ along $\mathbf{j}_0$; the tip of $\mathbf{j}_1$ moves to the side of $-\mathbf{i}_0$, hence $-\sin\Omega t$. A sits on the $\mathbf{i}_1$ axis at $OA = 2a$.

The rotation is about $\mathbf{k}_0$, so we look down from $+z$: $\mathbf{k}_0 = \mathbf{k}_1$ points out of the paper ($\odot$). With $\mathbf{i}_0$ to the right, $\mathbf{i}_0 \times \mathbf{j}_0 = \mathbf{k}_0$ requires the fingers to curl counter-clockwise from $\mathbf{i}_0$ to $\mathbf{j}_0$ for the thumb to point at the reader, so $\mathbf{j}_0$ points up the page. With $\Omega > 0$ the frame S_1 turns counter-clockwise, and after a time t:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of the shaded triangle: the leg adjacent to Ωt is $\cos\Omega t$ along $\mathbf{i}_0$ and the leg opposite to Ωt is $\sin\Omega t$ along $+\mathbf{j}_0$. So $\mathbf{i}_1 = \cos\Omega t\,\mathbf{i}_0 + \sin\Omega t\,\mathbf{j}_0$.
- $\mathbf{j}_1$ starts on $\mathbf{j}_0$ and turns the same angle counter-clockwise, so its tip moves to the side of $-\mathbf{i}_0$: leg $\cos\Omega t$ along $\mathbf{j}_0$ and leg $\sin\Omega t$ towards $-\mathbf{i}_0$. **That is the origin of the minus sign** in $\mathbf{j}_1 = -\sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{j}_0$.
- $\mathbf{k}_1 = \mathbf{k}_0$, the axis of the rotation.

$$\mathbf{i}_1 = \cos\Omega t\,\mathbf{i}_0 + \sin\Omega t\,\mathbf{j}_0, \qquad \mathbf{j}_1 = -\sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{j}_0, \qquad \mathbf{k}_1 = \mathbf{k}_0$$

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & -\sin\Omega t & 0 \\ \sin\Omega t & \cos\Omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1\cdot(\cos\Omega t\cos\Omega t - (-\sin\Omega t)\sin\Omega t) = \cos^2\Omega t + \sin^2\Omega t = 1$$

The rows are unit vectors and the first two are orthogonal ($\cos\Omega t\sin\Omega t - \sin\Omega t\cos\Omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.

Motion of S_1: $\mathbf{a}_0^O = \mathbf{0}$, $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{k}_1$ (simple rotation, Slide 16), $\boldsymbol{\alpha}_{10} = \mathbf{0}$. Check: by the chain rule $\frac{d\mathbf{i}_1}{dt}\big|_0 = -\Omega\sin\Omega t\,\mathbf{i}_0 + \Omega\cos\Omega t\,\mathbf{j}_0 = \Omega\,\mathbf{j}_1 = \Omega\,\mathbf{k}_1\times\mathbf{i}_1$. The attracting point is fixed in S_1: $\mathbf{r}_1^A = 2a\,\mathbf{i}_1$ (it lies on the dashed $\mathbf{i}_1$ axis of the figure).

The azimuth θ: view along OA

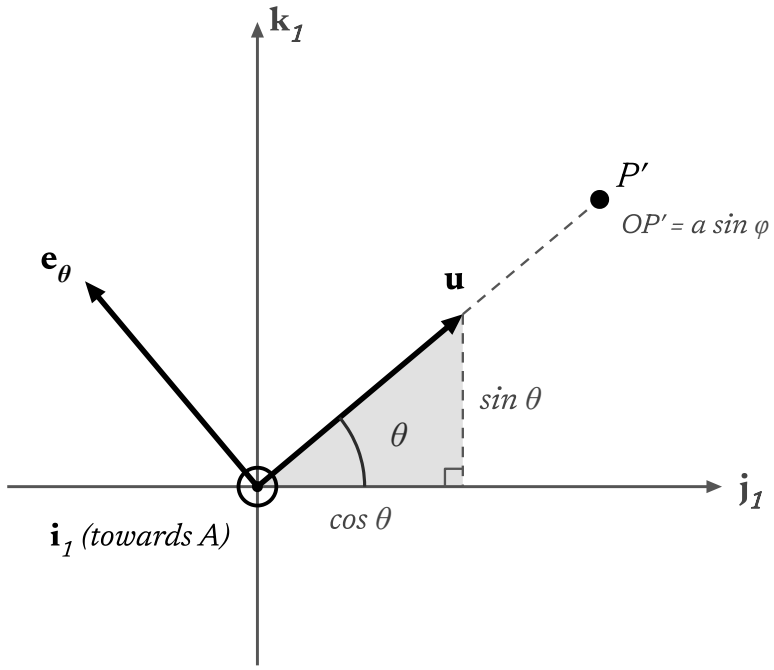


Figure 2. View along OA, with $\mathbf{i}_1$ out of the paper ($\odot$): then $\mathbf{j}_1$ points right and $\mathbf{k}_1 = \mathbf{i}_1 \times \mathbf{j}_1$ up. The azimuth θ turns counter-clockwise from $\mathbf{j}_1$: $\mathbf{u} = \cos\theta\,\mathbf{j}_1 + \sin\theta\,\mathbf{k}_1$ (legs of the shaded triangle) and $\mathbf{e}_\theta = d\mathbf{u}/d\theta$. P projects onto this plane at P', at distance $a\sin\phi$ from the axis.

θ turns about the axis OA, so we look along that axis with $\mathbf{i}_1$ out of the paper ($\odot$). With $\mathbf{j}_1$ to the right, $\mathbf{k}_1 = \mathbf{i}_1 \times \mathbf{j}_1$ points up (fingers from $\mathbf{i}_1$, towards the reader, to $\mathbf{j}_1$, right; thumb up). The azimuth starts at $\mathbf{j}_1$ ($\theta = 0$, equatorial plane) and grows counter-clockwise towards $\mathbf{k}_1$ ($\theta = \pi/2$, upper hemisphere), as in the reading of the figure. The unit vector $\mathbf{u}$ of the plane $\mathbf{j}_1\mathbf{k}_1$ in the direction θ is the hypotenuse of the shaded triangle, with the leg $\cos\theta$ along $\mathbf{j}_1$ (adjacent) and $\sin\theta$ along $\mathbf{k}_1$ (opposite):

$$\mathbf{u}(\theta) = \cos\theta\,\mathbf{j}_1 + \sin\theta\,\mathbf{k}_1, \qquad \mathbf{e}_\theta = \frac{d\mathbf{u}}{d\theta} = -\sin\theta\,\mathbf{j}_1 + \cos\theta\,\mathbf{k}_1$$

($\frac{d}{d\theta}\cos\theta = -\sin\theta$, $\frac{d}{d\theta}\sin\theta = \cos\theta$; $|\mathbf{e}_\theta| = 1$ and $\mathbf{u}\cdot\mathbf{e}_\theta = -\cos\theta\sin\theta + \sin\theta\cos\theta = 0$). The minus sign in $\mathbf{e}_\theta$ appears because turning $\mathbf{u}$ a further 90° counter-clockwise takes its tip to the side of $-\mathbf{j}_1$.

The polar angle ϕ: the meridian plane of P

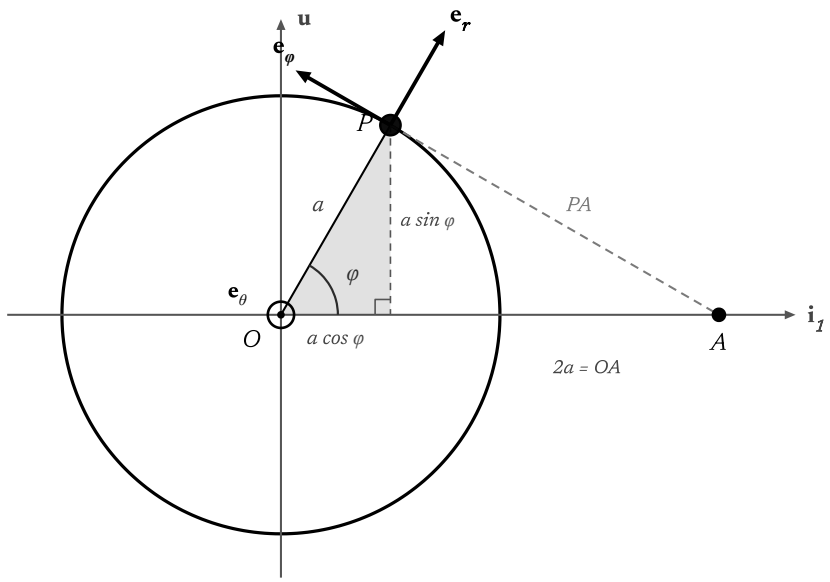


Figure 3. Meridian plane of P (it contains $\mathbf{i}_1$ and $\mathbf{u}$; $\mathbf{e}_\theta = \mathbf{i}_1 \times \mathbf{u}$ is out of the paper). ϕ is measured at O from OA: in the shaded triangle the hypotenuse is a, the leg adjacent to ϕ is $a\cos\phi$ along $\mathbf{i}_1$ and the opposite one $a\sin\phi$ along $\mathbf{u}$. $\mathbf{e}_r, \mathbf{e}_\phi$ in orange; the attraction acts along PA.

P lies in the plane that contains the axis OA and the direction $\mathbf{u}$ (the meridian plane of P); it is drawn with $\mathbf{i}_1$ to the right and $\mathbf{u}$ up, so $\mathbf{i}_1 \times \mathbf{u} = \cos\theta\,\mathbf{k}_1 - \sin\theta\,\mathbf{j}_1 = \mathbf{e}_\theta$ points out of the paper. In the shaded right triangle the hypotenuse is $|OP| = a$, the leg adjacent to ϕ is $a\cos\phi$ along $\mathbf{i}_1$ and the leg opposite to ϕ is $a\sin\phi$ along $\mathbf{u}$ (the distance OP' of the axis view). Hence

$$\mathbf{r}_1^P = \overrightarrow{OP} = a\cos\phi\,\mathbf{i}_1 + a\sin\phi\,\mathbf{u} = a\cos\phi\,\mathbf{i}_1 + a\sin\phi\cos\theta\,\mathbf{j}_1 + a\sin\phi\sin\theta\,\mathbf{k}_1 = a\,\mathbf{e}_r$$

Relation with the standard spherical coordinates. The usual spherical coordinates take the polar axis along $\mathbf{k}$ and the azimuth from $\mathbf{i}$ towards $\mathbf{j}$: $\mathbf{e}_r = \sin\phi\cos\theta\,\mathbf{i} + \sin\phi\sin\theta\,\mathbf{j} + \cos\phi\,\mathbf{k}$. Here the roles are permuted cyclically, $(\mathbf{i},\mathbf{j},\mathbf{k}) \to (\mathbf{j}_1,\mathbf{k}_1,\mathbf{i}_1)$, which keeps the triad right-handed ($\mathbf{j}_1 \times \mathbf{k}_1 = \mathbf{i}_1$); substituting gives exactly the $\mathbf{e}_r$ above. The polar axis is OA instead of the vertical because then $|AP|$ depends on ϕ only.

Spherical basis by differentiation. The radial vector is $\mathbf{e}_r = \cos\phi\,\mathbf{i}_1 + \sin\phi\,\mathbf{u}$. Its derivative with respect to ϕ ($\frac{d}{d\phi}\cos\phi = -\sin\phi$, $\frac{d}{d\phi}\sin\phi = \cos\phi$) already has unit length, $\sqrt{\sin^2\phi + \cos^2\phi} = 1$; with respect to θ, $\frac{\partial \mathbf{e}_r}{\partial\theta} = \sin\phi\,\frac{d\mathbf{u}}{d\theta} = \sin\phi\,\mathbf{e}_\theta$, whose length is $\sin\phi$:

$$\mathbf{e}_\phi = \frac{\partial \mathbf{e}_r}{\partial \phi} = -\sin\phi\,\mathbf{i}_1 + \cos\phi\cos\theta\,\mathbf{j}_1 + \cos\phi\sin\theta\,\mathbf{k}_1, \qquad \mathbf{e}_\theta = \frac{1}{\sin\phi}\frac{\partial \mathbf{e}_r}{\partial\theta} = -\sin\theta\,\mathbf{j}_1 + \cos\theta\,\mathbf{k}_1$$

with $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ given above in terms of $\mathcal{B}_0$. The triad is orthonormal and right-handed: $\mathbf{e}_r \times \mathbf{e}_\phi = \left(\cos^2\phi + \sin^2\phi\right)\mathbf{i}_1 \times \mathbf{u} = \cos\theta\,\mathbf{k}_1 - \sin\theta\,\mathbf{j}_1 = \mathbf{e}_\theta$, hence also $\mathbf{e}_\phi \times \mathbf{e}_\theta = \mathbf{e}_r$, $\mathbf{e}_\theta \times \mathbf{e}_r = \mathbf{e}_\phi$. Inverse relations used below (read from the meridian triangle): $\mathbf{u} = \sin\phi\,\mathbf{e}_r + \cos\phi\,\mathbf{e}_\phi$, so

$$\mathbf{i}_1 = \cos\phi\,\mathbf{e}_r - \sin\phi\,\mathbf{e}_\phi, \qquad \mathbf{k}_1 = \sin\theta\,\mathbf{u} + \cos\theta\,\mathbf{e}_\theta = \sin\phi\sin\theta\,\mathbf{e}_r + \cos\phi\sin\theta\,\mathbf{e}_\phi + \cos\theta\,\mathbf{e}_\theta$$

(the second from the axis-view triangle, $\mathbf{k}_1 = \sin\theta\,\mathbf{u} + \cos\theta\,\mathbf{e}_\theta$).

Kinematics

(a) Position, velocity and acceleration in S_1

Pedagogical Justification: $\mathbf{r}_1^P = a\,\mathbf{e}_r$ (Notes Eq. 3.16 with the origin O common to both frames). The relative derivative keeps $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ fixed, so the spherical vectors change only through $\phi(t)$ and $\theta(t)$ (chain rule).

Time derivatives of the basis in S_1:

$$\left.\frac{d\mathbf{e}_r}{dt}\right|_1 = -\sin\phi\,\dot\phi\,\mathbf{i}_1 + \cos\phi\,\dot\phi\,\mathbf{u} + \sin\phi\,\frac{d\mathbf{u}}{d\theta}\dot\theta = \dot\phi\,\mathbf{e}_\phi + \sin\phi\,\dot\theta\,\mathbf{e}_\theta$$

$$\left.\frac{d\mathbf{e}_\phi}{dt}\right|_1 = -\cos\phi\,\dot\phi\,\mathbf{i}_1 - \sin\phi\,\dot\phi\,\mathbf{u} + \cos\phi\,\dot\theta\,\mathbf{e}_\theta = -\dot\phi\,\mathbf{e}_r + \cos\phi\,\dot\theta\,\mathbf{e}_\theta$$

$$\left.\frac{d\mathbf{e}_\theta}{dt}\right|_1 = \dot\theta\,(-\cos\theta\,\mathbf{j}_1 - \sin\theta\,\mathbf{k}_1) = -\dot\theta\,\mathbf{u} = -\sin\phi\,\dot\theta\,\mathbf{e}_r - \cos\phi\,\dot\theta\,\mathbf{e}_\phi$$

Position and velocity:

$$\mathbf{r}_1^P = a\,\mathbf{e}_r, \qquad \mathbf{v}_1^P = a\left.\frac{d\mathbf{e}_r}{dt}\right|_1 = a\dot\phi\,\mathbf{e}_\phi + a\sin\phi\,\dot\theta\,\mathbf{e}_\theta$$

Acceleration, by the product rule on each term, $\frac{d}{dt}(a\dot\phi\,\mathbf{e}_\phi) = a\ddot\phi\,\mathbf{e}_\phi + a\dot\phi\left(-\dot\phi\,\mathbf{e}_r + \cos\phi\,\dot\theta\,\mathbf{e}_\theta\right)$ and $\frac{d}{dt}(a\sin\phi\,\dot\theta\,\mathbf{e}_\theta) = a\left(\cos\phi\,\dot\phi\dot\theta + \sin\phi\,\ddot\theta\right)\mathbf{e}_\theta + a\sin\phi\,\dot\theta\left(-\sin\phi\,\dot\theta\,\mathbf{e}_r - \cos\phi\,\dot\theta\,\mathbf{e}_\phi\right)$; collecting:

$$\mathbf{a}_1^P = -a\left(\dot\phi^2 + \sin^2\phi\,\dot\theta^2\right)\mathbf{e}_r + a\left(\ddot\phi - \sin\phi\cos\phi\,\dot\theta^2\right)\mathbf{e}_\phi + a\left(\sin\phi\,\ddot\theta + 2\cos\phi\,\dot\phi\dot\theta\right)\mathbf{e}_\theta$$

Forces and Equations of Motion

(b) Forces on the particle

> **Pedagogical Justification:** in S_1, $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63). The actual forces are the attraction and the reaction; the inertia forces (Notes Eq. 4.62) reduce to the centrifugal and Coriolis forces because $\mathbf{a}_0^O = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$.

Attraction (along the dashed segment PA of the meridian figure): $\mathbf{F}_k = k\left(\mathbf{r}_1^A - \mathbf{r}_1^P\right) = k\,(2a\,\mathbf{i}_1 - a\,\mathbf{e}_r)$; with $\mathbf{i}_1 = \cos\phi\,\mathbf{e}_r - \sin\phi\,\mathbf{e}_\phi$,

$$\mathbf{F}_k = ka\,(2\cos\phi - 1)\,\mathbf{e}_r - 2ka\sin\phi\,\mathbf{e}_\phi$$

Reaction: $\mathbf{N} = N\,\mathbf{e}_r$.

Centrifugal force. With the identity $\mathbf{a}\times(\mathbf{b}\times\mathbf{c}) = (\mathbf{a}\cdot\mathbf{c})\mathbf{b} - (\mathbf{a}\cdot\mathbf{b})\mathbf{c}$ (Slide 30): $\boldsymbol{\omega}_{10}\times\left(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P\right) = \Omega^2 z_1\,\mathbf{k}_1 - \Omega^2\mathbf{r}_1^P$, where $z_1 = \mathbf{r}_1^P\cdot\mathbf{k}_1 = a\sin\phi\sin\theta$. Hence $\mathbf{F}_C = m\Omega^2\left(\mathbf{r}_1^P - z_1\mathbf{k}_1\right)$, and with $\mathbf{k}_1$ in the spherical basis:

$$\mathbf{F}_C = m\Omega^2 a\left[\left(1-\sin^2\phi\sin^2\theta\right)\mathbf{e}_r - \sin^2\theta\sin\phi\cos\phi\,\mathbf{e}_\phi - \sin\phi\sin\theta\cos\theta\,\mathbf{e}_\theta\right]$$

Coriolis force. Using $\mathbf{k}_1 = \sin\phi\sin\theta\,\mathbf{e}_r + \cos\phi\sin\theta\,\mathbf{e}_\phi + \cos\theta\,\mathbf{e}_\theta$ and the cyclic products of the right-handed triad:

$$\mathbf{k}_1\times\mathbf{e}_\phi = \sin\phi\sin\theta\,\mathbf{e}_r\times\mathbf{e}_\phi + \cos\theta\,\mathbf{e}_\theta\times\mathbf{e}_\phi = \sin\phi\sin\theta\,\mathbf{e}_\theta - \cos\theta\,\mathbf{e}_r$$

$$\mathbf{k}_1\times\mathbf{e}_\theta = \sin\phi\sin\theta\,\mathbf{e}_r\times\mathbf{e}_\theta + \cos\phi\sin\theta\,\mathbf{e}_\phi\times\mathbf{e}_\theta = -\sin\phi\sin\theta\,\mathbf{e}_\phi + \cos\phi\sin\theta\,\mathbf{e}_r$$

$$\mathbf{F}_{COR} = -2m\Omega\,\mathbf{k}_1\times\left(a\dot\phi\,\mathbf{e}_\phi + a\sin\phi\,\dot\theta\,\mathbf{e}_\theta\right) = 2m\Omega a\left[\left(\cos\theta\,\dot\phi - \sin\theta\sin\phi\cos\phi\,\dot\theta\right)\mathbf{e}_r + \sin\theta\sin^2\phi\,\dot\theta\,\mathbf{e}_\phi - \sin\theta\sin\phi\,\dot\phi\,\mathbf{e}_\theta\right]$$

(b, continued) Vector equation and scalar equations

Vector equation of motion in S_1:

$$m\mathbf{a}_1^P = k\left(\mathbf{r}_1^A - \mathbf{r}_1^P\right) + N\,\mathbf{e}_r - m\boldsymbol{\omega}_{10}\times\left(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P\right) - 2m\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P$$

Scalar equations free of N: project on $\mathbf{e}_\phi$ and $\mathbf{e}_\theta$, which are perpendicular to $\mathbf{N}$:

$$\boxed{ma\left(\ddot\phi - \sin\phi\cos\phi\,\dot\theta^2\right) = -2ka\sin\phi - m\Omega^2 a\sin^2\theta\sin\phi\cos\phi + 2m\Omega a\sin\theta\sin^2\phi\,\dot\theta}$$

$$\boxed{ma\left(\sin\phi\,\ddot\theta + 2\cos\phi\,\dot\phi\dot\theta\right) = -m\Omega^2 a\sin\phi\sin\theta\cos\theta - 2m\Omega a\sin\theta\sin\phi\,\dot\phi}$$

Projection on $\mathbf{e}_r$ (it contains N):

$$-ma\left(\dot{\phi}^2+\sin^2\phi\,\dot{\theta}^2\right)=N+ka\left(2\cos\phi-1\right)+m\Omega^2a\left(1-\sin^2\phi\sin^2\theta\right)+2m\Omega a\left(\cos\theta\,\dot{\phi}-\sin\theta\sin\phi\cos\phi\,\dot{\theta}\right)$$

Answers and Checks

(a) Position, velocity and acceleration

$\mathbf{r}_1^P=a\,\mathbf{e}_r$, $\mathbf{v}_1^P=a\dot{\phi}\,\mathbf{e}_\phi+a\sin\phi\,\dot{\theta}\,\mathbf{e}_\theta$, $\mathbf{a}_1^P=-a\left(\dot{\phi}^2+\sin^2\phi\,\dot{\theta}^2\right)\mathbf{e}_r+$
$a\left(\ddot{\phi}-\sin\phi\cos\phi\,\dot{\theta}^2\right)\mathbf{e}_\phi+a\left(\sin\phi\,\ddot{\theta}+2\cos\phi\,\dot{\phi}\dot{\theta}\right)\mathbf{e}_\theta$, with the unit vectors defined above in terms of $\mathcal{B}_1$ and, through $[{}_0R_1]$, of $\mathbf{i}_0,\mathbf{j}_0,\mathbf{k}_0$.

(b) Equations of motion

The forces are $\mathbf{F}_k$, $\mathbf{N}$, $\mathbf{F}_C$ and $\mathbf{F}_{COR}$ above; the vector equation and the two boxed scalar equations for ϕ and θ are the answer.

(c) Conservation of mechanical energy in S_1

> **Pedagogical Justification:** yes, E_1^P is conserved. The sphere is fixed in S_1 and smooth, so $\mathbf{N}\cdot\mathbf{v}_1^P=0$ ($\mathbf{v}_1^P$ has no $\mathbf{e}_r$ component); $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P=0$ (Slide 31); the Euler force and $-m\mathbf{a}_0^O$ are zero; the attraction towards the point A, fixed in S_1, is conservative with $V_k=\frac{1}{2}k\left|\mathbf{r}_1^P-\mathbf{r}_1^A\right|^2$; and the centrifugal force has $V_C=-\frac{1}{2}m\Omega^2r_\perp^2$ because $\boldsymbol{\omega}_{10}$ is constant (Slide 30).

Potentials on the sphere:

$$\left|\mathbf{r}_1^P-\mathbf{r}_1^A\right|^2=a^2-4a^2\,\mathbf{e}_r\cdot\mathbf{i}_1+4a^2=5a^2-4a^2\cos\phi\;\Rightarrow\;V_k=\frac{5}{2}ka^2-2ka^2\cos\phi$$

$$r_\perp^2=x_1^2+y_1^2=a^2-z_1^2=a^2-a^2\sin^2\phi\sin^2\theta\;\Rightarrow\;V_C=-\frac{1}{2}m\Omega^2a^2+\frac{1}{2}m\Omega^2a^2\sin^2\phi\sin^2\theta$$

Dropping the constants, with $T_1^P=\frac{1}{2}m\left|\mathbf{v}_1^P\right|^2=\frac{1}{2}ma^2\left(\dot{\phi}^2+\sin^2\phi\,\dot{\theta}^2\right)$:

$$\boxed{E_1^P=\frac{1}{2}ma^2\left(\dot{\phi}^2+\sin^2\phi\,\dot{\theta}^2\right)-2ka^2\cos\phi+\frac{1}{2}m\Omega^2a^2\sin^2\phi\sin^2\theta=\text{const}}$$

Check. Differentiating with the chain and product rules ($\frac{d}{dt}\sin^2\phi = 2\sin\phi\cos\phi\,\dot\phi$, $\frac{d}{dt}\sin^2\theta = 2\sin\theta\cos\theta\,\dot\theta$) and grouping:

$$\frac{dE_1^P}{dt} = a\dot\phi\left[ma\left(\ddot\phi - \sin\phi\cos\phi\,\dot\theta^2\right) + 2ka\sin\phi + m\Omega^2 a\sin^2\theta\sin\phi\cos\phi\right] + a\sin\phi\,\dot\theta\left[ma\left(\sin\phi\,\ddot\theta + 2\cos\phi\,\dot\phi\dot\theta\right) + m\Omega^2 a\sin\phi\sin\theta\cos\theta\right]$$

By the equations of (b) the brackets equal $2m\Omega a\sin\theta\sin^2\phi\,\dot\theta$ and $-2m\Omega a\sin\theta\sin\phi\,\dot\phi$, so

$$\frac{dE_1^P}{dt} = 2m\Omega a^2\sin\theta\sin^2\phi\,\dot\phi\dot\theta - 2m\Omega a^2\sin\theta\sin^2\phi\,\dot\phi\dot\theta = 0$$

which is exactly $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P = 0$. The constant is fixed by initial conditions (not given).

(d) Equilibrium configurations

> **Pedagogical Justification:** a relative equilibrium has $\dot\phi = \dot\theta = \ddot\phi = \ddot\theta = 0$, so the right-hand sides of the equations of (b) must vanish (the Coriolis terms disappear). The points $\phi = 0, \pi$ are the poles of the coordinates (θ undefined there) and must be examined separately; the symbolic check confirms the list below with Cartesian coordinates and a Lagrange multiplier.

With $\Lambda = m\Omega^2/k$, the two conditions are

$$\sin\phi\left(2k + m\Omega^2\sin^2\theta\cos\phi\right) = 0 \iff \sin\phi\left(2 + \Lambda\sin^2\theta\cos\phi\right) = 0, \qquad \sin\phi\sin\theta\cos\theta = 0$$

- $\sin\phi = 0$**:** both are satisfied. **E1:** $\phi = 0$, $\mathbf{r}_1^P = a\,\mathbf{i}_1$, the point of the sphere nearest to A. **E2:** $\phi = \pi$, $\mathbf{r}_1^P = -a\,\mathbf{i}_1$, the point farthest from A. They exist for every Λ: at both, the attraction and the centrifugal force are radial (along $\pm\mathbf{i}_1$) and the reaction balances them.
- $\sin\phi \neq 0$**:** the second condition needs $\sin\theta = 0$ or $\cos\theta = 0$. If $\sin\theta = 0$, the first becomes $2 = 0$: impossible. If $\cos\theta = 0$ ($\theta = \pm\pi/2$, $\sin^2\theta = 1$), the first gives

$$\cos\phi^* = -\frac{2k}{m\Omega^2} = -\frac{2}{\Lambda}$$

which has a solution with $\sin\phi^* \neq 0$ only if $\Lambda > 2$. **E3, E4** ($\theta = \pm\pi/2$):

$$\mathbf{r}_1^P = -\frac{2a}{\Lambda}\,\mathbf{i}_1 \pm a\sqrt{1-\frac{4}{\Lambda^2}}\,\mathbf{k}_1$$

They lie in the meridian plane $\mathbf{i}_1\mathbf{k}_1$ that contains A, on the hemisphere opposite to A, symmetric above and below the equator.

Discussion in terms of $m\Omega^2/k$:

- $\Lambda = m\Omega^2/k \leq 2$: only E1 and E2.
- $\Lambda > 2$: E1, E2, E3 and E4. As $\Lambda \to 2^+$, $\cos\phi^* \to -1$ and E3, E4 merge into E2 (a pitchfork bifurcation); as $\Lambda \to \infty$, $\cos\phi^* \to 0$ and they tend to the poles $\pm a\,\mathbf{k}_1$.

Extension (stability, beyond what is asked). Near E1, $V_1 \approx \text{const} + k\left(y_1^2+z_1^2\right) + \frac{1}{2}m\Omega^2 z_1^2$, a strict minimum, so E1 is stable for every Λ (energy argument). Near E2, $V_1 \approx \text{const} - k y_1^2 + \left(\frac{1}{2}m\Omega^2 - k\right) z_1^2$; along $\mathbf{j}_1$ the linearised motion is $m\ddot{y}_1 = 2ky_1$ (no Coriolis coupling there, because $\mathbf{k}_1 \times \mathbf{v}_1^P$ is normal to the sphere at E2), so E2 is always unstable with growth rate $\sqrt{2k/m}$. E3 and E4 are maxima of V_1; the linearisation including the Coriolis terms (computed numerically for several values $\Lambda > 2$) has eigenvalues with positive real part, so they are unstable as well.

(e) Reaction of the sphere

> **Pedagogical Justification:** the projection of the vector equation on $\mathbf{e}_r$ (written at the end of the previous section) contains N and only known functions once $\phi(t), \theta(t)$ are known.

$$\boxed{\mathbf{N} = -\left[ma\left(\dot{\phi}^2 + \sin^2\phi\,\dot{\theta}^2\right) + ka\left(2\cos\phi - 1\right) + m\Omega^2 a\left(1-\sin^2\phi\sin^2\theta\right) + 2m\Omega a\left(\cos\theta\,\dot{\phi} - \sin\theta\sin\phi\cos\phi\,\dot{\theta}\right)\right]\mathbf{e}_r}$$

with $\mathbf{e}_r = \cos\phi\,\mathbf{i}_1 + \sin\phi\cos\theta\,\mathbf{j}_1 + \sin\phi\sin\theta\,\mathbf{k}_1$ and $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in terms of $\mathcal{B}_0$ from the change-of-basis section.

Checks

- **Competition of two effects:** the attraction pulls P towards the point nearest A (E1), and its tangential part has stiffness $\sim k$; the centrifugal force pushes P away from the vertical axis with stiffness $\sim m\Omega^2$. The ratio $\Lambda = m\Omega^2/k$ decides whether, on the far side of the sphere, the centrifugal push towards the equator can be balanced by the attraction: only for $\Lambda > 2$ do the extra equilibria E3, E4 appear.
- **Reaction at E1:** with $\phi = 0$ and no motion, $N = -ka - m\Omega^2 a$: the sphere pulls P inwards, since both the attraction (towards A, outside the sphere) and the centrifugal force push P outwards. The bilateral contact assumed among the hypotheses is needed for that.
- **Limit $\Omega \to 0$:** the inertia forces vanish, $\Lambda \to 0$, and only E1 (stable) and E2 (unstable) remain, as for a fixed sphere with a fixed attracting point.

- **Orthogonality:** $\mathbf{e}_r \cdot \mathbf{e}_\phi = -\cos\phi\sin\phi + \sin\phi\cos\phi(\cos^2\theta + \sin^2\theta) = 0$, $\mathbf{e}_r \cdot \mathbf{e}_\theta = \sin\phi(-\cos\theta\sin\theta + \sin\theta\cos\theta) = 0$, $\mathbf{e}_\phi \cdot \mathbf{e}_\theta = \cos\phi(-\cos\theta\sin\theta + \sin\theta\cos\theta) = 0$.
- **Dimensions:** in the $\mathbf{e}_\phi$ equation, $[ma\ddot{\phi}] = \mathrm{kg\,m\,s^{-2}} = \mathrm{N}$, $[ka] = \mathrm{N}$, $[m\Omega^2 a] = \mathrm{N}$, $[m\Omega a\dot{\theta}] = \mathrm{N}$; $[E_1^P] = [ka^2] = [m\Omega^2 a^2] = \mathrm{J}$; $\Lambda = \mathrm{kg\,s^{-2}/(N/m)}$ is dimensionless.
- **Symbolic check (sympy):** $\mathbf{a}_0^P$ from differentiating $[{}_0R_1]\,\mathbf{r}_1^P$ twice equals the Notes Eq. 3.27 decomposition; the force components, both equations of motion, the reaction and $dE_1^P/dt = 0$ are confirmed; the equilibria found with Cartesian coordinates and a Lagrange multiplier are exactly E1 to E4.
- **Check against the official key:** none for this problem (Problems.pdf, p. 76 lists no answer for 49), so the results rest on the independent checks above.

Source: Problems.pdf (PDF page 34); no official key

PROBLEM 51

Particle inside a Hollow Sphere in a Rotating Frame

Original statement · Problems.pdf, p. 36

Non-Inertial Frame + Surface Constraint

A heavy point particle P of mass m is constrained to move inside the surface of a hollow sphere of radius R and center C located at $\mathbf{r}_1^C = R(\mathbf{j}_1 + \mathbf{k}_1)$. The surface of the sphere is smooth. The reference frame S_1 shown in the figure is a non-inertial reference frame, because it rotates with a known angular velocity $\boldsymbol{\omega}_{10} = \Omega(t)\,\mathbf{k}_1$ with respect to inertial space. We introduce the two coordiantes [sic] θ and ϕ as shown in the figure, in order to characterize the configuration of P.

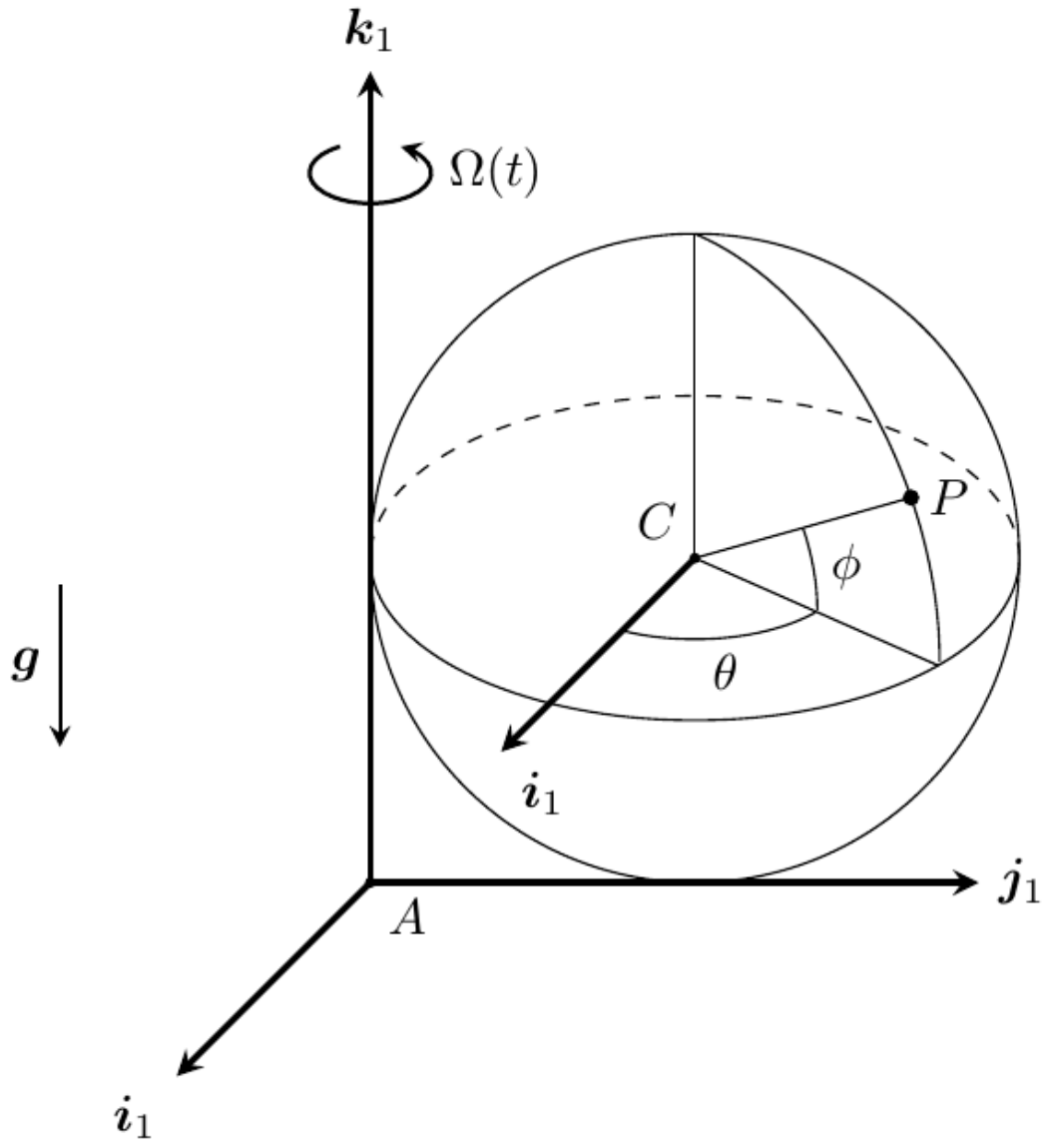


Figure: Rotating frame S1 with axes i1, j1, k1, hollow sphere of centre C at R(j1+k1), particle P located by the angles theta and phi, angular velocity Omega(t) about k1 (Problems.pdf, p. 36).

(a) Give the expressions for the position, velocity, and acceleration vectors of P with respect to S_1, as functions of θ, ϕ and their derivatives. Remember that you may use any unit vectors in your solution, provided that you define them first.

(b) Identify and write down all forces (actual and fictitious) on P as seen from S_1.

(c) Write down two differential equations for θ and ϕ, free of any other unknowns.

(d) Write down an expression for the reaction unknown. This expression may depend on $\theta, \dot{\theta}, \phi, \dot{\phi}$ and the known parameters of the problem. What is the mathematical condition for the particle to separate from the surface?

(e) In the case that $\Omega = \text{const}$, determine and discuss the value of θ, ϕ and the reaction unknown for the equilibrium configurations of P.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a heavy particle P (mass m) slides on the smooth inner surface of a sphere of radius R and centre C, $\mathbf{r}_1^C = R(\mathbf{j}_1 + \mathbf{k}_1)$. The sphere is fixed in the frame $S_1 = \{A; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, which turns about its vertical axis $A\mathbf{k}_1$ with the known rate $\Omega(t)$. The sphere touches the rotation axis at the point $(0, 0, R)$ and the plane $z_1 = 0$ at $(0, R, 0)$.
- **Why S_1.** The constraint (the sphere) is fixed in S_1, so in S_1 the particle simply moves on a fixed sphere described by two angles; all the effect of the rotation is collected in the inertia forces (Notes Eqs. 4.62–4.63). In S_0 the sphere would be a moving constraint.
- **Hypotheses:**
  1. The point A is fixed in an inertial frame S_0 (it lies on the rotation axis), so $\mathbf{v}_0^A = \mathbf{a}_0^A = \mathbf{0}$; $\mathbf{k}_1$ is the ascending vertical and gravity is $-g\,\mathbf{k}_1$ (figure).
  2. The surface is smooth: the reaction is normal to the sphere, along the radius CP. It is unilateral: the particle is inside, so the surface can only push it toward C.
  3. $\Omega(t)$ is a known function; $\dot{\Omega}(t)$ is then known too.
- **Frames.** $S_0 = \{A; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ inertial, with $\mathbf{k}_0 = \mathbf{k}_1$; S_1, basis $\mathcal{B}_1$, turns about $\mathbf{k}_1$ through the angle $\Psi(t)$, with $\dot{\Psi} = \Omega(t)$.

- **Coordinates (from the figure):** θ is the azimuth about the vertical through C, measured from the direction $\mathbf{i}_1$ toward $\mathbf{j}_1$; ϕ is the elevation of CP above the horizontal plane through C. Ranges: $\theta \in (-\pi, \pi]$, $\phi \in [-\pi/2, \pi/2]$ (the poles $\phi = \pm\pi/2$ are coordinate singularities).
- **Spherical basis at P:** $\{\mathbf{u}_r, \mathbf{u}_\theta, \mathbf{u}_\phi\}$, deduced in the next section.
- **Degrees of freedom:** 3 coordinates minus 1 constraint $|CP| = R$: **2 DOF**, (θ, ϕ). The reaction adds **1 unknown**, N.

Symbol	Meaning	SI unit
m	mass of P	kg
R	radius of the sphere	m
g	gravitational acceleration	m/s^2
$\Omega(t)$, $\dot{\Omega}(t)$	angular velocity and acceleration of S_1 about $\mathbf{k}_1$	rad/s, rad/s^2
θ, ϕ	azimuth and elevation of P seen from C	rad
N	magnitude of the reaction, $\mathbf{N} = -N\,\mathbf{u}_r$, $N \geq 0$ while in contact	N

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from above

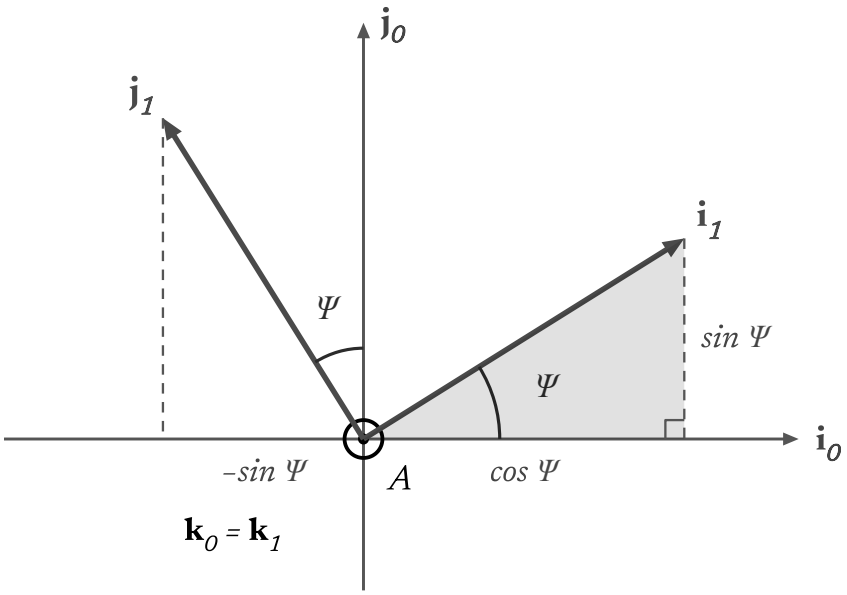


Figure 1. View from $+z$, with $\mathbf{k}_0 = \mathbf{k}_1$ out of the paper ($\odot$). S_1 has turned the angle $\Psi(t)$, $\dot{\Psi} = \Omega(t)$, counter-clockwise: $\mathbf{i}_1$ has the legs $\cos\Psi$ along $\mathbf{i}_0$ and $\sin\Psi$ along $\mathbf{j}_0$; the tip of $\mathbf{j}_1$ moves to the side of $-\mathbf{i}_0$, hence $-\sin\Psi$.

The rotation is about $\mathbf{k}_1 = \mathbf{k}_0$, so we look down from $+z$: $\mathbf{k}_0$ points out of the paper ($\odot$), $\mathbf{i}_0$ to the right and, since $\mathbf{i}_0 \times \mathbf{j}_0 = \mathbf{k}_0$ needs a counter-clockwise turn from $\mathbf{i}_0$ to $\mathbf{j}_0$ for the thumb to point at the reader, $\mathbf{j}_0$ points up. A positive Ω turns S_1 counter-clockwise by $\Psi(t)$:

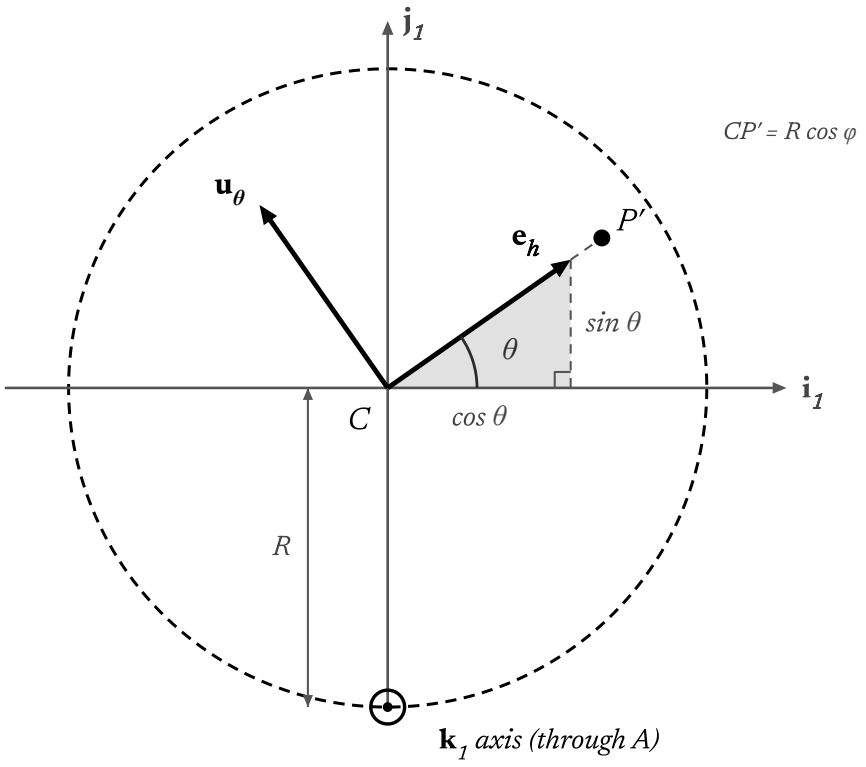
- $\mathbf{i}_1$ is the hypotenuse (length 1) of the shaded triangle: adjacent leg $\cos\Psi$ along $\mathbf{i}_0$, opposite leg $\sin\Psi$ along $+\mathbf{j}_0$.
- $\mathbf{j}_1$ starts on $\mathbf{j}_0$ and its tip moves to the side of $-\mathbf{i}_0$: leg $\cos\Psi$ along $\mathbf{j}_0$ and $\sin\Psi$ towards $-\mathbf{i}_0$, **hence the minus sign** in $\mathbf{j}_1 = -\sin\Psi\,\mathbf{i}_0 + \cos\Psi\,\mathbf{j}_0$.
- $\mathbf{k}_1 = \mathbf{k}_0$.

$$[{}_0R_1] = \begin{pmatrix} \cos\Psi & -\sin\Psi & 0 \\ \sin\Psi & \cos\Psi & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1 \cdot (\cos\Psi\cos\Psi - (-\sin\Psi)\sin\Psi) = \cos^2\Psi + \sin^2\Psi = 1$$

The columns have unit norm and the first two are orthogonal ($-\cos\Psi\sin\Psi + \sin\Psi\cos\Psi = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.

Motion of S_1: $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{k}_1$; since $\mathbf{k}_1$ is fixed in both frames, $\boldsymbol{\alpha}_{10} = \left.\frac{d\boldsymbol{\omega}_{10}}{dt}\right|_0 = \dot{\Omega}\,\mathbf{k}_1 + \Omega\,\boldsymbol{\omega}_{10} \times \mathbf{k}_1 = \dot{\Omega}\,\mathbf{k}_1$; $\mathbf{r}_0^A = \mathbf{v}_0^A = \mathbf{a}_0^A = \mathbf{0}$. Check: by the chain rule, $\left.\frac{d\mathbf{i}_1}{dt}\right|_0 = \dot{\Psi}\,(-\sin\Psi\,\mathbf{i}_0 + \cos\Psi\,\mathbf{j}_0) = \Omega\,\mathbf{j}_1 = \Omega\,\mathbf{k}_1 \times \mathbf{i}_1$.

The azimuth θ: horizontal plane through C



dashed circle: equator of the sphere (radius R)

Figure 2. Horizontal plane through C seen from above ($\mathbf{k}_1$ out of the paper, $\odot$). The azimuth θ is measured at C from $\mathbf{i}_1$ towards $\mathbf{j}_1$: $\mathbf{e}_h = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1$ (legs of the shaded triangle) and $\mathbf{u}_\theta = d\mathbf{e}_h/d\theta$. P projects to P' at distance $R\cos\phi$ from C; the axis of rotation is at distance R from C, on the side of $-\mathbf{j}_1$.

Seen from above ($\mathbf{k}_1$ out of the paper), with $\mathbf{i}_1$ to the right and $\mathbf{j}_1$ up, the centre C is at distance R from the rotation axis on the side of $+\mathbf{j}_1$ (its horizontal position is $R\,\mathbf{j}_1$). The azimuth θ is measured at C from $\mathbf{i}_1$ towards $\mathbf{j}_1$, counter-clockwise in this view. The horizontal unit vector in that direction is the hypotenuse of the shaded triangle, with the leg $\cos\theta$ along $\mathbf{i}_1$ (adjacent) and $\sin\theta$ along $\mathbf{j}_1$ (opposite), and its derivative with respect to θ ($\frac{d}{d\theta}\cos\theta = -\sin\theta$, $\frac{d}{d\theta}\sin\theta = \cos\theta$) is the horizontal unit vector of increasing θ:

$$\mathbf{e}_h = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1, \qquad \mathbf{u}_\theta = \frac{d\mathbf{e}_h}{d\theta} = -\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1$$

(the minus sign: turning $\mathbf{e}_h$ a further 90° counter-clockwise takes its tip to the side of $-\mathbf{i}_1$). Also $\mathbf{k}_1 \times \mathbf{e}_h = \mathbf{u}_\theta$ and $\mathbf{k}_1 \times \mathbf{u}_\theta = -\mathbf{e}_h$.

The elevation ϕ: vertical plane through C and P

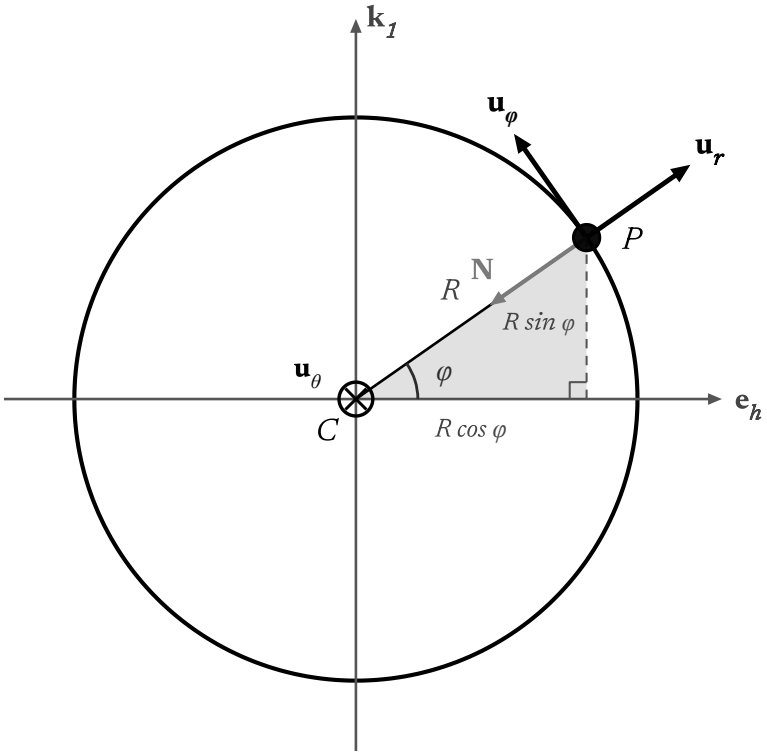


Figure 3. Vertical plane through C that contains $\mathbf{e}_h$ ($\mathbf{u}_\theta$ points into the paper, $\otimes$). The elevation ϕ is measured at C from the horizontal: in the shaded triangle the hypotenuse is R, the leg adjacent to ϕ is $R\cos\phi$ along $\mathbf{e}_h$ and the opposite one $R\sin\phi$ along $\mathbf{k}_1$. $\mathbf{u}_r, \mathbf{u}_\phi$ in orange; the reaction $\mathbf{N} = -N\,\mathbf{u}_r$ (pink) pushes towards C.

P lies in the vertical plane through C that contains $\mathbf{e}_h$; it is drawn with $\mathbf{e}_h$ to the right and $\mathbf{k}_1$ up, so $\mathbf{e}_h \times \mathbf{k}_1 = -\mathbf{u}_\theta$ points out of the paper and $\mathbf{u}_\theta$ points into it ($\otimes$). In the shaded right triangle the hypotenuse is $|CP| = R$, the leg adjacent to ϕ is $R\cos\phi$ along $\mathbf{e}_h$ (the distance CP' of the horizontal view) and the leg opposite to ϕ is $R\sin\phi$ along $\mathbf{k}_1$. The unit vector from C to P and its derivative with respect to ϕ are

$$\mathbf{u}_r = \cos\phi\,\mathbf{e}_h + \sin\phi\,\mathbf{k}_1 = \cos\phi\cos\theta\,\mathbf{i}_1 + \cos\phi\sin\theta\,\mathbf{j}_1 + \sin\phi\,\mathbf{k}_1 \quad (\text{from } C \text{ to } P)$$

$$\mathbf{u}_\phi = \frac{\partial \mathbf{u}_r}{\partial \phi} = -\sin\phi\,\mathbf{e}_h + \cos\phi\,\mathbf{k}_1 = -\sin\phi\cos\theta\,\mathbf{i}_1 - \sin\phi\sin\theta\,\mathbf{j}_1 + \cos\phi\,\mathbf{k}_1 \quad (\text{increasing } \phi)$$

while $\frac{\partial \mathbf{u}_r}{\partial \theta} = \cos\phi \, \frac{d\mathbf{e}_h}{d\theta} = \cos\phi \, \mathbf{u}_\theta$, so $\mathbf{u}_\theta = \frac{1}{\cos\phi} \frac{\partial \mathbf{u}_r}{\partial \theta}$ (horizontal, increasing θ).

Relation with the standard spherical coordinates. The usual polar angle ϑ is measured from the upward vertical $\mathbf{k}_1$ (colatitude), with $\mathbf{u}_r = \sin\vartheta\cos\theta\,\mathbf{i}_1 + \sin\vartheta\sin\theta\,\mathbf{j}_1 + \cos\vartheta\,\mathbf{k}_1$. The elevation is measured from the horizontal instead, so $\phi = \frac{\pi}{2} - \vartheta$, and

$$\sin\vartheta = \sin\left(\tfrac{\pi}{2} - \phi\right) = \sin\tfrac{\pi}{2}\cos\phi - \cos\tfrac{\pi}{2}\sin\phi = \cos\phi, \qquad \cos\vartheta = \cos\left(\tfrac{\pi}{2} - \phi\right) = \cos\tfrac{\pi}{2}\cos\phi + \sin\tfrac{\pi}{2}\sin\phi = \sin\phi$$

which turns the standard $\mathbf{u}_r$ into ours. Because ϕ grows upwards while ϑ grows downwards, $\mathbf{u}_\phi = -\mathbf{u}_\vartheta$.

Change-of-basis matrix (columns: $\mathbf{u}_r, \mathbf{u}_\theta, \mathbf{u}_\phi$ in $\mathcal{B}_1$):

$$[{}_1R_u] = \begin{pmatrix} \cos\phi\cos\theta & -\sin\theta & -\sin\phi\cos\theta \\ \cos\phi\sin\theta & \cos\theta & -\sin\phi\sin\theta \\ \sin\phi & 0 & \cos\phi \end{pmatrix}$$

Checks: each column has unit norm ($\cos^2\phi(\cos^2\theta + \sin^2\theta) + \sin^2\phi = 1$, etc.); $\mathbf{u}_r \cdot \mathbf{u}_\theta = -\cos\phi\cos\theta\sin\theta + \cos\phi\sin\theta\cos\theta = 0$, $\mathbf{u}_r \cdot \mathbf{u}_\phi = -\cos\phi\sin\phi + \sin\phi\cos\phi = 0$, $\mathbf{u}_\theta \cdot \mathbf{u}_\phi = \sin\phi\sin\theta\cos\theta - \sin\phi\sin\theta\cos\theta = 0$, so $[{}_1R_u]^T[{}_1R_u] = I$. The determinant is the triple product $\det[{}_1R_u] = \mathbf{u}_r \cdot (\mathbf{u}_\theta \times \mathbf{u}_\phi)$, and $\mathbf{u}_\theta \times \mathbf{u}_\phi = -\sin\phi\,(\mathbf{u}_\theta \times \mathbf{e}_h) + \cos\phi\,(\mathbf{u}_\theta \times \mathbf{k}_1) = \sin\phi\,\mathbf{k}_1 + \cos\phi\,\mathbf{e}_h = \mathbf{u}_r$ (using $\mathbf{u}_\theta \times \mathbf{e}_h = -\mathbf{k}_1$ and $\mathbf{u}_\theta \times \mathbf{k}_1 = \mathbf{e}_h$), hence $\det[{}_1R_u] = \mathbf{u}_r \cdot \mathbf{u}_r = 1$: the triad $\{\mathbf{u}_r, \mathbf{u}_\theta, \mathbf{u}_\phi\}$ is right-handed.

Useful inverse relations (read from the two triangles): $\mathbf{e}_h = \cos\phi\,\mathbf{u}_r - \sin\phi\,\mathbf{u}_\phi$, $\mathbf{k}_1 = \sin\phi\,\mathbf{u}_r + \cos\phi\,\mathbf{u}_\phi$, $\mathbf{i}_1 = \cos\theta\,\mathbf{e}_h - \sin\theta\,\mathbf{u}_\theta$, $\mathbf{j}_1 = \sin\theta\,\mathbf{e}_h + \cos\theta\,\mathbf{u}_\theta$; and $\mathbf{k}_1 \times \mathbf{e}_h = \mathbf{u}_\theta$, $\mathbf{k}_1 \times \mathbf{u}_\theta = -\mathbf{e}_h$, $\mathbf{k}_1 \times \mathbf{u}_\phi = -\sin\phi\,\mathbf{u}_\theta$.

Derivatives in S_1 (chain rule, θ and ϕ depend on t):

$$\left.\frac{d\mathbf{u}_r}{dt}\right|_1 = \frac{\partial \mathbf{u}_r}{\partial\theta}\dot\theta + \frac{\partial \mathbf{u}_r}{\partial\phi}\dot\phi = \dot\theta\cos\phi\,\mathbf{u}_\theta + \dot\phi\,\mathbf{u}_\phi$$

$$\left.\frac{d\mathbf{u}_\theta}{dt}\right|_1 = -\dot\theta\,\mathbf{e}_h = -\dot\theta\,(\cos\phi\,\mathbf{u}_r - \sin\phi\,\mathbf{u}_\phi), \qquad \left.\frac{d\mathbf{u}_\phi}{dt}\right|_1 = -\dot\phi\cos\phi\,\mathbf{e}_h - \sin\phi\,\dot\theta\,\mathbf{u}_\theta - \dot\phi\sin\phi\,\mathbf{k}_1 = -\dot\phi\,\mathbf{u}_r - \dot\theta\sin\phi\,\mathbf{u}_\theta$$

Position vector

To reach P from A we pass through the centre C, because the distance from C is the one that stays constant:

$$\mathbf{r}_1^P = \overrightarrow{AC} + \overrightarrow{CP} = R\,(\mathbf{j}_1 + \mathbf{k}_1) + R\,\mathbf{u}_r$$

In components of $\mathcal{B}_1$: $x_1 = R\cos\phi\cos\theta$, $y_1 = R(1 + \cos\phi\sin\theta)$, $z_1 = R(1 + \sin\phi)$.

Kinematics

(a) Position, velocity and acceleration of P relative to S_1

Pedagogical Justification: relative quantities are derivatives in S_1; $\mathbf{r}_1^C$ is constant in S_1 and only $\mathbf{u}_r$ changes, through $\theta(t)$ and $\phi(t)$.

$$\boxed{\mathbf{r}_1^P = \mathbf{r}_1^C + \overrightarrow{CP} = R\,(\mathbf{j}_1 + \mathbf{k}_1) + R\,\mathbf{u}_r}$$

$$\boxed{\mathbf{v}_1^P = R\,\frac{d\mathbf{u}_r}{dt}\bigg|_1 = R\dot\theta\cos\phi\,\mathbf{u}_\theta + R\dot\phi\,\mathbf{u}_\phi}$$

Differentiating again in S_1 (product rule, $\frac{d}{dt}(\dot\theta\cos\phi) = \ddot\theta\cos\phi - \dot\theta\dot\phi\sin\phi$, and the basis derivatives of the change-of-basis section):

$$\mathbf{a}_1^P = R\left(\ddot\theta\cos\phi - \dot\theta\dot\phi\sin\phi\right)\mathbf{u}_\theta + R\dot\theta\cos\phi\left[-\dot\theta\left(\cos\phi\,\mathbf{u}_r - \sin\phi\,\mathbf{u}_\phi\right)\right] + R\ddot\phi\,\mathbf{u}_\phi + R\dot\phi\left(-\dot\phi\,\mathbf{u}_r - \dot\theta\sin\phi\,\mathbf{u}_\theta\right)$$

Collecting components:

$$\boxed{\mathbf{a}_1^P = -R\left(\dot\phi^2 + \dot\theta^2\cos^2\phi\right)\mathbf{u}_r + R\left(\ddot\theta\cos\phi - 2\dot\theta\dot\phi\sin\phi\right)\mathbf{u}_\theta + R\left(\ddot\phi + \dot\theta^2\sin\phi\cos\phi\right)\mathbf{u}_\phi}$$

Forces and Equations of Motion

(b) Forces on P seen from S_1

Pedagogical Justification: in S_1, Notes Eq. 4.63 reads $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$, where $\mathbf{F}_i$ (Notes Eq. 4.62) has four terms (Slide 28). We evaluate each with $\mathbf{a}_0^A = \mathbf{0}$, $\boldsymbol{\omega}_{10} = \Omega\mathbf{k}_1$, $\boldsymbol{\alpha}_{10} = \dot{\Omega}\mathbf{k}_1$, and project on the spherical basis.

Actual forces.

- Weight: $m\mathbf{g} = -mg\,\mathbf{k}_1 = -mg\sin\phi\,\mathbf{u}_r - mg\cos\phi\,\mathbf{u}_\phi$.
- Reaction (smooth, unilateral, toward C, as drawn in the vertical view): $\mathbf{N} = -N\,\mathbf{u}_r$, $N \geq 0$ while in contact.

Fictitious forces. Only the part of $\mathbf{r}_1^P$ normal to $\mathbf{k}_1$ enters the cross products with $\mathbf{k}_1$. That part is $\mathbf{r}_\perp = x_1\mathbf{i}_1 + y_1\mathbf{j}_1$; substituting $\mathbf{i}_1 = \cos\theta\,\mathbf{e}_h - \sin\theta\,\mathbf{u}_\theta$ and $\mathbf{j}_1 = \sin\theta\,\mathbf{e}_h + \cos\theta\,\mathbf{u}_\theta$:

$$\mathbf{r}_\perp = R\left[\cos\phi\cos^2\theta + \sin\theta + \cos\phi\sin^2\theta\right]\mathbf{e}_h + R\left[-\cos\phi\cos\theta\sin\theta + \cos\theta + \cos\phi\sin\theta\cos\theta\right]\mathbf{u}_\theta = R\left(\cos\phi + \sin\theta\right)\mathbf{e}_h + R\cos\theta\,\mathbf{u}_\theta$$

- **Origin force:** $\mathbf{F}_A = -m\mathbf{a}_0^A = \mathbf{0}$.
- **Euler force** (Slide 31), with $\mathbf{k}_1 \times \mathbf{e}_h = \mathbf{u}_\theta$, $\mathbf{k}_1 \times \mathbf{u}_\theta = -\mathbf{e}_h$:

$$\mathbf{F}_E = -m\dot{\Omega}\,\mathbf{k}_1 \times \mathbf{r}_\perp = -m\dot{\Omega}R\left[\left(\cos\phi + \sin\theta\right)\mathbf{u}_\theta - \cos\theta\,\mathbf{e}_h\right]$$

$$\mathbf{F}_E = m\dot{\Omega}R\left[\cos\theta\cos\phi\,\mathbf{u}_r - \left(\cos\phi + \sin\theta\right)\mathbf{u}_\theta - \cos\theta\sin\phi\,\mathbf{u}_\phi\right]$$

- **Centrifugal force** (Slide 30), $\mathbf{F}_C = m\Omega^2\mathbf{r}_\perp$:

$$\mathbf{F}_C = m\Omega^2 R\left[\left(\cos\phi + \sin\theta\right)\cos\phi\,\mathbf{u}_r + \cos\theta\,\mathbf{u}_\theta - \left(\cos\phi + \sin\theta\right)\sin\phi\,\mathbf{u}_\phi\right]$$

- **Coriolis force** (Slide 31): $\mathbf{k}_1 \times \mathbf{v}_1^P = R\dot{\theta}\cos\phi\,(-\mathbf{e}_h) + R\dot{\phi}\,(-\sin\phi\,\mathbf{u}_\theta)$, so

$$\mathbf{F}_{COR} = -2m\Omega\,\mathbf{k}_1 \times \mathbf{v}_1^P = 2m\Omega R\left(\dot{\theta}\cos\phi\,\mathbf{e}_h + \dot{\phi}\sin\phi\,\mathbf{u}_\theta\right) = 2m\Omega R\left[\dot{\theta}\cos^2\phi\,\mathbf{u}_r + \dot{\phi}\sin\phi\,\mathbf{u}_\theta - \dot{\theta}\sin\phi\cos\phi\,\mathbf{u}_\phi\right]$$

As required, $\mathbf{F}_{COR} \cdot \mathbf{v}_1^P = 2m\Omega R^2 \left(\dot{\phi} \sin\phi \, \dot{\theta} \cos\phi - \dot{\theta} \sin\phi \cos\phi \, \dot{\phi} \right) = 0$.

(c) Two differential equations for θ and ϕ

> **Pedagogical Justification:** the reaction is along $\mathbf{u}_r$, so the projections of Notes Eq. 4.63 on $\mathbf{u}_\theta$ and $\mathbf{u}_\phi$ contain no unknown force.

Projection on $\mathbf{u}_\theta$ (weight and reaction have no $\mathbf{u}_\theta$ component):

$$mR \left(\ddot{\theta} \cos\phi - 2\dot{\theta}\dot{\phi} \sin\phi \right) = m\Omega^2 R \cos\theta - m\dot{\Omega} R \left(\cos\phi + \sin\theta \right) + 2m\Omega R \dot{\phi} \sin\phi$$

$$\boxed{\ddot{\theta} \cos\phi - 2\dot{\phi} \sin\phi \left(\dot{\theta} + \Omega \right) = \Omega^2 \cos\theta - \dot{\Omega} \left(\cos\phi + \sin\theta \right)}$$

Projection on $\mathbf{u}_\phi$:

$$mR \left(\ddot{\phi} + \dot{\theta}^2 \sin\phi \cos\phi \right) = -mg\cos\phi - m\Omega^2 R \left(\cos\phi + \sin\theta \right) \sin\phi - m\dot{\Omega} R \cos\theta \sin\phi - 2m\Omega R \dot{\theta} \sin\phi \cos\phi$$

$$\boxed{\ddot{\phi} + \dot{\theta} \left(\dot{\theta} + 2\Omega \right) \sin\phi \cos\phi + \frac{g}{R} \cos\phi + \Omega^2 \sin\phi \left(\cos\phi + \sin\theta \right) + \dot{\Omega} \cos\theta \sin\phi = 0}$$

These two coupled, non-linear ODEs with the known forcing $\Omega(t)$ determine $\theta(t)$, $\phi(t)$ from the initial conditions.

Projection on $\mathbf{u}_r$ (it contains the only unknown force, N):

$$-mR \left(\dot{\phi}^2 + \dot{\theta}^2 \cos^2\phi \right) = -N - mg\sin\phi + m\Omega^2 R \cos\phi \left(\cos\phi + \sin\theta \right) + m\dot{\Omega} R \cos\theta \cos\phi + 2m\Omega R \dot{\theta} \cos^2\phi$$

Answers and Checks

(a) Kinematics

$\mathbf{r}_1^P$, $\mathbf{v}_1^P$ and $\mathbf{a}_1^P$ are the three boxed expressions of the kinematics section, with $\mathbf{u}_r$, $\mathbf{u}_\theta$, $\mathbf{u}_\phi$ defined in the change-of-basis section.

(b) Forces

Weight $-mg\,\mathbf{k}_1$, reaction $-N\,\mathbf{u}_r$, origin force $\mathbf{0}$, and the Euler, centrifugal and Coriolis forces written above.

(c) Equations of motion

The two boxed equations for θ and ϕ.

(d) Reaction and separation condition

From the projection on $\mathbf{u}_r$:

$$\boxed{N = mR\left(\dot{\phi}^2 + \dot{\theta}^2\cos^2\phi\right) - mg\sin\phi + m\Omega^2 R\cos\phi\,(\cos\phi + \sin\theta) + m\dot{\Omega}R\cos\phi\cos\theta + 2m\Omega R\dot{\theta}\cos^2\phi}$$

Grouping $\dot{\theta}^2 + 2\Omega\dot{\theta} + \Omega^2 = (\dot{\theta}+\Omega)^2$ in the $\cos^2\phi$ terms gives the equivalent compact form

$$N = mR\dot{\phi}^2 + mR\cos^2\phi\left(\dot{\theta}+\Omega\right)^2 + m\Omega^2 R\cos\phi\sin\theta + m\dot{\Omega}R\cos\phi\cos\theta - mg\sin\phi$$

It depends only on $\theta, \dot{\theta}, \phi, \dot{\phi}$ and the known $m, R, g, \Omega(t), \dot{\Omega}(t)$. **Separation:** the surface can only push ($N \geq 0$). The particle leaves the surface at the first instant at which

$$\boxed{N\left(\theta, \dot{\theta}, \phi, \dot{\phi}, t\right) = 0 \ \text{ and would become negative}}$$

i.e. the motion predicted by (c) is valid only while $N \geq 0$.

(e) Equilibrium configurations for $\Omega = \text{const}$

> **Pedagogical Justification:** equilibrium in S_1 means θ, ϕ constant: $\dot{\theta} = \dot{\phi} = \ddot{\theta} = \ddot{\phi} = 0$, with $\dot{\Omega} = 0$. Then $\mathbf{F}_E = \mathbf{F}_{COR} = \mathbf{0}$ and the equations of (c) become algebraic. We take $\Omega \neq 0$ (the case $\Omega = 0$ is discussed at the end).

From the θ equation: $\Omega^2\cos\theta = 0 \Rightarrow \cos\theta = 0$, so $\theta = \pi/2$ or $\theta = -\pi/2$: P must lie in the vertical plane $x_1 = 0$ that contains the rotation axis and C (otherwise the centrifugal force has a component $m\Omega^2 R\cos\theta$ along $\mathbf{u}_\theta$ that nothing balances).

From the ϕ equation, multiplied by R: $g\cos\phi + \Omega^2 R\sin\phi(\cos\phi + \sin\theta) = 0$. The poles $\cos\phi = 0$ do not satisfy it ($\Omega^2 R\sin\phi\sin\theta = \pm\Omega^2 R \neq 0$), so we can divide by $g\cos\phi > 0$. With the dimensionless number $\lambda = \Omega^2 R/g > 0$:

Branch 1, $\theta = \pi/2$ (far side of the sphere from the axis, $y_1 = R(1+\cos\phi) \geq R$):

$$1 + \lambda\,\frac{\sin\phi\,(\cos\phi + 1)}{\cos\phi} = 0 \quad\Longleftrightarrow\quad \boxed{\tan\phi_1 + \sin\phi_1 = -\frac{g}{\Omega^2 R}}$$

The function $f(\phi) = \tan\phi + \sin\phi$ is strictly increasing on $(-\pi/2, \pi/2)$ ($f' = \sec^2\phi + \cos\phi > 0$), goes from $-\infty$ to $+\infty$ and vanishes at 0; so there is **exactly one root,** $\phi_1 \in (-\pi/2, 0)$, below the horizontal plane of C.

Branch 2, $\theta = -\pi/2$ (side of the sphere next to the axis, $y_1 = R(1-\cos\phi)$):

$$1 + \lambda\,\frac{\sin\phi\,(\cos\phi - 1)}{\cos\phi} = 0 \quad\Longleftrightarrow\quad \boxed{\tan\phi_2 - \sin\phi_2 = \frac{g}{\Omega^2 R}}$$

$h(\phi) = \tan\phi - \sin\phi$ has $h' = \sec^2\phi - \cos\phi \geq 0$ (zero only at $\phi = 0$), $h(0) = 0$ and $h \to +\infty$ as $\phi \to \pi/2$, so there is **exactly one root,** $\phi_2 \in (0, \pi/2)$, above the horizontal plane of C.

Reaction at equilibrium. With $\dot\theta = \dot\phi = \dot\Omega = 0$, (d) gives $N = m\Omega^2 R\cos\phi(\cos\phi + \sin\theta) - mg\sin\phi$. The equilibrium relation $\Omega^2 R(\cos\phi + \sin\theta) = -g\cos\phi/\sin\phi$ ($\sin\phi \neq 0$ at both roots) turns it into

$$N = -\frac{mg\cos^2\phi}{\sin\phi} - mg\sin\phi = -\frac{mg\left(\cos^2\phi + \sin^2\phi\right)}{\sin\phi} \quad\Longrightarrow\quad \boxed{N = -\frac{mg}{\sin\phi}}$$

This is simply the vertical balance: the centrifugal force is horizontal, so the vertical component of $\mathbf{N} = -N\mathbf{u}_r$, which is $-N\sin\phi$ (the leg $\sin\phi$ of the vertical-view triangle), must cancel the weight.

Discussion.

- **Branch 1** ($\theta = \pi/2$, $\phi_1 < 0$): $N_1 = mg/|\sin\phi_1| > 0$. **This is the only physically possible equilibrium.** The particle sits in the lower half of the sphere, on the side away from the axis, where the inward normal $-\mathbf{u}_r$ points up and toward the axis, balancing the weight and the outward centrifugal force. As Ω grows, ϕ_1 rises toward the equator of the sphere: for $\lambda \gg 1$, $\phi_1 \approx -g/(2\Omega^2 R)$ and $N_1 \approx 2m\Omega^2 R$ (the centrifugal force at distance $2R$ from the axis); for $\Omega \to 0$, $\phi_1 \to -\pi/2$ (bottom) and $N_1 \to mg$. Example ($R = 1$ m, $g = 9.81$ m/s^2): $\Omega = 3$ rad/s gives $\phi_1 = -30.3^\circ$ and $N_1 = 1.98\,mg$.

- **Branch 2** ($\theta = -\pi/2$, $\phi_2 > 0$): $N_2 = -mg/\sin\phi_2 < 0$. It solves the equations of (c), but the reaction would have to **pull** the particle away from C, which the inner surface cannot do: it is **not an admissible equilibrium** for the unilateral contact (it would be one for a particle held on both sides of the surface, or resting on the outside of the sphere). On that side the centrifugal force points toward the inside of the sphere, so it cannot help the surface to hold the particle up.
- **Stability of branch 1** (extension, using the energy in S_1). For constant Ω, A fixed and a smooth sphere fixed in S_1, the energy $E_1^P = T_1^P + V_1$ is conserved (Slides 29–31), with $V_1 = mgz_1 - \tfrac{1}{2}m\Omega^2(x_1^2 + y_1^2) = mgR(1+\sin\phi) - \tfrac{1}{2}m\Omega^2R^2\left(1+\cos^2\phi + 2\cos\phi\sin\theta\right)$. Its derivatives $\partial V_1/\partial\theta = -m\Omega^2R^2\cos\phi\cos\theta$ and $\partial V_1/\partial\phi = mgR\cos\phi + m\Omega^2R^2\sin\phi(\cos\phi+\sin\theta)$ reproduce the equilibrium equations. At branch 1, $\partial^2V_1/\partial\theta^2 = m\Omega^2R^2\cos\phi_1\sin\theta = m\Omega^2R^2\cos\phi_1 > 0$, $\partial^2V_1/\partial\theta\partial\phi = m\Omega^2R^2\sin\phi_1\cos\theta = 0$ and $\partial^2V_1/\partial\phi^2 = mgR\lambda\cos\phi_1\left(\sec^2\phi_1 + \cos\phi_1\right) > 0$: V_1 has a strict minimum, so the equilibrium is stable (the linearised equations give purely imaginary eigenvalues, see the verification).
- **Case $\Omega = 0$:** the θ equation is satisfied identically and the ϕ equation gives $\cos\phi = 0$: the bottom point $\phi = -\pi/2$ with $N = mg$ (the top point $\phi = \pi/2$ would need $N = -mg < 0$); θ is undefined there (pole of the coordinates).

Checks

- **Role of each inertia force.** The centrifugal force pushes P away from the axis and is responsible for the tilt of the equilibrium; the Euler force (only while Ω changes) drags P against the sense of the angular acceleration, along $-\mathbf{u}_\theta$-like directions; the Coriolis force couples the two angular motions (terms $2\Omega\dot\phi\sin\phi$ and $2\Omega\dot\theta\sin\phi\cos\phi$) and does no work.
- **Absolute reading of N.** In the compact form, $\dot\theta + \Omega$ is the absolute azimuthal rate of the particle about the vertical through C, so $mR\dot\phi^2 + mR\cos^2\phi(\dot\theta+\Omega)^2$ is the centripetal demand of the absolute motion on the sphere.
- **Limit $\Omega \to 0$:** the equations reduce to those of the spherical pendulum, $\ddot\theta\cos\phi - 2\dot\theta\dot\phi\sin\phi = 0$ and $\ddot\phi + \dot\theta^2\sin\phi\cos\phi + (g/R)\cos\phi = 0$, and $N = mR(\dot\phi^2 + \dot\theta^2\cos^2\phi) - mg\sin\phi$.
- **Dimensions:** in (c) every term is s^{-2} ($[\ddot\theta] = [\Omega^2] = [\dot\Omega] = [\Omega\dot\phi] = [g/R] = \mathrm{s}^{-2}$); in (d) every term is $\mathrm{kg\,m\,s^{-2}} = \mathrm{N}$; $\lambda = \Omega^2R/g$ is dimensionless.
- **Verification (sympy):** with a general $\Psi(t)$, $\dot\Psi = \Omega(t)$, the inertial double derivative of $[{}_0R_1]\{\mathbf{r}_1^P\}$ equals the Notes Eq. 3.27 decomposition; solving $m\mathbf{a}_0^P = -mg\mathbf{k}_1 - N\mathbf{u}_r$ in S_0 returns exactly the two equations of (c) and the N of (d); for $\Omega = 1, 3, 10$ rad/s ($R = 1$ m) both branches satisfy the equations with $N = -mg/\sin\phi$, branch 1 has $N > 0$ and purely imaginary eigenvalues, branch 2 has $N < 0$.

- **Check against the official key:** `Problems.pdf` has no entry for Problem 51 in its solution keys, so the results are verified by independent derivation (the sympy check recorded above).

Source: Problems.pdf (PDF page 36); no official key